

Advances in Quantum Interior-Point Methods for Sparse Problems: Conditioning Geometry, Query Lower Bounds, and Limits of Access Models

Abstract

A quantum interior-point method (QIPM) solves the Newton systems of an interior-point method with a quantum linear-system (QLS) solver. We ask when QIPMs beat classical ones on sparse linear and semidefinite programs (LPs and SDPs), counting queries to the input under explicit input-access and output models. (i) For every self-concordant barrier on a compact feasible region with a nonconstant linear objective, the barrier Hessian on the feasible affine subspace at the central point with objective gap g has condition number at least $c(D(g)/g)^2$, where $D(g)$ is the diameter of the feasible points with gap at most g and c depends only on the barrier parameter and the projection of the objective onto that subspace. Log and log-det barriers attain this bound up to instance-dependent factors. (ii) Near the optimum of a degenerate LP, the log-barrier Hessian H has two eigenvalue clusters, and conjugate gradients needs iterations polylogarithmic in their separation. Yet a known bound gives worst-case cost $\tilde{\Theta}(\kappa)$ on such spectra for a QLS solver given only a block encoding of H with normalization of order $\|H\|$, where κ is the condition number and the dimension is at least of order κ . Given instead such an encoding of a factor of H (a matrix whose Gram matrix is H), there is an even polynomial in the factor's singular values that approximates the inverse of H scaled to unit norm, with degree times maximum modulus $\tilde{O}(\sqrt{\kappa})$; a full solve at this cost also assumes that its two pseudoinverse steps have constant relative output amplitude. (iii) Explicit sparse LP and SDP families have Newton matrices of condition number one on the null space of the equality constraints, yet preparing the Newton direction or solution as a quantum state in the original coordinates needs a number of input queries linear in the problem size. Convex-mixture, cut-resistance, Walsh-row, Fourier, congruence, preconditioning, and state-interface theorems mark the limits of the classes of gadgets and access models that these lower bounds cover. (iv) Let G PSD blocks share one trace constraint, with $G - 1$ of them identical, one whose cost matrix has a strictly smaller least eigenvalue, and block dimension and spectral gaps fixed. As $G \rightarrow \infty$, the log-det central point at path parameter $\tau = \alpha/G$ keeps a constant fraction of its trace on that block exactly when α is below an explicit threshold set by the cost spectra. This trace controls the reduced-Hessian condition number, the trace distance of the central state from a reference state independent of the hidden input, and the query cost of preparing that state. For a sparse holonomy SDP we prove matching query bounds, and we give query bounds for state preparation as functions of the accuracy. (v) Conditional positive results include reuse of Newton directions checked by residuals, stable equilibration and face repair, dual output in compressed form, and a block-angular border solver whose $\sqrt{N}$ dependence on the number N of blocks is optimal in the block-oracle model. No unconditional end-to-end advantage or disadvantage follows; the costs left open are data loading, recovery of the solution or Newton direction, oracle updates between iterations, and output. We also correct the simulator-norm estimate and the clock analysis of the quantum central-path method (QCPM) preprint.

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1 Introduction

An interior-point method (IPM) for a linear or semidefinite program follows the central path and reduces optimization to a sequence of structured Newton linear systems. A quantum interior-point method (QIPM) performs each solve with a quantum linear-system (QLS) solver, which, in favorable access models, prepares a quantum state proportional to the Newton direction without writing down its coordinates. Sparse problems look like a natural target, because their data can be queried without reading a dense matrix. This paper asks:

Do quantum interior-point methods beat classical interior-point methods on sparse problems?

Sparsity alone does not decide the answer. A sparse input matrix can have dense normal equations, null-space basis, factorization, or inverse. Conversely, a matrix with a large condition number can have a clustered spectrum that a classical Krylov method exploits. A quantum state proportional to a direction is not a classical iterate, and access to a preconditioned matrix is not free just because that matrix is well conditioned. A fair comparison must therefore specify the Newton system, the input oracle, the normalization of the block encoding, the right-hand side, the required output, and the classical solver that is given the same structure.

We study the question at three levels: *geometry* (how much of the ill-conditioning of the central path is forced by the problem rather than by the barrier), *access models* (what classical and quantum linear solvers can extract from the same spectral structure, and what their inputs cost to build), and *interfaces* (whether the output of one solve can be read out, checked, and used to set up the next). The answer is neither a general quantum speedup nor a general impossibility theorem. We prove sharp bounds at each level, state the classes of methods they cover, and so show where a speedup or a lower bound would still have to be proved.

1.1 State of play

The first QIPMs inserted a QLS solver into a classical path-following method. Kerenidis and Prakash treated LPs and SDPs; Casares and Martin-Delgado developed a predictor–corrector LP method; Augustino et al. gave conic and semidefinite variants; and work from Terlaky’s group developed inexact feasible, infeasible, orthogonal-subspace, and iterative-refinement methods [11–14, 38, 83, 106, 107, 137]. Their cost analyses exposed the costs of conditioning, block-encoding normalization, tomography, QRAM, and classical updates, which Dalzell et al. also stress in their QIPM review and resource analysis [51, 52].

Two newer lines change the algorithm itself rather than only the linear solver. The quantum central-path method (QCPM) of Augustino, Leng, Nannicini, Terlaky, and Wu uses Hamiltonian evolution and potential evaluation instead of repeated QLS solves followed by tomography [15]. The self-concordant Schrödinger-operator approach of Gribling et al. likewise aims to optimize without a factor that depends on the Newton condition number [70]. Hence a conditioning theorem must name the Newton matrix it concerns: a lower bound on the condition number of a barrier Hessian is not a lower bound for every quantum optimization algorithm.

Apers and Gribling avoid the condition number in a different way. Their algorithms for tall LPs use quantum spectral approximation and local-norm estimation, return an explicit solution, and prove a separation in the number of row queries when the number of constraints is sufficiently larger than the number of variables [10]. Among the QIPM results compared here, this is the only proved asymptotic quantum–classical separation. It concerns queries to a succinct row oracle, not automatically running time against the best classical sparse solver whose cost scales with nnz, and its assumptions and high powers of the number of variables make it a specific favorable regime, not a general consequence of sparsity.

The QLS primitive itself has evolved from HHL through the improved precision dependence of Childs, Kothari, and Somma to pseudoinversion by quantum singular value transformation (QSVT) [43, 65, 76]. Orsucci and Dunjko proved upper and lower bounds for positive-definite systems that depend on the access model. Mori et al. proved an $\Omega(\kappa\sqrt{s})$ lower bound for sparse access at fixed constant error, and restated an $\Omega(\kappa\log(1/\epsilon))$ bound for constant sparsity from earlier unpublished work of Harrow and Kothari. Dalzell, Li, and Su (DLS) introduced truncation and filtering parameters that depend on the right-hand side and can be smaller than the worst-case condition number [53, 109, 118]. Somma and de Wolf’s related separation between plain and sum-of-squares access concerns guided ground-state energy estimation, not QLS (see section 4.3) [125]. Hence the cost of a QLS solve depends on the access (plain matrix, factor, sparse-entry, or a stronger oracle) and on how much of the right-hand side lies in the eigenspaces that are hard to invert.

Tomography is the second standard bottleneck. A path-following method usually needs a

classical direction to update all primal and dual coordinates, whereas a QLS solver returns only a normalized quantum state. Tomography with state-preparation unitaries improves the dependence on accuracy, but still scales with the dimension of the direction [105, 131]. Avoiding this cost requires a different kind of output and a way to build from it the next Newton matrix, whose diagonal scaling changes at every iteration.

Classical methods that exploit sparsity set a demanding baseline. Practical IPMs choose between normal equations and augmented systems, control fill-in, reuse symbolic factorizations and preconditioners, and adapt the tolerances of inner solves [67, 69, 136]. For problems of bounded treewidth, specialized IPMs with supplied elimination orders run in nearly linear time in the problem size [56, 72]. A claimed sparse QIPM speedup must therefore be compared with a structure-aware factorization or Krylov method, not with dense Gaussian elimination.

1.2 What a matched comparison counts

The QIPMs above use different Newton systems, state-preparation primitives, refinement schemes, and outputs, so dimension and sparsity alone do not compare them. Throughout the paper we count raw queries (calls to a precisely defined input oracle) and the QRAM writes needed to build or refresh QRAM, and we give qualitative end-to-end accounts of construction, reuse, recovery, and output. Gate and arithmetic costs are given only for selected results, so these accounts are not a uniform running-time analysis.

From sparse input to the solved matrix. For an LP, a diagonal scaling of a sparse constraint matrix A leads to the normal equations with matrix $AD^2A^\top$, to an augmented KKT system, or to an orthogonal-subspace system. Constant row sparsity of A does not imply constant sparsity of any of these matrices: one dense column of A creates a clique in the graph of the normal equations, and a null-space basis can be dense. Conversely, a well-chosen basis or augmented system can keep structure that the normal equations lose. Our notation therefore distinguishes $\text{nnz}(A)$, the sparsity s_H of the matrix H actually solved, and the access to a factor G when $H = G^\dagger G$ (sections 2.1.1 and 2.2). A claim that starts from a sparse A must prove the access properties of the matrix it actually solves.

Condition number and normalization. A block encoding of H is a unitary whose upper-left block is H/α_H (definition 2.13). The cost of inverting H through it is governed by $\alpha_H\|H^{-1}\|$, not by the spectral condition number alone, and rescaling H changes neither this product nor the problem. Multiplying separately encoded factors can multiply their normalizations, and forming a perfectly preconditioned matrix can require the same input information that the solve was meant to find (section 12). A classical preconditioner also costs something, but it can often be reused and can use explicitly stored entries. A matched comparison counts preconditioner costs on both sides and gives neither side a free inverse oracle.

Right-hand sides and solve accuracy. The Newton direction solves the system for one particular right-hand side (RHS), so worst-case QLS complexity is not always the right estimate. A solver may ignore a hard eigenspace only if the allowed output error permits losing the part of the solution in that eigenspace. A residual bound, a bound on the error of the normalized state, and a bound on the coordinatewise error of the direction are different requirements on the output, which we call *output requirements* (section 2.3). They are related by inequalities that involve the condition number, so a quantum state-error bound cannot be compared with a classical residual tolerance without converting one into the other. Section 5 evaluates these distinctions on exact IPM Newton systems.

State preparation and readout. A hybrid QIPM that updates a classical iterate must estimate the full solution vector from QLS output states, recover its norm and scale, and check that the step stays in the neighborhood of the central path required by the outer method. Each state preparation for tomography repeats every oracle call and RHS preparation whose result is not stored. A method that requests only an expectation value or a compact output avoids full readout, but must show that this output suffices for optimization and for setting up the next iteration. [Section 2.3](#) distinguishes classical-vector, compact, state, density, coordinate-oracle, and heralded outputs, so that a proof for one kind of output is not silently reused for another.

The classical comparator. For a sparse direct method, the cost per iteration is set by the actual fill-in, the numerical factorization, and the triangular solves; the symbolic ordering is normally reused, and several predictor or corrector right-hand sides can share one factorization. For a Krylov method, the cost is the number of matrix–vector products and preconditioner applications, times an iteration count that depends on the spectrum and the tolerance; clustered eigenvalues can make this count far smaller than the worst-case $\sqrt{\kappa}$ estimate ([proposition 4.2](#)). Graph, treewidth, block-angular, and low-rank structure can improve the classical baseline further. We therefore compare quantum query counts neither with dense cubic arithmetic nor with the cost of writing a dense classical direction alone.

Setup, refresh, and reuse along the path. One-step costs do not automatically multiply by the number of IPM iterations: a fixed sparse input can be read once, a symbolic factorization kept, and an oracle can combine stored data with the current scalar path parameter. Conversely, a QRAM or block encoding of the changing primal and slack diagonals may need new writes, and tomography may be repeated, at every step. [Definition 2.27](#) formalizes this accounting. Our lower bound for a sequence of solves ([theorem 12.14](#)) applies only when each round supplies new oracle information; the caching argument in [section 12.5](#) shows why, for a fixed LP, no general bound of the form T times the one-step cost follows.

Query separation versus time separation. Raw coefficient, sparse-entry, row, block, QRAM, and preconditioned-product queries reveal different amounts of information. Sublinearly many row queries can give a real query separation even when generating a queried row or the local arithmetic prevents a time advantage; conversely, an explicitly stored sparse matrix may have negligible query cost while repeated solves with it remain expensive. The Apers–Gribling result is positive in its row-query model; our parity and holonomy lower bounds are negative in their raw local-query models; neither carries over to another model without proof. This is why many later theorems bound the total of setup, solve, and recovery costs rather than give a single power of the condition number.

1.3 Main results

Let the feasible set P be compact and convex, let the objective $c^\top x$ be linear with optimal value OPT , and let F be a ν_F -self-concordant barrier for P . A central point $x = x_F(\mu)$ has objective gap $g = c^\top x - \text{OPT}$. Let W have orthonormal columns spanning the directions of the feasible affine subspace, and let c_V be the projection of c onto them. The reduced Hessian is $W^\top \nabla^2 F(x) W$, with condition number κ_{red} ([definition 2.9](#)), and $D(g)$ is the diameter of the sublevel set $\{y \in P : c^\top y \leq \text{OPT} + g\}$.

Our first result is that part of the ill-conditioning of the central path is forced by geometry, whatever the barrier. The sublevel chord law ([theorem 3.3](#)) states that at every exact central point $\kappa_{\text{red}} \geq (D(g)\|c_V\|_2/(2C_F g))^2$, where $C_F = \nu_F + 2\sqrt{\nu_F}$. [Theorem 3.5](#) shows that the canonical log and log-det barriers attain this bound at equal objective gap, up to explicit instance-dependent factors, provided the dual slack stays bounded on the tail of the path. Thus no choice of barrier

can in general remove the divergence. Its rate depends on the geometry of the sublevel sets: every barrier has $\kappa_{\text{red}} = \Omega(g^{-2})$ when the optimal face has positive dimension, whereas the canonical log barrier has $\kappa_{\text{red}} = \Theta(1)$ at a unique LP vertex, even a degenerate one (corollary 3.6). The result does not cover methods that avoid barrier Newton solves.

Our second result is that a large condition number does not determine the relative classical and quantum costs. Near the optimum of a degenerate LP, the eigenvalues of the reduced log-barrier Hessian lie in two intervals, each with a bounded ratio of endpoints, whose separation grows as $\Theta(\mu^{-2})$ (proposition 4.1). Conjugate gradients can place roots of its residual polynomial in the gap, so its iteration count is polylogarithmic in this separation (proposition 4.2). QSVT, by contrast, applies a polynomial to a normalized operator, and the polynomial must stay bounded on the whole interval allowed by the access model, not only at the eigenvalues. For block encodings normalized at the operator norm (norm-tight encodings), plain block-encoding access has worst-case QLS cost $\tilde{\Theta}(\kappa)$, even for two-point clusters, when $N_{\text{QLS}} \gtrsim \kappa$; the lower bound is due to Orsucci and Dunjko. With norm-tight factor access $H = G^\dagger G$, there is always a polynomial transformation whose degree times amplitude bound (its maximum absolute value on the allowed interval) is $\tilde{\Theta}(\sqrt{\kappa})$; only the corresponding end-to-end solve needs the overlap assumptions in theorem 4.3. The right-hand sides that occur on the central path are easier still: the exact central-path right-hand side has only an $O(\mu^2)$ fraction of its norm on the cluster of small eigenvalues (proposition 4.5 and corollary 4.6). Hence, for a fixed relative-residual tolerance η , once $\mu = O(\sqrt{\eta})$, a solve that inverts only the cluster of large eigenvalues meets this tolerance at a cost independent of κ . A solver should therefore be chosen by how the right-hand side couples to the small eigenvalues and how its spectral mass is distributed, not by κ alone (section 5).

Our third result is that the decisive lower bounds often concern loading and recovery rather than the linear solve. In the raw sparse coefficient model (definition 2.11), the positions of the nonzero coefficients are public and each query reveals one hidden sign. The LP families of section 9 include ones whose reduced Newton matrices have condition number one, yet the parity of the hidden signs determines either an affine offset of the solution or the Newton direction in the original coordinates, and preparing that direction as a quantum state needs a number of queries linear in the problem size. The SDP families of section 11 show that the optimal value, Schur-complement solves, access to the KKT matrix, and recovery of the solution can have very different query costs. By the product bounds of section 12, improving the spectral condition number can worsen the normalization of the factor or the sensitivity of recovering the physical state, and a state produced by one solve need not provide the coordinate oracle needed by the next.

Condensation, the concentration of the trace of the central point on a single block, has both a positive and a negative side. Consider G positive semidefinite blocks coupled by a single trace constraint, one winning block and $G - 1$ identical losing blocks with fixed dimension and spectral gaps, and the log-det central point at path parameter $\tau = \alpha/G$ (the μ of definition 2.3). As $G \rightarrow \infty$, the trace on the block whose cost matrix has the smallest eigenvalue (the winner mass) tends to a positive constant when α is below a critical value α_* , and to zero at and above α_* (theorem 6.2). The winner mass controls at once the reduced-Hessian condition number, the trace distance between the central state and a public reference state that does not depend on the hidden input, and the query cost of preparing the state. On a bounded-incidence holonomy SDP whose G groups each hide the parity of N signs, the transition comes with matching query bounds; for example, estimating the optimal value costs $\Theta(N\sqrt{G})$ quantum queries. For preparing a central state with k winners to trace distance ε , we prove query bounds, as functions of ε , that match up to logarithmic factors on the domain of corollary 8.9; the range where the margin $\delta = p - \varepsilon$ between the winner mass p and the allowed error is $o(k/G)$, and the effect of a discrepancy η_{int} between internal states of losing blocks, remain open (section 8.4). Output hardness is thus proved for an explicit output requirement, through a public measurement that

decodes the hidden information, not inferred from the condition number.

Finally, our impossibility results cover stated classes of constructions and methods; they are not universal. Convex mixtures, cut and resistance bounds, bounded-arity Walsh rows, symmetry work identities, scalar congruences, and products for state interfaces rule out the hardness constructions (gadgets) that satisfy their hypotheses, and [section 10.6](#) lists five ways to escape them. The positive modules of [section 13](#) take some of these routes, but they are conditional and apply per step: they need stable geometry, explicit access, compressed output, or a low-dimensional border. No section combines them into an end-to-end sparse-QIPM advantage.

1.4 Contributions

The technical contributions follow the order of the paper.

1. **Conditioning geometry for every barrier ([section 3](#)).** For every self-concordant barrier, we prove the lower bound $\lambda_{\min} \geq 1/\ell(x)^2$ on the reduced Hessian at every interior point x , where $\ell(x)$ is the length of the longest segment from x within its sublevel set ([theorem 3.2](#)), and the sublevel chord law ([theorem 3.3](#)). We refine them to two-sided bounds on every eigenvalue by directional widths of the sublevel set ([theorem 3.4](#)), and show that the canonical LP and product-SDP barriers attain the resulting lower bound up to an explicit factor at equal objective gap ([theorem 3.5](#)). Corollaries give the rates for positive-dimensional optimal faces, degenerate vertices, curved optima, SDPs with a given singularity degree, near-degenerate instances, and approximately central points ([corollaries 3.6, 3.8, 3.10 and 3.11](#)).
2. **Clustered spectra and the separation of solvers ([section 4](#)).** [Section 4](#) proves the results on clustered spectra, solver cost, and central-path right-hand sides described in [section 1.3](#) ([propositions 4.1, 4.2 and 4.5, theorem 4.3, and corollary 4.6](#)).
3. **Solver choice beyond κ ([section 5](#)).** For norm-tight encodings, exact block inversion at central points shows that DLS filtering improves on the worst-case condition number if and only if a limiting coupling vector (up to sign, the limit of the Newton solution's component on the small-eigenvalue subspace, divided by μ) vanishes ([theorem 5.1](#)). A truncation transition separates requirements on the solution state from requirements on the residual ([proposition 5.3](#)). If the spectral measure of the right-hand side has effective dimension D , meaning that its mass below t is of order $t^{D/2}$, then fixed-accuracy truncation becomes independent of the condition number for $D > 4$, and filtering for $D > 8$ ([theorem 5.4](#)). Given the orthogonal projector Q onto the active range, a stable equilibration removes the divergent part of the spectrum and exactly preserves the normalized Newton state for every $f \in \text{range } Q$ ([theorem 5.7](#)).
4. **Block log-det condensation ([section 6](#)).** We derive the exact transition and the conditioning phases, which depend on the multiplicity of the smallest eigenvalue of the winning block ([theorems 6.2 and 6.4](#)); prove bounds between winner mass and conditioning for finite G ([theorem 6.5](#)); and relate the winner mass to the trace distance from the no-winner state and to raw-query complexity ([theorems 6.6 and 6.7](#)). The transition extends to sublinearly many winners and to a stated model of approximate centrality ([theorems 6.9 and 6.10](#)). These statements concern the Frobenius-metric Hessian with equality constraints eliminated and the trace-normalized output, not every preconditioned KKT formulation.
5. **A sparse winner-take-all SDP ([section 7](#)).** A bounded-incidence family of qutrit-holonomy SDPs realizes the condensation law, with optimal-value estimation costing $\Theta(N\sqrt{G})$ quantum and $\Theta(NG)$ randomized queries and a public starting point where the Hessian in root coordinates has condition number one. A state within trace distance $1/100$ of an approximate optimizer reveals a winning group if one exists, so preparing it needs $\Omega(N\sqrt{G})$ queries ([theorem 7.2](#)); a search-then-solve algorithm with compact output matches this bound

- (theorem 7.3). Section 7.9 explains why bounded incidence and a condition-one ambient formulation are not claimed together.
6. **Accuracy curves for hidden-block state conversion (section 8).** A decoder that verifies sampled labels gives a lower bound for mixed-state outputs (theorem 8.1); detuned rejection preparation gives the upper bound (theorem 8.2); and parity collapse yields the tight curve for preparing, to trace distance ε , a state with winner mass p on k of G blocks (theorem 8.7 and corollary 8.9). For a batch of uncoupled instances, vector output cannot be amortized, whereas a single averaged scalar has the usual approximate-counting cost (theorems 8.12 and 8.13). The discrepancy between internal loser states, the very-small-success range, and logarithmic fixed-point costs remain explicit limits of these theorems.
 7. **LP loading and recovery hardness (section 9).** The gain-plateau family gives a linear lower bound for preparing any primal state with small residual (theorem 9.4). Prefix rigidification strengthens this to an LP whose reduced Newton geometry has condition number one while recovering the original coordinates costs linearly many queries (theorem 9.7). Variants with complete KKT systems and Schur-complement preconditioning show the tradeoffs among a public right-hand side, residual accuracy, bounded data, and scale (theorems 9.8 to 9.11 and 9.13). None of them is a lower bound for a condition-one QLS problem with free access to transformed oracles.
 8. **Limits of classes of gadgets (section 10).** A convex-mixture theorem for single-bit flips gives the default obstruction without any graph structure (theorem 10.1). Cut-resistance and Walsh-row theorems give the size or scale that signed-copy and bounded-arity constructions need (theorems 10.4 and 10.5); gauge-flow, work, and Fourier identities show how the cost appears in multiplier size or input incidence (theorems 10.6 to 10.8). The hypotheses that a construction must drop to escape these theorems, listed in section 10.6, are part of the result.
 9. **SDP-native loading and recovery hardness (section 11).** An intrinsic trace-normalized qutrit family has an optimal value that reveals a parity, and a public reduced Newton step of condition number one whose primal and complete-KKT solution states are parity hard (theorems 11.1 and 11.2). Schur-complement families that are not SOCP-representable give an exact tradeoff between loading and accuracy (theorems 11.3 and 11.4), and tree and group holonomy extend the construction to matrix-valued words (theorems 11.5 to 11.7). The query lower bound for the S_3 family is still proved on an abelian restriction; hardness for intrinsically nonabelian words remains open.
 10. **Preconditioning and state interfaces (section 12).** Whitening and reusable inverse access trade condition number against normalization (theorems 12.2 and 12.3); scalar congruence trades the transformed condition number against the cost of recovering the physical state (theorems 12.4 and 12.5); and for an explicit parity-hard Newton update, the queries for setup, RHS preparation, solve, and recovery together are at least linear in the problem size for every invertible preconditioner (theorems 12.7 and 12.8). The state-to-next-solve and lazy-oracle theorems distinguish a state that can be used once from a reusable coordinate oracle (theorems 12.9, 12.13, 12.15 and 12.16).
 11. **Conditional positive modules (section 13).** We prove a bound on the number of fresh solves when a Newton direction is reused along the path until a residual check fails (theorem 13.1); a constructive stable equilibration of the active range and an exact face repair (theorems 13.6 and 13.7); an implicit slack oracle and a conditional dual QIPM with compressed output (theorems 13.8 and 13.9); and a solver for block-angular systems through their border, whose $\sqrt{N}$ dependence on the number of blocks is optimal in the block-oracle model (section 13.5).

12. **Reusable structural tools (section 14).** Work identities for involutions and compact groups (theorems 14.1 and 14.2), a noncommutative sandwich for mixtures of central points (theorem 14.5), and lower bounds on the rank and support of Newton steps (theorems 14.8 and 14.9) explain several of the obstructions abstractly. Sparse OSS and treewidth-two KKT embeddings carry known QLS lower bounds over to these formulations (theorems 14.10 and 14.11), while value hardness at exact treewidth and a light-cone bound for central oracles separate query, output, and physical-locality models (theorems 14.12 and 14.13).
13. **Corrections and dead ends (section 15).** We show that the global potential supremum in the QCPM simulator-norm analysis cannot scale as asserted (proposition 15.1), correct the separate clock (time-dilation) error (the correct clock is equation (693)), derive the corrected norm scale and its tradeoff with speed (theorem 15.3 and corollary 15.4), and give explicit counterexamples for densification and hidden starting points (theorems 15.5 and 15.6). The recorded dead ends show why norms of the first inverse, sparse metrics, exact Schur products, or one-step bounds cannot be reused as claims they do not prove.

1.5 Related work and positioning

Quantum interior-point methods. We do not put the QIPMs of section 1.1 into a single complexity table [10, 14, 15, 38, 52, 70, 83, 107, 137], because each uses a different access model, condition parameter, precision, and output requirement. Each technical section compares its theorems with the nearest prior work.

Quantum optimization lower bounds. Van Apeldoorn et al. established general quantum SDP upper and lower bounds and constructions that embed a Boolean function in the optimum; Brandão and Svore initiated the line of quantum SDP solvers; and Theorem 8.4 of Apers and Gribling gives a value lower bound for sparse LPs that is already linear for balanced dimensions [10, 30, 130]. We do not claim parity hardness, adversary composition, or the linear exponent as new for our parity and holonomy families. What is new is their combination with exact central paths, formulations of condition number one or bounded condition number, robust output requirements, and complete accounting of loading and recovery (sections 7, 9 and 11).

Classical interior-point structure. Wright’s work on logarithmic-barrier Hessians and numerical error, the standard primal–dual theory, and the survey of Forsgren, Gill, and Wright provide the background on conditioning and implementation [63, 134–136]. Gondzio and collaborators document preconditioning, the use of structure, and the evolution of practical IPMs [67, 69]. The sublevel chord law differs from classical conditioning analyses of the log barrier in holding for all self-concordant barriers, and section 4 uses, rather than ignores, the classical advantage of Krylov methods on clustered spectra.

Established ingredients. Many ingredients used below are established: self-concordant containment, weak-sharp and singularity-degree error bounds, matrix-Burg resolvents, Rayleigh–Jeans condensation, parity and search, group synchronization, Kantorovich inequalities, effective resistance, and sparse elimination. We credit each where it enters. The detailed statements of novelty and scope in sections 3.1, 6.11, 7.9, 11.5, 12.6, 13.7 and 14.6 are part of the claims.

1.6 Conventions and corrections

The distinctions of section 2 hold throughout: input sparsity differs from Newton-system sparsity, κ_H from the block-encoding normalization, reduced from ambient KKT conditioning, and value, classical-vector, quantum-state, coordinate-oracle, and heralded outputs are different output requirements. Query bounds count the setup and transformations named by their access model;

a stronger oracle supplied at no cost defines a stronger problem, not an implementation of raw sparse access.

Several results correct earlier versions of our own analysis and two independent errors in the QCPM preprint. The corrections that matter for reading the paper are:

1. [Section 15.1](#) corrects the simulator-norm estimate of QCPM v2, because a global supremum on a fixed truncation box is not controlled by concentration near the current center, and its separate clock error, because the time-varying kinetic coefficient requires a corrected clock. This corrects that derivation; it is not an impossibility result for Hamiltonian optimization.
2. [Section 4.3](#) corrects an overbroad conclusion about factor access in our earlier analysis: a square-root polynomial cost, or a two-stage solve with explicit overlap, does not imply a generic $\tilde{\Theta}(\sqrt{\kappa})$ QLS solver with factor access, as the Orsucci–Dunjko instance shows.
3. [Section 4.1](#) replaces a coarse three-scale description of the degree-two SDP witness by its four-window spectrum; this changes the intermediate spectral description, not the exponent of the condition number.
4. [Section 5.3](#) anchors the effective spectral dimension at the smallest eigenvalue that the right-hand side sees, repairing an earlier formulation by spectral shells; the phase laws of [theorem 5.4](#) need this.
5. [Sections 8.1](#), [8.2](#) and [8.4](#) replace the collision shortcut by a decoder that verifies sampled labels, restrict the classical law that is linear in the success probability to its actual domain, leave the very-small-success range open, and attribute the small-success search bound with k marked items to Dohotaru–Høyer rather than to Zalka’s theorem for one marked item alone.
6. [Section 13.2](#) corrects the earlier claim that two convergent leakage sums alone bound the projective variation: a counterexample with an oscillating face displacement has convergent sums, satisfies an additional $O(\mu_t)$ tail bound, and has linearly growing realized variation. Finite realized variation therefore remains a separate hypothesis.

The list includes only changes that affect theorem statements or their interpretation; other rejected shortcuts and their valid replacements are in [section 15.4](#).

1.7 Organization and reading paths

[Section 2](#) fixes the conventions. [Sections 3](#) to [15](#) follow the order of the contributions above, and [section 16](#) gives the overall conclusions and open problems.

A reader interested only in conditioning can read [sections 2](#) to [5](#). A reader interested only in lower bounds can read [sections 2](#) and [6](#) to [12](#). A reader interested only in algorithmic modules can read [sections 2](#), [5](#) and [13](#).

2 Preliminaries

This section fixes the optimization problems, access models, and output requirements used throughout the paper. Three distinctions are part of the problem specification: input sparsity is not the sparsity of the Newton system, the spectral condition number is not a block-encoding normalization, and a normalized quantum state is not an explicit solution vector.

2.1 Conic programs and central paths

For a linear program (LP) in standard form, the primal–dual pair is

$$\begin{aligned} (\text{P}_{\text{LP}}) \quad & \min_{x \in \mathbb{R}^n} \quad c^\top x \quad \text{subject to} \quad Ax = b, \quad x \geq 0, \\ (\text{D}_{\text{LP}}) \quad & \max_{y \in \mathbb{R}^m, s \in \mathbb{R}^n} \quad b^\top y \quad \text{subject to} \quad A^\top y + s = c, \quad s \geq 0, \end{aligned} \quad (1)$$

Throughout the LP discussion, $A \in \mathbb{R}^{m \times n}$ has full row rank unless stated otherwise. We write $X = \text{diag}(x)$, $S = \text{diag}(s)$, and e for the all-ones vector. A feasible primal–dual point has gap

$$c^\top x - b^\top y = x^\top s. \quad (2)$$

The product-cone form of a semidefinite program (SDP) used below has q matrix blocks. Put $\mathcal{K} = \prod_{j=1}^q \mathbb{S}_+^{d_j}$ and equip it with $\langle X, S \rangle = \sum_j \text{tr}(X_j S_j)$. For symmetric data $C_j, A_{ij} \in \mathbb{S}^{d_j}$, define $\mathcal{A}(X)_i = \sum_j \text{tr}(A_{ij} X_j)$ and $(\mathcal{A}^* y)_j = \sum_i y_i A_{ij}$. The pair is

$$\begin{aligned} (\text{P}_{\text{SDP}}) \quad & \min_{X \in \mathcal{K}} \quad \langle C, X \rangle \quad \text{subject to} \quad \mathcal{A}(X) = b, \\ (\text{D}_{\text{SDP}}) \quad & \max_{y \in \mathbb{R}^m, S \in \mathcal{K}} \quad b^\top y \quad \text{subject to} \quad \mathcal{A}^* y + S = C. \end{aligned} \quad (3)$$

Its primal–dual gap is $\langle C, X \rangle - b^\top y = \langle X, S \rangle$. The one-block formulation is the special case $q = 1$; scalar nonnegative variables can equivalently be included as 1×1 blocks. These are the standard conic dual pairs under the trace inner product [28].

Definition 2.1 (Strict feasibility). An LP is *strictly primal–dual feasible* if there are points satisfying (1) with $x > 0$ and $s > 0$. An SDP is strictly primal–dual feasible if there are feasible points with $X_j \succ 0$ and $S_j \succ 0$ for every block. Unless one side is named, *strict feasibility* means this two-sided condition. Both problems are then feasible, so their optimal values are finite; the two-sided Slater condition gives zero duality gap and attainment of both optima [28].

We next recall the barrier terminology in the normalization used by Nesterov and Nemirovskii [113]. For a C^3 convex function F on an open convex domain, define the local norm

$$\|h\|_x = \sqrt{D^2 F(x)[h, h]}. \quad (4)$$

On an affine feasible slice, derivatives and Hessians in (4) are restricted to its tangent space.

Definition 2.2 (ν -self-concordant barrier). A function F is a ν -self-concordant barrier for a convex set Q if $\text{dom } F = \text{relint } Q$, it tends to $+\infty$ at the relative boundary, its Hessian is positive definite on the tangent space, and, for all $x \in \text{relint } Q$ and tangent directions h ,

$$|D^3 F(x)[h, h, h]| \leq 2\|h\|_x^3, \quad |DF(x)[h]| \leq \sqrt{\nu} \|h\|_x. \quad (5)$$

We call the declared value ν the barrier parameter; it need not be the least value for which (5) holds.

The logarithmic barrier for the nonnegative orthant and the log-determinant barrier for the product PSD cone are, respectively,

$$F_{\log}(x) = -\sum_{i=1}^n \log x_i, \quad F_{\log \det}(X) = -\sum_{j=1}^q \log \det X_j. \quad (6)$$

They have parameters $\nu = n$ and $\nu = \sum_j d_j$. Both are logarithmically homogeneous: $F(tx) = F(x) - \nu \log t$ in the appropriate cone notation. The symbol ν always denotes the parameter of the cone barrier; for these canonical barriers it equals the degree of logarithmic homogeneity. Restricted to an affine feasible slice, the cone barrier remains self-concordant with parameter at most ν , possibly less; we do not redefine ν after restriction. The standard path-following iteration bound depends on this cone barrier parameter, not on the number of stored coefficients [113].

Definition 2.3 (Central path and its two parametrizations). Let F be a self-concordant barrier on the relative interior of the primal feasible region, suppose OPT is finite and attained, and suppose the barrier subproblem has a unique minimizer. The μ -parametrized primal central path is

$$x_F(\mu) = \arg \min \{c^\top x + \mu F(x) : Ax = b, x \in \text{dom } F\}, \quad \mu > 0. \quad (7)$$

Its primal objective gap is

$$g_F(\mu) = c^\top x_F(\mu) - \text{OPT}. \quad (8)$$

On any interval where g_F is one-to-one, the same curve indexed by $g > 0$ is the g -parametrized central path, denoted $x_F(g)$. For the canonical LP barrier, the primal–dual central path additionally satisfies

$$Ax = b, \quad A^\top y + s = c, \quad x, s > 0, \quad XSe = \mu e. \quad (9)$$

For the product-PSD log-det barrier it satisfies

$$\mathcal{A}(X) = b, \quad \mathcal{A}^*y + S = C, \quad X_j, S_j \succ 0, \quad S_j = \mu X_j^{-1} \quad (1 \leq j \leq q). \quad (10)$$

Thus the primal–dual gap on either canonical path is exactly $\nu\mu$, with ν the cone barrier parameter fixed above.

If the gradient of F is identified with a cone dual variable, the first-order condition of (7) gives the dual slack relation $s = -\mu \nabla F(x)$.

Definition 2.4 (Trace-normalized log-det center). On the affine slice $\sum_j \text{tr } X_j = 1$, the *trace-normalized log-det center* at parameter μ is the minimizer of

$$\langle C, X \rangle - \mu \sum_j \log \det X_j, \quad X_j \succ 0, \quad \sum_j \text{tr } X_j = 1. \quad (11)$$

With the Lagrangian sign convention $\langle C, X \rangle - \mu \sum_j \log \det X_j - \lambda(\sum_j \text{tr } X_j - 1)$, stationarity reads $X_j = \mu(C_j - \lambda I)^{-1}$. This expression is valid only for $\lambda < \min_j \lambda_{\min}(C_j)$, which makes every inverse positive definite. The corresponding log-det Hessian is $D^2F(X)[U, V] = \sum_j \text{tr}(X_j^{-1}U_j X_j^{-1}V_j)$ [28].

This is the product-PSD central path with a single trace constraint, not a different optimization model. Trace normalization makes $\bigoplus_j X_j$ a density operator, which links the conic formulation to the quantum outputs below.

Remark 2.5 (The convention for g). Section 3 uses g only for the primal objective gap (8); other sections that use g for another object define it locally. An objective gap is sometimes informally called a “duality gap”; here *primal–dual gap* means $c^\top x - b^\top y = x^\top s$, which equals $\nu\mu$ on the canonical LP central path. For a general self-concordant barrier, $g_F(\mu) \leq \nu\mu$, and equality need not hold [113].

Definition 2.6 (Analytic center). If F attains a minimum over the relative interior of a feasible convex set, its minimizer is the F -analytic center. With the canonical barriers we simply say *analytic center*. For a bounded feasible region, the large- μ limit of (7), when it exists, is this center. For a face on the boundary of the original domain, “analytic center of the face” means the minimizer of a barrier defined on that face’s relative interior; it does not mean evaluating the original barrier at a boundary point [113].

For an LP point satisfying the two feasibility equations, put $\bar{\mu} = x^\top s/n$. Short-step analyses use the following standard narrow neighborhood [136].

Definition 2.7 ($N_2(\theta)$ neighborhood). For $0 < \theta < 1$, the $N_2(\theta)$ neighborhood of the LP central path is

$$N_2(\theta) = \left\{ (x, y, s) : \begin{array}{l} Ax = b, \quad A^\top y + s = c, \quad x, s > 0, \quad \bar{\mu} = x^\top s/n, \\ \|XSe - \bar{\mu}e\|_2 \leq \theta \bar{\mu} \end{array} \right\}. \quad (12)$$

Unless an infeasible or matrix-valued analogue is explicitly named, $N_2(\theta)$ always means (12).

A short-step path-following method keeps a fixed small θ , multiplies the target μ by a factor $1 - \Theta(1/\sqrt{\nu})$, and takes a full Newton step back into the neighborhood. Under standard initialization and scaling conventions, this takes $O(\sqrt{\nu} \log(\nu/\epsilon_{\text{opt}}))$ Newton steps. The bound counts iterations only, not the cost of solving linear systems, loading data, or reading out the result [113, 136].

2.1.1 Newton formulations and conditioning

At a positive, possibly infeasible LP iterate, define the residuals

$$r_p = b - Ax, \quad r_d = c - A^\top y - s, \quad r_c = \hat{\mu}e - XSe, \quad (13)$$

where typically $\hat{\mu} = \sigma \bar{\mu}$ for a centering parameter $\sigma \in [0, 1]$. Linearizing primal feasibility, dual feasibility, and complementarity gives the full primal–dual KKT system

$$\begin{bmatrix} A & 0 & 0 \\ 0 & A^\top & I \\ S & 0 & X \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta s \end{bmatrix} = \begin{bmatrix} r_p \\ r_d \\ r_c \end{bmatrix}. \quad (14)$$

We call the solution of this system, with the residual signs in (13), the *standard infeasible-start Newton correction*; a feasible-start direction has $r_p = r_d = 0$ [136]. Because the feasibility equations are affine, a full step removes their residuals exactly. Complementarity is matched only to first order: after a full step satisfying the third block row exactly, $(X + \Delta X)(S + \Delta S)e = \hat{\mu}e + \Delta X \Delta Se$. The linear system alone does not control the quadratic term, so the method also needs a step-size rule and must keep the iterates in a central-path neighborhood.

Eliminating Δx and Δs gives the normal-equation coefficient matrix

$$H_{\text{NE}} = AXS^{-1}A^\top \quad (15)$$

and the equation

$$H_{\text{NE}} \Delta y = r_p - AS^{-1}r_c + AXS^{-1}r_d. \quad (16)$$

Under the full-row-rank assumption, H_{NE} is positive definite for $x, s > 0$. Even when A is sparse, H_{NE} can be dense, because rows of A that share a nonzero column are coupled. We therefore treat $\text{nnz}(A)$ and s_H , the maximum number of nonzeros in a row or column of the matrix passed to a solver, as separate parameters.

Two reformulations used later put the residual of an inexact solve in the complementarity block, so that the feasibility equations still hold exactly. At a feasible iterate, if $V \in \mathbb{R}^{n \times (n-m)}$ has full column rank and $\text{range } V = \ker A$, a feasible direction can be represented as $\Delta x = V\lambda$ and $\Delta s = -A^\top \Delta y$. Substitution in the third row of (14) gives the *orthogonal-subspaces system (OSS)*

$$\underbrace{\begin{bmatrix} -XA^\top & SV \end{bmatrix}}_{H_{\text{OSS}}} \begin{bmatrix} \Delta y \\ \lambda \end{bmatrix} = r_c. \quad (17)$$

For any approximate solution $(\Delta y, \lambda)$ of (17), the resulting directions satisfy $A\Delta x = 0$ and $A^\top \Delta y + \Delta s = 0$ exactly. The OSS was introduced by Mohammadisiahroudi et al. [107].

The *modified normal-equation system (MNES)* instead chooses an index set B for which $A_B \in \mathbb{R}^{m \times m}$ is nonsingular. With $D = X^{1/2}S^{-1/2}$, D_B its principal submatrix indexed by B , $\hat{A} = A_B^{-1}A$, and $E = D_B^{-1}\hat{A}D$, the MNES coefficient matrix and right-hand side are

$$H_{\text{MNES}} = EE^\top = D_B^{-1}A_B^{-1}H_{\text{NE}}(D_B^{-1}A_B^{-1})^\top, \quad h_{\text{MNES}} = D_B^{-1}A_B^{-1}h_{\text{NE}}, \quad (18)$$

where h_{NE} denotes the right-hand side of (16). If an approximate solution has residual $\hat{r}$, the reconstruction uses the correction v with $v_B = D_B\hat{r}$ and $v_{\bar{B}} = 0$, where $\bar{B}$ is the complement of B ; the feasibility rows are then exact, while the residual $-Sv$ is confined to the complementarity row [106]. The names OSS and MNES always mean the formulations (17) and (18). Neither formulation guarantees that V , $\hat{A}$, or the resulting system is sparse; sparsity must be proved for each instance.

Definition 2.8 (OSS/MNES inexact Newton residual convention). Let $\eta \in [0, 1]$ be a stated forcing tolerance. An inexact OSS solve returns $\tilde{z}$ whose linear-system residual $r_{\text{OSS}} = H_{\text{OSS}}\tilde{z} - r_c$ satisfies $\|r_{\text{OSS}}\|_2 \leq \eta\bar{\mu}$. An inexact MNES solve returns $\tilde{z}$ whose residual $\hat{r} = H_{\text{MNES}}\tilde{z} - h_{\text{MNES}}$ satisfies the $\sqrt{\bar{\mu}}$ -scaled bound

$$\|\hat{r}\|_2 \leq \eta\sqrt{\bar{\mu}/n}, \quad \text{which ensures} \quad \|Sv\|_\infty \leq \eta\bar{\mu}. \quad (19)$$

These are residuals of the OSS and MNES systems, not the global RHS-relative feasibility residual of definition 2.26. The dependence of each bound on $\bar{\mu}$ is part of the definition and follows the cited analyses [106, 107].

To measure the conditioning of the primal barrier along the central path, let $x_F(\mu)$ lie in the affine slice $x^0 + \ker A$, and let W have orthonormal columns spanning $\ker A$. The reduced Hessian of F at $x_F(\mu)$ is

$$H_{\text{red}}(\mu) = W^\top \nabla^2 F(x_F(\mu)) W. \quad (20)$$

Definition 2.9 (Reduced central-path Hessian condition number). The *reduced central-path Hessian condition number* is

$$\kappa_{\text{red}}(\mu; F) = \frac{\lambda_{\max}(H_{\text{red}}(\mu))}{\lambda_{\min}(H_{\text{red}}(\mu))}. \quad (21)$$

The definition uses the Euclidean metric and an orthonormal tangent basis. The value does not depend on which orthonormal basis W is used, but it can change under a nonorthogonal reparametrization. For the LP log barrier, $\nabla^2 F(x) = X^{-2}$.

Definition 2.10 (Ambient and KKT condition numbers). For an ambient cone barrier F , the *ambient barrier-Hessian condition number* is $\kappa_{\text{amb}}(x; F) = \kappa_2(\nabla^2 F(x))$, taken before restriction to the feasible tangent space. It depends on the representation: changing the ambient cone representation or its coordinates changes F and may change κ_{amb} . For a given nonsingular Newton or saddle-point matrix K in a specified scaling, its *KKT condition number* is $\kappa_2(K) = \sigma_{\max}(K)/\sigma_{\min}(K)$. We write κ_H for the condition number of the actual linear-system matrix H being solved.

These quantities are not interchangeable. Eliminating variables, changing row scales, choosing a nonorthogonal null-space basis, or passing from an indefinite KKT matrix to normal equations can change singular values and can square a condition number. In particular, a statement about κ_{red} says nothing by itself about $\kappa_2(H_{\text{NE}})$, $\kappa_2(H_{\text{OSS}})$, or the full saddle system.

2.2 Quantum computation and access models

Algorithms are uniform quantum circuits with arbitrary input-independent gates, intermediate measurements, and classical control. Queries and gates are counted separately. Calls to an oracle, its inverse, or a controlled version count as oracle calls unless a result explicitly states otherwise.

Definition 2.11 (Raw sparse coefficient (sign) access). In the *raw sparse coefficient (sign) access model*, the dimensions and a fixed list of nonzero positions of the LP or SDP data are public. The hidden input is a string $\varsigma \in \{-1, +1\}^{n_{\text{bit}}}$ that determines the signs of specified coefficients (or specified entries with values in a fixed finite alphabet). With enc a fixed binary encoding, the counted oracle is

$$O_{\varsigma}|i, z\rangle = |i, z \oplus \text{enc}(\varsigma_i)\rangle. \quad (22)$$

A sparse-row, sparse-column, or coefficient query is allowed only if it can be simulated coherently with the stated number of calls to O_{ς} ; a query does not reveal several hidden signs merely because a coefficient is repeated in a derived matrix. The phase convention $|i\rangle \mapsto \varsigma_i|i\rangle$ is equivalent: it and the XOR convention (22) simulate each other with one query in either direction, using phase kickback and the controlled oracle, which we assume is available.

Remark 2.12. Definition 2.11 is a deliberately restrictive specialization, defined for this paper, of the standard quantum query model [20]. Because the support is public and only the signs are hidden, the oracle calls used by each step of a reduction can be counted exactly. Unit-cost access to a table of prefix parities, to a matrix obtained by eliminating variables, or to an arbitrary input-dependent unitary completing a block encoding is a stronger model, which we never assume implicitly.

Definition 2.13 (Block encoding). Let M act on ℓ qubits. An (α, a, ϵ) *block encoding* of M is an $(\ell + a)$ -qubit unitary U such that

$$\|M - \alpha(|0^a\rangle\langle 0^a| \otimes I)U(|0^a\rangle\langle 0^a| \otimes I)\|_2 \leq \epsilon, \quad \alpha > 0. \quad (23)$$

The scalar α is the *block-encoding normalization* (also called subnormalization), a is the ancilla count, and ϵ is the absolute error in operator norm [65].

Normalizations multiply: the product of block encodings with normalizations α and β has normalization $\alpha\beta$, with correspondingly propagated error. The cost of polynomial inversion of a block encoding of H depends on the smallest singular value of the encoded matrix H/α_H , which is $1/(\alpha_H\|H^{-1}\|_2)$. For invertible H , we therefore define the access-normalized condition parameter

$$\kappa_{\text{BE}}(H; \alpha_H) = \alpha_H\|H^{-1}\|_2, \quad (24)$$

whereas $\kappa_H = \|H\|_2\|H^{-1}\|_2$; the two coincide only when the normalization equals the operator norm. In sparse-access constructions, α_H can be controlled by the row and column sparsity of H rather than by $\text{nnz}(A)$, and a product of block encodings can have a larger normalization than a direct encoding of the same matrix [65].

Definition 2.14 (State-preparation access). For a nonzero vector $v \in \mathbb{C}^N$, *state-preparation access* supplies a unitary U_v with

$$U_v|0\rangle = |v\rangle := \frac{1}{\|v\|_2} \sum_{i=1}^N v_i|i\rangle, \quad (25)$$

together with controlled access to U_v and $U_v^\dagger$. If $\|v\|_2$ is needed by the task, it must be supplied or estimated separately; when signed amplitudes matter, the oracle promise also fixes a phase convention for the controlled circuit. For a mixed state, the analogous oracle prepares a specified purification. This model is stronger than receiving independent copies, because inverse access enables amplitude-estimation procedures [131].

State-preparation access provides one normalized state. By itself it gives no coherent access to the values v_i , to $\text{diag}(v)$, or to a reusable update oracle; such interfaces are defined as outputs below.

Definition 2.15 (Plain block-encoding access and factor access). *Plain block-encoding access* to a Hermitian positive-definite matrix H supplies a controlled $(\alpha_H, a_H, \epsilon_H)$ block encoding of H and its adjoint. *Factor access* supplies instead a controlled $(\alpha_G, a_G, \epsilon_G)$ block encoding of a possibly rectangular G (padded to square dimensions in the standard way) and its adjoint, together with the promise

$$H = G^\dagger G. \quad (26)$$

These are different input models. The standard product construction using two factor-oracle calls gives an $(\alpha_G^2, 2a_G, 2\alpha_G\epsilon_G + \epsilon_G^2)$ block encoding of H (the ϵ_G^2 term is needed because [definition 2.13](#) does not require $\alpha_G \geq \|G\|$). Thus a QLS solver ([definition 2.16](#)) applied to this composed encoding uses normalization $\alpha_H = \alpha_G^2$ and access-normalized condition parameter $\alpha_G^2 \|H^{-1}\|_2$. A solver that acts on G directly must instead state how its cost depends on α_G and on the relevant singular-value cutoff of G , and may not assume plain access to H without saying so.

Normal equations and reduced log-barrier Hessians often have natural factors, so later results state which model of [definition 2.15](#) they assume. A factorization that exists algebraically is not thereby free.

Definition 2.16 (Quantum linear-system solver). A *quantum linear-system (QLS) solver* for a nonsingular Hermitian matrix H is a procedure that, given the stated matrix access and a preparation oracle for $|b\rangle$, produces a state ρ satisfying

$$D_{\text{tr}}\left(\rho, \frac{H^{-1}|b\rangle\langle b|H^{-1}}{\|H^{-1}|b\rangle\|_2^2}\right) \leq \epsilon_{\text{ls}}. \quad (27)$$

A cost bound for such a solver must state the linear-system condition number κ_H , the block-encoding normalization α_H (or sparse-oracle normalization), the preparation cost of $|b\rangle$, the success probability, and the precision ϵ_{ls} .

HHL introduced quantum linear-system algorithms with quantum-state output [\[76\]](#). The Childs–Kothari–Somma algorithm achieves polylogarithmic precision dependence in sparse access, variable-time amplitude amplification reduces the amplification overhead, and QSVT implements inversion as a polynomial of the singular values [\[7, 43, 65\]](#). Discrete-adiabatic and adiabatic solvers provide other implementations [\[47, 128\]](#). These families differ in constants and parameter dependence. For each, a cost bound must specify the matrix access and the output, and none converts the state in [\(27\)](#) into a classical vector at no cost.

Definition 2.17 (Charged QRAM access). A *QRAM access assumption* supplies a coherent lookup unitary for an explicit classical table, for example $|i, z\rangle \mapsto |i, z \oplus \text{enc}(d_i)\rangle$ [\[66\]](#). In *charged QRAM access*, raw-input oracle calls and QRAM writes used to build or update that table are counted as two separate resources; later lookups have the stated QRAM query cost.

Once data are loaded, QRAM can reduce the depth of a coherent lookup, but it does not remove the cost of obtaining the information in an input-dependent table. Our lower bounds therefore count the queries and writes used to build the table. A QRAM that contains all hidden parities, Newton entries, or solution coordinates has already been given information that raw coefficient access is designed to hide.

Tomography shows why output can be a bottleneck. For an N -dimensional pure state, a classical estimate of the amplitude vector with ℓ_2 error ϵ requires $\tilde{\Theta}(N/\epsilon)$ controlled calls in the state-preparation unitary model, while sampling-based approaches have quadratic dependence on $1/\epsilon$; mixed-state costs also depend on rank and the requested matrix norm [\[131\]](#). Thus a QLS solver may prepare a solution state at a cost polylogarithmic in the dimension, while classical readout of that state costs a number of calls linear in N .

2.3 Output requirements and cost accounting

An algorithm meets only the output requirement it is asked to meet. The following output requirements have fixed meanings throughout the paper.

Definition 2.18 (Value-estimation output). For a scalar target v , an *additive ϵ value-estimation output* is a classical $\tilde{v}$ such that $|\tilde{v} - v| \leq \epsilon$. A *target-relative ϵ value-estimation output* satisfies $|\tilde{v} - v| \leq \epsilon|v|$ and is used only under a nonzero-target promise. An unqualified “relative value estimate” always means this target-relative output. Both requirements include a required success probability.

Definition 2.19 (Reference-scaled value-estimation output). For a stated scale $s_{\text{ref}} > 0$, a *reference-scaled ϵ value-estimation output* satisfies $|\tilde{v} - v| \leq \epsilon s_{\text{ref}}$. We use it when the target may be zero or the problem has a natural external scale. The scale is always stated, and we do not call this error relative.

Definition 2.20 (Classical vector output). A *classical vector output* for $x \in \mathbb{R}^N$ is an explicit classical list $\tilde{x} \in \mathbb{R}^N$ satisfying a stated error bound, with its norm and scale specified. Unless stated otherwise, the bound is $\|\tilde{x} - x\|_2 \leq \epsilon$. Producing only $\tilde{x}/\|\tilde{x}\|_2$, samples from its coordinates, or an implicit circuit is a different output requirement.

Definition 2.21 (Compact optimizer output). A *compact (sparse-support) optimizer output* is a classical description of an identified component or coordinate subset I together with the nonzero coordinates of a solution supported on I , satisfying stated optimality and feasibility tolerances. It is an explicit sparse solution description, not a full dense vector and not merely a quantum state or sampling oracle.

Definition 2.22 (Normalized solution-state output). For a nonzero solution vector x in the original problem coordinates, an *ϵ normalized solution-state output* (also called an *amplitude-encoded solution state*, or, for an LP primal candidate, an *amplitude-encoded primal-state output*) is a density operator ρ such that

$$D_{\text{tr}}(\rho, |x/\|x\|_2\rangle\langle x/\|x\|_2|) \leq \epsilon, \quad D_{\text{tr}}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1. \quad (28)$$

The output does not include $\|x\|_2$, coordinate values, or a preparation unitary unless explicitly requested. A state that encodes coefficients in a reduced or null-space basis is a different output; our later lower bounds for states in the original coordinates do not automatically apply to it.

Definition 2.23 (Trace-normalized density output). For a nonzero PSD solution X , an *ϵ trace-normalized density output* is ρ satisfying

$$D_{\text{tr}}(\rho, X/\text{tr } X) \leq \epsilon. \quad (29)$$

For a product-PSD solution this means the direct-sum density operator $\bigoplus_j X_j/(\sum_j \text{tr } X_j)$, including its block-label register. This output omits the total trace; when needed, the trace must be output separately.

Definition 2.24 (Coordinate-oracle output). An *ϵ coordinate-oracle output* for $x \in \mathbb{R}^N$ is a reusable, controlled circuit $O_{\tilde{x}}$ and its adjoint acting as

$$O_{\tilde{x}}|i, z\rangle = |i, z \oplus \text{enc}(\tilde{x}_i)\rangle, \quad |\tilde{x}_i - x_i| \leq \epsilon \quad (30)$$

for every i , with a stated fixed-point encoding and failure convention. The cost of one call and any one-time compilation cost must both be stated. A normalized solution state is not a coordinate-oracle output.

Definition 2.25 (Unconditional and heralded preparation). An *unconditional preparation* is a channel whose output state itself satisfies the promised error bound after all internal randomness and flags are discarded. A *heralded preparation* also outputs a success flag in a classical or computational-basis register; conditioned on success, the state obeys the error bound, the success probability is at least a stated $p > 0$ on every allowed input, and the cost includes the repetitions or amplification needed to obtain success. An unobserved or unspecified “good branch” satisfies neither definition.

Definition 2.26 (Global RHS-relative residual). For $b \neq 0$, the *global RHS-relative ℓ_2 primal residual* of an LP candidate x is

$$\text{res}_p(x) = \frac{\|Ax - b\|_2}{\|b\|_2}. \quad (31)$$

For an SDP candidate X , replace Ax by $\mathcal{A}(X)$. A statement that the residual is at most η always refers to this ratio unless a different residual is named. Per-row relative residuals, coefficientwise residuals, normal-equation residuals, local-norm residuals, and backward error under perturbed data are different and do not imply one another without additional scale assumptions.

Definition 2.27 (First-preparation and amortized total-query accounting). For one fixed input oracle and R requested outputs, *setup* is the reusable input-dependent work performed before the first output is produced. Let Q_{setup} and W_{setup} be, respectively, its raw-input query count and QRAM-write count, and let Q_t and W_t be the corresponding nonreusable costs attributable to output t . Raw queries and QRAM writes are never added together. The *first-preparation cost* is the pair $(Q_{\text{setup}} + Q_1, W_{\text{setup}} + W_1)$. The total costs are

$$Q_{\text{total}}(R) = Q_{\text{setup}} + \sum_{t=1}^R Q_t, \quad W_{\text{total}}(R) = W_{\text{setup}} + \sum_{t=1}^R W_t, \quad (32)$$

and the amortized costs are obtained by dividing each total by R . In a pure query model, if setup stores the complete fixed instance, later Q_t may be zero. When advice or additional state copies are consumed, every raw query or QRAM write used to prepare them is counted, in setup or in the call that consumes them.

[Definition 2.27](#) prevents two opposite accounting errors: counting the same static information again each time it is used, and treating an input-dependent reusable data structure as free. Consequently, in the pure query model with a fixed, finite, cacheable hidden input, a raw-query lower bound that applies to each of the R outputs and adds up over them needs an explicit assumption about fresh inputs in each round, memory, communication, or consumable resources.

2.4 Lower-bound toolkit

Parity. For n_{bit} oracle bits, bounded-error quantum computation of their parity needs $\Omega(n_{\text{bit}})$ queries. This follows from the polynomial method because a T -query algorithm has an acceptance probability represented by a real multilinear polynomial of degree at most $2T$, whereas parity has linear approximate degree [20]. Our reductions from raw sign access therefore show that each query to the optimization data can be simulated by $O(1)$ sign queries, and then recover the parity from the required output.

Adversary bounds and composition. Ambainis’s adversary method assigns nonnegative weights to pairs of inputs with different answers and bounds the progress one query can make [6]. Its spectral formulation with positive weights composes: the bound for a composed function is the product of the outer and inner bounds. This already covers an OR of parities on disjoint sets of input bits [77]. The more general negative-weight bound $\text{Adv}^\pm$ composes for Boolean outer functions and characterizes bounded-error Boolean function evaluation up to constants [92, 122]. We use neither composition statement when inner instances share input bits or when the algorithm has direct access to a function of several inner instances.

Polynomial method and symmetrization. More generally, every measurement probability after T standard input queries is a degree- $2T$ polynomial in the input bits. Averaging over inputs of equal Hamming weight turns it into a univariate polynomial without increasing degree. Approximation constraints at the promised weights can then be combined with classical polynomial inequalities to lower-bound T [20]. Later applications state the promised weights and the error interval explicitly; symmetrization does not justify adding inputs outside the promise.

Trace distance and Helstrom decoding. Trace distance is contractive under channels, and the statistical distance between the outcome distributions of any measurement on ρ and σ is at most $D_{\text{tr}}(\rho, \sigma)$. Conversely, for equal priors the optimal binary discrimination success probability is $\frac{1}{2}(1 + D_{\text{tr}}(\rho, \sigma))$, the Helstrom bound [116]. Hence a fixed measurement that decodes a hidden bit from an ideal solution state with bias β still has positive bias from any output within trace distance less than β .

State conversion. In the state-conversion problem, an input oracle indexed by x must transform a prescribed family of initial states $\{|\rho_x\rangle\}$ into target states $\{|\sigma_x\rangle\}$ to bounded error. The filtered γ_2 query distance compares the two Gram matrices $[\langle\rho_x|\rho_y\rangle]$ and $[\langle\sigma_x|\sigma_y\rangle]$ through the matrices that record which input symbols a query can distinguish. For general state conversion it gives lower bounds (the direction used here) and upper bounds that match them up to a gap in the error parameters. In the special case of function evaluation, the resulting general adversary bound characterizes bounded-error query complexity up to constants [92]. Later uses specify the action of U_x on all input states, because an algorithm may apply U_x to states other than $|0\rangle$.

2.5 Recurring notation

Table 1 lists notation used across the technical sections.

3 The Geometry of Central-Path Conditioning

3.1 Question, answer, and relation to prior work

Quantum interior-point methods that call a quantum linear-system solver, and classical interior-point methods that solve the Newton equations inexactly, have costs that grow with the condition number of a Newton matrix. A familiar estimate is $\kappa = \Theta(\mu^{-2})$, but its classical derivations concern the logarithmic barrier, and often a primal–dual Schur complement rather than the reduced primal Hessian. This section asks whether a different self-concordant barrier can remove the ill-conditioning.

For the reduced barrier Hessian (definition 2.9) the answer is no in the minimax sense: no self-concordant barrier improves on the canonical barriers by more than an instance-dependent factor. The obstruction is the Euclidean geometry of the objective sublevel sets. We prove a lower bound on the condition number for every self-concordant barrier (theorem 3.3), bound every eigenvalue between directional widths of the sublevel set (theorem 3.4), and show that the canonical log and log-det barriers attain the lower bound up to an instance-dependent factor (theorem 3.5). Hölder error bounds with attained exponents then give explicit growth rates (corollaries 3.6 and 3.8).

Unlike earlier conditioning analyses, our lower bounds and eigenvalue bounds hold for every self-concordant barrier. Earlier analyses concern the log barrier, including Wright’s analysis of logarithmic barrier Hessians and the survey of Forsgren, Gill, and Wright [63, 134]. The μ^{-2} -type estimates in Augustino et al. [14, Section 7.4], for example, are for primal–dual Schur systems. Conversely, Allamigeon et al. [5] prove a lower bound uniform over self-concordant barriers, but for the number and geometry of central-path iterations, not Hessian conditioning. The error-bound

Table 1: Notation used across the technical sections. Several sections give G , N , P , α , and η a local meaning, stated there.

Symbol	Meaning
m, n	Number of LP equality constraints and primal variables.
q, d_j	Number of PSD blocks and dimension of block j .
$n_{\text{bit}}, \varsigma$	Number of hidden signs and the sign string in a reduction from raw coefficient access.
N	Dimension of a vector required by an output requirement.
N_H	Dimension of the actual Newton, normal-equation, OSS, or MNES system sent to a linear solver.
$A, b, c; \mathcal{A}, b, C$	LP data; product-form SDP data.
X, S	Diagonal LP matrices $\text{diag}(x), \text{diag}(s)$, or primal and dual PSD variables when the context is semidefinite.
F, ν	Ambient cone barrier and its declared barrier parameter.
μ	Central-path parameter.
$\bar{\mu}$	Average LP complementarity $x^\top s/n$ at an iterate.
$\sigma; \sigma_{\min}, \sigma_{\max}$	Newton centering parameter; minimum and maximum singular values.
g	Primal objective gap $c^\top x_F - \text{OPT}$, not automatically the primal-dual gap.
$N_2(\theta)$	Feasible Euclidean complementarity neighborhood in (12).
W, V	Orthonormal tangent basis used in H_{red} ; a general full-rank null-space basis used in OSS.
$H_{\text{red}}, H_{\text{NE}}, H_{\text{OSS}}, H_{\text{MNES}}$	Reduced barrier Hessian and the Newton-system matrices defined above.
κ_{red}	Condition number of H_{red} in an orthonormal Euclidean tangent basis.
κ_{amb}	Condition number of the ambient cone-barrier Hessian before restriction to the tangent space.
κ_H	Spectral condition number of the linear-system matrix H actually solved; not $\kappa_2(A)$ unless proved equal.
s_H	Maximum row/column sparsity of H ; not $\text{nnz}(A)$.
$(\alpha_H, a_H, \epsilon_H)$	Normalization, ancilla count, and operator error of a block encoding of H .
$(\alpha_G, a_G, \epsilon_G)$	Corresponding factor block-encoding data for $H = G^\dagger G$.
κ_{BE}	Access-normalized condition parameter $\alpha_H \ H^{-1}\ _2$.
$\epsilon_{\text{opt}}, \epsilon_{\text{ls}}$	Optimization accuracy; trace-distance accuracy of one linear solve's output state.
η	Feasibility or inexact-Newton residual tolerance; the applicable definition specifies which.
D_{tr}	Trace distance $\frac{1}{2} \ \rho - \sigma\ _1$.
$Q_{\text{setup}}, Q_{\text{total}}$	Setup raw queries and total raw queries including setup.
$W_{\text{setup}}, W_{\text{total}}$	Setup QRAM writes and total QRAM writes including setup.
$\text{nnz}(A)$	Number of stored nonzeros of A ; it does not measure fill-in or the sparsity of derived systems.

consequences below connect to weak sharp minima [33] and to Sturm's singularity-degree bounds for semidefinite systems [127].

Remark 3.1 (Novelty and textbook ingredients). The Dikin-ellipsoid and asymmetric-containment facts used in the proofs, as well as the centrality identity $\Pi_{\mathcal{V}}\nabla F(x_{\mu}) = -\Pi_{\mathcal{V}}c/\mu$, are standard self-concordance tools [113, 123]. To our knowledge, what is new here is the conditioning statement that holds for every barrier, together with its geometric consequences. We do not claim that these proof ingredients are new.

3.2 Sublevel chords and two containment facts

Let $P \subset \mathbb{R}^N$ be compact and convex, with affine hull $x^0 + \mathcal{V}$ and nonempty relative interior P° . Let W have orthonormal columns spanning the d -dimensional tangent space $\mathcal{V}$. We minimize the nonconstant linear functional $c^\top x$, so $c_{\mathcal{V}} := \Pi_{\mathcal{V}}c \neq 0$. Define

$$\text{OPT} = \min_{y \in P} c^\top y, \quad \Phi = \{y \in P : c^\top y = \text{OPT}\}. \quad (33)$$

For $t \geq 0$, define the objective sublevel set, its diameter, and the longest sublevel chord from an interior point by

$$L(t) = \{y \in P : c^\top y \leq \text{OPT} + t\}, \quad D(t) = \text{diam } L(t), \quad \ell(x) = \sup_{y \in L(c^\top x - \text{OPT})} \|y - x\|_2. \quad (34)$$

Thus D is nondecreasing. If x has objective gap g , then $x \in L(g)$ and the triangle inequality gives

$$\frac{D(g)}{2} \leq \ell(x) \leq D(g). \quad (35)$$

Indeed, for any $y, z \in L(g)$, at least one of $\|x - y\|_2, \|x - z\|_2$ is at least $\|y - z\|_2/2$.

Let F be a ν_F -self-concordant barrier for P in the sense of [definition 2.2](#). We use two standard consequences of self-concordance. For $x \in P^\circ$,

$$\begin{aligned} y \in x^0 + \mathcal{V}, \quad \|y - x\|_x < 1 &\implies y \in P^\circ, & (\text{Dikin}) \\ y \in P, \quad \langle \nabla F(x), y - x \rangle \geq 0 &\implies \|y - x\|_x \leq C_F, \quad C_F := \nu_F + 2\sqrt{\nu_F}. & (\text{asymmetric}) \end{aligned}$$

The first implication is understood for points in the affine hull; its contrapositive says that a boundary or exterior point has local distance at least one. We also use semiboundedness,

$$\langle \nabla F(x), y - x \rangle \leq \nu_F \quad (y \in P). \quad (36)$$

These facts follow from the standard containment theorems for self-concordant barriers; restriction to an affine subspace preserves them and does not increase the barrier parameter [113, 123].

At an exact central point $x = x_F(\mu)$, first-order optimality on the affine slice is

$$W^\top (c + \mu \nabla F(x)) = 0. \quad (37)$$

Consequently, for every $y \in P$,

$$c^\top (x - y) = \mu \langle \nabla F(x), y - x \rangle \leq \nu_F \mu. \quad (38)$$

Taking $y \in \Phi$ proves the bound stated in [remark 2.5](#):

$$0 < g_F(\mu) \leq \nu_F \mu. \quad (39)$$

The strict inequality holds because a nonconstant linear functional has no minimizer in the relative interior. The upper bound is generally not an equality: the value $\nu\mu$ in [definition 2.3](#) is the canonical *primal-dual* gap, not the primal objective gap g used here.

3.3 Rigidity of the Hessian spectrum

We first bound the smallest eigenvalue from below; centrality is not needed.

Theorem 3.2 (Universal minimum-eigenvalue rigidity). *Under the setup of [section 3.2](#), for every ν_F -self-concordant barrier F for P and every $x \in P^\circ$,*

$$\lambda_{\min}\left(W^\top \nabla^2 F(x) W\right) \geq \frac{1}{\ell(x)^2}. \quad (40)$$

Here $\ell(x)$ is defined with the objective value of x itself, whether or not x is central.

Proof. Fix a Euclidean unit vector $h \in \mathcal{V}$ and replace it by $-h$ if necessary so that $c^\top h \leq 0$. Compactness implies that the ray $x + th$ leaves P° in finite time. More precisely, if

$$\tau = \sup\{t \geq 0 : x + sh \in P^\circ \text{ for } 0 \leq s < t\},$$

then $0 < \tau < \infty$ and $y = x + \tau h \in P$. The objective does not increase along the ray, so $y \in L(c^\top x - \text{OPT})$ and hence $\tau \leq \ell(x)$. Since $y \notin P^\circ$, the contrapositive of [\(Dikin\)](#) gives

$$1 \leq \|\tau h\|_x^2 = \tau^2 \langle \nabla^2 F(x) h, h \rangle.$$

Thus the Rayleigh quotient is at least $\tau^{-2} \geq \ell(x)^{-2}$. The Hessian form is even in h , so the initial sign choice loses no direction. Minimizing over the Euclidean unit sphere in $\mathcal{V}$ proves [\(40\)](#). $\square$

At a central point, the largest eigenvalue is bounded below in terms of the objective gap. Combining this bound with [theorem 3.2](#) gives a lower bound on the condition number.

Theorem 3.3 (Sublevel chord law). *Under the setup of [section 3.2](#), let $x = x_F(\mu)$ be an exact central point of any ν_F -self-concordant barrier, and set $g = g_F(\mu)$. Then*

$$\lambda_{\max}\left(W^\top \nabla^2 F(x) W\right) \geq \frac{\|c_{\mathcal{V}}\|_2^2}{g^2}, \quad (41)$$

$$\frac{1}{\ell(x)^2} \leq \lambda_{\min}\left(W^\top \nabla^2 F(x) W\right) \leq \frac{C_F^2}{\ell(x)^2}, \quad (42)$$

$$\kappa_{\text{red}}(\mu; F) \geq \left(\frac{\ell(x)\|c_{\mathcal{V}}\|_2}{C_F g}\right)^2 \geq \left(\frac{D(g)\|c_{\mathcal{V}}\|_2}{2C_F g}\right)^2. \quad (43)$$

In particular, using only the μ parametrization,

$$\kappa_{\text{red}}(\mu; F) \geq \left(\frac{D(g_F(\mu))\|c_{\mathcal{V}}\|_2}{2C_F \nu_F \mu}\right)^2. \quad (44)$$

The theorem holds for every $\mu > 0$ and assumes no polyhedrality, nondegeneracy, or strict complementarity.

Proof. The gap statement [\(39\)](#) was proved above. For the largest eigenvalue, let

$$\hat{h} = -\frac{c_{\mathcal{V}}}{\|c_{\mathcal{V}}\|_2}, \quad t_* = \frac{g}{\|c_{\mathcal{V}}\|_2}.$$

The affine point $z = x + t_* \hat{h}$ has $c^\top z = \text{OPT}$. If it were in P° , it would be a relative-interior minimizer of a linear functional, forcing $c_{\mathcal{V}} = 0$. Therefore $z \notin P^\circ$, and [\(Dikin\)](#) gives

$$\langle \nabla^2 F(x) \hat{h}, \hat{h} \rangle \geq t_*^{-2} = \frac{\|c_{\mathcal{V}}\|_2^2}{g^2}.$$

This proves [\(41\)](#).

For the upper bound in (42), take any $y \in L(g)$. By (37),

$$\langle \nabla F(x), y - x \rangle = -\frac{c^\top (y - x)}{\mu} \geq 0.$$

Asymmetric containment (asymmetric) therefore gives $\|y - x\|_x \leq C_F$. If $y \neq x$, the Euclidean unit vector $v = (y - x)/\|y - x\|_2$ satisfies

$$\langle \nabla^2 F(x)v, v \rangle \leq \frac{C_F^2}{\|y - x\|_2^2}.$$

The smallest eigenvalue is no larger than this Rayleigh quotient. Taking a sequence of y whose distance from x tends to $\ell(x)$ proves the upper bound; the lower bound is theorem 3.2. Dividing (41) by the upper bound in (42), then using (35), proves (43). Finally $g \leq \nu_F \mu$ proves (44). $\square$

The constant in (41) can be attained in the limit. For the simplex problem $\min x_3$ subject to $x_1 + x_2 + x_3 = 1$, numerical central paths of the two barriers we tested give $\lambda_{\max} g^2 / \|c_V\|_2^2 \rightarrow 1$; for the ordinary log barrier, $g/\mu \rightarrow 1$ and $\lambda_{\max} \mu^2 \rightarrow 2/3 = \|c_V\|_2^2$.

The same argument bounds every eigenvalue, not only the extreme ones. For a line $[h] \subset \mathcal{V}$ through the origin, choose the unit vector $\hat{h}$ on it with $c^\top \hat{h} < 0$. If $c^\top h = 0$, choose the sign with the longer exit time, defined next, breaking a tie arbitrarily. Define

$$\tau_x([h]) = \sup\{t \geq 0 : x + s\hat{h} \in P^\circ \text{ for } 0 \leq s < t\}. \quad (45)$$

The exit point $x + \tau_x([h])\hat{h}$ lies in $L(c^\top x - \text{OPT})$, so $0 < \tau_x([h]) \leq \ell(x)$. The sign convention makes τ_x well defined on lines; continuity is not needed.

For $j = 1, \dots, d$, define the two width profiles

$$w_j^-(x) = \inf_{\substack{U \subseteq \mathcal{V} \\ \text{codim } U = j-1}} \sup_{\substack{h \in U \\ \|h\|_2=1}} \tau_x([h]), \quad (46)$$

$$w_j^+(x) = \sup_{\substack{S \subseteq \mathcal{V} \\ \dim S = j}} \inf_{\substack{h \in S \\ \|h\|_2=1}} \tau_x([h]). \quad (47)$$

We use infima and suprema because the sign choice can make the exit-time function discontinuous on $c^\perp$.

Theorem 3.4 (Width-spectrum rigidity). *In the setting of theorem 3.3, list the eigenvalues of $W^\top \nabla^2 F(x)W$ in nondecreasing order. For every $j = 1, \dots, d$,*

$$\frac{1}{(w_j^-(x))^2} \leq \lambda_j(W^\top \nabla^2 F(x)W) \leq \frac{C_F^2}{(w_j^+(x))^2}. \quad (48)$$

Moreover $w_j^+(x) \leq w_j^-(x)$ for every j , and $w_1^+(x) = \sup_{[h]} \tau_x([h]) \geq \ell(x)$.

Proof. For every unit h , Dikin containment at the exit point gives

$$\langle \nabla^2 F(x)h, h \rangle \geq \tau_x([h])^{-2}. \quad (49)$$

For the selected orientation, centrality gives

$$\langle \nabla F(x), \tau_x([h])\hat{h} \rangle = -\frac{\tau_x([h])c^\top \hat{h}}{\mu} \geq 0.$$

Asymmetric containment (asymmetric) applies directly to the exit point and gives

$$\langle \nabla^2 F(x)h, h \rangle \leq C_F^2 \tau_x([h])^{-2}. \quad (50)$$

The Rayleigh quotient is unchanged by orientation.

The two Courant–Fischer formulas for eigenvalues in nondecreasing order are

$$\begin{aligned}\lambda_j &= \sup_{\text{codim } U=j-1} \inf_{h \in U, \|h\|_2=1} R(h), \\ \lambda_j &= \inf_{\text{dim } S=j} \sup_{h \in S, \|h\|_2=1} R(h),\end{aligned}$$

where $R(h) = \langle \nabla^2 F(x)h, h \rangle$. Applying (49) in the first formula yields

$$\lambda_j \geq \sup_U \frac{1}{(\sup_{h \in U} \tau_x([h]))^2} = \frac{1}{(w_j^-(x))^2}.$$

Applying (50) in the second formula gives

$$\lambda_j \leq C_F^2 \inf_S \frac{1}{(\inf_{h \in S} \tau_x([h]))^2} = \frac{C_F^2}{(w_j^+(x))^2}.$$

For any j -dimensional S and codimension- $(j-1)$ subspace U , the dimension formula gives a nonzero line $[h] \subseteq S \cap U$. Hence

$$\inf_{v \in S, \|v\|=1} \tau_x([v]) \leq \tau_x([h]) \leq \sup_{v \in U, \|v\|=1} \tau_x([v]).$$

Taking the supremum over S and infimum over U proves $w_j^+ \leq w_j^-$. Finally, if $y \in L(g)$, the segment $[x, y)$ lies in P° . The line through $y - x$ is oriented toward y when the objective strictly decreases, and our longer-exit convention gives the same conclusion when it is unchanged. Thus $\tau_x([y - x]) \geq \|y - x\|_2$. Taking the supremum over y proves $w_1^+ \geq \ell(x)$. $\square$

Thus every eigenvalue is constrained by directional widths of the sublevel set, up to the factor C_F^2 and the gap between w_j^- and w_j^+ . Section 4 studies the consequences for eigenvalue clustering.

3.4 Canonical barriers attain the lower bound

We next specialize to compact affine slices of the nonnegative orthant and of the product PSD cone in section 2.1. To distinguish parameters when two barriers are compared, write $\nu_{\text{can}} = n$ for the LP log barrier and $\nu_{\text{can}} = \sum_j d_j$ for the product log-det barrier. This is the cone parameter denoted by ν in section 2.1.

Theorem 3.5 (Canonical achievability and gap-matched comparison). *Consider either of the following strictly primal–dual feasible problems with a compact primal feasible region and nonconstant objective.*

- (i) *An LP in (1), with canonical path $(x(\mu), y(\mu), s(\mu))$ and $M(\mu) = \|s(\mu)\|_\infty$.*
- (ii) *A product-form SDP in (3). Its canonical path is $(X(\mu), y(\mu), S(\mu))$, and $M(\mu) = \max_j \|S_j(\mu)\|_2$.*

Fix $\bar{\mu} > 0$, call the part of the canonical path with $0 < \mu \leq \bar{\mu}$ its tail, and suppose the tail constant

$$M_{\text{tail}} = \sup_{0 < \mu \leq \bar{\mu}} M(\mu) \tag{51}$$

is finite. Use the Euclidean metric for the LP and the product Frobenius metric for the SDP. Write $g_{\text{can}}(\bar{\mu}) = g_{F_{\text{can}}}(\bar{\mu})$ for the canonical primal objective gap at the endpoint of the tail. We write $\kappa_{\text{red}}^{\text{can}}(g)$ and $\kappa_{\text{red}}^F(g)$ for the reduced condition number at the central point with gap g of the

canonical barrier and of F , using the g -parametrization in [definition 2.3](#) on an interval where that barrier's gap map g_F is one-to-one. For a canonical central point with $0 < \mu \leq \bar{\mu}$ and primal objective gap g ,

$$\lambda_{\min}(H_{\text{red}}) \geq \ell(x)^{-2}, \quad (52)$$

$$\lambda_{\max}(H_{\text{red}}) \leq M_{\text{tail}}^2 \mu^{-2} \leq M_{\text{tail}}^2 \nu_{\text{can}}^2 g^{-2}, \quad (53)$$

$$\kappa_{\text{red}}^{\text{can}}(g) \leq \left(\frac{M_{\text{tail}} \nu_{\text{can}} D(g)}{g} \right)^2. \quad (54)$$

For every gap $0 < g \leq g_{\text{can}}(\bar{\mu})$ attained on the canonical tail, any ν_F -self-concordant barrier F on the same feasible region, and any exact F -central point having that gap,

$$\kappa_{\text{red}}^{\text{can}}(g) \leq \left(\frac{2M_{\text{tail}} \nu_{\text{can}} (\nu_F + 2\sqrt{\nu_F})}{\|c_V\|_2} \right)^2 \kappa_{\text{red}}^F(g). \quad (55)$$

The comparison is between points of equal objective gap; their values of μ need not agree.

Proof. The first bound is [theorem 3.2](#). For the LP, canonical complementarity gives $x_i = \mu/s_i$, and therefore, for every Euclidean unit tangent vector h ,

$$\langle \nabla^2 F_{\log}(x) h, h \rangle = \sum_i \frac{h_i^2}{x_i^2} \leq \frac{M_{\text{tail}}^2}{\mu^2}.$$

For the SDP, $X_j^{-1} = S_j/\mu$, and for a unit product-Frobenius tangent direction H ,

$$\begin{aligned} D^2 F_{\log \det}(X)[H, H] &= \sum_j \|X_j^{-1/2} H_j X_j^{-1/2}\|_F^2 \\ &\leq \max_j \|X_j^{-1}\|_2^2 \sum_j \|H_j\|_F^2 \leq \frac{M_{\text{tail}}^2}{\mu^2}. \end{aligned}$$

This proves the first inequality in [\(53\)](#). The canonical primal–dual gap is $\nu_{\text{can}}\mu$, whereas its primal objective gap satisfies $g \leq \nu_{\text{can}}\mu$ by [\(39\)](#); hence $\mu^{-1} \leq \nu_{\text{can}}/g$. Combining the largest-eigenvalue bound with [\(52\)](#) and $\ell(x) \leq D(g)$ proves [\(54\)](#). Finally divide that upper bound by the universal lower bound [\(43\)](#) for F to obtain [\(55\)](#). $\square$

Finiteness of M_{tail} is an assumption on the instance; it holds in the standard setting where the canonical path converges. The restriction to a tail is necessary because the supremum can diverge as $\mu \rightarrow \infty$. By [\(55\)](#), no other barrier improves the canonical reduced condition number by more than a factor that depends on the instance but not on g ; it does not rule out that other barriers are worse.

3.5 From error bounds to conditioning rates

By [theorems 3.3](#) and [3.5](#), the reduced condition number is governed by $D(g)/g$. Error bounds give upper estimates of $D(g)$. If, for some $\theta \in [0, 1]$ and $\gamma > 0$,

$$\text{dist}(y, \Phi) \leq \gamma^{-1} (c^\top y - \text{OPT})^\theta \quad (y \in P), \quad (56)$$

then comparing two points of $L(g)$ through their nearest optimal points gives

$$D(g) \leq \text{diam } \Phi + 2\gamma^{-1} g^\theta. \quad (57)$$

For lower estimates, note that $\Phi \subseteq L(g)$ and that a central point x with gap g lies in $L(g)$. For every $x^* \in \Phi$, $g = c_V^\top (x - x^*) \leq \|c_V\|_2 \|x - x^*\|_2$, so

$$D(g) \geq \text{diam } \Phi, \quad D(g) \geq \frac{g}{\|c_V\|_2} \quad \text{at every attained central gap } g. \quad (58)$$

Corollary 3.6 (Conditioning regimes). *Assume the hypotheses of [theorem 3.3](#) and, for the canonical upper bounds, those of [theorem 3.5](#); the canonical upper bounds hold on the canonical tail.*

- (i) *If $\dim \Phi \geq 1$, then every fixed ν_F -self-concordant barrier satisfies $\kappa_{\text{red}}^F(g) = \Omega(g^{-2})$. The canonical barrier satisfies $\kappa_{\text{red}}^{\text{can}}(g) = \Theta(g^{-2})$.*
- (ii) *If P is a polytope with a unique optimum, including a degenerate vertex, then $D(g) = \Theta(g)$ as $g \downarrow 0$. Consequently the canonical log barrier has $\kappa_{\text{red}}^{\text{log}}(g) = \Theta(1)$.*
- (iii) *If the optimum is unique and $D(g) = \Theta(\sqrt{g})$, as under two-sided quadratic growth of the objective gap, then every fixed barrier has $\kappa_{\text{red}}^F(g) = \Omega(g^{-1})$ and the canonical barrier has $\kappa_{\text{red}}^{\text{can}}(g) = \Theta(g^{-1})$.*

The constants may depend on the instance and, for a noncanonical barrier, on ν_F . Whenever the canonical objective gap is $\Theta(\mu)$, as in the standard strictly complementary tail, the same three canonical rates are $\Theta(\mu^{-2})$, $\Theta(1)$, and $\Theta(\mu^{-1})$, respectively.

Proof. If $\dim \Phi \geq 1$, the face contains two distinct points and hence $\text{diam } \Phi > 0$; compactness makes this diameter finite and attained. Substitution of $D(g) \geq \text{diam } \Phi$ in [\(43\)](#) proves the universal lower bound. Since $D(g) \leq \text{diam } P$, [\(54\)](#) gives the matching canonical upper bound.

For a polytope, the solution set of a feasible bounded linear program is a weak sharp minimum [\[33, 99\]](#). With a unique optimum x^* , this gives [\(56\)](#) with $\theta = 1$, hence $D(g) \leq 2g/\gamma$. For the reverse bound, choose a feasible $z \neq x^*$ and set $\Delta = c^\top z - \text{OPT} > 0$. For every $0 < g \leq \Delta$, convexity makes

$$y_g = \left(1 - \frac{g}{\Delta}\right) x^* + \frac{g}{\Delta} z$$

feasible, and $c^\top y_g = \text{OPT} + g$. Thus

$$D(g) \geq \|y_g - x^*\|_2 = \frac{\|z - x^*\|_2}{\Delta} g.$$

Combining these two bounds with [theorems 3.3](#) and [3.5](#) proves (ii). Part (iii) follows by substituting $D(g) = \Theta(\sqrt{g})$ in the same two theorems. The final conversion is immediate under the separately stated relation $g = \Theta(\mu)$. $\square$

Remark 3.7. Only the lower bounds in [corollary 3.6](#) are proved for every barrier; the matching upper bounds are proved only for the canonical barrier. Thus, for a positive-dimensional optimal face, the best rate over all barriers is $\Theta(g^{-2})$.

Part (ii) includes degenerate vertices, because weak sharpness does not require nondegeneracy. Part (iii) explains why the reduced primal log-det Hessian at a generic curved unique SDP optimum can have condition number $\Theta(\mu^{-1})$, even though its largest eigenvalue and a primal-dual Schur complement can be of order μ^{-2} . This is the conic analogue of the μ^{-1} behavior of structured logarithmic-barrier Hessians that Wright [\[134\]](#) identified for nonlinear programs.

Other growth rates of $D(g)$ give fractional exponents. Sturm proves an $O(\epsilon^{2-d})$ distance bound when the relevant semidefinite feasibility system has singularity degree d [\[127\]](#). An upper error bound alone gives no lower bound on the condition number; that also requires a matching lower bound on $D(g)$. When the exponent is attained in this sense, the result is immediate.

Corollary 3.8 (Attained Hölder and singularity-degree laws). *Suppose that, along central gaps $g \downarrow 0$, $D(g) \geq ag^\theta$ for constants $a > 0$ and $\theta \in [0, 1]$. Then every ν_F -self-concordant barrier satisfies*

$$\kappa_{\text{red}}^F(g) \geq \frac{a^2 \|c_V\|_2^2}{4C_F^2} g^{2\theta-2}. \quad (59)$$

If, in addition, the problem is a canonical LP or product-form SDP satisfying all hypotheses of [theorem 3.5](#) and $D(g) = O(g^\theta)$, then the canonical barrier attains $\Theta(g^{2\theta-2})$ on its tail. In particular, when the Sturm exponent $\theta = 2^{-d}$ is attained, the exponents are

$$d = 0 : 0, \quad d = 1 : -1, \quad d = 2 : -\frac{3}{2}, \quad 2\theta - 2 = 2^{1-d} - 2 \longrightarrow -2. \quad (60)$$

Proof. Insert $D(g) \geq ag^\theta$ into (43). If the matching upper bound on D holds, combine the resulting lower bound with (54). Substituting $\theta = 2^{-d}$ gives (60). $\square$

The following 3×3 SDP shows that the $d = 2$ law occurs in a concrete problem.

Proposition 3.9 (An explicit $g^{-3/2}$ witness). *Consider*

$$\begin{aligned} & \min_{X \in \mathbb{S}^3} X_{22} \\ & \text{subject to} \quad \text{tr } X = 1, \quad X_{33} = X_{12}, \quad X \succeq 0. \end{aligned} \quad (61)$$

Its unique optimum is $X^* = e_1 e_1^\top$, its optimality system has singularity degree two, and

$$D(g) = \Theta(g^{1/4}). \quad (62)$$

Consequently every self-concordant barrier obeys $\kappa_{\text{red}}(g) = \Omega(g^{-3/2})$, while the log-det barrier, under the hypotheses of [theorem 3.5](#), has $\kappa_{\text{red}}(g) = \Theta(g^{-3/2})$ on its canonical tail.

Proof. Write a feasible matrix as

$$X = \begin{bmatrix} a & b & t \\ b & d & e \\ t & e & b \end{bmatrix}, \quad a + d + b = 1, \quad X \succeq 0.$$

Here $d = X_{22}$ is the objective and $b = X_{12} = X_{33} \geq 0$. If $d = 0$, positive semidefiniteness makes the second row and column zero, hence $b = e = 0$. Then the third diagonal entry is zero, so the third row and column vanish and $t = 0$; the trace constraint gives $a = 1$. The optimum is therefore unique.

For $X \in L(g)$, the 2×2 principal minors give

$$b^2 \leq ad \leq d \leq g, \quad t^2 \leq ab \leq b \leq \sqrt{g}, \quad e^2 \leq db \leq g^{3/2}.$$

Also $|a - 1| = d + b \leq g + \sqrt{g}$. Thus $\|X - X^*\|_F = O(g^{1/4})$, and the triangle inequality gives $D(g) = O(g^{1/4})$.

For the reverse bound, let $0 < g \leq 1/16$, set

$$d = g, \quad b = \frac{\sqrt{g}}{2}, \quad a = 1 - g - \frac{\sqrt{g}}{2}, \quad e = 0, \quad t = \pm \frac{g^{1/4}}{2\sqrt{2}}.$$

Then $a \geq 13/16$. The one- and two-dimensional principal minors are positive, and

$$\det X = agb - b^3 - gt^2 = g^{3/2} \left(\frac{a}{2} - \frac{1}{4} \right) > 0.$$

Both sign choices therefore give positive-definite feasible points in $L(g)$. Their Frobenius distance is $2\sqrt{2}|t| = g^{1/4}$, proving the lower bound in (62).

For completeness, the two facial reductions can be read directly from the optimality equations. The first exposing equation $X_{22} = 0$ removes the e_2 row and column; then $X_{12} = 0$ turns the affine equation $X_{33} = X_{12}$ into $X_{33} = 0$, which removes the e_3 row and column. No one-step positive-semidefinite exposing combination can do both: a PSD linear combination of the trace, $X_{33} - X_{12}$, and X_{22} equations whose $(1, 1)$ entry is zero must have zero first row, forcing the coefficient of $X_{33} - X_{12}$ to vanish. Thus two steps are necessary and sufficient. Finally apply [corollary 3.8](#) with $\theta = 1/4$. $\square$

For the log-det barrier, the leading constant can also be computed exactly. For sufficiently small gap $g > 0$, the central point and its parameter μ are

$$X_{\log}(g) = \begin{pmatrix} a & b & 0 \\ b & g & 0 \\ 0 & 0 & b \end{pmatrix}, \quad b = \frac{-g + \sqrt{3g - 2g^2}}{3}, \quad a = 1 - g - b, \quad \mu = \frac{ag - b^2}{1 - b - 2g}.$$

Indeed, sign symmetry forces the two displayed zero entries, and stationarity of $g/\mu - \log b - \log(g(1 - g - b) - b^2)$ gives $3b^2 + 2gb + g^2 - g = 0$ and the stated μ . Strict convexity makes this interior stationary point the unique center. The spectral calculation in [section 4](#) gives

$$\kappa_{\text{red}}\mu^{3/2} \longrightarrow \frac{2\sqrt{2}}{7}, \quad \kappa_{\text{red}}g^{3/2} \longrightarrow \frac{3\sqrt{3}}{7}.$$

The first constant is approximately 0.40406. The exact center, the Frobenius-metric spectrum, and these limits are verified in Lean; the report `formal/FRACTIONAL_SDP.md` states the verification scope. The proof of [proposition 3.9](#) is not part of the formalization.

A nearly optimal face makes the condition number level off at a finite height instead of diverging.

Corollary 3.10 (Near-degeneracy plateau). *Suppose a family of compact instances has a scale $\vartheta > 0$ and constants independent of g, ϑ such that*

$$D(g) = \begin{cases} \Theta(g/\vartheta), & 0 < g \lesssim \vartheta, \\ \Theta(1), & \vartheta \lesssim g \leq g_0. \end{cases} \quad (63)$$

Assume also that $\|c_V\|_2$ is bounded below and that the barrier parameters under comparison are bounded above uniformly in ϑ ; for the canonical upper bound, assume the same of $M_{\text{tail}}\nu_{\text{can}}$ and assume $g_0 \leq g_{\text{can}}(\bar{\mu})$ uniformly in ϑ . Then every self-concordant barrier satisfies the lower bounds

$$\kappa_{\text{red}}^F(g) = \begin{cases} \Omega(\vartheta^{-2}), & 0 < g \lesssim \vartheta, \\ \Omega(g^{-2}), & \vartheta \lesssim g \leq g_0, \end{cases} \quad (64)$$

with constants that depend only on C_F and the constants in the hypotheses, not on g or ϑ . Under the hypotheses of [theorem 3.5](#), the canonical barrier attains both orders on the tail. Thus the g^{-2} growth stops near $g \asymp \vartheta$, at height $\Theta(\vartheta^{-2})$.

Proof. In the first range, $D(g)/g = \Theta(\vartheta^{-1})$; in the second it is $\Theta(g^{-1})$. Apply the lower bound [\(43\)](#) and the canonical upper bound [\(54\)](#). $\square$

For the simplex family $\min\{x_3 + \vartheta x_2 : x_1 + x_2 + x_3 = 1, x \geq 0\}$, the height of the log-barrier plateau can be computed exactly. For each fixed $\vartheta > 0$,

$$\lim_{\mu \downarrow 0} \kappa_{\text{red}}(\mu) = \frac{1 + \vartheta^2 + \sqrt{1 - \vartheta^2 + \vartheta^4}}{1 + \vartheta^2 - \sqrt{1 - \vartheta^2 + \vartheta^4}}, \quad \lim_{\vartheta \downarrow 0} \vartheta^2 \left(\lim_{\mu \downarrow 0} \kappa_{\text{red}}(\mu) \right) = \frac{4}{3}. \quad (65)$$

To verify the inner limit, write the equality multiplier as $\alpha_\mu = \mu/x_1$. Centrality gives $x_2 = \mu/(\vartheta + \alpha_\mu)$, $x_3 = \mu/(1 + \alpha_\mu)$, and $\alpha_\mu \rightarrow 0$. On the tangent plane, put $h = (-u - v, u, v)$. After multiplication by μ^2 , the limiting Rayleigh quotient is

$$\frac{\vartheta^2 u^2 + v^2}{2(u^2 + uv + v^2)}.$$

Its generalized eigenvalues are $(1 + \vartheta^2 \pm \sqrt{1 - \vartheta^2 + \vartheta^4})/3$, whose ratio is the first expression in [\(65\)](#); expansion at $\vartheta = 0$ gives the second limit. Computed values of $\vartheta^2 \lim_{\mu \downarrow 0} \kappa_{\text{red}}(\mu)$ at $\vartheta = 10^{-2}$ and 10^{-3} round to $4/3$.

3.6 Approximately central points

The lower bound holds, up to a constant factor, in the short-step neighborhood of the central path. All norms in the next statement are restricted to the tangent space.

Corollary 3.11 (Lower bound at approximately central points). *Fix $\mu > 0$, let $x_* = x_F(\mu)$, and suppose $x \in P^\circ$ satisfies*

$$\lambda_\mu(x) := \|c/\mu + \nabla F(x)\|_x^* \leq \rho \leq \frac{1}{6}. \quad (66)$$

Then, with $H_ = W^\top \nabla^2 F(x_*) W$ and $H = W^\top \nabla^2 F(x) W$,*

$$\|x - x_*\|_{x_*} \leq \frac{\rho}{(1 - \rho)^2} \leq \frac{1}{4}, \quad \kappa(H) \geq \left(\frac{3}{4}\right)^4 \kappa(H_*). \quad (67)$$

Consequently $\kappa(H)$ is at least $(3/4)^4$ times the right-hand sides of (43) and (44), evaluated at the exact center x_ .*

Proof. Restrict $f = F + c^\top(\cdot)/\mu$ to the affine slice. It is standard self-concordant, has Hessian $\nabla^2 F$, and is minimized at x_* . With

$$\omega(t) = t - \log(1 + t), \quad \omega_*(t) = -t - \log(1 - t),$$

the self-concordant Newton estimates give

$$\omega(\|x - x_*\|_{x_*}) \leq f(x) - f(x_*) \leq \omega_*(\lambda_\mu(x)) \leq \omega_*(\rho).$$

The elementary scalar inequality $\omega^{-1}(\omega_*(\rho)) \leq \rho/(1 - \rho)^2$ follows by setting $t = \rho/(1 - \rho)^2$: the difference $\omega(t) - \omega_*(\rho)$ vanishes at $\rho = 0$, and differentiation in ρ shows that it stays nonnegative. This proves the distance bound, and $\rho \leq 1/6$ gives $\rho/(1 - \rho)^2 \leq 6/25 < 1/4$.

If $r = \|x - x_*\|_{x_*} < 1$, self-concordant Hessian comparison gives

$$(1 - r)^2 H_* \preceq H \preceq (1 - r)^{-2} H_*.$$

It follows that $\lambda_{\max}(H) \geq (1 - r)^2 \lambda_{\max}(H_*)$ and $\lambda_{\min}(H) \leq (1 - r)^{-2} \lambda_{\min}(H_*)$; hence $\kappa(H) \geq (1 - r)^4 \kappa(H_*) \geq (3/4)^4 \kappa(H_*)$. $\square$

3.7 Implications and open geometric questions

For QIPMs that solve barrier Newton systems, changing the barrier cannot remove the lower bound of [theorem 3.3](#). At a central point with target primal objective gap ϵ , the reduced Hessian of every barrier has condition number at least

$$\left(\frac{D(\epsilon) \|c_V\|_2}{2(\nu_F + 2\sqrt{\nu_F})\epsilon} \right)^2$$

For canonical LPs or product-form SDPs with finite M_{tail} and $0 < \epsilon \leq g_{\text{can}}(\bar{\mu})$ attained on the canonical tail, the canonical barrier is worse than any other barrier by at most the instance-dependent factor in [\(55\)](#). In particular, a positive-dimensional optimal face forces growth like ϵ^{-2} , a curved unique optimum can force growth like ϵ^{-1} , and attained singularity-degree exponents force fractional power laws.

These conclusions concern only κ_{red} ([definition 2.9](#)). They say nothing about condition-number-free QIPM approaches such as Apers and Gribling [\[10\]](#), about preconditioners that are not barrier Hessians, or about scalar congruences of the Newton system. Later sections, in particular [section 12](#), treat these alternatives.

Open Problem 3.12 (Degenerate-vertex constants). For a polytope with a unique degenerate optimal vertex, characterize the finite limiting constant (or limiting spectrum) of the canonical reduced log Hessian directly from the active-set geometry.

Open Problem 3.13 (Facial characterization of chord growth). Characterize the exact growth of $D(g)$ for semidefinite programs from facial data beyond the generic strict-complementarity and attained singularity-degree cases. In particular, determine when Sturm’s 2^{-d} exponent is sharp for objective sublevel diameters rather than only an upper error-bound exponent.

4 Clustered Spectra and Solver Cost

[Theorem 3.3](#) says when the condition number must diverge, but not how the eigenvalues between the extremes are distributed. This section answers that question for the logarithmic barrier of a degenerate LP and then asks what the resulting spectrum costs classical and quantum linear solvers. Near the optimum the spectrum consists of two intervals, each with endpoint ratio bounded independently of μ ([proposition 4.1](#)), so conjugate gradients (CG) needs only $\text{polylog}(\kappa)$ matrix–vector products ([proposition 4.2](#)). Quantum polynomial solvers cannot exploit the empty gap between the intervals in the same way ([theorem 4.3](#)): in sufficiently large dimension, plain block-encoding access costs $\tilde{\Theta}(\kappa)$ in the worst case, whereas the factor access of [definition 2.15](#) admits a polynomial whose degree times amplitude bound is $\tilde{\Theta}(\sqrt{\kappa})$ under norm-tight encodings. Clustering thus improves neither quantum access model beyond its usual dependence on κ , while it separates CG from these quantum solvers on the Newton systems themselves. There is an important qualification. At exact central points, the path’s own Newton right-hand sides are asymptotically orthogonal to the *weak modes* (the bottom-cluster eigenvectors), so once the gap opens, path-following steps can solve on the top cluster alone at a cost independent of the condition number ([proposition 4.5](#) and [corollary 4.6](#)).

4.1 The two-cluster tail

Consider the standard-form LP (1) and suppose that its primal feasible polytope P is compact with $P^\circ \neq \emptyset$. Let $Z \in \mathbb{R}^{n \times p}$ have orthonormal columns spanning $\ker A$, assume as in [section 3](#) that the objective is nonconstant on the affine hull, and follow the exact log-barrier central path $(x(\mu), y(\mu), s(\mu))$. Let B be the maximal primal optimal support and $N = [n] \setminus B$; equivalently, this is the Goldman–Tucker optimal partition. Write P_B and P_N for coordinate projections, and put $D_P = \text{diam } P$.

Choose a fixed central-path tail $0 < \mu \leq \bar{\mu}$ on which

$$S_* = \sup_{0 < \mu \leq \bar{\mu}} \max_j s_j(\mu) < \infty, \quad \beta = \inf_{0 < \mu \leq \bar{\mu}} \min_{j \in B} x_j(\mu) > 0. \quad (68)$$

Strict complementarity of the central-path limit also gives $\underline{s}_N := \inf_{0 < \mu \leq \bar{\mu}} \min_{i \in N} s_i(\mu) > 0$. These constants depend on the instance but not on μ .

Proposition 4.1 (Two-cluster log-barrier Hessian). *Let*

$$H_\mu = Z^\top X(\mu)^{-2} Z = H_{\text{red}}(\mu), \quad r = \text{rank}(P_N Z), \quad (69)$$

where the equality uses the reduced-Hessian notation of [section 2.1.1](#) with $W = Z$. If $r > 0$, let $\sigma > 0$ be the smallest positive singular value of $P_N Z$. The largest r eigenvalues of H_μ then lie in

$$\left[\frac{\sigma^2 \min_{i \in N} s_i(\mu)^2}{\mu^2}, \frac{\max_{i \in N} s_i(\mu)^2}{\mu^2} + \beta^{-2} \right]. \quad (70)$$

If $p - r > 0$, the remaining $p - r$ eigenvalues lie in

$$[D_P^{-2}, \beta^{-2}]. \quad (71)$$

Each nonempty interval has endpoint ratio bounded independently of μ . When both intervals are nonempty, their separation grows as $\Theta(\mu^{-2})$; when $r = 0$ the statement about the top cluster is empty.

Proof. Complementarity gives $x_i(\mu)^{-2} = s_i(\mu)^2/\mu^2$. Split

$$H_\mu = H_{\mu,N} + H_{\mu,B}, \quad H_{\mu,N} = Z^\top P_N^\top X_N^{-2} P_N Z, \quad H_{\mu,B} = Z^\top P_B^\top X_B^{-2} P_B Z. \quad (72)$$

The first summand is positive semidefinite of rank r . When $r > 0$, its nonzero eigenvalues are the squared nonzero singular values of $X_N^{-1} P_N Z$ and therefore lie in

$$\left[\frac{\sigma^2 \min_{i \in N} s_i(\mu)^2}{\mu^2}, \frac{\max_{i \in N} s_i(\mu)^2}{\mu^2} \right].$$

The tail bound on x_B gives $0 \preceq H_{\mu,B} \preceq \beta^{-2} I$. Weyl monotonicity now gives the lower edge in (70), and Weyl's upper inequality gives its upper edge. Since $\ker H_{\mu,N} = \ker(P_N Z)$ has dimension $p - r$, the min-max principle applied to this subspace gives $\lambda_{p-r}^\uparrow(H_\mu) \leq \beta^{-2}$ when $p - r > 0$, where eigenvalues are ordered increasingly. Hence every one of the lowest $p - r$ eigenvalues has the claimed upper bound.

It remains to bound their lower edge. For every Euclidean unit tangent direction, one of its two signs exits the compact polytope within distance D_P . The Dikin containment argument in theorem 3.2 therefore gives $H_\mu \succeq D_P^{-2} I$. Finally, (68) and $\underline{s}_N > 0$ bound each nonempty interval's endpoint ratio independently of μ . If both clusters are nonempty, the lower endpoint of (70) is $\Theta(\mu^{-2})$ while (71) is fixed. $\square$

The two extreme cases agree with the geometry in section 3. At a unique nondegenerate optimum—equivalently here, a singleton optimal face with a nondegenerate optimal basis— $r = p$: if $h \in \ker A$ and $h_N = 0$, then the optimal basis gives $A_B h_B = 0$ and hence $h = 0$. The bottom cluster is empty and the condition number remains bounded. If the optimal face has positive dimension, $\ker(P_N Z)$ contains its tangent space and the bottom cluster is nonempty; its fixed scale and the μ^{-2} top scale recover the chord-law divergence.

The two-cluster structure is specific to the log barrier of an LP near the optimum. On the log-det central path of the SDP (61) from proposition 3.9, the four reduced-Hessian eigenvalues, in increasing order and in the Frobenius metric, satisfy

$$\lambda_1 \sim \sqrt{2} \mu^{-1/2}, \quad \lambda_2 \sim \mu^{-1}, \quad \lambda_3 \sim \sqrt{2} \mu^{-3/2}, \quad \lambda_4 \sim \frac{4}{7} \mu^{-2}.$$

To derive these, use tangent coordinates (b, g, t, e) for the symmetric matrix with diagonal $(1 - g - b, g, b)$ and upper entries (b, t, e) . At the center $t = e = 0$, write $q = ag - b^2$ and $A_2 = \begin{pmatrix} a & b \\ b & g \end{pmatrix}$. On the coordinates (t, e) , the reduced Hessian acts in the Frobenius metric as A_2^{-1}/b . This gives the first and third scales because $b \sim \sqrt{\mu/2}$ and $q \sim \mu$. In the coordinates (b, g) , the Frobenius metric has Gram matrix $G = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}$, so the reduced Hessian on this block is $G^{-1}K$, where K is the coordinate Hessian. Direct differentiation gives $\mu K_{bb} \rightarrow 6$, $\mu^{3/2} K_{bg} \rightarrow -\sqrt{2}$, and $\mu^2 K_{gg} \rightarrow 1$. Thus the larger eigenvalue has coefficient $(G^{-1})_{22} = 4/7$, while $\mu^3 \det K \rightarrow 4$ and $\det G = 7$ give coefficient one for the smaller eigenvalue. The four distinct powers of μ fix the order for small μ . In particular, no eigenvalue stays bounded, unlike the bottom cluster of proposition 4.1. The complete calculation, including the actual second derivative, the characteristic polynomial with multiplicities, and the full central-path tail, is verified in Lean, as documented in `formal/FRACTIONAL_SDP.md`.

4.2 Why conjugate gradients see the clusters

Convergence bounds for CG on clustered spectra are classical [16]. The following elementary construction suffices here and makes the logarithmic dependence on the gap between the clusters explicit.

Proposition 4.2 (CG needs few iterations on two intervals). *Let $H \succ 0$ have*

$$\text{spec}(H) \subseteq [a_1, b_1] \cup [a_2, b_2], \quad 0 < a_1 \leq b_1 < a_2 \leq b_2,$$

and set $\rho_i = b_i/a_i$ and $\Lambda = b_2/a_1$. Thus $\kappa(H) \leq \Lambda$, with equality only when a_1 and b_2 are eigenvalues of H . For every $0 < \epsilon \leq 1/2$ there is a residual polynomial q , with $q(0) = 1$ and $\max_{\lambda \in \text{spec}(H)} |q(\lambda)| \leq \epsilon$, of degree

$$O(\sqrt{\rho_1 \rho_2} \log \Lambda \log(1/\epsilon) + \sqrt{\rho_2} \log(1/\epsilon)). \quad (73)$$

Consequently CG reaches relative H -norm error ϵ in $O_\rho(\log(\Lambda/\epsilon) \log(1/\epsilon))$ iterations, where O_ρ hides factors depending on ρ_1 and ρ_2 .

Proof. If $a_2 < b_2$, let p_2 be the shifted-Chebyshev residual for $[a_2, b_2]$, normalized by $p_2(0) = 1$, of degree $m_2 = \lceil \sqrt{\rho_2} \rceil$. The standard Chebyshev estimate gives

$$\max_{[a_2, b_2]} |p_2| \leq 2e^{-2m_2/\sqrt{\rho_2}} \leq \frac{1}{2}.$$

On $[0, a_2]$ its magnitude is at most one: after the affine map from $[a_2, b_2]$ to $[-1, 1]$, the normalized Chebyshev quotient is monotone in magnitude between the image of zero and the left endpoint. At the edge $a_2 = b_2$, instead take $p_2(x) = 1 - x/a_2$ and $m_2 = 1$; the same two bounds hold.

If $a_1 = b_1$, take the exact factor $r_1(x) = 1 - x/a_1$ and $m_1 = 1$. Otherwise let r_1 be the shifted-Chebyshev residual for $[a_1, b_1]$ of degree $m_1 = O(\sqrt{\rho_1} \log(1/\epsilon))$, with $r_1(0) = 1$ and $\max_{[a_1, b_1]} |r_1| \leq \epsilon$. In its explicit quotient, the denominator is $T_{m_1}(\eta_1)$, where $\eta_1 = (a_1 + b_1)/(b_1 - a_1) > 1$, and $T_{m_1}(\eta_1) \geq \eta_1^{m_1}$. Combining this denominator bound with $|T_m(t)| \leq (2|t|)^m$ for the numerator outside the approximation interval gives, also for the exact-factor edge case,

$$\max_{[a_2, b_2]} |r_1| \leq (4\Lambda)^{m_1}. \quad (74)$$

Choose

$$j = \lceil m_1 \log_2(4\Lambda) + \log_2(1/\epsilon) \rceil, \quad q = r_1 p_2^j.$$

Then $q(0) = 1$. On the first interval, $|q| \leq \epsilon$ because $|p_2| \leq 1$; on the second, (74) and the choice of j again give $|q| \leq \epsilon$. Its degree is $m_1 + jm_2$, which is (73). CG minimizes the H -norm error over residual polynomials of its degree, so its error is no larger. $\square$

The accompanying analysis scripts implement the float64 CG test below, the Remez computation, the right-hand-side experiments, the adaptive damped-Newton experiment, and the Netlib survey with its `adlittle` follow-up; the 80-bit check below is not included. We checked [proposition 4.2](#) numerically on the *40-variable simplex LP* $\min\{x_{39}/2 + x_{40} : \sum_{i=1}^{40} x_i = 1, x \geq 0\}$, whose optimal face, where $x_{39} = x_{40} = 0$, has dimension 37; [section 4.4](#) uses the same instance. From $\mu = 10^{-2}$ to 10^{-8} , κ grew from about 10 to 7×10^{12} , the bottom-cluster ratio stayed below 1.006, and the top ratio approached 4.004 (from 2.44 at the shallowest point). For random right-hand sides, CG reduced the recursively updated residual below 10^{-8} in 3–8 iterations. At $\mu = 10^{-8}$, however, the recursive residual after eight steps hides a true relative residual $\|b - Hx\|_2 / \|b\|_2 \approx 10^{-4}$ and a relative H -norm error of about 6×10^{-6} , caused by loss of conjugacy in float64. A separate 80-bit check needed six iterations, so the small iteration count is not a precision artifact. The unstructured $\sqrt{\kappa}$ estimate is about 2.6×10^6 iterations.

4.3 Quantum polynomial solvers in two access models

CG needs its residual polynomial to be small only at the eigenvalues. QSVT requires its normalized polynomial to be bounded on all of $[-1, 1]$, including the gap between the clusters and the negative half of the interval, so the composite polynomial of [proposition 4.2](#) cannot be

used directly. For representative intervals, a rough evaluation of that polynomial inside the gap gives about 10^{85} ; we do not use this estimate below.

Normalize $\|H\|_2 = 1$ and write

$$\text{spec}(H) \subseteq [1/\Lambda, \rho_1/\Lambda] \cup [1/\rho_2, 1], \quad \Lambda = \kappa(H), \quad (75)$$

where $\rho_1, \rho_2 \geq 1$ are fixed. Cost statements below assume norm-tight encodings, $\alpha_H = \Theta(\|H\|)$ and $\alpha_G = \Theta(\|G\|)$. Without this hypothesis the plain-access parameter is $\kappa_{\text{BE}} = \alpha_H \|H^{-1}\|$ from equation (24), and the factor parameter is $\kappa_G = \alpha_G \|H^{-1}\|^{1/2}$.

Theorem 4.3 (Clustered solves in the two access models). *For the two-cluster spectra (75):*

- (i) *In the one-sided polynomial model, where polynomials need to be bounded only on $[0, 1]$, a residual polynomial q of degree d bounded by M on $[0, 1]$, with $q(0) = 1$ and $|q(1/\Lambda)| \leq \epsilon \leq 1/2$, has degree*

$$d \geq \sqrt{\frac{(1-\epsilon)\Lambda}{2M}}. \quad (76)$$

Thus the product dM of degree and amplitude bound satisfies $dM = \Omega(\sqrt{\Lambda})$.

- (ii) *Let $\rho_1 \geq 2$. In the fixed-polynomial plain-access model, let p be a single polynomial of degree d that approximates $1/y$, in the rescaled variable $y = \Lambda\lambda$, with relative error at most $1/8$ on a bottom interval containing $y = 1$ and $y = 2$, and that is bounded by M on the symmetric interval $[-\Lambda, \Lambda]$ required by QSVT. Then*

$$dM \geq \frac{5}{16} \sqrt{\Lambda^2 - 4} = \Omega(\Lambda). \quad (77)$$

This bounds the polynomial cost dM for a right-hand side supported on the bottom cluster; it does not by itself bound the cost of a general QLS.

- (iii) *With factor access $H = G^\dagger G$, there is a one-sided construction, implemented as an even polynomial in the singular values of G , with $dM = \tilde{O}(\sqrt{\Lambda})$. Item (i), applied to the residual of this construction, gives the matching lower bound $\Omega(\sqrt{\Lambda})$ on the degree.*
- (iv) *Let N_{QLS} denote the Hilbert-space dimension, distinct from the optimal-partition index set N above. In the full query model, plain access has $\tilde{\Theta}(\kappa)$ worst-case QLS cost for $N_{\text{QLS}} \gtrsim \kappa$, even for two-point clusters. More precisely, Orsucci and Dunjko [118, Proposition 6] prove an $\Omega(\min\{\kappa, N_{\text{QLS}}\})$ normalized-block-encoding lower bound for all algorithms. Their Proposition 7 gives the sparse/Grover lower bound $\Omega(\min\{\kappa, \sqrt{N_{\text{QLS}}}\})$, which is linear in κ only when $N_{\text{QLS}} \geq \kappa^2$.*

Proof. For (i), the mean-value theorem gives a point $\xi \in (0, 1/\Lambda)$ with $|q'(\xi)| \geq (1-\epsilon)\Lambda$. Markov's inequality, rescaled from $[-1, 1]$ to $[0, 1]$, gives $\max_{[0,1]} |q'| \leq 2d^2 M$. Rearranging proves (76). Since $M \geq |q(0)| = 1$, multiplying the bound by M gives $dM = \Omega(\sqrt{\Lambda})$.

For (ii), use the rescaled eigenvalue $y = \Lambda\lambda$. Relative error $\delta' \leq 1/8$ for the target $1/y$ implies

$$p(1) \geq 1 - \delta', \quad p(2) \leq \frac{1 + \delta'}{2}, \quad p(1) - p(2) \geq \frac{5}{16}.$$

The mean-value theorem therefore gives a $\xi \in (1, 2)$ with $|p'(\xi)| \geq 5/16$. Bernstein's inequality on the symmetric interval states

$$|p'(y)| \leq \frac{dM}{\sqrt{\Lambda^2 - y^2}}.$$

Since $|\xi| < 2$, this yields (77).

The quantity dM measures cost only once the support of the right-hand side is fixed. For a right-hand side in the bottom cluster, normalizing a polynomial bounded by M reduces the amplitude of the useful output by $\Theta(M)$, so with amplitude amplification dM is a proxy for the cost. For a general right-hand side the success amplitude also depends on its spectral overlap. Indeed, on a two-point spectrum the degree-one separator $p(y) = 1 - (y - 1)/\Lambda$ has $dM = O(1)$ even though the all-algorithms result in item (iv) still gives an $\Omega(\kappa)$ worst case. Thus item (ii) is not the basis for the plain-access row of the cost table; item (iv) is.

For (iii), the matching lower bound is item (i) applied to the residual of the construction below. Writing $q(y) = 1 - y p_y(y)$ in the rescaled variable $y = \Lambda\lambda$, one has $q(0) = 1$, $|q(1)| \leq e^{-T} + O(\epsilon) = O(\epsilon)$, and $|q(y)| \leq e^{-Ty} + y |p_y(y) - f_T(y)| \leq 1 + O(\epsilon)$ on $[0, \max\{L_G, \Lambda\}] \supseteq [0, \Lambda]$, the approximation interval of the construction below. In λ units this is a residual polynomial of degree $d + 1$ bounded by $1 + O(\epsilon)$ on $[0, 1]$, so item (i) forces $d = \Omega(\sqrt{\Lambda})$; since $M \geq |p_y(1)| \geq 1 - O(\epsilon)$, also $dM = \Omega(\sqrt{\Lambda})$. Here M is the amplitude bound of the inverse polynomial in y units, whose target is $1/y$; an inverse approximant in λ units necessarily has bound at least $(1 - \epsilon)\Lambda$, so the two normalizations are not interchangeable.

Now include the encoding normalization explicitly. A block encoding of G presents singular values $s = \sigma/\alpha_G$ in $[-1, 1]$. Put $L_G = \Lambda\alpha_G^2$. If p_y is bounded on $[0, L_G]$, then

$$q_G(s) = p_y(\Lambda\alpha_G^2 s^2) \quad (78)$$

is even and bounded on $[-1, 1]$. On a singular value of G , its argument is $\Lambda\sigma^2$, exactly the one-sided eigenvalue variable for $G^\dagger G$. Norm-tightness gives $\alpha_G = \Theta(\|G\|) = \Theta(1)$ under (75), so $L_G = \Theta(\Lambda)$; it does not set α_G equal to one.

It remains to construct the one-sided polynomial with $dM = \tilde{O}(\sqrt{\Lambda})$. In $y = \Lambda\lambda$ units define

$$f_T(y) = \frac{1 - e^{-Ty}}{y} = \int_0^T e^{-ty} dt, \quad T = \Theta(\log(1/\epsilon)).$$

For $y \geq 1$, its relative error as an approximation to $1/y$ is at most e^{-T} , and $|f_T(y)| \leq T$ on $[0, L'_G]$, where $L'_G = \max\{L_G, \Lambda\} = \Theta(\Lambda)$. The shifted-Chebyshev expansion of e^{-ty} on this interval, truncated uniformly for $0 \leq t \leq T$, has degree

$$O\left(\sqrt{L'_G T \log(L'_G T/\epsilon)} + \log(L'_G T/\epsilon)\right).$$

Integrating its coefficients over t produces a polynomial $p_y(y)$ approximating f_T to absolute error $O(\epsilon/\Lambda)$, hence to relative error $O(\epsilon)$ for every $y \in [1, \Lambda]$. Its bound is $M = O(T)$, so $dM = \tilde{O}(\sqrt{\Lambda})$ under norm-tight access. The even polynomial q_G in (78) has only twice the degree. This makes the construction admissible in the factor-access polynomial model and proves item (iii). The corresponding solve through the factor is algebraically simple. For a full-column-rank factor, first form $w = (G^\dagger)^+ b$ and then $x = G^+ w$; an SVD verifies $x = (G^\dagger G)^{-1} b$. Define the two relative output amplitudes

$$\zeta_1 = \frac{\|(G^\dagger)^+ b\|}{\|(G^\dagger)^+ \| \|b\|}, \quad \zeta_2 = \frac{\|G^+ w\|}{\|G^+ \| \|w\|}.$$

Each pseudoinverse application costs $\tilde{O}(\kappa_G)$ to constant relative amplitude. The end-to-end $\tilde{O}(\kappa_G) = \tilde{O}(\sqrt{\kappa})$ solve follows only under the explicit overlap hypothesis $\zeta_1, \zeta_2 = \Omega(1)$; otherwise amplitude amplification multiplies the cost by the inverse overlaps.

Factor access does not give a general QLS algorithm of cost $\tilde{O}(\sqrt{\kappa})$. On the Orsucci–Dunjko two-point instance, the square-root factor is diagonal and its oracle is implementable with $O(1)$ hidden-input queries. A generic square-root-cost factor solver would therefore contradict their $\Omega(\kappa)$ query lower bound. The factor statement here is a bound on the polynomial cost dM together with a solve under the overlap hypothesis, not an unrestricted QLS upper bound.

Item (iv) is the prior lower bound, valid for all algorithms, on which the plain-access worst case rests. Together with the standard $\tilde{O}(\kappa_{\text{BE}})$ block-encoding QLS upper bound, it establishes the norm-tight $\tilde{\Theta}(\kappa)$ worst-case row below. $\square$

When $\rho_1 = 1$, item (ii) is vacuous because its two interpolation points are absent; this does not make the case algorithmically easy. Orsucci–Dunjko’s all-algorithms lower bound in item (iv) already covers exact two-point clusters.

Boundedness on the symmetric interval is essential. The polynomial in the last proof is well behaved on $[0, \Lambda]$ but grows on the negative half; it is not a plain-access QSVT polynomial. The value $f_T(0) = T \neq 0$ also prevents the odd extension used for the usual $1/x$ target. A discretized Remez linear program shows the difference: it gives the minimum degree of a polynomial with $p(1/\Lambda) \leq 0.1$ and $p(2/\Lambda) \geq 0.9$ that is bounded by one on $[-1, 1]$ or on $[0, 1]$.

Λ	$d_{[-1,1]}$	$d_{[-1,1]}/\Lambda$	$d_{[0,1]}$	$d_{[0,1]}/\sqrt{\Lambda}$
50	52	1.04	12	1.70
100	102	1.02	17	1.70
200	205	1.02	24	1.70
400	376	0.94	33	1.65

Thus the minimum degree is 0.94–1.04 times Λ on the symmetric interval and about $1.7\sqrt{\Lambda}$ on $[0, 1]$.

The following table compares two-cluster solve costs, suppressing logarithmic factors in the precision.

solver	matrix access	two-cluster cost
classical CG	matrix–vector products	$\text{polylog}(\kappa)$
factor polynomial route ¹	norm-tight $H = G^\dagger G$	$\tilde{\Theta}(\sqrt{\kappa})$
QLS worst case, $N_{\text{QLS}} \gtrsim \kappa$	norm-tight plain encoding of H	$\tilde{\Theta}(\kappa)$

With the convention $D = X^{1/2}S^{-1/2}$ of [section 2.1.1](#), the normal-equations matrix is $H = AD^2A^\top$, and the factor $G = DA^\top$ satisfies $H = G^\dagger G$. This factor is available from the IPM data when an oracle for G can be implemented. A solver should then act on G directly; composing it into an encoding of H would square its condition number.

The plain-access lower bound for all algorithms is due to Orsucci and Dunjko. Somma and de Wolf’s sum-of-squares separation is also prior work, but it concerns guided ground-state energy estimation, under their dimension condition $\dim \geq \log(1/\epsilon)/\gamma_{\text{guide}}^2$, where γ_{guide} is their guide overlap, and their stated error assumptions [125]. It separates $\Omega(1/\delta)$ resolution with plain access from $\Omega(1/\sqrt{\delta})$ resolution with sum-of-squares access; we do not claim that this result applies to QLS. Within the polynomial model, the matching lower bound for the even (one-sided) factor polynomials is item (i). To our knowledge, what is new is the observation that CG solves the hard two-cluster quantum instances in $O(1)$ iterations when the clusters are points, together with the resulting separation on central-path Newton spectra. We do not claim the quantum lower bounds as new.

Open Problem 4.4 (Exact-subnormalization shift). The shift $H \mapsto I - H$ moves the bottom cluster to an endpoint of the spectrum, where distinguishing eigenvalues at scale $1/\Lambda$ would need polynomial degree only of order $\sqrt{\Lambda}$. Can a subnormalization-one block encoding of $I - H$ be constructed from plain access to H without paying the apparent $\Theta(\Lambda)$ uniform amplification cost near the new spectral edge?

¹The factor row is a bound on the polynomial cost; it is an end-to-end solve cost only under the overlap hypothesis $\zeta_1, \zeta_2 = \Omega(1)$; it is not a generic factor-access QLS guarantee.

Orsucci and Dunjko raise the closely related normalization problem for $B = I - \eta A$ (in their notation) in their Section 4.3 [118]. They observe that a linear combination of I and a normalized block encoding of A yields only the subnormalized B/β for some $\beta > 1$, and that any black-box amplification of B/β to B is in general inefficient, by the lower bound of [65, Theorem 73] applied to $f(x) = \beta x$. They leave open whether a normalized block encoding of a matrix with operator norm at most one can be built from a sparse-matrix oracle for it, and construct normalized block encodings of B for diagonally dominant A and for sums of positive semidefinite local Hamiltonians. [Open Problem 4.4](#) is a promise version of their black-box obstruction: the subnormalization to remove is the fixed factor two, the spectrum of H is promised to satisfy (75), and the question is whether this promise lowers the $\Theta(\Lambda)$ uniform-amplification degree at the shifted edge, not whether a general normalization procedure exists.

4.4 The central path does not use a worst-case right-hand side

The lower bounds above concern worst-case right-hand sides. On the exact log-barrier path, the Newton right-hand side is much more special. Let $P_{\text{bot}}(\mu)$ denote the spectral projector of H_μ onto the bottom cluster in [proposition 4.1](#).

Proposition 4.5 (Universal Newton right-hand side and weak-mode alignment). *In the setting of [proposition 4.1](#), at an exact central point define $v_\mu = Z^\top X^{-1}e$. For a centering target $\mu_2 = \gamma\mu$ with $0 < \gamma < 1$, the exact reduced Newton equations are*

$$\mu_2 H_\mu d_c = r_c, \quad r_c = -(1 - \gamma)\mu v_\mu, \quad \mu H_\mu d_a = r_a, \quad r_a = -\mu v_\mu. \quad (79)$$

Here d_a is the affine-scaling direction computed in the barrier metric at the current point. We write the centering parameter as γ because σ denotes the singular value in [proposition 4.1](#). If the right-hand side is nonzero, then

$$\frac{\|P_{\text{bot}}(\mu)r\|_2}{\|r\|_2} = O(\mu^2) \quad (r = r_c \text{ or } r_a). \quad (80)$$

For either equation in (79), if the bottom component of its exact solution is dropped, the resulting relative Euclidean residual is exactly the fraction in (80).

Proof. At exact centrality, $Z^\top(c - \mu X^{-1}e) = 0$. Negating the reduced gradient for the target μ_2 gives

$$-Z^\top(c - \mu_2 X^{-1}e) = -(1 - \gamma)\mu Z^\top X^{-1}e,$$

and setting the target barrier term to zero gives $-Z^\top c = -\mu Z^\top X^{-1}e$. This proves the right-hand sides in (79). The reduced Hessian of the target barrier objective is $\mu_2 H_\mu$; the affine direction uses the current local metric μH_μ . Thus the scalar factors μ_2 and μ in (79) are also exact.

For the alignment claim, recall the splitting $H_\mu = H_{\mu,N} + H_{\mu,B}$ in (72). The bottom eigenspace of $H_{\mu,N}$ is exactly

$$\mathcal{T}_\Phi = \ker(P_N Z), \quad (81)$$

the reduced tangent space of the primal optimal face. The positive spectrum of $H_{\mu,N}$ begins at $\Theta(\mu^{-2})$, whereas $\|H_{\mu,B}\|_2 = O(1)$. The Davis–Kahan theorem therefore gives

$$\|P_{\text{bot}}(\mu) - P_{\mathcal{T}_\Phi}\|_2 = O(\mu^2) \quad (82)$$

[54].

Let x^a be the analytic center of the primal optimal face. The log-barrier path converges to x^a , and strict complementarity gives the tail expansion $x_B(\mu) = x_B^a + O(\mu)$ [101, 134]. For every reduced unit vector $u \in \mathcal{T}_\Phi$, the ambient direction Zu has $(Zu)_N = 0$, and first-order optimality of x^a on the face says

$$\langle (Zu)_B, (X_B^a)^{-1}e_B \rangle = 0.$$

Since the coordinates of x_B^a are positive, the tail expansion then gives $\langle u, Z^\top X^{-1} e \rangle = O(\mu)$. On the other hand $\|X^{-1} e\|_2 = O(1/\mu)$ because $x_N = \mu S_N^{-1} e_N$. Combining these facts with (82) yields

$$\|P_{\text{bot}}(\mu) v_\mu\|_2 = O(\mu). \quad (83)$$

Centrality also gives $v_\mu = Z^\top c/\mu$. The nonconstant-objective hypothesis makes $Z^\top c \neq 0$, so $\|v_\mu\|_2 = \Theta(1/\mu)$; dividing (83) proves (80).

Finally, write $d = d_{\text{top}} + d_{\text{bot}}$ for either system $\tau H_\mu d = r$, where $\tau = \mu_2$ for centering and $\tau = \mu$ for the affine direction, and use $\tilde{d} = d_{\text{top}}$. Then

$$\tau H_\mu \tilde{d} - r = -\tau H_\mu d_{\text{bot}} = -P_{\text{bot}}(\mu) r.$$

Taking Euclidean norms proves the exact residual identity. $\square$

Corollary 4.6 (Condition-number-free top-window solve). *Suppose an inexact Newton step only requires a relative linear-system residual at most η , with η independent of μ . On the exact log-barrier path, once $\mu = O(\sqrt{\eta})$, it suffices to invert the top cluster. Plain-access QSVT can do so with degree*

$$O(\rho_2 \log(\rho_2/\eta)), \quad (84)$$

independent of $\kappa(H_\mu)$.

Proof. Dropping the bottom solution component contributes $O(\mu^2)$ relative residual by [proposition 4.5](#); choose the tail constant so that this is at most $\eta/2$. After scaling the top interval to $[1/\rho_2, 1]$, the standard bounded QSVT approximation to $1/x$ has degree $O(\rho_2 \log(\rho_2/\eta))$ and can make the remaining residual at most $\eta/2$. The interval stays a fixed distance from zero, so this polynomial is admissible with plain access and has no dependence on the remote bottom cluster. $\square$

The accuracy requirement in this corollary is the relative residual of the reduced primal system, $\|\tau H_\mu \tilde{d} - r\|/\|r\|$, with $\tau = \mu_2$ for centering and $\tau = \mu$ for the affine direction. Here r_c denotes the reduced primal right-hand side in (79), not the KKT complementarity residual denoted by r_c in [section 2.1.1](#). This requirement differs from the absolute residual conventions for OSS and MNES in [definition 2.8](#): OSS requires an $O(\bar{\mu})$ residual, while MNES requires an $O(\sqrt{\bar{\mu}/n})$ system residual chosen to produce an $O(\bar{\mu})$ complementarity residual. Once the two residual vectors are identified, a relative tolerance η_{rel} implies an absolute requirement $\|\text{res}\| \leq \tau_{\text{abs}}$ only when $\eta_{\text{rel}}\|r\| \leq \tau_{\text{abs}}$. Here $r_a = -Z^\top c$ and $r_c = -(1 - \gamma)Z^\top c$ have norms independent of μ , so a constant relative tolerance does not by itself imply either absolute requirement, both of which shrink with $\bar{\mu}$.

In experiments, the $O(\mu^2)$ law holds with a small constant. On Netlib **afiro**, the dropped-bottom residual was 6.7×10^{-6} , 6.4×10^{-10} , and 6.7×10^{-14} at $\mu = 10^{-1}, 10^{-3}, 10^{-5}$, respectively: an instance constant of about 7×10^{-4} multiplying μ^2 . The centering and affine right-hand sides gave the same values, as (79) predicts; the top-cluster fraction of the right-hand side rounded to 1.0000, and that of the solution was at least 0.9929. For the 40-variable simplex LP, the weak-mode weight was zero to machine precision; by the symmetry among $x_1, \dots, x_{38}$ it is exactly zero on the central path.

We also tested the window solve away from exact central points, in a complete damped-Newton run with shrink factor 0.85 for μ , up to four corrections per value of μ , and fraction-to-boundary steps. The window solver dropped the bottom modes only after detecting a spectral gap of more than two decades; otherwise it solved the full system. The window and exact runs needed the same number of corrections and reached the same final gap: the 40-variable simplex LP finished at gap 1.758×10^{-8} with 205 corrections in both modes, and **afiro** finished at relative gap 7.4×10^{-10} with 393 corrections in both. Intermediate iterates can differ. In a separate experiment, a naive geometric-mean threshold without gap detection stalled on **afiro** at relative

gap 0.98, because it split the early continuous spectrum arbitrarily. The practical rule is therefore to use full solves until a clear gap between clusters opens, and only then drop the weak modes.

4.5 A Netlib conditioning survey

To test how early the asymptotic picture becomes visible, we surveyed 14 small presolved standard-form Netlib LPs. For each instance, a damped Newton method followed the log-barrier path in $\ker A$. We fitted $\kappa_{\text{red}}(g) \asymp g^{-\alpha}$ over the last four grid points, spanning about three gap decades, counted clusters using gaps larger than one decade, and ran CG to 10^{-8} tolerance with a 4000-iteration cap at the deepest point. A second computation estimated the diameter of the objective sublevel polytope $L(g)$ by solving directional LPs over it in 30 fixed random directions. The resulting $\widehat{D}(g)$ only bounds the diameter from below, so fitting $\widehat{D}(g) \asymp g^\theta$ and comparing with $\alpha = 2 - 2\theta$ is a consistency check, computed independently of the Hessian eigenvalues along the path.

instance	m	n	α	κ_{end}	clusters	CG its
adlittle	55	136	2.07	1.7×10^{13}	2	4000 cap
afiro	9	18	2.00	3.8×10^{12}	2	20
beaconfd	11	30	0.02	1.3×10^6	3	28
bore3d	36	65	0.03	3.4×10^{11}	3	521
kb2	46	71	0.36	9.6×10^9	4	198
sc105	34	65	0.03	9.9×10^3	1	55
sc205	67	125	0.03	7.2×10^5	1	286
sc50a	17	33	0.01	1.5×10^3	1	20
sc50b	15	29	0.00	5.8×10^2	1	16
scagr7	93	147	1.30	8.1×10^{12}	2	4000 cap
scorpion	77	136	0.12	1.1×10^5	1	160
seba	10	17	0.00	1.3×10^3	2	7
brandy	107	211	numerical failure			
recipelp	63	107	numerical failure			

Table 2: Tail conditioning fits and deepest-point solver diagnostics. The cluster count uses a one-decade heuristic and should not be read as an exact spectral partition.

Most instances in [table 2](#) fall into two extreme classes. The positive-dimensional-face instances **afiro** and **adlittle** have $\alpha \approx 2$, while the vertex-like **sc** instances, **seba**, **scorpion**, **beaconfd**, and **bore3d** have exponents $\alpha \approx 0$, although their condition numbers can still be large. For **afiro**, $\widehat{D}(g)$ levels off at 126.84, so $\theta \rightarrow 0$ predicts $\alpha = 2.000$, matching the path fit 2.00. For **sc50b**, $\widehat{D}/g$ levels off at 126, so $\theta \rightarrow 1$ predicts $\alpha = 0$, matching its condition-number plateau near 582.

The intermediate **kb2** slope is a crossover, not evidence for a third asymptotic class. Its random-direction estimate of θ climbs from 0.08 to 0.99 across the available gap decades, which is the near-degeneracy plateau of [corollary 3.10](#) appearing in a benchmark instance; the measured $\alpha = 0.36$ lies in the transition and will tend to zero. For **scagr7**, local fits of θ range from 0.1 to 0.6, which may reflect a slower crossover across several scales. We claim no fractional asymptotic exponent for a Netlib instance; the proved fractional example remains the constructed SDP in [proposition 3.9](#).

The instance **adlittle** shows that the split can appear late. We examined it at three values of μ at which float64 eigenvalue computations are reliable. At the largest, its 81 reduced eigenvalues form one cluster with spread ratio $\kappa = 4.3 \times 10^4$ (about 4.6 decades). At the middle value, $\kappa = 1.85 \times 10^7$, they again form one cluster, with spread ratio 1.9×10^7 (about 7.3 decades), and at the smallest value the spectrum spans nine decades at $\kappa = 1.31 \times 10^9$. No one-decade gap appears at any of these three values. CG correspondingly takes 250, 985, and 2935 iterations;

counts above 81 reflect finite-precision loss of conjugacy. The deeper survey row heuristically reports two clusters at $\kappa = 1.7 \times 10^{13}$, where float64 spectral diagnostics are already fragile. This does not contradict [proposition 4.1](#): the asymptotic split exists, but the reliable computations show only that it has not opened by $\kappa \approx 1.3 \times 10^9$.

The limits of float64 arithmetic are visible in `brandy` and `recipelp`: at computed condition numbers above roughly 10^{15} , roundoff produces negative eigenvalues for a mathematically positive-definite reduced Hessian. This is consistent with the classical finite-precision analysis of interior methods [\[135\]](#). Presolve, rank reduction, and Phase-I choices affect the face approached by the path, so every exponent in [table 2](#) belongs to the particular presolved representation. The table reports one deterministic run on cached instances.

4.6 Scope of the right-hand-side result

The two-cluster theorem and the $O(\mu^2)$ right-hand-side fraction use the LP logarithmic barrier, strict complementarity on a sufficiently small tail, and exact centrality. For a general barrier, it is plausible that the first-order optimality condition at the barrier’s center of the optimal face produces an analogous alignment, perhaps only an $O(\mu)$ fraction, but this is a conjecture: neither the two-cluster structure of the Hessian nor the required expansion of the path near its limit has been proved in that generality. Off-path iterates can acquire weak-mode components. Nothing here establishes the same rate for predictor–corrector methods, infeasible-start variants, or unreduced primal–dual systems. Accordingly, [corollary 4.6](#) concerns the exact central path, and the damped-Newton experiment is evidence for the adaptive rule above; neither is a convergence theorem for those variants.

5 Beyond the Condition Number

The condition number is a worst-case spectral parameter: a solver whose cost scales with it pays for a small eigenvalue even when the right-hand side has almost no component in its eigenspace. Dalzell, Li, and Su (DLS) recently gave two quantum linear-system solvers whose costs depend instead on truncation and filtering parameters set by both the matrix and the right-hand side [\[53\]](#); their filtering solver improves an earlier construction of Li [\[94\]](#). The classical idea of effective conditioning is older [\[41\]](#), but the DLS parameters had not been evaluated on IPM Newton systems. We do so here.

[Section 4](#) showed that, for an LP whose optimal face has positive dimension, the reduced Hessian has two spectral clusters whose separation grows along the central path. The normal matrix H_{NE} used here has two clusters with the same normalized scales, $\Theta(1)$ and $\Theta(\mu^2)$, when the LP is primal degenerate ([sections 5.1](#) and [5.2](#)). We call the cluster of small eigenvalues the slow cluster and the other the fast cluster. They correspond to the bottom and top clusters of [section 4](#), but they belong to a different matrix and need not occur for the same LPs. This section asks when the Newton right-hand side lets a solver exploit that separation. Exact block identities give a necessary and sufficient condition, in terms of the coupling between the active and inactive blocks, for filtering to beat the condition number ([section 5.1](#)), and locate a truncation threshold ([section 5.2](#)). A basis-invariant effective spectral dimension gives more general rules for choosing between truncation and filtering ([section 5.3](#)). Finally, when the active range is supplied, a directly assembled operator stays well conditioned as $\mu \rightarrow 0$ and has the same normalized solution as the Newton system ([section 5.4](#)).

In the first part of the section, $H \succ 0$ is given through a block encoding of H/α_H . For a nonzero right-hand side f , set $x = H^{-1}f$ and define

$$\xi(H, f) = \alpha_H \frac{\|H^{-1}f\|}{\|f\|}, \quad \rho(H, f) = \alpha_H \frac{\|H^{-2}f\|}{\|H^{-1}f\|}. \quad (85)$$

The second quantity is the ideal DLS filtering parameter $\|(H/\alpha_H)^{-1}x\|/\|x\|$, not the one-inverse quantity ξ . We write ρ rather than ν , which [section 2.1](#) reserves for the barrier parameter. More generally, for a nonsingular, possibly non-Hermitian matrix A encoded as A/α_A , filtering uses

$$\rho(A, f) = \alpha_A \frac{\|A^{-\dagger}x\|}{\|x\|}, \quad x = A^{-1}f. \quad (86)$$

The DLS truncation parameter instead measures how far toward zero the solver must invert the spectrum to keep all but a prescribed fraction of the normalized solution mass [\(100\)](#).

5.1 Exact-center coupling law

Consider a standard-form LP with full-row-rank A , and suppose its exact central path converges to a strictly complementary solution. Let

$$B = \{i : x_i^* > 0\}, \quad N = \{i : s_i^* > 0\}, \quad U = \text{range}(A_B), \quad K = U^\perp,$$

and assume $0 < \dim U < m$. This is the mixed, primal-degenerate case; if $U = \mathbb{R}^m$, the normalized normal matrix has no eigenvalue cluster tending to zero. At an exact center, complementarity gives

$$H_\mu = H_{\text{NE}}(\mu) := AXS^{-1}A^\top = \mu^{-1}C_\mu + \mu D_\mu, \quad C_\mu = A_B X_B^2 A_B^\top, \quad D_\mu = A_N S_N^{-2} A_N^\top. \quad (87)$$

Here $C_\mu \rightarrow C_0$, $D_\mu \rightarrow D_0$, and, in $U \oplus K$ blocks,

$$D_\mu = \begin{pmatrix} D_\mu^U & F_\mu^\top \\ F_\mu & E_\mu \end{pmatrix}.$$

The restrictions $C = C_0|_U$ and $E = E_0|_K$ are positive definite. Indeed, C is a positive-weight Gram matrix with range U , and if $z \in K$ obeys $A_N^\top z = 0$, then also $A_B^\top z = 0$; full row rank of A forces $z = 0$, proving positivity of E on K . At the optimal limit, $b = Ax^* = A_B x_B^*$, so $b \in U$. A short step from the exact center with target $\sigma_\mu \mu$, $\sigma_\mu \neq 1$, solves $H_\mu \Delta y = (1 - \sigma_\mu)b$; the scalar does not affect normalized QLS parameters, so the next theorem uses b . Here $b \neq 0$: indeed $x^* \neq 0$; if $b = 0$, optimality gives $c^\top x^* = 0$, while a strictly feasible dual slack s would give $s^\top x^* = c^\top x^* > 0$, a contradiction.

Theorem 5.1 (Exact-center coupling criterion). *Define*

$$\begin{aligned} R_\mu &= D_\mu^U - F_\mu^\top E_\mu^{-1} F_\mu, \\ p_\mu &= (C_\mu|_U + \mu^2 R_\mu)^{-1} b, & g_\mu &= E_\mu^{-1} F_\mu p_\mu, \\ q_\mu &= (C_\mu|_U + \mu^2 R_\mu)^{-1} (p_\mu + F_\mu^\top E_\mu^{-1} g_\mu). \end{aligned} \quad (88)$$

Then the following identities hold exactly:

$$H_\mu^{-1} b = \mu \begin{pmatrix} p_\mu \\ -g_\mu \end{pmatrix}, \quad (89)$$

$$H_\mu^{-2} b = \begin{pmatrix} \mu^2 q_\mu \\ -E_\mu^{-1} g_\mu - \mu^2 E_\mu^{-1} F_\mu q_\mu \end{pmatrix}. \quad (90)$$

For a norm-tight encoding, $\alpha_H = \Theta(\|H_\mu\|) = \Theta(\mu^{-1})$ and

$$\xi(H_\mu, b) = \Theta(1), \quad \rho(H_\mu, b) = \Theta\left(1 + \frac{\|g_\mu\|}{\mu^2}\right). \quad (91)$$

In particular, with

$$g = E^{-1} F C^{-1} b, \quad (92)$$

the filtering parameter is asymptotically smaller than the condition number if and only if $g = 0$. If $g \neq 0$, then $\rho(H_\mu, b) = \Theta(\mu^{-2}) = \Theta(\kappa(H_\mu))$; if $g = 0$, then $\rho = O(\mu^{-1}) = O(\sqrt{\kappa(H_\mu)})$, and ρ is bounded if and only if $g_\mu = O(\mu^2)$.

Proof. Write a vector in $U \oplus K$ coordinates. Multiply $H_\mu(u, v)^\top = b$ by μ ; the lower block equation then gives $v = -E_\mu^{-1}F_\mu u$. Substitution in the upper equation gives $(C_\mu|_U + \mu^2 R_\mu)u = \mu b$, proving (89).

Apply the same elimination to $H_\mu(u, v)^\top = H_\mu^{-1}b$. Its lower equation is $F_\mu u + E_\mu v = -g_\mu$, so $v = -E_\mu^{-1}g_\mu - E_\mu^{-1}F_\mu u$. The upper equation then becomes

$$(C_\mu|_U + \mu^2 R_\mu)u = \mu^2(p_\mu + F_\mu^\top E_\mu^{-1}g_\mu),$$

which proves (90).

Continuity and positive definiteness give $p_\mu \rightarrow p = C^{-1}b \neq 0$. Moreover, $q_\mu \rightarrow C^{-1}(p + F^\top E^{-1}g)$, and the vector in parentheses is nonzero because its inner product with p is $\|p\|^2 + \|E^{-1}Fp\|^2 > 0$. Equations (89)–(90) therefore imply

$$\|H_\mu^{-1}b\| = \Theta(\mu), \quad \|H_\mu^{-2}b\| = \Theta(\|g_\mu\| + \mu^2).$$

For the lower bound, write the second inverse as (a_μ, v_μ) , where $a_\mu = \mu^2 q_\mu$. Its lower block equation gives $g_\mu = -E_\mu v_\mu - F_\mu a_\mu$. Boundedness of E_μ, F_μ controls $\|g_\mu\|$ by $\|(a_\mu, v_\mu)\|$, and $q_\mu \rightarrow q \neq 0$ controls μ^2 by the same norm. This also covers cancellation between the two terms in the lower block. Substitution in (85) proves (91). Finally $g_\mu \rightarrow g$. Thus $g \neq 0$ gives the full μ^{-2} scale. Under strict complementarity the LP central path, and hence $C_\mu, D_\mu, F_\mu, E_\mu$ and g_μ , is analytic at $\mu = 0$. Halická proves this boundary analyticity for LP and the corresponding result for SDP [73, 74]. If $g = 0$, differentiability alone gives $g_\mu = O(\mu)$, hence $\rho = O(\mu^{-1})$. More precisely, let g'_0 denote the derivative of the coupling at zero. If $g'_0 \neq 0$, then $\|g_\mu\| = \Theta(\mu)$ and $\rho = \Theta(\mu^{-1})$. If $g'_0 = 0$, the analytic Taylor remainder gives $g_\mu = O(\mu^2)$ and bounded ρ . Thus boundedness is equivalent to $g_0 = g'_0 = 0$ for this analytic family. Equation (91) also shows directly that ρ is bounded exactly when $g_\mu = O(\mu^2)$. $\square$

Equivalently, filtering fails to improve on the condition number exactly when $\Pi_K D_0(C|_U)^{-1}b \neq 0$, with Π_K the orthogonal projector onto K . The limiting right-hand sides without this obstruction form $C(\ker F)$, and the obstruction vanishes for every $b \in U$ exactly when $F = 0$, that is, when D_0 is block diagonal in $U \oplus K$. Analyticity also limits the rates: when $g = 0$, the only divergent case is a first-order term $g_\mu = \Theta(\mu)$, which gives $\rho = \Theta(\mu^{-1})$; vanishing to any higher integer order gives bounded ρ . For an encoding that is not norm-tight, the proof gives the exact scale

$$\rho(H_\mu, b) = \Theta\left(\alpha_H \left(\mu + \frac{\|g_\mu\|}{\mu}\right)\right). \quad (93)$$

Proposition 5.2 (Sparse sharpness witnesses). *There are two LPs whose constraint matrices have at most two nonzeros per row and column, with the same $\kappa(H_\mu) \sim (2\mu^2)^{-1}$ and $\xi = \Theta(1)$, but with $\rho = \Theta(\mu^{-2})$ and $\rho = 1$, respectively.*

Proof. For the coupled instance, take

$$\min x_2 + x_3 \quad \text{subject to} \quad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix} x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x \geq 0. \quad (94)$$

Its primal optimum and optimal partition are unique. The optimal duals have $y_1 = 0, y_2 \in [-1, 1]$; their strictly complementary members have $y_2 \in (-1, 1)$. The displayed slack $s^* = (0, 1, 1)$ corresponds to $y^* = 0$, the dual analytic center and hence the central-path limit, while $x^* = e_1$. The central point is $x = (1 - t, t, t)$, where

$$t = \frac{2 + 3\mu - \sqrt{4 - 4\mu + 9\mu^2}}{4} = \mu + O(\mu^2).$$

With $\theta_1 = (1 - t)^2/\mu$ and $\theta = t^2/\mu$,

$$H_\mu = \begin{pmatrix} \theta_1 + \theta & \theta \\ \theta & 2\theta \end{pmatrix}, \quad \kappa(H_\mu) \sim \frac{1}{2\mu^2}.$$

Here $C = 1$, $E = 2$, $F = 1$, and the centering right-hand side is $b = (1, 0)^\top$, so $p = 1$ and $g = 1/2$. Direct inversion gives

$$H_\mu^{-1}b = \frac{(2, -1)^\top}{2\theta_1 + \theta}, \quad H_\mu^{-2}b \longrightarrow (0, -1/4)^\top.$$

For $\alpha_H = \|H_\mu\|$, $\xi \rightarrow \sqrt{5}/2$ and $\mu^2\rho \rightarrow 1/(2\sqrt{5})$. The normalized solution is exactly $(2, -1)/\sqrt{5}$. Since the slow eigenspace tends to the second coordinate axis, the solution's component in it has norm tending to $1/\sqrt{5}$.

For the decoupled instance, replace the constraint matrix by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

and keep the same objective and right-hand side. This LP has the same unique primal optimum, optimal partition, dual-optimal interval, central-path dual limit, and sparsity bound, but for $0 < \mu \leq 1/\sqrt{2}$ and $\alpha_H = \|H_\mu\|$,

$$H_\mu = \text{diag}(\mu^{-1}, 2\mu), \quad \kappa(H_\mu) = \frac{1}{2\mu^2}, \quad \xi = \rho = 1.$$

Thus neither κ nor ξ distinguishes the two instances; the coupling g does. $\square$

Formal verification and scope. The Lean development in `formal/QipmFormal/Coupling/` checks the split of the LP normal matrix, the centering right-hand side, the block elimination, the inverse identities, the lower bound that rules out cancellation, and the space $C(\ker F)$. It derives (91) from continuous dependence of the blocks on μ , and checks the rate consequences under differentiability and analyticity. The sparse witnesses are checked from their primal and dual equations, with Euclidean norms and the normalizations stated in the proof. `formal/COUPLING.md` lists the checked claims and the reproduction command. The development uses only the axioms `propext`, `Classical.choice`, and `Quot.sound`. The existence of the LP central path and its boundary analyticity are inputs, not formalized; only their consequences for the blocks are checked. The development does not verify the quantum solver, construct its block encoding, or prove the later truncation and spectral-dimension claims.

5.2 Truncation threshold for the slow cluster

Let $P_s(\mu)$ be the spectral projector of H_μ onto its slow cluster, the eigenvalues of order $\Theta(\mu)$, or $\Theta(\mu^2)$ after division by $\|H_\mu\| = \Theta(\mu^{-1})$. For a general nonzero f , define its slow right-hand-side amplitude and slow solution mass by

$$q_s = \frac{\|P_s f\|}{\|f\|}, \quad p_s = \frac{\|P_s H_\mu^{-1} f\|^2}{\|H_\mu^{-1} f\|^2}.$$

Proposition 5.3 (Two-cluster truncation transition). *If the fast-cluster component of f is bounded below, then*

$$p_s = \Theta\left(\frac{q_s^2}{\mu^4 + q_s^2}\right). \quad (95)$$

Consequently, p_s passes from $o(1)$ to $\Theta(1)$ at $q_s = \Theta(\mu^2)$. There, DLS truncation with any fixed target error below the slow mass must use a cutoff $\Theta(\mu^{-2})$, the order of the full normalized

condition number. If $q_s = o(\mu^2)$, truncation can eventually discard the slow cluster at fixed accuracy, although the filtering parameter need not yet be bounded. Under a norm-tight encoding, $q_s = O(\mu^4)$ suffices for a bounded ideal filtering parameter. At exact centering, $q_s = \Theta(\mu^2)$ whenever the limiting coupling is nonzero.

Proof. On the unnormalized fast and slow clusters, H_μ^{-1} has scales $\Theta(\mu)$ and $\Theta(\mu^{-1})$, respectively. The lower bound on the fast part of f therefore gives fast solution norm $\Theta(\mu\|f\|)$, while the slow solution norm is $\Theta(\mu^{-1}q_s\|f\|)$. Dividing the squared slow norm by the sum proves (95).

For the exact-centering vector $b \in U$, standard perturbation theory for separated invariant subspaces, or direct substitution in the block eigenvector equation, gives in the leading K component

$$P_s(\mu)b = -\mu^2 D_{0,KU} C^{-1}b + o(\mu^2). \quad (96)$$

It is therefore $\Theta(\mu^2)$ when the limiting coupling is nonzero. Equation (95) proves the truncation claims. One more inverse multiplies the relative weight of the slow part by another μ^{-2} , so its contribution is bounded only at $q_s = O(\mu^4)$, which is the filtering statement. $\square$

A comparison with [proposition 4.5](#) and [corollary 4.6](#) must distinguish two formulations. This section uses the m -dimensional normal matrix

$$H_{\text{NE}}(\mu) = AXS^{-1}A^\top, \quad \text{spec}(H_{\text{NE}}(\mu)) \subset \Theta(\mu^{-1}) \cup \Theta(\mu),$$

whereas [section 4.4](#) uses the $(n - m)$ -dimensional reduced Hessian

$$H_{\text{red}}(\mu) = Z^\top X^{-2}Z, \quad \text{spec}(H_{\text{red}}(\mu)) \subset \Theta(\mu^{-2}) \cup \Theta(1).$$

When the corresponding clusters are nonempty, both have fast normalized scale $\Theta(1)$ and slow normalized scale $\Theta(\mu^2)$, but they act on different variables and right-hand sides. [Section 4](#) proves an $O(\mu^2)$ slow right-hand-side fraction for $H_{\text{red}}(\mu)$. For $H_{\text{NE}}(\mu)$, the present coupling calculation instead gives $q_s = \Theta(\mu^2)$ only when the limiting coupling is nonzero; the decoupled witness has $P_s b = 0$.

For the normal equation with nonzero limiting coupling, the slow cluster can be discarded when only a fixed relative residual is required, but not when the normalized solution state must be prepared. Let $z = H_{\text{NE}}(\mu)^{-1}b$ and discard $z_s = P_s z$. Because P_s is a spectral projector,

$$\frac{\|H_{\text{NE}}(\mu)(z - z_s) - b\|}{\|b\|} = \frac{\|P_s b\|}{\|b\|} = q_s = \Theta(\mu^2), \quad (97)$$

while (95) gives $\|z_s\|^2/\|z\|^2 = \Theta(1)$, so preparing the state to an error below this constant requires the slow cluster. Thus the two formulations have the same normalized spectral separation, but whether the slow cluster must be resolved depends on both the formulation and the required output; in this sense the result is consistent with [section 4](#).

5.3 Effective spectral dimension

Sharp rules for choosing between truncation and filtering do not require two clusters. Normalize $\|H\| = 1$ and $\|b\| = 1$, let

$$\mathbf{m}_b(S) = \langle b, \mathbf{1}_S(H)b \rangle, \quad (98)$$

and let λ_* be the smallest eigenvalue visible to b , that is, the smallest point in the support of $\mathbf{m}_b$, with $\kappa_* = \lambda_*^{-1}$. We say that (H, b) has effective spectral dimension D if constants $c, C > 0$, independent of λ_* , satisfy

$$ct^{D/2} \leq F_b(t) := \mathbf{m}_b([0, t]) \leq Ct^{D/2} \quad (\lambda_* \leq t \leq 1). \quad (99)$$

This condition is invariant under simultaneous orthogonal changes of basis $(H, b) \mapsto (VHV^\top, Vb)$, so it is not coordinate sparsity in disguise. Because it bounds the cumulative mass from λ_* upward, rather than the mass of separate spectral shells, it controls the mass at the smallest visible eigenvalue while allowing fixed gaps between the first few eigenvalues.

Theorem 5.4 (Spectral-dimension phase laws). *Assume the norm-tight unit normalization: with $\|H\| = 1$, the block-encoding normalization is $\alpha_H = \Theta(1)$, and truncation cutoffs are measured in the same normalization. Define the effective cutoff by*

$$\kappa_{\text{eff}}(\epsilon_{\text{ls}}) := \inf \left\{ K \in [1, \kappa_*] : \frac{\langle b, H^{-2} \mathbf{1}_{[0, 1/K)}(H)b \rangle}{\langle b, H^{-2}b \rangle} \leq \epsilon_{\text{ls}}^2 \right\}. \quad (100)$$

Under (99), this quantity obeys

$$\kappa_{\text{eff}}(\epsilon_{\text{ls}}) = \begin{cases} \Theta_{\epsilon_{\text{ls}}}(\kappa_*), & D < 4, \ 0 < \epsilon_{\text{ls}} < 1, \\ \kappa_*^{1 - \Theta(\epsilon_{\text{ls}}^2)}, & D = 4, \\ \Theta(\min\{\kappa_*, \epsilon_{\text{ls}}^{-4/(D-4)}\}), & D > 4. \end{cases} \quad (101)$$

For $D < 4$, the hidden constant may depend on the fixed error and on the constants c, C . A sublinear power of κ_* is possible only when the allowed error tends to one as $\kappa_* \rightarrow \infty$. At $D = 4$, if F_b has a definite leading coefficient, the exponent is $1 - \epsilon_{\text{ls}}^2 + o(1)$; under (99) alone, the exponent $1 - \Theta(\epsilon_{\text{ls}}^2)$ is sharp, and the cutoff saturates at $\Theta(\kappa_*)$ when $\epsilon_{\text{ls}}^2 \log \kappa_* = O(1)$. The ideal filtering parameter is

$$\rho = \begin{cases} \Theta(\kappa_*), & D < 4, \\ \Theta(\kappa_*/\sqrt{\log \kappa_*}), & D = 4, \\ \Theta(\kappa_*^{(8-D)/4}), & 4 < D < 8, \\ \Theta(\sqrt{\log \kappa_*}), & D = 8, \\ \Theta(1), & D > 8. \end{cases} \quad (102)$$

Thus, at fixed accuracy, the truncation cutoff is independent of κ_* for $D > 4$, whereas the filtering parameter is so only for $D > 8$. For an encoding that is not norm-tight, the effective cutoff scale and every case of the filtering law acquire a factor α_H ; in particular the $D > 8$ filtering cost is $\Theta(\alpha_H)$.

Proof. For $p > 0$, put $Z_p = \langle b, H^{-p}b \rangle$ and write $s = D/2$. Abel summation, equivalently Stieltjes integration by parts, gives the exact layer-cake identity

$$Z_p = 1 + p \int_{\lambda_*}^1 t^{-p-1} F_b(t) dt. \quad (103)$$

Using $F_b(t) = \Theta(t^s)$ in this integral gives

$$Z_p = \begin{cases} \Theta(\kappa_*^{p-D/2}), & D < 2p, \\ \Theta(\log \kappa_*), & D = 2p, \\ \Theta(1), & D > 2p. \end{cases} \quad (104)$$

The squared norm of $H^{-1}b$ is Z_2 . If a cutoff discards eigenvalues strictly below $u = 1/K$, as in (100), its discarded squared solution norm is

$$T_2(u) = u^{-2} F_b(u^-) + 2 \int_{\lambda_*}^u t^{-3} F_b(t) dt, \quad (105)$$

where $F_b(u^-) = \mathfrak{m}_b([0, u))$ is the left limit. For $D < 4$, fix a sufficiently small constant $u_0 > 0$. Dyadically summing the retained moment on $[u, 1]$ and comparing it with Z_2 gives

$$\frac{\langle b, H^{-2} \mathbf{1}_{[u, 1]}(H)b \rangle}{Z_2} = \Theta\left(\left(\frac{\lambda_*}{u}\right)^{2-D/2}\right) \quad (\lambda_* \leq u \leq u_0). \quad (106)$$

Indeed, the upper bound follows by geometric summation, while a shell $[u, \gamma_{\text{sh}}u]$ with a fixed $\gamma_{\text{sh}} > 1$ chosen from c, C has mass $\Theta(u^{D/2})$ and gives the matching lower bound. Requiring the retained fraction to be at least $1 - \epsilon_{\text{ls}}^2$ therefore gives

$$u = \Theta\left(\lambda_*(1 - \epsilon_{\text{ls}}^2)^{-2/(4-D)}\right) = \Theta_{\epsilon_{\text{ls}}}(\lambda_*), \quad (107)$$

for every fixed $\epsilon_{\text{ls}} < 1$, proving the first line. A cutoff $u = \lambda_*^{1-\beta}$ for fixed $\beta > 0$ has retained fraction tending to zero, so it is possible only when $\epsilon_{\text{ls}} \rightarrow 1$. For $D = 4$, both (103) and (105) are logarithmic. Discarding an ϵ_{ls}^2 fraction yields $K = \kappa_*^{1-\Theta(\epsilon_{\text{ls}}^2)}$, with the stated refinement when the leading cumulative coefficient exists. For $D > 4$, the discarded tail is $\Theta(K^{-(D-4)/2})$ relative to the constant total. Setting it to ϵ_{ls}^2 and capping at κ_* proves (101).

For filtering, $x = H^{-1}b$ and hence

$$\rho^2 = \alpha_H^2 \frac{\|H^{-1}x\|^2}{\|x\|^2} = \alpha_H^2 \frac{Z_4}{Z_2} = \Theta\left(\frac{Z_4}{Z_2}\right).$$

Applying (104) with $p = 4$ in the numerator and $p = 2$ in the denominator gives all five cases in (102). $\square$

In the remaining witnesses, e_0, e_{e_1}, e_f are coordinate basis vectors, whereas e_k in the marked-diagonal construction is the k -dimensional all-ones vector; $\delta_c, \delta_Q, \delta_{\mathcal{G}}$ denote, respectively, the corrector perturbation, the projector error, and the expander weight contrast.

Proposition 5.5 (Constant-degree torus witnesses). *Let L be even and T_L^d the periodic d -dimensional grid of side L . With its combinatorial Laplacian $\Delta_{T_L^d}$, set*

$$H_{L,d} = \frac{L^{-2}I + \Delta_{T_L^d}}{L^{-2} + 4d}. \quad (108)$$

Then $H_{L,d}$ is $(2d+1)$ -sparse, has norm one, and has $\kappa = 1 + 4dL^2 = \Theta(L^2)$. The localized right-hand side e_0 has $D = d$, and the dipole $(e_0 - e_{e_1})/\sqrt{2}$ has $D = d + 2$. The dipole regimes are:

physical d	effective D	κ_{eff}	ρ
3	5	$\Theta(\min\{\kappa, \epsilon_{\text{ls}}^{-4}\})$	$\Theta(\kappa^{3/4})$
4	6	$\Theta(\min\{\kappa, \epsilon_{\text{ls}}^{-2}\})$	$\Theta(\kappa^{1/2})$
5	7	$\Theta(\min\{\kappa, \epsilon_{\text{ls}}^{-4/3}\})$	$\Theta(\kappa^{1/4})$
6	8	$\Theta(\min\{\kappa, \epsilon_{\text{ls}}^{-1}\})$	$\Theta(\sqrt{\log \kappa})$
≥ 7	> 8	<i>mesh-independent at fixed ϵ_{ls} mesh-independent</i>	

Proof. Fourier modes diagonalize the torus Laplacian, with

$$\lambda_k = \frac{L^{-2} + 4 \sum_{j=1}^d \sin^2(\pi k_j / L)}{L^{-2} + 4d}.$$

For the localized source, every squared Fourier overlap is L^{-d} . Lattice-point counting bounds, applied first at low frequency, where $\lambda_k \asymp L^{-2}(1 + \|k\|^2)$, and then, in product form, on the remaining fixed spectral range, give

$$\mathbf{m}_{e_0}([0, t]) = \Theta(t^{d/2}) \quad (\lambda_* \leq t \leq 1).$$

Thus (99), including the lower end $t = \lambda_*$, holds with $D = d$. Equivalently, Abel summation gives

$$\sum_k \lambda_k^{-p} = \begin{cases} \Theta(L^{2p}), & d < 2p, \\ \Theta(L^{2p} \log L), & d = 2p, \\ \Theta(L^d), & d > 2p, \end{cases}$$

The dipole Fourier symbol is $1 - e^{2\pi i k_1/L}$. It vanishes on modes with $k_1 = 0$, so no pointwise comparison with λ_k is valid. Instead sum over permutation-symmetric spectral balls. Symmetry gives

$$\sum_{\lambda_k \leq t} \sin^2(\pi k_1/L) = \frac{1}{d} \sum_{\lambda_k \leq t} \sum_{j=1}^d \sin^2(\pi k_j/L).$$

Together with these lattice-point counting bounds, this aggregated identity yields

$$\mathfrak{m}_{(e_0 - e_{e_1})/\sqrt{2}}([0, t]) = \Theta(t^{(d+2)/2})$$

from the smallest visible eigenvalue through $t = 1$. At the bottom, the first visible modes have squared overlap $\Theta(L^{-(d+2)})$, exactly $\Theta(\lambda_*^{(d+2)/2})$; the constant mode has zero overlap and $\lambda_* = \Theta(L^{-2})$. Hence (99) holds with $D = d + 2$. The table follows by substituting $D = d + 2$ in [theorem 5.4](#). $\square$

Both right-hand sides arise in actual QIPM correctors.

Proposition 5.6 (Positive full-step feasible corrector). *Let B_T be an oriented incidence matrix of T_L^d and set*

$$\mathcal{A} = [B_T, I], \quad W = \text{diag}(I_E, L^{-2}I_V), \quad \mathcal{A}W\mathcal{A}^\top = \Delta_{T_L^d} + L^{-2}I. \quad (109)$$

For $d \geq 2$, choose perpendicular torus edges f, g meeting at a vertex, fix $\tau > 0$ and $0 < \delta_c \leq 1/4$, and keep $x_i/s_i = w_i$, where w_i is the corresponding diagonal entry of W . Set

$$p_i = x_i s_i = \begin{cases} \tau(1 + \delta_c), & i = f, \\ \tau(1 - \delta_c), & i = g, \\ \tau, & \text{otherwise,} \end{cases} \quad x_i = \sqrt{p_i w_i}, \quad s_i = \sqrt{p_i / w_i}. \quad (110)$$

With $b_{\text{LP}} = \mathcal{A}x$, $c = s$, and $y = 0$, this is primal and dual feasible, has average complementarity τ , and has centrality $\|XSe - \tau e\|/\tau = \sqrt{2}\delta_c$. The full feasible centering step with $\sigma = 1$ keeps x and s positive, has a normal-equation right-hand side with $D = d + 2$, preserves average complementarity, and satisfies

$$\frac{\|X^+ S^+ e - \tau e\|}{\tau} \leq \frac{\delta_c^2}{1 - \delta_c^2}. \quad (111)$$

Proof. The definitions make feasibility and the centrality calculation immediate. The feasible corrector equations are

$$\mathcal{A}\Delta x = 0, \quad \mathcal{A}^\top \Delta y + \Delta s = 0, \quad S\Delta x + X\Delta s = \tau e - XSe.$$

Eliminating $\Delta x, \Delta s$ gives

$$(\mathcal{A}W\mathcal{A}^\top)\Delta y = -\mathcal{A}S^{-1}(\tau e - XSe). \quad (112)$$

After normalization, its right-hand side is proportional to

$$\frac{\mathcal{A}_f}{\sqrt{1 + \delta_c}} - \frac{\mathcal{A}_g}{\sqrt{1 - \delta_c}}.$$

Its low-frequency Fourier symbol vanishes linearly and has nonzero gradient, so its cumulative spectral weight is $\Theta(t^{(d+2)/2})$. Explicitly, for any nonzero linear form ℓ , lattice-ball comparison gives $\sum_{\|k\| \leq r} |\ell(k)|^2 = \Theta(r^{d+2})$; applying this to the linear part of the symbol gives the claimed cumulative spectral weight.

It remains to prove positivity, preservation of average complementarity, and the centrality bound. Put $u = X^{-1}\Delta x$, $v = S^{-1}\Delta s$, and $P = \text{diag}(p)$. The Newton equations imply

$$P(u + v) = \tau e - p, \quad u^\top P v = \Delta x^\top \Delta s = 0. \quad (113)$$

Thus $P^{1/2}u$ and $P^{1/2}v$ are orthogonal and sum to $P^{-1/2}(\tau e - p)$, whose squared norm is $2\tau\delta_c^2/(1 - \delta_c^2)$. Since $p_i \geq \tau(1 - \delta_c)$, for either $z = u, v$,

$$|z_i|^2 \leq \frac{2\delta_c^2}{(1 - \delta_c^2)(1 - \delta_c)} < 1 \quad (\delta_c \leq 1/4).$$

Hence $x + \Delta x > 0$ and $s + \Delta s > 0$. Moreover

$$p_i^+ = p_i(1 + u_i)(1 + v_i) = \tau + p_i u_i v_i.$$

Equation (113) gives $\sum_i p_i u_i v_i = 0$, so the average remains τ . Finally,

$$\|P(u \circ v)\| \leq \|P^{1/2}u\| \|P^{1/2}v\| \leq \frac{1}{2} \|P^{-1/2}(\tau e - p)\|^2 = \frac{\tau\delta_c^2}{1 - \delta_c^2},$$

which proves (111). $\square$

The localized right-hand side also arises in a QIPM, through an infeasible corrector. At complementarity τ , set $x_i = \sqrt{\tau w_i}$, $s_i = \sqrt{\tau/w_i}$, and $b_{\text{LP}} = \mathcal{A}x$. Give this exactly centered ($x_i s_i = \tau > 0$), primal-feasible point a one-coordinate dual residual $r_d = \eta e_f$. The $\sigma = 1$ infeasible corrector then solves

$$(\mathcal{A}W\mathcal{A}^\top)\Delta y = \eta \mathcal{A}W e_f. \quad (114)$$

If f is the ground variable at vertex zero (a column of the I block of $\mathcal{A}$), the normalized right-hand side is e_0 and $D = d$; if f is a torus edge, it is a dipole and $D = d + 2$.

These witnesses illustrate solver selection, not end-to-end speedups: FFT and multigrid solvers already handle the torus. The constructions concern a single corrector, not an IPM trajectory, and do not count block encoding, norm estimation, state or observable readout, or classical iterate recovery.

5.4 Stable active-subspace equilibration

The obstruction in [theorem 5.1](#) can be removed when the orthogonal projector Q onto the active range is available; put $R = I - Q$. The next theorem assembles a left-equilibrated operator directly from its terms. This direct assembly, not multiplication of separately normalized encodings ([proposition 5.8](#)), is the source of the gain.

Theorem 5.7 (Stable split equilibration). *Suppose*

$$H_\mu = \mu^{-1}C_\mu + \mu D_\mu, \quad 0 < \mu \leq 1,$$

where, for constants independent of μ ,

$$C_\mu = QC_\mu Q, \quad c_-Q \preceq C_\mu \preceq c_+Q, \quad D_\mu \succeq 0, \quad \|D_\mu\| \leq d_+, \quad RD_\mu R \succeq d_-R.$$

Let $P_\mu = \mu Q + \mu^{-1}R$ and assemble

$$M_\mu = C_\mu + RD_\mu + \mu^2 QD_\mu = C_\mu + D_\mu - (1 - \mu^2)QD_\mu. \quad (115)$$

Then $M_\mu = P_\mu H_\mu$ exactly and $\|M_\mu\| + \|M_\mu^{-1}\| = O(1)$. For every $f \in U = \text{range } Q$,

$$M_\mu^{-1}f = \mu^{-1}H_\mu^{-1}f, \quad (116)$$

so the normalized Newton state is preserved exactly, without recovery or postselection. If C_μ, D_μ, Q have block-encoding normalizations $\alpha_C, \alpha_D, 1$, direct LCU assembly has normalization

$$\Gamma_\mu \leq \alpha_C + (2 - \mu^2)\alpha_D. \quad (117)$$

Thus, if α_C and α_D are bounded, a conventional QLS applied to the Hermitian dilation below solves a system whose condition number and normalization are bounded uniformly in μ , and the ideal DLS filtering parameter is also $O(\Gamma_\mu)$.

The left-equilibrated M_μ need not be Hermitian. To apply the Hermitian QLS of [definition 2.16](#), encode its standard dilation

$$\mathcal{H}(M_\mu) = \begin{pmatrix} 0 & M_\mu \\ M_\mu^\dagger & 0 \end{pmatrix}. \quad (118)$$

The dilation has the same block-encoding normalization Γ_μ and the same singular-value condition number as M_μ . Moreover, $\mathcal{H}(M_\mu)^{-1}(f, 0)^\top = (0, M_\mu^{-1}f)^\top$, so selecting the second register preserves the solution normalization and requires no postselection beyond the fixed register label.

Proof. Since $RC_\mu = 0$,

$$P_\mu H_\mu = C_\mu + \mu^2 Q D_\mu + R D_\mu = M_\mu.$$

In $U \oplus U^\perp$ blocks,

$$M_\mu = \begin{pmatrix} C_\mu + \mu^2 D_{UU} & \mu^2 D_{UK} \\ D_{KU} & D_{KK} \end{pmatrix}.$$

The Schur complement of D_{KK} is

$$G_\mu = C_\mu + \mu^2(D_{UU} - D_{UK}D_{KK}^{-1}D_{KU}) \succeq c_- I_U,$$

because the parenthesized Schur complement is positive semidefinite. Also $D_{KK} \succeq d_- I_{U^\perp}$. The block inverse formula therefore bounds every block of M_μ^{-1} using only c_-, d_-, d_+ , while $\|M_\mu\| \leq c_+ + 2d_+$.

For state preservation, let $x_\mu = H_\mu^{-1}f$. Because $f \in U$, $P_\mu f = \mu f$, and hence $M_\mu x_\mu = P_\mu H_\mu x_\mu = \mu f$. This proves (116). Finally, applying LCU to the second form in (115) gives coefficient weight $\alpha_C + \alpha_D + (1 - \mu^2)\alpha_D$, proving (117). Since $\|M_\mu^{-1}\| = O(1)$, the condition number $\Gamma_\mu \|M_\mu^{-1}\|$ of the encoded system, and hence the filtering parameter, is $O(\Gamma_\mu)$, proving the two solver claims. $\square$

The split is also stable under errors in the projector. If $\|\widehat{Q} - Q\| \leq \delta_Q$ and one assembles

$$\widehat{M}_\mu = C_\mu + D_\mu - (1 - \mu^2)\widehat{Q}D_\mu,$$

then $\|\widehat{M}_\mu - M_\mu\| \leq \delta_Q d_+$. For a sufficiently small constant δ_Q , the Neumann perturbation bound and [theorem 5.7](#) give

$$\frac{\|\widehat{M}_\mu^{-1}f - M_\mu^{-1}f\|}{\|M_\mu^{-1}f\|} = O(\delta_Q), \quad (119)$$

uniformly in μ . Thus $\delta_Q = O(\epsilon_{\text{ls}})$ suffices. In contrast, forming $\widehat{P}_\mu H_\mu$, with $\widehat{P}_\mu = \mu\widehat{Q} + \mu^{-1}(I - \widehat{Q})$, from the raw normal matrix loses this stability. For $H_\mu = \text{diag}(\mu^{-1}, \mu)$ and a projector rotated by angle θ , its $(2, 1)$ entry is $(1 - \mu^{-2})\sin\theta\cos\theta$; uniform normalization requires $\sin\theta = O(\mu^2)$. Given a block encoding of A_B/α_B , QSVT can construct Q to the needed accuracy in

$$O\left(\frac{\alpha_B}{\sigma_{\min}^+(A_B)} \log(1/\epsilon_{\text{ls}})\right) \quad (120)$$

queries, which depend on the conditioning of A_B but not on μ .

Proposition 5.8 (Separately normalized composition cannot help). *Let H/α_H and an invertible P/α_P be separately encoded, with $\alpha_H \geq \|H\|$ and $\alpha_P \geq \|P\|$, and compose their encodings. For $x = H^{-1}f$, the DLS parameter of the resulting system $PHx = Pf$ satisfies*

$$\rho_{\text{prod}} = \alpha_H \alpha_P \frac{\|P^{-\dagger} H^{-1} x\|}{\|x\|} \geq \alpha_H \frac{\|H^{-1} x\|}{\|x\|} = \rho_H. \quad (121)$$

Proof. Equation (121) uses the general non-Hermitian parameter (86), because $(PH)^{-\dagger} = P^{-\dagger} H^{-1}$. Since $\alpha_P \geq \|P\|$, every vector y satisfies $\alpha_P \|P^{-\dagger} y\| \geq \|y\|$. Apply this with $y = H^{-1}x$. The displayed expression is precisely the filtering parameter for normalization $\alpha_H \alpha_P$, proving the result. $\square$

This normalization obstruction is known in quantum preconditioning: Lapworth and S nderhauf show explicitly how multiplying separately normalized block encodings can erase a classical gain, while Nie, Lai, and An show how algebraic regrouping can restore it for Pauli structure [90, 114]. Here separate composition has $\alpha_H, \alpha_P = \Theta(\mu^{-1})$, whereas the direct split has $\Gamma_\mu = O(1)$, a normalization smaller by a factor $\Theta(\mu^{-2})$.

Proposition 5.9 (Constant-sparse expander example). *There is a family of constant-sparse standard-form LPs for which the unpreconditioned exact-center Newton system has $\rho = \Theta(\mu^{-2})$, while the operator (115) has $O(1)$ singular-value condition number and $O(1)$ block-encoding normalization and yields exactly the same normalized Newton state at every μ . The uniform bounds hold on $0 < \mu \leq \mu_0$ with μ_0 and all constants depending only on the degree d , the Laplacian gap, and $\delta_{\mathcal{G}}$, not on m .*

Proof. Let $\mathcal{G} = (\mathcal{L}, \mathcal{R}, \mathcal{E})$ be a connected balanced d -regular bipartite expander on m vertices, with constant d and a constant Laplacian gap. Orient each edge from $\mathcal{L}$ to $\mathcal{R}$, let $B_{\mathcal{G}}$ be its signed incidence matrix, and put

$$r = \mathbf{1}_{\mathcal{R}}, \quad \zeta = 2r - e, \quad u_0 = m^{-1/2}e.$$

Here r is the right-part indicator and e is the all-ones vector of the dimension required by context; these symbols are local to this witness and are distinct from the coupling variables above. Then $B_{\mathcal{G}}^\top r = e$ and $B_{\mathcal{G}} e = L_{\mathcal{G}} r = d\zeta$. Fix $0 < \delta_{\mathcal{G}} < 1$, set $\omega_i = 1 + \delta_{\mathcal{G}} \zeta_i$ and $a_i = \sqrt{2/\omega_i}$, and consider

$$\begin{aligned} \min_{x, u, v \geq 0} \quad & \sum_{i=1}^m a_i (u_i + v_i) \\ \text{subject to} \quad & B_{\mathcal{G}} x + u - v = B_{\mathcal{G}} e. \end{aligned} \quad (122)$$

Its constraint matrix $[B_{\mathcal{G}}, I, -I]$ has row and column sparsity $O(d)$.

The point $x = e, u = v = te$ is strictly primal feasible for $t > 0$. Taking $y = -\varepsilon r$ for sufficiently small $\varepsilon > 0$ gives strictly positive dual slacks. Nonnegativity of the objective shows that

$$x^* = e, \quad u^* = v^* = 0, \quad y^* = 0, \quad s_x^* = 0, \quad s_u^* = s_v^* = a$$

is optimal and strictly complementary. Its active face is bounded: if $h \geq 0$ and $B_{\mathcal{G}} h = 0$, any left-vertex row forces all incident components of h to vanish. Also e is the primal face's analytic center because $B_{\mathcal{G}}^\top r = e$.

For completeness, the dual limit is also unique even though the dual optimal face is not a singleton. Dual feasibility and optimality imply $B_{\mathcal{G}}^\top y \leq 0$ and $e^\top B_{\mathcal{G}}^\top y = 0$, hence $B_{\mathcal{G}}^\top y = 0$. Connectedness gives $y = ce$, and the remaining dual inequalities give

$$\{y : y \text{ is dual optimal}\} = \{ce : |c| \leq \min_i a_i\}.$$

On its relative interior the dual analytic-center objective is $\sum_i \log(a_i^2 - c^2)$, which is strictly maximized at $c = 0$. Thus $y(\mu) \rightarrow 0$, and the displayed primal–dual point is the central-path limit.

Along the path,

$$\begin{aligned} H_\mu &= \mu^{-1}C_\mu + \mu D_\mu, \\ C_\mu &= B_{\mathcal{G}} \operatorname{diag}(x_{\mathcal{E}}(\mu)^2) B_{\mathcal{G}}^\top \rightarrow L_{\mathcal{G}}, \\ D_\mu &= \operatorname{diag}\left(\frac{1}{(a_i - y_i(\mu))^2} + \frac{1}{(a_i + y_i(\mu))^2}\right) \rightarrow \operatorname{diag}(\omega_i). \end{aligned} \quad (123)$$

Here $U = \operatorname{range} B_{\mathcal{G}} = e^\perp$ and $Q = I - |u_0\rangle\langle u_0|$. The constants are uniform in m . Write $s_x = -B_{\mathcal{G}}^\top y$, $s_u = a - y$, $s_v = a + y$; the central equations are $x_e s_{x,e} = \mu$, $u_i s_{u,i} = \mu$, $v_i s_{v,i} = \mu$, and $B_{\mathcal{G}}x + u - v = B_{\mathcal{G}}e$. Seek a solution constant on edges and on each side: $x_e = \bar{x}$, $y_i = y_{\mathcal{L}}$ on $\mathcal{L}$, $y_i = y_{\mathcal{R}}$ on $\mathcal{R}$, with $\bar{x} = 1 + \mu^2 w$, $y_{\mathcal{L}} = \mu z_{\mathcal{L}}$, $y_{\mathcal{R}} = \mu z_{\mathcal{R}}$. Since $\omega = 1 \mp \delta_{\mathcal{G}}$ on $\mathcal{L}$ and $\mathcal{R}$, the system reduces to three scalar equations independent of m :

$$dw + \frac{2z_{\mathcal{R}}}{2/(1 + \delta_{\mathcal{G}}) - \mu^2 z_{\mathcal{R}}^2} = 0, \quad -dw + \frac{2z_{\mathcal{L}}}{2/(1 - \delta_{\mathcal{G}}) - \mu^2 z_{\mathcal{L}}^2} = 0, \quad z_{\mathcal{L}} - z_{\mathcal{R}} = \frac{1}{1 + \mu^2 w}.$$

At $\mu = 0$ they have the solution $w = (1 - \delta_{\mathcal{G}}^2)/(2d)$, $z_{\mathcal{L}} = (1 + \delta_{\mathcal{G}})/2$, $z_{\mathcal{R}} = -(1 - \delta_{\mathcal{G}})/2$, with Jacobian determinant $-2d \neq 0$. The implicit function theorem gives a branch on $0 < \mu \leq \mu_0(d, \delta_{\mathcal{G}})$ with $\bar{x} > 0$, $y_{\mathcal{L}} - y_{\mathcal{R}} = \mu/\bar{x} > 0$, and $|y_i| < a_i$; the ansatz therefore satisfies the full central-path system, and uniqueness of the central point identifies it as the central path. Hence $C_\mu = \bar{x}^2 L_{\mathcal{G}}$ exactly, D_μ takes two side-constant values, and the expander gap together with $\omega_i \in [1 - \delta_{\mathcal{G}}, 1 + \delta_{\mathcal{G}}]$ gives the hypotheses of [theorem 5.7](#) with constants depending only on d , the Laplacian gap, and $\delta_{\mathcal{G}}$.

For the exact-centering right-hand side $f = b = B_{\mathcal{G}}e$, $L_{\mathcal{G}}(Qr) = b$ and $Qr = \zeta/2$. The limiting coupling between U and $K = \operatorname{span}\{u_0\}$ is nonzero:

$$u_0^\top \operatorname{diag}(\omega) Qr = \frac{\delta_{\mathcal{G}} \sqrt{m}}{2} = \delta_{\mathcal{G}} \|Qr\|. \quad (124)$$

The coupling theorem therefore gives $\rho(H_\mu, f) = \Theta(\mu^{-2})$. In contrast, direct assembly gives

$$M_\mu = C_\mu + \mu^2 D_\mu + (1 - \mu^2) |u_0\rangle\langle u_0| D_\mu. \quad (125)$$

It has uniformly bounded singular-value condition number and $O(d)$ normalization: C_μ is a bounded-weight degree- d Laplacian, D_μ is diagonal, u_0 is the uniform state, and the last term is rank one. Since $b \in U$, $M_\mu^{-1}b = \mu^{-1}H_\mu^{-1}b$ at every μ , completing the witness. $\square$

[Theorem 5.7](#) assumes a supplied active partition, coherent access to the current weights, and direct access to C_μ, D_μ, Q . Its cost also depends on $\sigma_{\min}^+(A_B)$, right-hand-side preparation, norm estimation, and the requested output. [Section 13](#) removes the supplied-partition assumption on the tail of the central path under strict complementarity; that construction is not needed here. A classical Krylov method already applies [\(125\)](#) in linear time per iteration, so the quantum result is useful only when the output is a state or a few observables; tomography or recovery of a classical iterate brings back a dimension-dependent cost.

5.5 An exact-center marked-diagonal specialization

The sparse search lower bound of Orsucci and Dunjko [\[118, Proposition 7\]](#) can also be realized at an exact LP center. This gives a QIPM specialization; its marked-search core is prior work.

Choose an integer $T \geq 2$, set the local dimension $N_{\text{md}} = T^8$ and $\mu = T^{-2} = N_{\text{md}}^{-1/4}$, and fix a hidden index $j \in [N_{\text{md}}]$. The subscript distinguishes this dimension from the optimal-partition

set N above. Let e_k denote the k -dimensional all-ones vector in this construction. Row i of $A_j \in \mathbb{R}^{N_{\text{md}} \times 2N_{\text{md}}}$ has disjoint support $(2i-1, 2i)$ and values

$$(\mu/2, \mu/2) \quad (i = j), \quad (1, \mu - 1) \quad (i \neq j). \quad (126)$$

Because μ , $\sqrt{\mu}$, and $\mu^{3/2}$ are rational, all entries have magnitude at most one and $O(\log N_{\text{md}})$ -bit rational descriptions; row sparsity is two and column sparsity one. Set

$$b = \mu^{3/2}e_{N_{\text{md}}}, \quad c = \sqrt{\mu}e_{2N_{\text{md}}}, \quad x^0 = s^0 = \sqrt{\mu}e_{2N_{\text{md}}}, \quad y^0 = 0.$$

Then $A_j x^0 = b$, $(A_j)^\top y^0 + s^0 = c$, and $X^0 S^0 e = \mu e$, so this is an exact μ -center. Since $X^0 (S^0)^{-1} = I$,

$$H_j = A_j A_j^\top = \text{diag}(h, \dots, \mu^2/2, \dots, h), \quad h = 1 + (1 - \mu)^2, \quad (127)$$

and the exact short-step right-hand side is $f = (1 - \sigma)\mu^{3/2}e_{N_{\text{md}}}$. The normalized right-hand side is the uniform state, independent of j . The squared solution weight on the marked coordinate is at least $4/5$, while

$$\begin{aligned} \kappa(H_j) &= \Theta(\mu^{-2}), \quad \xi(H_j, f) = \Theta(1), \quad \rho(H_j, f) = \Theta(\mu^{-2}), \\ \kappa_{\text{eff}}(\epsilon_{\text{ls}}) &= \frac{2h}{\mu^2} = \Theta(\mu^{-2}) \quad \text{for every fixed } \epsilon_{\text{ls}}^2 < 4/5. \end{aligned}$$

After norm-tight normalization the only eigenvalues are 1 and $\mu^2/(2h)$, so κ_{eff} drops to 1 only once ϵ_{ls}^2 reaches the marked weight, which is at least $4/5$. Consider access to A_j or H_j in which the positions of the nonzero entries are public and a query returns the value of an entry, as in the sparse-matrix access of Orsucci and Dunjko [118, Definition 4]. Each entry is determined by its public position and by whether its row index equals j , so such a query, or a query to the canonical diagonal block encoding of H_j , can be simulated with $O(1)$ queries to an oracle that marks j . Consequently, by the quantum search lower bound, preparing, to trace error $1/10$, the normalized state proportional to the exact multiplier step $H_j^{-1}f$ (so that measuring the marked coordinate succeeds with probability at least $4/5 - 1/10 = 7/10$) requires $\Omega(\sqrt{N_{\text{md}}}) = \Omega(\mu^{-2}) = \Omega(\kappa)$ queries.

With $1 - \sigma = \mu^2/4 = \Theta(N_{\text{md}}^{-1/2})$, the short step is feasible and positive, and its endpoint stays in a fixed two-norm neighborhood of the central path. The hard instance therefore arises inside an IPM, subject to three caveats. First, it is not representation invariant: row equilibration turns (127) into the identity and makes the scaled multiplier state easy (the primal/slack direction itself is invariant). Second, omitting the marked component leaves a relative normal-equation residual of only $\Theta(\mu^2)$, so residual-based inexact IPMs can avoid the state-preparation requirement. Third, for each fixed instance the hard scale occurs at one point of the path, not as $\mu \rightarrow 0$. The lower bound therefore applies to preparing the exact normalized multiplier state in the original representation, not to every QIPM or equivalent representation, consistent with the cautions about representation and finite precision in classical IPM analysis [135]. The $\Omega(\kappa)$ bound holds in the full query model, including with factor access. It does not contradict theorem 4.3(iii), which bounds the polynomial cost dM and yields an end-to-end factor-access solve only under its explicit overlap hypothesis.

5.6 Novelty and scope

To our knowledge, the following are new: the application of the 2026 DLS solvers to IPM Newton systems, the exact coupling law, the $D = 4$ versus $D = 8$ rule for choosing between truncation and filtering, together with its QIPM correctors, and the stable state-preserving split operator M_μ . Effective conditioning, active-set preconditioning, Green-function moment thresholds, and

the generic marked-diagonal search bound are prior work. None of the results in this section proves an end-to-end quantum LP speedup or a new universal QLS lower bound. Any such claim must also account for constructing the block encodings, finding the active set where needed, finite precision, maintaining the iterate, and output.

6 Block Log-Det Condensation

This section studies the log-det center of an SDP whose variable is a tuple of PSD blocks with one shared trace constraint $\sum_g \text{tr } Z_g = 1$. Suppose one block, the winner, has a cost whose least eigenvalue is smaller than that of every other block. As the path parameter decreases, the trace multiplier approaches this eigenvalue, where the winner's resolvent has a pole. The losing blocks can hold only a bounded amount of trace near that eigenvalue, so the remaining trace condenses on the winner. We show that the winner's trace mass controls three quantities: the condition number of the equality-reduced Hessian, the trace distance between the centers with and without a winner (the visibility of the winner), and the query cost of preparing the center as a quantum state or extracting the winner from it.

We work with an abstract two-cost block model. [Section 7](#) gives a sparse SDP that realizes it, and [section 8](#) proves a stronger lower bound for mixed-state output. In this section only, G is the number of blocks, d their common dimension, m the multiplicity of the least eigenvalue of the winning cost (not the number of equality constraints), τ the path parameter (called μ in [definition 2.3](#)), $\alpha = \tau G$, and p the trace of the winning block (the winner mass). The C - and Γ -symbols and d, b, c are also local, and the resolvent trace $A(x)$ is unrelated to the constraint matrix A of [section 2.1](#).

6.1 Two-cost model and exact center

Fix real symmetric costs $C_*, C_L \in \mathbb{S}^d$. There is one winning block with cost C_* and $G - 1$ identical losing blocks with cost C_L . After subtracting the winning spectral minimum v_* from both costs, suppose

$$\text{spec}(C_* - v_* I) = \{\underbrace{0, \dots, 0}_m, \gamma_{m+1}, \dots, \gamma_d\}, \quad \gamma_i > 0, \quad (128)$$

$$\text{spec}(C_L - v_* I) = \{\delta_1, \dots, \delta_d\}, \quad \delta_j > 0. \quad (129)$$

Here $1 \leq m \leq d$, and the winning block is unique. Unless otherwise stated, d, m , and all positive gaps are fixed while G grows.

At path parameter $\tau > 0$, consider

$$\min_{\substack{Z_g \succ 0 \\ \sum_g \text{tr } Z_g = 1}} \left\{ \langle C_*, Z_* \rangle + \sum_{g=2}^G \langle C_L, Z_g \rangle - \tau \sum_{g=1}^G \log \det Z_g \right\}. \quad (130)$$

For $x > 0$, define the winner and loser resolvent traces

$$A(x) = \frac{m}{x} + \sum_{i=m+1}^d \frac{1}{x + \gamma_i}, \quad B(x) = \sum_{j=1}^d \frac{1}{x + \delta_j}. \quad (131)$$

An empty regular sum is zero. Also put

$$B_0 = B(0) = \sum_{j=1}^d \delta_j^{-1}, \quad \beta = -B'(0) = \sum_{j=1}^d \delta_j^{-2}, \quad \alpha_* = B_0^{-1}. \quad (132)$$

Proposition 6.1 (Exact block center and finite concentration bounds). *The unique minimizer of (130) is*

$$Z_* = \tau(C_* - \lambda I)^{-1}, \quad Z_g = \tau(C_L - \lambda I)^{-1} \quad (g > 1), \quad (133)$$

where $\lambda = v_* - x$ and x is the unique positive solution of

$$1 = \tau\{A(x) + (G-1)B(x)\}. \quad (134)$$

The winner mass $p_G = \text{tr } Z_* = \tau A(x)$ satisfies, with $\delta_{\max} = \max_j \delta_j$,

$$p_G \geq 1 - \tau(G-1)B_0, \quad (135)$$

$$p_G \geq p_0 > 0 \implies x \leq \frac{d\tau}{p_0}, \quad 1 - p_G \geq \frac{d\tau(G-1)}{\delta_{\max} + d\tau/p_0}. \quad (136)$$

Consequently, for $G \geq 2$ and $0 < \varepsilon < 1$,

$$\tau \leq \frac{\varepsilon}{B_0(G-1)} \implies p_G \geq 1 - \varepsilon, \quad (137)$$

$$p_G \geq 1 - \varepsilon \implies \tau \leq \frac{\varepsilon \delta_{\max}}{d\{G-1-\varepsilon/(1-\varepsilon)\}}, \quad (138)$$

whenever the denominator in (138) is positive. For fixed G and $\tau \downarrow 0$, if $A_{\text{reg}}(0) = \sum_{i=m+1}^d \gamma_i^{-1}$, then

$$x = m\tau + m\{A_{\text{reg}}(0) + (G-1)B_0\}\tau^2 + O_G(\tau^3), \quad (139)$$

$$p_G = 1 - (G-1)B_0\tau + m(G-1)\beta\tau^2 + O_G(\tau^3). \quad (140)$$

Proof. The objective is strictly convex on the affine slice and diverges at its relative boundary, so it has a unique minimizer. With the multiplier sign from definition 2.4, stationarity gives $C_g - \lambda I = \tau Z_g^{-1}$. Positive definiteness requires $\lambda < v_*$, and taking traces gives (134). Its right-hand side is continuous and strictly decreasing from infinity to zero, which proves existence and uniqueness of x .

The loser mass is $1 - p_G = \tau(G-1)B(x)$. Since $B(x) \leq B_0$, this proves (135). Since $A(x) \leq d/x$, $p_G \geq p_0$ implies $x \leq d\tau/p_0$. The bound $B(x) \geq d/(x + \delta_{\max})$ then proves (136). Equations (137)–(138) follow by substituting $p_0 = 1 - \varepsilon$ and rearranging.

Finally, Taylor expansion gives $A(x) = m/x + A_{\text{reg}}(0) + O(x)$ and $B(x) = B_0 - \beta x + O(x^2)$. Substitute the ansatz $x = m\tau + c\tau^2 + O_G(\tau^3)$ into (134). Subtracting one, dividing by τ , and taking the constant term gives $c = m\{A_{\text{reg}}(0) + (G-1)B_0\}$. Substitution into $1 - p_G = \tau(G-1)B(x)$ gives (140). $\square$

Thus the winner mass stays bounded away from zero only when $\tau = O(1/G)$, and it is close to one when τ is a small enough multiple of $1/G$. The next result resolves this scale.

6.2 Capacity transition

With $\tau = \alpha/G$, the losers hold total trace $\tau(G-1)B(x) < \alpha B_0$. We call $\alpha_* = 1/B_0$ the capacity and the cases $\alpha < \alpha_*$, $\alpha = \alpha_*$, and $\alpha > \alpha_*$ subcritical, critical, and supercritical.

Theorem 6.2 (Block log-det condensation). *Set $\tau = \alpha/G$ for fixed $\alpha > 0$. Under (128)–(129), the following limits hold as $G \rightarrow \infty$:*

$$0 < \alpha < \alpha_* : \quad \frac{x}{\tau} \longrightarrow \frac{m}{1 - \alpha B_0}, \quad p_G \longrightarrow 1 - \alpha B_0; \quad (141)$$

$$\alpha = \alpha_* : \quad \sqrt{G}x \longrightarrow \sqrt{m/\beta}, \quad \sqrt{G}p_G \longrightarrow \alpha_* \sqrt{m\beta}; \quad (142)$$

$$\alpha > \alpha_* : \quad x \longrightarrow x_\alpha, \quad Gp_G \longrightarrow \alpha A(x_\alpha), \quad (143)$$

where $x_\alpha > 0$ is the unique solution of $B(x_\alpha) = 1/\alpha$.

Proof. Write the secular equation as

$$1 = \frac{\alpha}{G}A(x) + \alpha(1 - G^{-1})B(x). \quad (144)$$

If $\alpha < \alpha_*$, x must tend to zero: along any subsequence bounded away from zero, the first term vanishes and the second has limit at most $\alpha B_0 < 1$. Using the expansions following (140) in (144) gives

$$1 = \frac{\alpha m}{Gx} + \alpha B_0 + o(1),$$

so $Gx \rightarrow \alpha m / (1 - \alpha B_0)$ and $p_G = (\alpha/G)A(x) \rightarrow 1 - \alpha B_0$.

At $\alpha = \alpha_*$ the same compactness argument gives $x \rightarrow 0$. After cancelling $\alpha_* B_0 = 1$, the Taylor expansions give

$$0 = \frac{\alpha_* m}{Gx} - \alpha_* \beta x + O(G^{-1}) + O(x^2), \quad (145)$$

where the bounded regular part of A is absorbed in $O(G^{-1})$. Monotonicity first sandwiches x between constant multiples of $G^{-1/2}$; dividing (145) by $G^{-1/2}$ then yields $\sqrt{G}x \rightarrow \sqrt{m/\beta}$. Since $A(x) = m/x + O(1)$, the mass limit in (142) follows.

If $\alpha > \alpha_*$, x cannot tend to zero because the loser term alone would tend to $\alpha B_0 > 1$. It also cannot diverge. Every convergent subsequence therefore has a positive limit, at which the winner term in (144) vanishes and $B(x) = 1/\alpha$. Strict monotonicity of B makes that limit unique. Multiplying $p_G = (\alpha/G)A(x)$ by G proves (143). $\square$

Below capacity, the winner's pole term m/x supplies the trace the losers cannot hold; at capacity, it balances the first-order decrease βx of the loser trace; above capacity, the loser equation alone determines x_α . The assumption $\delta_j > 0$ is essential: a zero loser gap makes B_0 infinite, and there is no finite capacity.

6.3 Equality-reduced Hessian and conditioning

All condition numbers below are those of [definition 2.9](#): the block Hessian restricted to the tangent space of the trace constraint, in Frobenius coordinates. They say nothing about preconditioned systems or the saddle-point KKT matrix. At the center put $D_* = C_* - \lambda I$ and $D_L = C_L - \lambda I$. On the tangent space

$$\mathcal{T} = \left\{ Y = (Y_1, \dots, Y_G) : \sum_{g=1}^G \text{tr } Y_g = 0 \right\}, \quad (146)$$

the barrier-scaled log-det Hessian has quadratic form

$$\mathcal{Q}(Y) = \frac{1}{\tau} \left\{ \text{tr}(D_* Y_* D_* Y_*) + \sum_{g=2}^G \text{tr}(D_L Y_g D_L Y_g) \right\}. \quad (147)$$

This $\mathcal{Q}$ is the quadratic form of $\tau \nabla^2 F_{\log \det}$ at the center, so $\kappa(\mathcal{Q}|\mathcal{T}) = \kappa_{\text{red}}$; the factor τ does not change curvature ratios.

Let $\delta_{\min} = \min_j \delta_j$, let r be its multiplicity, and, when $m < d$, let $\gamma_{\min} = \min_{i>m} \gamma_i$. Define

$$b(x) = \min(\{x + \delta_{\min}\} \cup \{x + \gamma_{\min} : m < d\}), \quad \Lambda = \max(\{\delta_j\} \cup \{\gamma_i : i > m\}), \quad (148)$$

with the second set omitted when $m = d$.

Lemma 6.3 (Finite Hessian sandwich). *For $m \geq 2$,*

$$\lambda_{\min}(\mathcal{Q}|\mathcal{T}) = \frac{x^2}{\tau}. \quad (149)$$

For $m = 1$, let λ_{diag} be the least curvature on the diagonal part of $\mathcal{T}$ (directions diagonal in the cost eigenbases). Then

$$\frac{(dG - 1)x^2 + b(x)^2}{dG\tau} \leq \lambda_{\text{diag}} \leq \frac{r(G - 1)x^2 + (x + \delta_{\min})^2}{\tau\{r(G - 1) + 1\}}. \quad (150)$$

If $m = 1 < d$, the ground-excited winner direction, which couples the ground eigenvector with an eigenvector of gap $\gamma_{\min}$, has curvature $x(x + \gamma_{\min})/\tau$. Consequently, with constants depending only on the fixed spectral data,

$$\lambda_{\min}(\mathcal{Q}|\mathcal{T}) \asymp \frac{1}{\tau} \begin{cases} \min\{x(x + \gamma_{\min}), x^2 + (x + \delta_{\min})^2/G\}, & m = 1 < d, \\ x^2 + (x + \delta_1)^2/G, & m = d = 1. \end{cases} \quad (151)$$

For $G \geq 3$,

$$\frac{(x + \delta_{\max})^2}{\tau} \leq \lambda_{\max}(\mathcal{Q}|\mathcal{T}) \leq \frac{(x + \Lambda)^2}{\tau}. \quad (152)$$

Proof. Use eigenbases of the two costs. Diagonal and symmetric off-diagonal coordinates are orthogonal invariant subspaces. An off-diagonal (i, j) coordinate in a block with shifted eigenvalues d_i, d_j has curvature $d_i d_j / \tau$, and a diagonal coordinate has curvature d_i^2 / τ . Only diagonal coordinates are coupled by (146).

If $m \geq 2$, the difference of two winner ground diagonals, or their off-diagonal coordinate, is tangent and has curvature x^2 / τ . No ambient coordinate has smaller curvature, proving (149). For $m = 1$, write the winner's ground diagonal coefficient as a and all other $dG - 1$ diagonal coefficients as z . The trace constraint gives $\mathbf{1}^\top z = -a$, hence $\|z\|_2^2 \geq a^2 / (dG - 1)$; every corresponding curvature is at least $b(x)^2 / \tau$. Minimizing the resulting two-term quotient gives the lower bound in (150). For the upper bound, cancel a equally among the $r(G - 1)$ loser coordinates with gap $\delta_{\min}$ and calculate its Rayleigh quotient.

The stated winner off-diagonal coordinate supplies the other upper bound in (151); all remaining off-diagonal curvatures are bounded below by the same expression up to fixed spectral constants. Together with the diagonal sandwich this proves (151), including the scalar case where no within-block off-diagonal exists. Finally, opposite maximum-gap diagonal directions in two loser blocks are tangent and give the lower bound in (152); its upper bound is the largest curvature of an ambient coordinate. $\square$

Theorem 6.4 (Multiplicity-dependent conditioning phases). *For fixed $\alpha > 0$, $\tau = \alpha/G$, and fixed spectral data,*

$$\kappa_{\text{red}} = \begin{cases} \Theta(G^2), & m \geq 2, 0 < \alpha < \alpha_*, \\ \Theta(G), & m = 1, 0 < \alpha \leq \alpha_*, \\ \Theta(G), & m \geq 2, \alpha = \alpha_*, \\ \Theta(1), & \alpha > \alpha_* \text{ for every } m. \end{cases} \quad (153)$$

Proof. In the subcritical phase, theorem 6.2 gives $x = \Theta(G^{-1})$. By (152), the largest curvature is $\Theta(G)$. For $m \geq 2$, (149) gives minimum curvature $\Theta(G^{-1})$; for $m = 1$, (151) gives $\Theta(1)$. At criticality $x = \Theta(G^{-1/2})$, and either minimum formula gives $\Theta(1)$, while the maximum remains $\Theta(G)$. Above capacity x is bounded away from zero, so every curvature is $\Theta(G)$. $\square$

The ground-excited direction needs $m < d$. For $m = d = 1$, the diagonal direction that spreads the winner's trace across the losers plays its role, and (151) gives the same orders of κ_{red} .

6.4 Finite mass-conditioning laws

The next theorem bounds κ_{red} at finite G in terms of the winner mass p , without fixing the schedule $\tau = \alpha/G$ of [theorem 6.4](#).

Theorem 6.5 (Finite mass-conditioning law). *Let $G \geq 3$ and let $p = p_G \in (0, 1)$ be the winner mass at an exact center. Define*

$$R_G(p) = \frac{(G-1)p}{1-p}. \quad (\text{MC1})$$

Then

$$m \geq 2 : \quad \kappa_{\text{red}} \geq R_G(p)^2, \quad (\text{MC2})$$

$$m = 1 < d : \quad \kappa_{\text{red}} \geq \min \left\{ 1, \frac{\delta_{\max}}{\gamma_{\min}} \right\} R_G(p). \quad (\text{MC3})$$

If $m = d = 1$, writing $\delta = \delta_1$, the exact closed form is

$$\boxed{\kappa_{\text{red}} = \frac{G}{1 + (1-p)^2 / \{(G-1)p^2\}}}. \quad (\text{MC4})$$

For scalar blocks with $G = 2$, the tangent space is one-dimensional and $\kappa_{\text{red}} = 1$, so [\(MC4\)](#) is asserted only for $G \geq 3$.

Proof. The exact mass equations are

$$p = \tau A(x), \quad 1 - p = \tau(G-1)B(x). \quad (154)$$

Because $A(x) \leq d/x$ and $B(x) \geq d/(x + \delta_{\max})$,

$$\frac{1-p}{p} = (G-1) \frac{B(x)}{A(x)} \geq (G-1) \frac{x}{x + \delta_{\max}}, \quad \frac{x + \delta_{\max}}{x} \geq R_G(p). \quad (155)$$

The tangent direction with opposite unit entries on the $\delta_{\max}$ diagonal coordinates of two loser blocks gives $\lambda_{\max} \geq (x + \delta_{\max})^2/\tau$. If $m \geq 2$, divide this by [\(149\)](#) and use [\(155\)](#).

If $m = 1 < d$, the winner ground-excited direction gives $\lambda_{\min} \leq x(x + \gamma_{\min})/\tau$, while

$$\frac{x + \delta_{\max}}{x + \gamma_{\min}} \geq \min \left\{ 1, \frac{\delta_{\max}}{\gamma_{\min}} \right\}.$$

Factoring the curvature ratio proves [\(MC3\)](#).

For $d = m = 1$, put $a = x^2/\tau$ and $b = (x + \delta)^2/\tau$. On the scalar trace-zero space, $G - 2$ loser-difference directions have curvature b , and the direction balancing the winner against equal loser changes has curvature $\varrho = \{(G-1)a + b\}/G$. Since $b > a$, these are the largest and smallest eigenvalues. The mass equations reduce to

$$\frac{x}{x + \delta} = \frac{1-p}{(G-1)p}.$$

Thus $\kappa_{\text{red}} = b/\varrho = G/\{1 + (G-1)a/b\}$, which is [\(MC4\)](#). $\square$

In particular, $p \geq p_0 > 0$ forces $\kappa_{\text{red}} = \Omega(G^2)$ when $m \geq 2$ and $\kappa_{\text{red}} = \Omega(G)$ when $m = 1$. This also follows from $x \leq d\tau/p_0$, the bound $\tau = O(G^{-1})$ from the loser mass, and the test directions above; the exact laws add the explicit dependence on p .

6.5 Visibility, collision, and the conditioning dichotomy

The next result uses only the symmetry of the two-cost model, not the spectral assumptions of [section 6.1](#). Let $C_W, C_L \in \mathbb{S}^d$, $G \geq 2$, and $\tau > 0$. We compare the trace-one center ρ_0 where every block has the loser cost with the center ρ_1 where block 1 has cost C_W :

$$\rho_0 = \arg \min_{\substack{Z_g \succ 0 \\ \sum_g \text{tr } Z_g = 1}} \left\{ \sum_{g=1}^G \langle C_L, Z_g \rangle - \tau \sum_g \log \det Z_g \right\}, \quad (156)$$

$$\rho_1 = \arg \min_{\substack{Z_g \succ 0 \\ \sum_g \text{tr } Z_g = 1}} \left\{ \langle C_W, Z_1 \rangle + \sum_{g=2}^G \langle C_L, Z_g \rangle - \tau \sum_g \log \det Z_g \right\}. \quad (157)$$

Identify each tuple with its block-diagonal density matrix and write

$$\rho_0 = \bigoplus_{g=1}^G L_0, \quad \rho_1 = W_1 \oplus \bigoplus_{g=2}^G L_1, \quad p = \text{tr } W_1. \quad (158)$$

Permutation symmetry gives $\text{tr } L_0 = 1/G$ and $\text{tr } L_1 = (1-p)/(G-1)$.

Theorem 6.6 (Visibility and label collision). *For the centers above,*

$$\left| p - \frac{1}{G} \right| \leq D_{\text{tr}}(\rho_0, \rho_1) \leq \max \left\{ p, \frac{1}{G} \right\}. \quad (\text{O1})$$

If the block labels of two independent copies are measured and compared, the probabilities c_0, c_1 that they agree are

$$c_0 = \frac{1}{G}, \quad c_1 = p^2 + \frac{(1-p)^2}{G-1}, \quad \boxed{c_1 - c_0 = \frac{(Gp-1)^2}{G(G-1)}}. \quad (\text{O2})$$

The measurement is public (independent of the input) and acts only on the two label registers, so the identity holds regardless of degeneracy inside the blocks and is unchanged by purifying and then discarding auxiliary registers.

Proof. Measuring whether the label is 1 gives the lower bound in (O1). Let λ_0, λ_1 be the scalar trace multipliers of the two centers. Their loser blocks satisfy

$$L_b = \tau(C_L - \lambda_b I)^{-1} \quad (b = 0, 1),$$

and are therefore Loewner ordered. If $\lambda_1 \leq \lambda_0$, then $L_1 \preceq L_0$ and the trace identities imply $p \geq 1/G$. Hence

$$(G-1)\|L_1 - L_0\|_1 = p - \frac{1}{G}, \quad \|W_1 - L_0\|_1 \leq p + \frac{1}{G}.$$

Adding the trace norms of the direct-sum blocks and dividing by two gives $D_{\text{tr}}(\rho_0, \rho_1) \leq p$. If $\lambda_1 \geq \lambda_0$, the order and inequality reverse, and the same calculation gives the upper bound $1/G$. Finally the two label distributions are

$$(1/G, \dots, 1/G), \quad \left(p, \frac{1-p}{G-1}, \dots, \frac{1-p}{G-1} \right).$$

Summing their squared entries and subtracting proves (O2). $\square$

Only the upper trace-distance bound uses the resolvent form of the centers; the lower bound and the collision identity need only a uniform label distribution for ρ_0 and exchangeable losers. The condition $p \geq 1/G$ holds whenever $C_W \preceq C_L$, since $X \mapsto (X - \lambda I)^{-1}$ is operator antitone on $X \succ \lambda I$, and for the family of [section 7.6](#) by [\(225\)](#). Then the trace-distance bounds reduce to $p - 1/G \leq D_{\text{tr}} \leq p$.

For $C_W = C_*$, under the spectral assumptions of [section 6.1](#) and $G \geq 3$, [theorems 6.5](#) and [6.6](#) give the following finite laws relating conditioning and visibility. With $c = \min\{1, \delta_{\max}/\gamma_{\min}\}$,

$$p \leq \frac{\sqrt{\kappa_{\text{red}}}}{G - 1 + \sqrt{\kappa_{\text{red}}}} \quad (m \geq 2), \quad (\text{CV1})$$

$$p \leq \frac{\kappa_{\text{red}}}{c(G - 1) + \kappa_{\text{red}}} \quad (m = 1 < d), \quad (\text{CV2})$$

$$p = \frac{1}{1 + \sqrt{G - 1} \sqrt{G/\kappa_{\text{red}} - 1}} \quad (m = d = 1). \quad (\text{CV3})$$

By [\(O1\)](#), the maximum of each right-hand side and $1/G$ bounds $D_{\text{tr}}(\rho_0, \rho_1)$. Conversely, if $D_{\text{tr}}(\rho_0, \rho_1) \geq v > 1/G$, then $p \geq v$ and

$$\kappa_{\text{red}} \geq \left(\frac{(G - 1)v}{1 - v} \right)^2 \quad (m \geq 2), \quad \kappa_{\text{red}} \geq c \frac{(G - 1)v}{1 - v} \quad (m = 1 < d). \quad (\text{CV4})$$

For $m = d = 1$, [\(CV3\)](#) gives the same linear order. Thus constant visibility forces $\kappa_{\text{red}} = \Omega(G^2)$ when $m \geq 2$ and $\kappa_{\text{red}} = \Omega(G)$ when $m = 1$. A constant collision gap also forces $p = \Omega(1)$ and hence the same dichotomy.

Approximate outputs and reduced symmetry. Let $\tilde{\rho}_b$ satisfy $D_{\text{tr}}(\tilde{\rho}_b, \rho_b) \leq \epsilon$. Contractivity and the tensor-product triangle inequality give $D_{\text{tr}}(\tilde{\rho}_b^{\otimes 2}, \rho_b^{\otimes 2}) \leq 2\epsilon$. Thus the observed two-copy collision gap is at least

$$\Delta_{\text{out}} \geq d(p, G) - 4\epsilon, \quad d(p, G) := \frac{(Gp - 1)^2}{G(G - 1)}. \quad (159)$$

By standard concentration bounds, $O(\Delta_{\text{out}}^{-2})$ repetitions distinguish the two cases with constant success probability, and an extra $\log(1/\delta)$ factor gives failure probability δ .

The collision lower bound does not need equal loser masses. Suppose a public label measurement is uniform on the no-winner input and, on every unique-winner input, gives label masses w_g with winner mass $q \geq p \geq 1/G$. Cauchy–Schwarz gives

$$\sum_{g=1}^G w_g^2 \geq q^2 + \frac{(1 - q)^2}{G - 1} \geq p^2 + \frac{(1 - p)^2}{G - 1}, \quad (160)$$

where the last inequality holds because $q \mapsto q^2 + (1 - q)^2/(G - 1)$ is increasing for $q \geq 1/G$. The collision gap is therefore at least $d(p, G)$, with equality exactly at winner mass p and uniform loser mass. If the no-winner collision probability is only bounded above by b , replace the gap by $p^2 + (1 - p)^2/(G - 1) - b$. In particular, if $p \geq p_0 > 0$ and $G \geq 2/p_0$, then $d(p, G) \geq p_0^2/4$.

6.6 Unique-OR query lower bounds

To turn [theorem 6.6](#) into a query lower bound, we use a composed problem. Let $G \geq 2$ and let a possibly partial Boolean predicate $f : \mathcal{X}_0 \sqcup \mathcal{X}_1 \rightarrow \{0, 1\}$ have general adversary value $\text{Adv}_f = \text{Adv}^\pm(f)$. For G independently queried blocks, set

$$\mathcal{P}_0 = \mathcal{X}_0^G, \quad \mathcal{P}_1 = \bigsqcup_{a=1}^G \mathcal{X}_0^{a-1} \times \mathcal{X}_1 \times \mathcal{X}_0^{G-a}, \quad (161)$$

and let F be the problem of distinguishing $\mathcal{P}_0$ (no winner) from $\mathcal{P}_1$ (exactly one winner).

Theorem 6.7 (Unique-OR adversary and preparation). *In the raw-query model where one query addresses one coordinate of one block,*

$$\text{Adv}^\pm(F) \geq \text{Adv}_f \sqrt{G}. \quad (\text{O3})$$

Consequently, suppose that fixed public projectors onto the blocks give the uniform label distribution on every no-winner target state and winner mass at least $p \geq 1/G$ on every unique-winner target state; loser masses may be arbitrary. Any circuit that prepares, on every promised input, a state within trace distance ϵ of such a target uses $\Omega(\text{Adv}_f \sqrt{G})$ raw queries whenever $d(p, G) - 4\epsilon = \Omega(1)$. By tightness of the general adversary bound, this is equivalently $\Omega(Q(f)\sqrt{G})$ up to universal constants.

Proof. Take the rectangular zero-to-one block B , with rows indexed by $\mathcal{X}_0$ and columns by $\mathcal{X}_1$, of a feasible adversary matrix for f , normalized so that

$$\|B\| = \text{Adv}_f, \quad \max_j \|B \circ \Delta_j\| \leq 1, \quad (162)$$

where $\Delta_j[x, y] = 1$ exactly when query j has different answers on x, y , and let I_0 be the identity on $\ell_2(\mathcal{X}_0)$. For the sector of $\mathcal{P}_1$ whose winner is block a , define

$$C_a = I_0^{\otimes(a-1)} \otimes B \otimes I_0^{\otimes(G-a)}, \quad C = [C_1 \ C_2 \ \cdots \ C_G],$$

and use the Hermitian adversary matrix

$$\Gamma = \begin{pmatrix} 0 & C \\ C^* & 0 \end{pmatrix}.$$

The winner sectors are disjoint, and

$$CC^* = \sum_{a=1}^G I_0^{\otimes(a-1)} \otimes BB^* \otimes I_0^{\otimes(G-a)}.$$

The summands commute. Tensoring a top eigenvector of BB^* across all G blocks gives their common top eigenvector, so $\|\Gamma\| = \text{Adv}_f \sqrt{G}$. Filtering by raw query (a, j) kills every winner sector except a , since row and column inputs agree in block a in all other sectors. The remaining block is $B \circ \Delta_j$ and has norm at most one by (162). Thus Γ is feasible and proves (O3).

If $d(p, G) - 4\epsilon$ is a positive constant, (159) turns a constant number of fresh preparations and public collision measurements into a bounded-error algorithm for F . The adversary lower bound therefore applies to the preparation circuit. Tightness and composition of the negative-weight adversary bound give the equivalent $Q(f)$ statement [77, 92, 122]. $\square$

The theorem assumes that all product inputs in (161) are valid, both fibers of f are nonempty, and the block projectors are public. It needs no clean evaluator, candidate verification, inverse preparation, or restriction on the internal states of the blocks. If one coefficient-oracle call can be simulated with at most s raw queries, the theorem gives a lower bound of

$$\Omega\left(\frac{\text{Adv}_f \sqrt{G}}{s}\right) \quad (163)$$

coefficient-oracle calls. An oracle that returns a winner bit, an aggregate flag, or an input-dependent projector is stronger and may remove the inner factor Adv_f .

No such lower bound holds when the output error is large compared with the visibility: by theorem 6.6, outputting the public all-loser center ρ_0 uses no queries and is valid on every input with zero or one winner when

$$\epsilon \geq \max\{p, 1/G\}. \quad (164)$$

Hence a constant primal objective gap (8) need not imply a lower bound for trace-normalized primal-density output at constant error. This concerns the direct-sum density of definition 2.23, not a normalized vectorization, whose label probabilities are Frobenius rather than trace weights.

6.7 Preparation, extraction, and the conservation law

For the upper bounds we assume exactly one winner w and a clean coherent evaluator of the predicate

$$V_f|g, 0\rangle = |g, f_g\rangle, \quad f_g = 1 \iff g = w, \quad (\text{A1})$$

that costs T_{eval} raw queries, as does its inverse. The two cost matrices, G , and τ are public, so the center has the form

$$\rho_p = p|w\rangle\langle w| \otimes \rho_W + \frac{1-p}{G-1} \sum_{g \neq w} |g\rangle\langle g| \otimes \rho_L, \quad (\text{A2})$$

where the normalized block resolvents ρ_W, ρ_L are public. Put

$$a_W = p, \quad a_L = \frac{1-p}{G-1}, \quad a_{\max} = \max\{a_W, a_L\}. \quad (\text{A3})$$

Proposition 6.8 (Coherent rejection preparation). *There is a purification circuit for (A2) using*

$$\tilde{O}\left(T_{\text{eval}}\sqrt{Ga_{\max}} + T_{\text{loc}}\right) \quad (\text{A4})$$

operations, where T_{loc} is the gate cost (not a raw-query cost) of the one public preparation of ρ_W or ρ_L conditioned on the predicate. When $p \geq 1/G$, the raw-query part is

$$\tilde{O}(T_{\text{eval}}\sqrt{Gp}). \quad (\text{A5})$$

Proof. Prepare $G^{-1/2} \sum_g |g\rangle$, compute f_g , and rotate an acceptance qubit with amplitude $\sqrt{a_{f_g}/a_{\max}}$, but do not yet copy the label or prepare the internal state. This trial succeeds with probability

$$P_{\text{acc}} = \frac{1}{G} \left(\frac{a_W}{a_{\max}} + (G-1) \frac{a_L}{a_{\max}} \right) = \frac{1}{Ga_{\max}}. \quad (\text{A6})$$

This probability is known because the secular equation is solved classically, so exact amplitude amplification with matched phases applies to this trial, acting on the label, predicate, and acceptance registers [31]. After it succeeds, the label has probability a_{f_g} ; only then coherently copy the label to an environment register and use the retained predicate once to prepare a purification of ρ_W or ρ_L . Tracing out the copied label, predicate, acceptance, and local purification registers gives exactly (A2). Only the evaluator calls are repeated by the amplification, and the local preparation runs once, proving both bounds; the tilde hides only polylogarithmic factors in the requested implementation accuracy. $\square$

The secular equation and the fixed-size resolvents are computed classically; if d varies, one must add $\text{poly}(d, \log(1/\epsilon))$ costs for arithmetic and state synthesis. A bounded-error evaluator must first have its error reduced coherently below the final state error.

Suppose independent copies, each costing Q_ρ raw queries, satisfy $D_{\text{tr}}(\tilde{\rho}, \rho_p) \leq \epsilon < p$. Measuring the label proposes the winner with probability at least $p - \epsilon$, so verifying and repeating finds the winner with failure probability δ using

$$O\left(\frac{Q_\rho + T_{\text{eval}}}{p - \epsilon} \log \frac{1}{\delta}\right) \quad (\text{A7})$$

raw queries. For exact copies take $\epsilon = 0$; if $\epsilon \leq p/2$, this has the usual $O(1/p)$ repetition factor. Given instead a coherent preparation unitary U_ρ costing Q_U raw queries, its inverse, and a clean winner reflection, amplitude amplification gives

$$O\left(\frac{Q_U + T_{\text{eval}}}{\sqrt{p}} \log \frac{1}{\delta}\right). \quad (\text{A8})$$

This bound needs U_ρ and its inverse as operators; an accurate output density matrix says nothing about how the unitary acts on inputs other than $|0\rangle$.

Combining (A5) and (A8) gives the preparation–extraction conservation law

$$\tilde{O}(T_{\text{eval}}\sqrt{Gp}) O(p^{-1/2}) = \tilde{O}(T_{\text{eval}}\sqrt{G}). \quad (\text{A9})$$

Stronger condensation moves work from extraction to preparation without changing the end-to-end number of raw queries up to logarithmic factors. The three regimes give the following costs for exact output ($\epsilon \leq p/2$ suffices in the copy-extraction column):

regime	central preparation	copy extraction	coherent extraction
subcritical	$\tilde{O}(T_{\text{eval}}G^{1/2})$	$O(1)$	$O(1)$
critical	$\tilde{O}(T_{\text{eval}}G^{1/4})$	$O(G^{1/2})$	$O(G^{1/4})$
supercritical	$\tilde{O}(T_{\text{eval}})$	$O(G)$	$O(G^{1/2})$

(A10)

The supercritical preparation bound matters only for exact output or for accuracy fine enough to resolve a winner mass of order $1/G$; at fixed error, the zero-query output of (164) may suffice.

Alternatively, quantum search finds the winner at cost

$$\tilde{O}(T_{\text{eval}}\sqrt{G}), \quad (\text{A11})$$

and solving and recovering the selected block at costs T_{solve} and T_{recover} gives the end-to-end bound

$$\tilde{O}(T_{\text{eval}}\sqrt{G}) + T_{\text{solve}} + T_{\text{recover}}. \quad (\text{A12})$$

These bounds use standard quantum search and minimum finding [29, 57]. This is search followed by crossover, not a new interior-point method. If the local predicate has bounded-error raw-query complexity q and the blocks are independent, composition of the general adversary bound [77, 92, 122] gives the total bounded-error raw-query cost

$$Q(\text{UniqueOR}_G \circ f) = \Theta(q\sqrt{G}). \quad (\text{A13})$$

Combining this lower bound with the coherent extraction bound (A8) gives

$$Q_U + T_{\text{eval}} = \Omega(q\sqrt{Gp}), \quad (\text{A14})$$

and, for ϵ -accurate independent copies with $\epsilon < p$, the copy extraction bound gives the weaker

$$Q_\rho + T_{\text{eval}} = \Omega((p - \epsilon)q\sqrt{G}). \quad (\text{A15})$$

For exact output this is $\Omega(pq\sqrt{G})$, and the same asymptotic bound holds when $\epsilon \leq p/2$. These statements require independent block oracles; if one query can touch several blocks, or an aggregate winner flag is available, the composition argument fails.

For comparison, the block product barrier has parameter dG , so short-step path following from constant τ to $\tau = \Theta(1/G)$ takes $O(\sqrt{dG} \log G)$ Newton iterations. This count excludes the linear solves, whose cost depends on a condition number, and state preparation and recovery. It must not be multiplied by (153) unless an actual solver and preconditioner are chosen and analyzed.

6.8 Sublinear multiwinner capacity

Now let exactly k blocks have cost C_W , with $v = \lambda_{\min}(C_W)$ of multiplicity m , and suppose $C_L - vI \succ 0$. In this subsection, redefine

$$A(x) = \text{tr}(C_W - vI + xI)^{-1}, \quad B(x) = \text{tr}(C_L - vI + xI)^{-1}. \quad (\text{165})$$

Then $A(x) = m/x + O(1)$, $B_0 = B(0) < \infty$, and $\beta = -B'(0) > 0$. The exact equations are

$$1 = \tau\{kA(x) + (G - k)B(x)\}, \quad p_{G,k} = \tau kA(x), \quad 1 - p_{G,k} = \tau(G - k)B(x). \quad (\text{166})$$

Theorem 6.9 (Multiwinner capacity law). *Let $1 \leq k = k_G = o(G)$, put $\tau = \alpha/G$, and keep $\alpha > 0$ fixed. With $\alpha_* = 1/B_0$,*

$$p_{G,k} \longrightarrow 1 - \alpha B_0, \quad x \sim \frac{\alpha m k}{G(1 - \alpha B_0)} \quad (0 < \alpha < \alpha_*), \quad (167)$$

$$p_{G,k} \sim \alpha_* \sqrt{m \beta k / G}, \quad x \sim \sqrt{m k / (\beta G)} \quad (\alpha = \alpha_*), \quad (168)$$

$$p_{G,k} \sim \alpha A(x_\alpha) k / G, \quad B(x_\alpha) = 1/\alpha \quad (\alpha > \alpha_*). \quad (169)$$

Proof. Substitute $\tau = \alpha/G$ into the first equation in (166). Below criticality $x \rightarrow 0$: otherwise the winner term vanishes because $k/G \rightarrow 0$, while the loser term is at most $\alpha B_0 < 1$. The loser mass then tends to αB_0 , proving the first mass law, and the pole $A(x) \sim m/x$ gives the formula for x .

At criticality put $q = k/G$. Taylor expansion of the loser equation and the winner pole give two expressions for the total winner mass:

$$p_{G,k} = q + \frac{\beta}{B_0} x + o(q + x) = \frac{q m}{B_0 x} + O(q). \quad (170)$$

As in the one-winner proof, monotonicity first bounds x by constant multiples of $\sqrt{q}$; balancing the leading terms then gives $x \sim \sqrt{m q / \beta}$ and the critical mass. Above criticality the same compactness and strict-monotonicity argument as in theorem 6.2 gives $x \rightarrow x_\alpha > 0$; the winner equation gives the final law. $\square$

The assumption $k = o(G)$ is needed. If $k/G \rightarrow q \in (0, 1]$, the limit instead solves $1 = \alpha\{qA(x) + (1 - q)B(x)\}$ with $x > 0$, and there is no singular capacity transition at fixed α .

Nonidentical losers. Let every losing cost satisfy $C_{\ell,G} - vI \succ 0$, and define

$$B_{\ell,G}(x) = \text{tr}(C_{\ell,G} - vI + xI)^{-1}$$

and their average

$$\overline{B}_G(x) = \frac{1}{G - k} \sum_{\ell \text{ losing}} B_{\ell,G}(x).$$

If $k = o(G)$ and $\overline{B}_G \rightarrow B$ locally uniformly on $x \geq 0$, with $B(0) = B_0 \in (0, \infty)$, the subcritical and supercritical conclusions and the subcritical formula for x remain valid. The critical law also holds if, in addition,

$$\overline{B}_G(0) - B_0 = o(\sqrt{k/G}), \quad \overline{B}_G(x) = \overline{B}_G(0) - \beta x + r_G(x), \quad (171)$$

where $\beta > 0$ and, for every fixed $C > 0$,

$$\sup_{0 < x \leq C\sqrt{k/G}} \frac{|r_G(x)|}{x} \longrightarrow 0.$$

Indeed, replacing B by $\overline{B}_G$ in (166) leaves the subcritical and supercritical limit arguments unchanged. Under (171), the finite-size error is of lower order than both leading terms in (170), which proves the critical law. These hypotheses control only the winner mass; they do not give exchangeable losers, a common public internal state, or the exact collision identity.

For identical blocks and a fixed winner set S of size $k < G$, let $q = k/G$. The Loewner-order argument in the proof of theorem 6.6 gives

$$|p_{G,k} - q| \leq D_{\text{tr}}(\rho_0, \rho_S) \leq \max\{p_{G,k}, q\}, \quad (172)$$

and direct calculation gives the two-copy label-collision gap

$$c_S - c_0 = \frac{p_{G,k}^2}{k} + \frac{(1 - p_{G,k})^2}{G - k} - \frac{1}{G} = \boxed{\frac{(Gp_{G,k} - k)^2}{Gk(G - k)}}. \quad (173)$$

Thus a constant total winner mass gives a collision gap of only $O(1/k)$ when k grows, and a trace-distance bound for each fixed S does not give one measurement for an unknown S . [Section 8](#) avoids this $1/k$ loss for its structured family by the composed small-success bound ([theorem 8.7](#)); that stronger conclusion is not available in the abstract two-cost model.

6.9 Exactly- k coherent preparation

Assume now that $1 \leq k < G$ is known, that there are exactly k winners, and that a clean coherent membership marker costs T_{mark} raw queries. The two costs, G , k , τ , the trace multiplier, and the normalized internal resolvents are public, and controlled marker calls and the inverse preparation circuit are available. Put, for $p = p_{G,k}$,

$$a_W = \frac{p}{k}, \quad a_L = \frac{1 - p}{G - k}, \quad a_{\max} = \max\{a_W, a_L\}. \quad (\text{AP0})$$

The construction in [proposition 6.8](#), with the membership bit in place of the unique-winner bit, succeeds in one trial with probability $1/(Ga_{\max})$. This probability is known exactly, so exact amplitude amplification with matched phases uses

$$O(\sqrt{Ga_{\max}} + 1) \quad (\text{AP1})$$

marker calls [\[31\]](#). After it succeeds, we copy the label to an environment register and, controlled on the membership bit, prepare once a public purification of the normalized winner or loser block, where W and L are the common unnormalized winner and loser blocks of the center and

$$\omega_W = \frac{W}{p/k}, \quad \omega_L = \frac{L}{(1 - p)/(G - k)}. \quad (\text{AP2})$$

The resulting circuit satisfies

$$\boxed{Q_U = O\left(T_{\text{mark}}[\sqrt{Ga_{\max}} + 1]\right)}. \quad (\text{AP3})$$

When $p \geq k/G$, this is $O(T_{\text{mark}}[\sqrt{Gp/k} + 1])$; when $p < k/G$, it is $O(T_{\text{mark}}[\sqrt{G(1 - p)/(G - k)} + 1])$. Preparing to error $\epsilon > 0$ rather than exactly adds the standard factor logarithmic in $1/\epsilon$. Costs of loading nonpublic eigenvectors, resolvents, or multipliers must be added.

Conversely, amplification from the prepared purification finds a winner in

$$O\left(\frac{Q_U + T_{\text{mark}}}{\sqrt{p}}\right) \quad (\text{AP4})$$

raw queries. If the local predicate has bounded-error raw-query complexity q_f and independent composition gives the lower bound $\Omega(q_f \sqrt{G/k})$ for finding a winner, then ([AP4](#)) implies

$$\boxed{Q_U + T_{\text{mark}} = \Omega\left(q_f \sqrt{Gp/k}\right)}. \quad (\text{AP5})$$

This requires the preparation unitary and its inverse as operators; for mixed-state output, independent copies give only the weaker tradeoff from repetition.

If f is the parity of N input bits, then $T_{\text{mark}} = \Theta(N)$ and adversary composition gives winner-finding complexity $\Theta(N\sqrt{G/k})$. For $k = o(G)$, [theorem 6.9](#) gives the parity profile

$$Q_U = \begin{cases} \Theta(N\sqrt{G/k}), & 0 < \alpha < \alpha_*, \\ \Theta(N(G/k)^{1/4}), & \alpha = \alpha_*, \\ O(N), & \alpha > \alpha_*. \end{cases} \quad (\text{AP6})$$

The first two lines match [\(AP5\)](#) when the marker is optimal and the amplification factor diverges. The supercritical upper bound follows from $a_{\text{max}} = \Theta(1/G)$ even if $p < k/G$. At criticality $p \rightarrow 0$, so this operator-level lower bound need not hold for density output at fixed error.

This circuit does not solve the problem under the promise of zero or k winners: with no winner, the center has a different trace multiplier and loser resolvent. Under that promise, exact existence search followed by conditional preparation costs $O(T_{\text{mark}}\sqrt{G/k})$, while returning the public all-loser center costs zero queries and has error at most $\max\{p_{G,k}, k/G\}$ by [\(172\)](#). The two-copy collision measurement gives a matching lower bound only when the gap [\(173\)](#), reduced by the output error, stays constant.

6.10 Robust approximate centrality

The mass and conditioning bounds also hold, with modified constants, at approximately central points whose blocks need not commute with the costs. Keep the one-winner notation and, for positive definite $\tilde{Z}_g$ and $\tilde{\lambda} < v_*$, put

$$x = v_* - \tilde{\lambda}, \quad D_* = C_* - \tilde{\lambda}I, \quad D_L = C_L - \tilde{\lambda}I, \quad \tilde{T} = \sum_g \text{tr} \tilde{Z}_g, \quad \tilde{p} = \frac{\text{tr} \tilde{Z}_*}{\tilde{T}}. \quad (174)$$

Write $D_g = D_*$ for the winner and $D_g = D_L$ for every loser. Assume a dimensionless stationarity residual in the barrier norm and a trace residual:

$$E_g = \tau^{-1} \tilde{Z}_g^{1/2} D_g \tilde{Z}_g^{1/2} - I, \quad \max_g \|E_g\|_{\text{op}} \leq \eta < 1, \quad |\tilde{T} - 1| \leq \zeta < 1. \quad (175)$$

Theorem 6.10 (Robust capacity, mass, and conditioning). *Every point satisfying [\(174\)](#)–[\(175\)](#) satisfies the noncommutative Loewner sandwich*

$$(1 - \eta)\tau D_g^{-1} \preceq \tilde{Z}_g \preceq (1 + \eta)\tau D_g^{-1}. \quad (\text{R1})$$

Consequently,

$$\tilde{p} \geq 1 - \frac{(1 + \eta)\tau(G - 1)B_0}{1 - \zeta} \geq 1 - \frac{(1 + \eta)\alpha B_0}{1 - \zeta}. \quad (\text{R2})$$

Hence a uniform margin $(1 + \eta)\alpha B_0 \leq 1 - \zeta - c$ with a constant $c > 0$ forces a positive, G -independent lower bound on the winner mass. Conversely, if $G \rightarrow \infty$, $\alpha \rightarrow \bar{\alpha}$, and $x \rightarrow 0$, then

$$(1 - \eta)\bar{\alpha}B_0 \leq 1 + \zeta. \quad (\text{R3})$$

Thus $(1 + \eta)\alpha B_0 = 1 - \zeta$ and $(1 - \eta)\alpha B_0 = 1 + \zeta$ bracket the critical surface allowed by these residual bounds.

For the conditioning statement, suppose $G \geq 2$, $d \geq 2$, and $\tilde{p} \geq p_0 \in (0, 1)$. Set

$$q_0 = p_0(1 - \zeta), \quad \Gamma_* = \lambda_{\text{max}}(C_* - v_*I), \quad \delta_{\text{max}} = \lambda_{\text{max}}(C_L - v_*I). \quad (176)$$

Then

$$x \leq \frac{d(1 + \eta)\tau}{q_0}, \quad (\text{R4})$$

and, whenever the denominator is positive,

$$\tau \leq \frac{(1-p_0)(1+\zeta)\delta_{\max}}{d\{(1-\eta)(G-1) - (1-p_0)(1+\zeta)(1+\eta)/q_0\}}. \quad (\text{R5})$$

For the log-det Hessian at the approximate point,

$$\tilde{\mathcal{H}}[Y]_g = \tau \tilde{Z}_g^{-1} Y_g \tilde{Z}_g^{-1} \quad \text{on } \mathcal{T}, \quad (177)$$

we have

$$\kappa(\tilde{\mathcal{H}}|_{\mathcal{T}}) \geq \frac{\delta_{\min}^2(1-\eta)q_0}{d(1+\eta)^2\tau(x+\Gamma_*)} = \Omega(G). \quad (\text{R6})$$

If $m \geq 2$, the stronger bound is

$$\kappa(\tilde{\mathcal{H}}|_{\mathcal{T}}) \geq \frac{\delta_{\min}^2(1-\eta)^2}{(1+\eta)^2x^2} = \Omega(G^2). \quad (\text{R7})$$

For fixed $p_0, \eta, \zeta, C_*, C_L$, all asymptotic constants are uniform over such points.

Proof. The residual bound is equivalent to

$$(1-\eta)I \preceq \tau^{-1} \tilde{Z}_g^{1/2} D_g \tilde{Z}_g^{1/2} \preceq (1+\eta)I.$$

Congruence by $\tilde{Z}_g^{-1/2}$, inversion, and multiplication by τ prove (R1). Since $D_L \succeq C_L - v_*I$,

$$\sum_{g=2}^G \text{tr } \tilde{Z}_g \leq (1+\eta)\tau(G-1) \text{tr } D_L^{-1} \leq (1+\eta)\tau(G-1)B_0.$$

Divide by $\tilde{T} \geq 1-\zeta$ to prove (R2). The lower sandwich similarly gives $\tilde{T} \geq (1-\eta)\tau(G-1) \text{tr } D_L^{-1}$; taking the limit in the theorem proves (R3).

Winner mass and the upper sandwich give

$$q_0 \leq \text{tr } \tilde{Z}_* \leq (1+\eta)\tau \text{tr } D_*^{-1} \leq \frac{d(1+\eta)\tau}{x},$$

which is (R4). Every eigenvalue of D_L is at most $x + \delta_{\max}$. The lower sandwich on loser blocks and $(1-\tilde{p})\tilde{T} \leq (1-p_0)(1+\zeta)$ give

$$(1-p_0)(1+\zeta) \geq \frac{(1-\eta)\tau(G-1)d}{x + \delta_{\max}}.$$

Substitute (R4) and rearrange to obtain (R5), hence $\tau = O(G^{-1})$.

No commutation is needed for the curvature tests. In an eigenbasis of any loser block $\tilde{Z}_g$, a Frobenius-unit symmetric off-diagonal matrix is tangent. The upper sandwich and $D_L \succeq \delta_{\min}I$ bound every eigenvalue of this block by $(1+\eta)\tau/\delta_{\min}$, so this direction has curvature at least $\delta_{\min}^2/\{(1+\eta)^2\tau\}$. The largest eigenvalue of the winner block is at least q_0/d , while its other eigenvalues are at least $(1-\eta)\tau/(x+\Gamma_*)$. An off-diagonal direction between a largest eigenvector and an orthogonal eigenvector therefore has curvature at most $d(x+\Gamma_*)/\{(1-\eta)q_0\}$. Taking the quotient proves the finite bound in (R6); (R4)–(R5) give its asymptotic form.

If $m \geq 2$, the min-max principle and the lower sandwich show that the two largest winner eigenvalues are at least $(1-\eta)\tau/x$. Their off-diagonal direction has curvature at most $x^2/\{(1-\eta)^2\tau\}$. Comparing with the loser off-diagonal direction above proves (R7). $\square$

Scalar blocks, excluded above by $d \geq 2$, also have a robust linear lower bound. When $d = 1$ and $G \geq 3$, (R4)–(R5) remain valid. A difference of two loser coordinates has curvature at least $\delta_1^2 / \{(1 + \eta)^2 \tau\}$. Balancing a unit winner coordinate by $-1/(G - 1)$ in every loser yields a tangent Rayleigh quotient at most

$$\frac{\tau}{q_0^2} + \frac{(x + \delta_1)^2}{(1 - \eta)^2 \tau (G - 1)}. \quad (178)$$

The ratio of these two curvature bounds, together with (R4)–(R5), is $\Omega(G)$. The residual in (175) must be measured relative to the barrier: a small unscaled stationarity residual need not imply a Loewner sandwich near the spectral edge, where D_* becomes singular.

6.11 Relation to prior work and open questions

The resolvent form (133) of the center is classical: it is the density associated with the matrix Burg entropy, and Ishihara derives it from a maximum-entropy problem with trace and energy constraints [78]. For individual eigenmodes, the precise statistical-mechanics analogue is Rayleigh–Jeans condensation, where inverse-gap occupancies, saturation of excited modes, and a macroscopic ground-mode mass are established [18, 143]. Inverse-gap capacity and condensation onto several states also appear in spherical models [48, 97]. We therefore call our setting a matrix-Burg / Rayleigh–Jeans analogue and claim neither a Bose–Einstein distribution nor a new form of the center.

Square-root rates on degenerate SDP central paths and generic ill-conditioning of Newton systems near low-rank optima are also known [4, 49, 147], but those works do not study the reduced Frobenius Hessian as the number G of blocks grows. Our contributions are the critical coefficients as G grows, the dependence of the conditioning order (G versus G^2) on m , the finite mass-conditioning laws, their combination with visibility and raw-query bounds, and the conservation law. The quantum search, amplification, and adversary tools are standard.

Identical losers are needed for a sharp common capacity, the exact collision identity, and coherent preparation from public resolvents. The average-resolvent hypotheses (171) extend the mass law, but visibility for nonidentical losers requires a separate public statistic such as (160). It is open how the bound η on the barrier-norm residual in (175) relates to the algorithm-specific inexact Newton residuals of definition 2.8, whose forcing tolerance is a separate η ; we claim no conversion.

7 Winner-Take-All Global-Trace Holonomy SDP

This section gives an explicit SDP family with bounded scalar incidence to which the condensation theory of section 6 applies. It couples G groups, each a copy of the qutrit-holonomy component of section 11.1. Each group has $N = n_{\text{bit}}$ private signs, and its holonomy is their parity. A single trace constraint shared by all groups makes the optimal value the least minimum eigenvalue of the group costs, and a chain of free scalar accumulators keeps every row and column of the equality matrix sparse.

quantity	conclusion
value queries	$Q = \Theta(N\sqrt{G})$, $R = \Theta(NG)$ for error $< 1/20$
one-odd central path	$p = \Theta(1), \Theta(G^{-1/2}), \Theta(G^{-1})$ below, at, and above $\alpha_* = 2/45$
reduced-Hessian conditioning	$\Theta(G), \Theta(G), \Theta(1)$ in the same phases
coherent central-state preparation	$\Theta(N\sqrt{G}), \Theta(NG^{1/4}), O(N)$
end-to-end algorithm	search for the winner, then solve only its group

Here Q and R are the quantum and randomized query complexities of the optimal value, p is the central-path trace mass of the odd group when exactly one group is odd, and the phases refer to $\alpha = \tau G$ (section 7.6). The preparation costs assume exactly one winner. At the public starting point the root Hessian has condition number one, but the Newton-direction state does not reveal the winner (remark 7.1).

7.1 A bounded-incidence product SDP

Set

$$K = 16N, \quad P = 2K + N = 33N, \quad u = \frac{1}{10}, \quad q = \frac{K}{P} = \frac{16}{33}, \quad \gamma = \frac{u}{q} = \frac{33}{160}. \quad (179)$$

For $a < b$, write $H_{ab} = e_a e_b^\top + e_b e_a^\top$, and put $D_s = \text{diag}(s, 1, 1)$ for $s \in \{-1, +1\}$. Group g has private signs $\sigma_{g,1}, \dots, \sigma_{g,N}$ and holonomy

$$h_g = \prod_{j=1}^N \sigma_{g,j}. \quad (180)$$

Group g has P blocks $X_{g,0}, \dots, X_{g,P-1} \in \mathbb{S}_+^3$, joined in a path. For the edge from $i-1$ to i , let $G_{g,i} = I$ except that $G_{g,K+j-1} = D_{\sigma_{g,j}}$ for $1 \leq j \leq N$, and impose the copy equations, each as six isometric-svec scalar equations:

$$X_{g,i} = G_{g,i} X_{g,i-1} G_{g,i}^\top \quad (1 \leq i < P). \quad (181)$$

If $R_{g,0} = I$ and $R_{g,i} = G_{g,i} \cdots G_{g,1}$, these equations state that

$$X_{g,i} = R_{g,i} Z_g R_{g,i}^\top, \quad Z_g = X_{g,0} \succeq 0. \quad (182)$$

The first K matrices $R_{g,i}$ equal I , and the last K equal D_{h_g} .

The public block costs are

$$C_{g,i} = \begin{cases} 3I + uH_{23} + \gamma H_{12}, & 0 \leq i < K, \\ 3I + uH_{23}, & K \leq i < K + N, \\ 3I + uH_{23} + \gamma H_{13}, & K + N \leq i < P. \end{cases} \quad (183)$$

Because $\sqrt{u^2 + \gamma^2} < 1/4$, every $C_{g,i}$ has spectrum in $(11/4, 13/4)$. Introduce free public scalars $t_1, \dots, t_G$ and add

$$\begin{aligned} t_1 - \text{tr } X_{1,0} &= 0, \\ t_g - t_{g-1} - \text{tr } X_{g,0} &= 0 \quad (2 \leq g \leq G), \\ t_G &= 1. \end{aligned} \quad (184)$$

For $G = 1$, the first and last rows impose $t_1 = \text{tr } X_{1,0} = 1$. The SDP minimizes

$$f(X) = \frac{1}{P} \sum_{g=1}^G \sum_{i=0}^{P-1} \langle C_{g,i}, X_{g,i} \rangle \quad (185)$$

subject to (181)–(184) and $X_{g,i} \succeq 0$.

There are

$$L = GP = 33GN \quad (186)$$

blocks in $\mathbb{S}_+^3$. A copy row has two scalar nonzeros. The first accumulator row has four, an interior accumulator row has five, and the final row $t_G = 1$ has one. Every matrix coordinate and every

free-scalar column occurs in at most two rows. Thus the equality matrix has scalar row sparsity at most five and scalar column sparsity at most two. Its support, right-hand side, and all costs are public; only the 12 and 13 copy coefficients on the N signed edges of each path depend on the input.

Eliminating the path and accumulator variables leaves the roots Z_g , whose feasible set is exactly

$$\mathcal{B}_G = \left\{ (Z_1, \dots, Z_G) : Z_g \succeq 0, \sum_{g=1}^G \text{tr } Z_g = 1 \right\}. \quad (187)$$

The public point

$$X_{g,i}^0 = \frac{1}{3G} I, \quad t_g^0 = \frac{g}{G} \quad (188)$$

is strictly primal feasible. With all equality multipliers zero, the dual PSD slacks are $C_{g,i}/P \succ 0$ and stationarity in every free scalar is zero. Hence the formulation is strictly primal–dual feasible in the sense of [definition 2.1](#).

7.2 Exact spectral minimum

In terms of the roots, the objective (185) becomes

$$\bar{C}_h = 3I + u(H_{12} + H_{23} + hH_{13}), \quad \min_{Z \in \mathcal{B}_G} \sum_g \langle \bar{C}_{h_g}, Z_g \rangle. \quad (189)$$

Indeed, the first and last K blocks of each path contribute $K\gamma/P = u$ to H_{12} and H_{13} , respectively, and conjugation by D_h changes H_{13} to hH_{13} . The two effective costs have spectra

$$\text{spec}(\bar{C}_+) = \left\{ \frac{16}{5}, \frac{29}{10}, \frac{29}{10} \right\}, \quad \text{spec}(\bar{C}_-) = \left\{ \frac{31}{10}, \frac{31}{10}, \frac{14}{5} \right\}. \quad (190)$$

These follow either by diagonalizing the two signed triangle adjacency matrices, whose eigenvalues are $(2, -1, -1)$ and $(1, 1, -2)$, or from the intrinsic component analysis in [section 11.1](#).

For any feasible roots, Rayleigh–Ritz gives

$$\sum_g \langle \bar{C}_{h_g}, Z_g \rangle \geq \sum_g \lambda_{\min}(\bar{C}_{h_g}) \text{tr } Z_g \geq \min_g \lambda_{\min}(\bar{C}_{h_g}). \quad (191)$$

Equality holds when all trace is placed on a minimum eigenvector of a minimizing group. Therefore

$$\text{OPT} = \begin{cases} 29/10, & h_g = +1 \text{ for every } g, \\ 14/5, & h_g = -1 \text{ for at least one } g. \end{cases} \quad (192)$$

The gap is $1/10$, so the optimal value encodes $\text{OR}_G \circ \text{PARITY}_N$: whether some $h_g = -1$.

The root feasible set (187) has no lift over a finite product of second-order cones. Its conic hull is $(\mathbb{S}_+^3)^G$. If (187) had such a lift, homogenizing the compact lift and projecting to one factor would give such a lift of $\mathbb{S}_+^3$, contrary to Fawzi’s theorem [61]. The path-and-accumulator formulation projects linearly onto the root set, so it has no finite SOC lift either.

7.3 Sharp value-query complexity

We use the raw sparse coefficient access of [definition 2.11](#). In the canonical oracle, which returns isometric-svec coefficients at fixed positions, a public reversible address map sends a requested input-dependent position to (g, j) and a public position to a dummy zero. The value oracle XORs the sign of one of the two hidden coefficients $\pm\sigma_{g,j}$ directly into its target register. Thus

one raw-sign query, which returns one $\sigma_{g,j}$, simulates one coefficient query, coherently and also under inverse or external control. Conversely, one fixed input-dependent copy coefficient reveals $\sigma_{g,j}$. The two models therefore have the same query counts. A clean value oracle can instead be used as a phase oracle by compute–phase–uncompute, at two calls per query.

We instantiate [theorem 6.7](#) for parity blocks. Let $E, O \subseteq \{0, 1\}^N$ be the even- and odd-parity strings and let

$$B_{xy} = \mathbf{1}[|x - y|_1 = 1] \quad (x \in E, y \in O). \quad (193)$$

The constant vectors show that $\|B\| = N$, while each $B \circ \Delta_j$ is a permutation matrix and has norm one. Hence the inner adversary value in [\(162\)](#) is exactly N . Applying [\(O3\)](#) on the all-even versus exactly-one-odd promise gives

$$Q_{1/3} = \Omega(N\sqrt{G}). \quad (194)$$

The hypercube matrix [\(193\)](#) is thus an explicit adversary witness for [section 6.6](#).

If $|\tilde{v} - \text{OPT}| \leq \epsilon < 1/20$, thresholding at $57/20$ distinguishes the two cases in [\(192\)](#). The same threshold works for target-relative error $1/100$, since

$$\frac{101}{100} \frac{14}{5} < \frac{57}{20} < \frac{99}{100} \frac{29}{10}. \quad (195)$$

At $\epsilon = 1/20$, the midpoint $57/20$ is a valid answer for both values and uses no query, so the additive threshold is sharp.

For the randomized lower bound, let $\mathcal{D}_0$ make all groups independent and uniform conditional on even parity; let $\mathcal{D}_1$ choose a uniform group J , make it uniform conditional on odd parity, and leave the others as in $\mathcal{D}_0$. Couple adaptive decision trees by returning common fair answers to the first $N - 1$ distinct positions in each group and using the final unseen sign to enforce the promised parity. The transcripts can differ only if the algorithm queries all N signs of group J . A depth- r tree queries all signs of at most r/N groups, so

$$\text{TV}(\text{Tr}_{\mathcal{D}_0}, \text{Tr}_{\mathcal{D}_1}) \leq \frac{r}{NG}. \quad (196)$$

Yao’s principle therefore gives $r \geq NG/3$ for worst-case success $2/3$; reading every sign is a matching upper bound. Consequently, for every $\epsilon < 1/20$,

$$\boxed{Q_\epsilon(\text{OPT}) = \Theta(N\sqrt{G}), \quad R_\epsilon(\text{OPT}) = \Theta(NG).} \quad (197)$$

7.4 The Newton step at the public point

Use the log-det barrier on the PSD blocks, but no barrier on the free accumulators:

$$\Phi_\mu(X) = f(X) - \mu \sum_{g,i} \log \det X_{g,i}, \quad \mu = \frac{1}{GP}. \quad (198)$$

At [\(188\)](#), the PSD-block gradient and Hessian in isometric-svec coordinates are

$$g_{g,i} = \frac{1}{P}(C_{g,i} - 3I), \quad \nabla_X^2 \Phi_\mu(X^0) = \frac{9G}{P}I. \quad (199)$$

These and all equality residuals are public; the input enters the unreduced Newton system only through the signed copy constraints in its matrix.

After exact elimination of the copies and accumulators, a tangent direction in root coordinates is $Y = (Y_1, \dots, Y_G)$ with $\sum_g \text{tr } Y_g = 0$. The copy maps are orthogonal congruences and each root has P path copies, so

$$\boxed{\nabla_{\text{root}}^2 \Phi_\mu(X^0) = 9GI_{6G-1}.} \quad (200)$$

This root-coordinate Hessian has condition number one, including in directions that move trace between groups. With

$$A_h = H_{12} + H_{23} + hH_{13}, \quad (201)$$

the pulled-back gradient is uA_{h_g} , which is trace zero, and the exact Newton directions are

$$\Delta Z_g = -\frac{u}{9G}A_{h_g}, \quad \Delta t_g = 0. \quad (202)$$

Since $\|A_h\|_F^2 = 6$, the decrement is

$$\lambda^2 = \sum_g \langle uA_{h_g}, (9G)^{-1}uA_{h_g} \rangle = \frac{1}{150}, \quad \lambda = \frac{1}{\sqrt{150}}. \quad (203)$$

The full step remains strictly feasible. The eigenvalues inside the parentheses below are $14/45, 31/90, 31/90$ for $h = +1$ and $29/90, 29/90, 16/45$ for $h = -1$. Consequently,

$$Z_g^0 + \Delta Z_g = \frac{1}{G} \left(\frac{1}{3}I - \frac{u}{9}A_{h_g} \right) \succeq \frac{14}{45G}I. \quad (204)$$

On the root slice the barrier weight is $\tau = \mu P = 1/G$ (see (219)), which is below one for every $G \geq 2$; then Φ_μ is not standard self-concordant there, and the usual full-step decrement estimate does not apply to λ . We instead bound the next decrement directly. With $B_h = \frac{1}{3}I - \frac{u}{9}A_h$, the full step is $\hat{Z}_g = B_{h_g}/G$ and its slice gradient is $\bar{C}_{h_g} - B_{h_g}^{-1}$. On the trace slice the Newton decrement is the minimum over the trace multiplier of the Hessian-inverse norm of the shifted gradient, so taking that multiplier to be zero gives an upper bound:

$$\lambda_{\text{next}}^2 \leq \frac{1}{G} \sum_g \text{tr}[(B_{h_g}\bar{C}_{h_g} - I)^2] = \frac{u^4}{81} \text{tr} A_h^4 = \frac{2u^4}{9}, \quad \lambda_{\text{next}} \leq \frac{\sqrt{2}}{300}, \quad (205)$$

because $\bar{C}_h = 3I + uA_h$ gives $B_h\bar{C}_h - I = -\frac{u^2}{9}A_h^2$ and $\text{tr} A_h^4 = 18$ for both parities. This bound is uniform in G and in the parity pattern.

Remark 7.1 (The Newton-direction state hides the winner). The normalized root-direction state is

$$|\Delta(h)\rangle = \frac{1}{\sqrt{G}} \sum_{g=1}^G |g\rangle |a_{h_g}\rangle, \quad |a_h\rangle = \frac{|\text{svec}(A_h)\rangle}{\sqrt{6}}. \quad (206)$$

Looping once over the N raw signs and applying phase kickback only to the 13 coordinate prepares this state exactly with N queries. Conversely, the density-matrix entry between group 1's 12 and 13 coordinates is $h_1/(3G)$. Since each density-matrix entry of a q -query output has degree at most $2q$ in the input signs, exact preparation needs $q \geq N/2$; the same holds at trace error $\delta < 1/(6G)$. Hence exact preparation costs $\Theta(N)$.

Moreover, the all-even and one-odd states have overlap $1 - 2/(3G)$ and trace distance

$$\frac{2\sqrt{3G-1}}{3G} = \Theta(G^{-1/2}). \quad (207)$$

Therefore a constant-error Newton state cannot decide the unique-winner promise with constant bias. The $\Omega(N\sqrt{G})$ lower bound applies to the value and to optimizer outputs that reveal the winner, not to this state.

The condition number one in (200) holds in root coordinates only. A feasible root direction lifts to accumulator changes equal to prefix sums of traces, which is not an isometry, and the free scalars have zero barrier Hessian. We therefore do not claim condition number one for the ambient augmented Hessian or the saddle-point KKT matrix.

7.5 Approximate optimizers

For exactly feasible roots on an instance with at least one odd group, put $m_- = \sum_{g:h_g=-1} \text{tr } Z_g$. If the objective is at most $14/5 + \delta_{\text{obj}}$, (190) and (191) give

$$\frac{14}{5} + \frac{1-m_-}{10} \leq \sum_g \langle \bar{C}_{h_g}, Z_g \rangle \leq \frac{14}{5} + \delta_{\text{obj}}, \quad \boxed{m_- \geq 1 - 10\delta_{\text{obj}}}. \quad (208)$$

When the instance contains both an odd and an even group and $0 \leq \delta_{\text{obj}} \leq 1/10$, the constant is sharp: put trace $1 - 10\delta_{\text{obj}}$ in an odd and $10\delta_{\text{obj}}$ in an even minimum eigenspace.

A similar bound holds for approximate solutions of the sparse formulation, which may violate its equalities. Define the copy residuals

$$E_{g,i} = X_{g,i} - G_{g,i} X_{g,i-1} G_{g,i}^\top \quad (209)$$

and let $r_1, \dots, r_G, r_{\text{top}}$ be the residuals in (184). Suppose $X_{g,i} \succeq 0$ and

$$\sum_{g,i \geq 1} \|E_{g,i}\|_F^2 + \sum_{g=1}^G r_g^2 + r_{\text{top}}^2 \leq \epsilon^2, \quad f(X) \leq \text{OPT} + \delta_{\text{obj}}. \quad (210)$$

Put $Z_{g,i} = R_{g,i}^\top X_{g,i} R_{g,i}$ (each block in its root frame) and telescope along the paths and the accumulator chain. Cauchy–Schwarz and the spectral bound on $C_{g,i}$ give

$$\left| \sum_g \text{tr } Z_{g,0} - 1 \right| \leq \sqrt{G+1} \epsilon, \quad (211)$$

$$\left| \frac{1}{P} \sum_{g,i} \text{tr } X_{g,i} - 1 \right| \leq \sqrt{GP+1} \epsilon, \quad (212)$$

$$\left| f(X) - \sum_g \langle \bar{C}_{h_g}, Z_{g,0} \rangle \right| < \frac{13}{4} \sqrt{GP} \epsilon. \quad (213)$$

Combining these estimates with (190) gives, for the even root mass m_+ on a yes-instance,

$$m_+ \leq 10\delta_{\text{obj}} + 28\sqrt{G+1} \epsilon + \frac{65}{2} \sqrt{GP} \epsilon. \quad (214)$$

Theorem 7.2 (Robust winner recovery). *Suppose an algorithm prepares a state within trace distance $1/100$ of the trace-normalized physical density*

$$\rho(X) = \frac{\bigoplus_{g,i} X_{g,i}}{\sum_{g,i} \text{tr } X_{g,i}} \quad (215)$$

for PSD blocks and free accumulators satisfying (210), where

$$\delta_{\text{obj}} \leq \frac{1}{100}, \quad \epsilon \leq \frac{1}{1000\sqrt{GP}}. \quad (216)$$

Then a fixed input-independent measurement decides whether any group is odd with constant bias. On every yes-instance, measuring the group register returns an actual odd group with probability greater than $5/6$. Consequently any such first preparation (in the sense of definition 2.27) uses $\Omega(N\sqrt{G})$ raw or canonical coefficient queries.

Proof. Set $s = \sqrt{GP}\epsilon \leq 1/1000$. Since $P \geq 33$, $\sqrt{G+1}\epsilon < s/4$. Equations (212) and (214) put the physical trace of the even groups below $1/10 + (81/2)s$ and the total trace above $1 - (33/32)s$. Their ratio is less than $1/7$; trace error $1/100$ leaves odd-group probability above $5/6$. For existence alone, use the input-independent observable

$$M = \frac{1}{2/5} \left(\bigoplus_{g,i} C_{g,i} - \frac{57}{20} I \right) \quad (217)$$

which has norm at most one. The same three telescoping estimates put $f(X)/\bar{T} < 2.82$ on yes-instances and > 2.89 on no-instances, where $\bar{T} = P^{-1} \sum_{g,i} \text{tr } X_{g,i}$. Equivalently, $\text{tr}(M\rho(X)) < -3/40$ and $> 1/10$, respectively. The POVM $(I \pm M)/2$, repeated a constant number of times, therefore decides the promise. Now apply (194). $\square$

The order $1/\sqrt{GP}$ of the residual tolerance in (216) is necessary. On an all-even base input, flip each of the GN raw signs in turn, take the neighboring instance's exact odd optimizer, and average the resulting physical solutions. This PSD mixture has objective $14/5$, exact accumulator rows, and only the GN flipped-edge copy residuals. With $r_- = (1, -1, 1)^\top / \sqrt{3}$ and $Z_-^* = r_- r_-^\top$, each nonzero residual before averaging has Frobenius norm $\|D_- Z_-^* D_- - Z_-^*\|_F = 4/3$. The total Euclidean residual is therefore

$$\frac{4}{3\sqrt{GN}} = \frac{4\sqrt{33}}{3\sqrt{GP}}. \quad (218)$$

Yet on the all-even input its density makes (217) report a winner. Feasibility alone is also insufficient: the public point (188) is exactly feasible for every input. Hence both conditions in theorem 7.2 are needed.

7.6 Exact central path and condensation

Eliminating a length- P path multiplies the physical barrier coefficient by P . Thus, with $\tau = \mu P$, the root problem is

$$Z_g \succ 0, \min_{\sum_g \text{tr } Z_g = 1} \left\{ \sum_g \langle \bar{C}_{h_g}, Z_g \rangle - \tau \sum_g \log \det Z_g \right\}. \quad (219)$$

Its exact center is

$$Z_g(\tau) = \tau(\bar{C}_{h_g} - \lambda I)^{-1}, \quad 1 = \tau \sum_g \text{tr}(\bar{C}_{h_g} - \lambda I)^{-1}, \quad (220)$$

with the unique $\lambda < \min_g \lambda_{\min}(\bar{C}_{h_g})$. This instantiates proposition 6.1.

On the all-even input, every group has trace exactly $1/G$. With $y = 29/10 - \lambda$ and $\alpha = \tau G$, the scalar equation and its positive solution are

$$1 = \alpha \left(\frac{2}{y} + \frac{1}{y + 3/10} \right), \quad (221)$$

$$y_\alpha = \frac{3\alpha - 3/10 + \sqrt{(3/10 - 3\alpha)^2 + (12/5)\alpha}}{2}. \quad (222)$$

Now suppose group 1 is the unique odd group and put $x = 14/5 - \lambda$. The winner and loser resolvents are

$$A(x) = \frac{1}{x} + \frac{2}{x + 3/10}, \quad B(x) = \frac{1}{x + 2/5} + \frac{2}{x + 1/10}. \quad (223)$$

The exact central path is determined by

$$\boxed{1 = \tau \{A(x) + (G-1)B(x)\}}, \quad \boxed{p_G = \tau A(x), \quad 1 - p_G = \tau(G-1)B(x)}. \quad (224)$$

Direct algebra gives

$$A(x) - B(x) = \frac{3/250}{x(x+1/10)(x+3/10)(x+2/5)} > 0 \quad (x > 0). \quad (225)$$

Together with (224), this yields $p_G/(1-p_G) > 1/(G-1)$, and hence $p_G \geq 1/G$. This implication is special to the present spectra and is not part of the general two-cost model of [section 6.1](#). Since B is decreasing on $x > 0$ and $B(0) = 45/2$, (224) also gives

$$p_G \geq 1 - \frac{45}{2}\tau(G-1). \quad (226)$$

The constants of [theorem 6.2](#) are

$$B_0 = B(0) = \frac{45}{2}, \quad \beta = -B'(0) = \frac{825}{4}, \quad \boxed{\alpha_* = B_0^{-1} = \frac{2}{45}}. \quad (227)$$

Applying [theorem 6.2](#) gives, for fixed $\alpha > 0$ and $G \rightarrow \infty$,

$$0 < \alpha < 2/45 : \quad p_G \longrightarrow 1 - \frac{45}{2}\alpha, \quad \frac{x}{\tau} \longrightarrow \frac{1}{1 - (45/2)\alpha}, \quad (228)$$

$$\alpha = 2/45 : \quad \sqrt{G} p_G \longrightarrow \frac{\sqrt{33}}{9}, \quad \sqrt{G} x \longrightarrow \frac{2}{\sqrt{825}}, \quad (229)$$

$$\alpha > 2/45 : \quad G p_G \longrightarrow \alpha A(x_\alpha), \quad B(x_\alpha) = \frac{1}{\alpha}. \quad (230)$$

In the supercritical case, (225) implies the strict coefficient inequality $\alpha A(x_\alpha) > \alpha B(x_\alpha) = 1$. The coefficient $\sqrt{33}/9$ follows from $\alpha_* \sqrt{\beta} = (2/45)(\sqrt{825}/2)$.

At the center, the reduced Hessian in the Frobenius inner product is

$$\mathcal{H}_\tau[Y]_g = \frac{1}{\tau} D_g Y_g D_g, \quad D_g = \overline{C}_{h_g} - \lambda I, \quad \sum_g \text{tr } Y_g = 0. \quad (231)$$

For one odd group, the spectral gaps are $(x, x+3/10, x+3/10)$ in the winner and $(x+1/10, x+1/10, x+2/5)$ in every loser. For $G \geq 3$, [lemma 6.3](#) specializes to

$$\kappa_{\text{red}} \asymp \frac{(x+2/5)^2}{\min\{x(x+3/10), x^2 + (x+1/10)^2/G\}}. \quad (232)$$

Therefore [theorem 6.4](#) gives

$$\kappa_{\text{red}} = \begin{cases} \Theta(G), & 0 < \alpha \leq 2/45, \\ \Theta(1), & \alpha > 2/45. \end{cases} \quad (233)$$

On the all-even input and for $G \geq 2$, by contrast,

$$\kappa_{\text{even}} = \left(\frac{y_\alpha + 3/10}{y_\alpha} \right)^2 = \Theta_\alpha(1). \quad (234)$$

The bound ([MC3](#)) simplifies here because $\min\{1, (2/5)/(3/10)\} = 1$: for $G \geq 3$,

$$\kappa_{\text{red}} \geq \frac{(G-1)p_G}{1-p_G}. \quad (235)$$

Thus a constant winner mass forces $\kappa_{\text{red}} = \Omega(G)$ without preconditioning. This concerns the central path only and gives no lower bound for methods with an input-dependent preconditioner. The corresponding visibility bounds are [equations \(CV2\)](#) and [\(CV4\)](#).

For approximate centers, we measure centrality by the dimensionless local residual of [theorem 6.10](#), not an unscaled stationarity error. If positive definite $\tilde{Z}_g$ and a multiplier $\tilde{\lambda} < 14/5$ satisfy

$$\left\| \tau^{-1} \tilde{Z}_g^{1/2} (\bar{C}_{h_g} - \tilde{\lambda} I) \tilde{Z}_g^{1/2} - I \right\|_{\text{op}} \leq \eta < 1, \quad \left| \sum_g \text{tr} \tilde{Z}_g - 1 \right| \leq \zeta < 1, \quad (236)$$

put $\tilde{p}_G = \text{tr}(\tilde{Z}_1) / \sum_g \text{tr}(\tilde{Z}_g)$. The denominator need not equal one under (236). Then (R1)–(R3) specialize to

$$\tilde{p}_G \geq 1 - \frac{(1 + \eta)(45/2)\alpha}{1 - \zeta}. \quad (237)$$

The approximate transition lies between $(1 + \eta)(45/2)\alpha = 1 - \zeta$ and $(1 - \eta)(45/2)\alpha = 1 + \zeta$. Equation (R6) also gives $\kappa(\tilde{\mathcal{H}}|\tau) = \Omega(G)$ whenever $\tilde{p}_G$ is bounded below. Trace error δ lowers the probability of observing the winner label by at most δ .

7.7 Preparation, visibility, and extraction

The root and physical central densities are

$$\rho_\tau = \bigoplus_g Z_g(\tau), \quad \rho_\tau^{\text{phys}} = \frac{1}{P} \bigoplus_{g,i} R_{g,i} Z_g(\tau) R_{g,i}^\top. \quad (238)$$

Both return the unique winner with probability p_G when the group label is measured. The ordering (225) proves $p_G \geq 1/G$, so the general upper bound $\max\{p_G, 1/G\}$ in (O1) becomes p_G . Thus [theorem 6.6](#) gives

$$p_G - \frac{1}{G} \leq D_{\text{tr}}(\rho_0, \rho_1) \leq p_G, \quad (239)$$

$$c_1 - c_0 = \frac{(Gp_G - 1)^2}{G(G - 1)}, \quad c_0 = \frac{1}{G}. \quad (240)$$

These formulas bound the cost of the first preparation through a collision test on two independent preparations, with no query to verify a candidate winner. If $\tau \leq 1/(90G)$, (226) gives $p_G \geq 3/4$, and (240) gives $c_1 - c_0 \geq 1/8$ for every $G \geq 2$. Two independently prepared states of trace error ϵ change the collision gap $c_1 - c_0$ by at most 4ϵ . Thus every fixed $\epsilon < 1/32$ leaves a constant gap. At $\tau \leq 1/(135G)$, one has $p_G \geq 5/6$; trace error $1/20$ leaves gap at least $2/9 - 1/5 = 1/45$. [Theorem 6.7](#) then implies

$$\boxed{Q_\rho = \Omega(N\sqrt{G})} \quad (241)$$

for a first root or physical central-density preparation at either stated schedule and tolerance. At the first schedule, the physical duality gap is $3G\tau \leq 1/30$. Under [definition 2.27](#), (241) includes setup queries; it does not cover an algorithm given a parity table or a central-state oracle, which already contains the aggregated input.

Under the promise of exactly one winner, there is a matching coherent upper bound. A clean parity evaluator costs $T_{\text{eval}} = N$ raw queries. Instantiating (A4)–(A5) in [proposition 6.8](#) gives

$$Q_U = O\left(N(\sqrt{Gp_G} + 1)\right), \quad (242)$$

because the normalized 3-dimensional winner and loser resolvents are public functions of (G, τ) . The acceptance probability is classically known, so exact phase-matched amplitude amplification removes any logarithmic overhead from the raw-query count; the conditional local preparation

cost T_{loc} is gate cost only. With the lower bound (A14), this is tight when the amplification factor $\sqrt{Gp_G}$ diverges and the error resolves the winner mass p_G . Hence

$$Q_U = \begin{cases} \Theta(N\sqrt{G}), & 0 < \alpha < 2/45, \\ \Theta(NG^{1/4}), & \alpha = 2/45, \\ O(N), & \alpha > 2/45. \end{cases} \quad (243)$$

The extension to k promised winners, (AP3)–(AP6) in section 6.9, is not needed here.

Given an exact central-state preparation, (A10) gives these costs for extracting the winner label:

regime	p_G	verified copies	coherent calls
$0 < \alpha < 2/45$	$\Theta(1)$	$\Theta(1)$	$\Theta(1)$
$\alpha = 2/45$	$\Theta(G^{-1/2})$	$\Theta(G^{1/2})$	$\Theta(G^{1/4})$
$\alpha > 2/45$	$\Theta(G^{-1})$	$\Theta(G)$	$\Theta(G^{1/2})$

Verifying a label costs another N raw queries. Coherent amplification needs a preparation unitary, its inverse, and a clean marker; one trace-close output state does not suffice (section 8 treats that case). The table counts uses of an existing preparation, whereas (241) bounds the first preparation.

Visibility also shows when a fixed error allows a zero-query answer. If the allowed root trace error is at least p_G , then, under the promise of zero or one winner, the public all-even root center is a valid output, by (225) and (O1). At criticality this happens for $G = \Omega(\epsilon^{-2})$, and for fixed supercritical α it happens for $G = \Omega_\alpha(\epsilon^{-1})$. In physical coordinates, lifting that state through the input's prefix products $R_{g,i}$ costs exactly N phase queries and preserves trace distance, so

$$p_G \leq \epsilon \implies Q_\epsilon(\rho_{\text{central}}^{\text{root}}) = 0, \quad Q_\epsilon(\rho_{\text{central}}^{\text{phys}}) \leq N. \quad (244)$$

For sufficiently large fixed α , even the public maximally mixed physical state is zero-query: with y_α from (222), it suffices that

$$\frac{1}{3} - \frac{\alpha}{y_\alpha + 3/10} + p_G \leq \epsilon. \quad (245)$$

No $\Omega(N)$ physical-state lower bound is claimed above the transition.

For algorithm design, condensation reveals the winner only in central states whose preparation already costs as much as search, and a constant winner mass forces κ_{red} linear in G . The conservation law (A9) and the direct-search bounds equations (A11) and (A12) therefore favor search-then-crossover (find the winner, then solve only its group), at cost

$$O(N\sqrt{G}) + T_{\text{component solve}} + O(N)_{\text{recovery}}. \quad (246)$$

This family does not exhibit a QIPM speedup.

7.8 Search then solve

Write $\sigma_{g,j} = (-1)^{z_{g,j}}$. With the XOR-oracle answer register in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, query the public indices $j = 1, \dots, N$ in order while retaining the group address. Phase kickback implements

$$|g\rangle \mapsto \prod_{j=1}^N (-1)^{z_{g,j}} |g\rangle = h_g |g\rangle \quad (247)$$

with exactly N raw queries and no parity or prefix garbage.

Theorem 7.3 (Search-then-solve upper bound). *For failure probability δ , the value of the SDP can be returned using $O(N\sqrt{G}\log(1/\delta))$ raw queries. The same procedure returns a verified winning label, or ‘NONE’ on an all-even input, with a compact description of the corresponding exact root optimizer; another $O(N)$ queries prepare that optimizer’s physical state instead. It needs no QRAM.*

Proof. Apply bounded-error quantum search to the clean marker (247), and verify a returned label by evaluating its parity. Output 14/5 after successful verification and 29/10 when the search reports ‘NONE’. Standard reversible addressing uses $O(N\sqrt{G}\log(GP)\log(1/\delta))$ elementary routing gates and $O(\log G + \log P)$ work qubits, apart from raw-query implementation.

If g_* is odd, take $Z_{g_*} = r_- r_-^\top$ with $r_- = (1, -1, 1)^\top / \sqrt{3}$; if no odd group exists, take $g_* = 1$ and $Z_{g_*} = (I - r_+ r_+^\top)/2$ with $r_+ = (1, 1, 1)^\top / \sqrt{3}$. Set every other root to zero. This is a compact optimizer output in the sense of definition 2.21. Streaming the N prefix signs of g_* lifts it through (182) and prepares the normalized physical direct-sum state. Listing its $P = 33N$ nonzero blocks instead costs $\Theta(P)$ output work. $\square$

Together, theorem 7.3 and (194) show that bounded-error value estimation and winner recovery cost $\Theta(N\sqrt{G})$, using standard search [29, 57].

Exact quantum query complexity depends on the promise. Under the promise of zero or one winner, phase-matched exact amplification followed by exact parity verification gives

$$Q_E(\text{OR}_G^{0,1} \circ \text{PARITY}_N) = \Theta(N\sqrt{G}). \quad (248)$$

Without that promise, the unique multilinear output polynomial has a nonzero degree- NG monomial, so the polynomial method gives an $NG/2$ lower bound; computing every parity gives

$$Q_E(\text{OR}_G \circ \text{PARITY}_N) = \Theta(NG). \quad (249)$$

If exactly one winner is promised, the decision is trivial, but finding the winner’s label still costs $\Theta(N\sqrt{G})$ for $G \geq 2$.

Remark 7.4 (Stronger access and setup costs). An oracle for h_g removes the factor N only because it is a stronger input. Parallel query ports may reduce depth but not the total number of queries. The accumulator chain is public. The end-to-end bound requires compact optimizer output; writing out G arrays costs their output size. If preprocessing uses S raw queries and later calls use q_t , any procedure that outputs the value or a decodable winner obeys

$$S + \sum_t q_t = \Omega(N\sqrt{G}). \quad (250)$$

In particular, the queries needed to build an input-dependent block encoding, QRAM parity table, prefix table, null-space basis, or the cost matrix obtained by eliminating the path variables count toward S .

7.9 Formulation tradeoff and novelty

A single trace row on the roots has $3G$ nonzeros, but its lift from root directions to the matrix variables is an isometry. The accumulator chain (184) reduces row sparsity to five and column sparsity to two without changing the root SDP or the root-coordinate Hessian, but its lift is not an isometry (section 7.4). Replacing the free scalars by differences of nonnegative variables, or adding scalar log barriers, would change the geometry. Thus we do not claim a formulation with both bounded incidence and ambient condition number one.

The query tools are prior work: adversary composition is due to Høyer–Lee–Špalek, Belovs–Lee, and Reichardt; the randomized composition framework includes Ben-David–Kothari; search and minimum finding are due to BBHT and Dürr–Høyer; and fixed-point search is standard

[22, 24, 29, 57, 77, 122, 141]. Embeddings of Boolean functions into conic optimal values, and sparsity-dependent coefficient-query lower bounds, already appear in van Apeldoorn et al., Theorem 29 and Corollary 30, and Apers–Gribling, Theorem 8.4 [10, 130]. Fawzi’s non-SOC theorem and the Rayleigh–Ritz trace identity are likewise established tools.

To our knowledge, what is new is only their combination: a bounded-incidence public accumulator chain for the trace normalization, an exact spectral minimum that encodes $\text{OR} \circ \text{PARITY}$ with a constant gap, a public start whose root Hessian has condition number one, the exact constants in the condensation transition, and the preparation costs in each phase. None of Grover search, parity, composition, the Rayleigh–Ritz trace identity, generic SDP ill-conditioning, or log-det resolvents is claimed as new by itself.

8 Hidden-Block State Conversion

The central paths of [sections 6.8](#) and [7.6](#) concentrate trace on a hidden set of winning blocks. This section studies the query cost of preparing such a central state to trace-distance accuracy ε when the number of winners is promised to be either zero or k . The two extremes already appear in [section 6.9](#): a sufficiently large error allows the public all-loser output, whereas a sufficiently accurate coherent preparation costs as much as search. In between, the collision statistic in (173) loses a factor $1/k$, and a bound on pairwise trace distance gives no single measurement that works for every unknown winner set. Instead, we measure the label of the prepared state and verify it with fresh queries. This verified-sample decoder accepts a zero-winner input only through verifier error and applies directly to trace-close mixed outputs.

For parity blocks, the polynomial method then gives the query cost at every nontrivial accuracy. The key step is that averaging over inputs with fixed block parities removes every character supported on a proper nonempty part of a block, which divides the polynomial degree by the block length. We also compare the coupled centers with a batch of uncoupled qutrit-holonomy SDPs. Returning all component values admits no amortization, whereas estimating their average has the usual approximate-counting query complexity.

8.1 The zero-versus- k conversion problem

Let $f : \mathcal{X}_0 \sqcup \mathcal{X}_1 \rightarrow \{0, 1\}$ be a possibly partial Boolean predicate on one block. Its bounded-error query complexity and negative-weight adversary value satisfy

$$Q(f) = \Theta(A), \quad A = \text{Adv}^\pm(f).$$

There are G independently queried blocks with inputs $x_1, \dots, x_G$; in [section 7](#), block g is group g , whose input is its sign string $(\sigma_{g,1}, \dots, \sigma_{g,N})$, not one of its PSD blocks $X_{g,i}$. The winner set $S = \{g : f(x_g) = 1\}$ is promised to have size zero or k , where $1 \leq k < G$. One raw query addresses one coordinate of one block. The target on a k -winner input is

$$\rho_S = \frac{p}{k} \sum_{g \in S} |g\rangle\langle g| \otimes \rho_W + \frac{1-p}{G-k} \sum_{g \notin S} |g\rangle\langle g| \otimes \rho_L, \quad \rho_0 = \frac{1}{G} \sum_{g=1}^G |g\rangle\langle g| \otimes \rho_L^0, \quad (251)$$

where $p = p_{G,k}$ is the winner mass and the constant-dimensional internal states ρ_W , ρ_L , and ρ_L^0 are public. Write

$$\eta_{\text{int}} = D_{\text{tr}}(\rho_L, \rho_L^0), \quad \delta = p - \varepsilon. \quad (252)$$

Here η_{int} is the local internal-state discrepancy, not the residual tolerance η used elsewhere. The task is to output, on every promised input, a state within trace distance ε of its target. The upper bounds below use a clean coherent evaluator as in (A1), which computes the membership bit for S , at cost T_{eval} per call. An exact evaluator costs $\Theta(Q_E(f))$ raw queries, where Q_E

denotes exact quantum query complexity. A bounded-error evaluator may be used instead after coherent error reduction to operator-norm error below the per-call budget of the amplification, at cost $O(Q(f) \log(G/\min\{\delta, \varepsilon\}))$ per call; the tildes absorb this factor. For $f = \text{PARITY}_N$, $Q_E(f) = \lceil N/2 \rceil$, and $Q(f)$ and $\text{Adv}^\pm(f)$ are both $\Theta(N)$. As in the condensation applications, we assume $p \geq k/G$ when invoking rejection preparation. For classical algorithms we count only queries; preparing the output state from the classical transcript is free.

Consider the test that measures the label of an ε -accurate output and then evaluates f on the sampled block. On a k -winner input, it accepts with probability at least $p - \varepsilon = \delta$, ignoring verifier error. On a zero-winner input, every proposed block is a loser, so only verifier error can cause acceptance, whatever state the circuit produces. Unlike the collision statistic, this test is a single measurement that works for every winner set S .

Theorem 8.1 (Verified-sample lower bound). *Fix $\delta \in (0, 1)$. Suppose a T -query circuit solves (251) with $\varepsilon \leq p - \delta$. If*

$$\sqrt{G/k} \geq C\delta^{-1} \log(e/\delta) \quad (253)$$

for a sufficiently large universal constant C , then

$$T = \Omega\left(\delta A \sqrt{G/k}\right).$$

For $f = \text{PARITY}_N$, this is $T = \Omega(\delta N \sqrt{G/k})$.

Proof. First reduce the zero-versus-one problem of theorem 6.7 to zero-versus- k . Put $G' = \lfloor G/k \rfloor$. Replace each of the G' input blocks by k addressable copies and pin the remaining $G - kG' < k$ blocks to a fixed public element of $\mathcal{X}_0$. A query to a copy is simulated by one query to its original. Zero winners remain zero, and a unique winner produces exactly k winners. The Cartesian-product promise is preserved. Since (253) forces G/k to be at least a large constant, $G' = \Theta(G/k)$, and (O3) gives a decision lower bound $\Omega(A\sqrt{G/k})$. This construction also works when k does not divide G .

Now repeat the following trial $R = \Theta(1/\delta)$ times. Run the preparation circuit afresh, measure its label g , and evaluate $f(x_g)$ with error at most $\beta = \delta/8$. Standard error reduction costs $O(Q(f) \log(e/\delta))$ queries per verification. On a k -winner input, contractivity of trace distance under the two-outcome measurement $\{\sum_{g \in S} |g\rangle\langle g| \otimes I, I - \dots\}$ gives

$$\Pr[g \in S] \geq p - \varepsilon \geq \delta, \quad \Pr[\text{accept}] \geq \delta(1 - \beta) \geq \delta/2.$$

On a zero-winner input, every label fails the true predicate, regardless of the prepared state, and $\Pr[\text{accept}] \leq \beta = \delta/8$. A multiplicative Chernoff bound distinguishes these two rates with constant error using $R = \Theta(1/\delta)$ trials. Its total cost

$$O(\delta^{-1}[T + Q(f) \log(e/\delta)])$$

is therefore $\Omega(A\sqrt{G/k})$. Since $Q(f) = \Theta(A)$, (253) absorbs the verification term and proves the claim. $\square$

The domain condition (253) is restrictive. At constant δ it requires only $G/k = \Omega(1)$; corollary 8.3 assumes the stronger uniform condition $k = O(G/\log^2 G)$. In general it requires $G/k = \Omega(\delta^{-2})$ up to logarithms. In exchange, theorem 8.1 holds for a general inner predicate f and assumes no access to a purification or to the inverse of the preparation.

Theorem 8.2 (Detuned rejection preparation). *Assume $p \geq k/G$, $1 - p \geq c_0 > 0$, and $0 < \varepsilon < p$. With $\delta = p - \varepsilon$, if $4\eta_{\text{int}} \leq \varepsilon$, then*

$$Q_\varepsilon = O\left(T_{\text{eval}} \left[1 + \sqrt{\delta G/k}\right] c_0^{-1/2} \log \frac{e}{\min\{\delta, \varepsilon\}}\right). \quad (254)$$

If $0 < \varepsilon < 4\eta_{\text{int}}$, then

$$Q_\varepsilon = O\left(T_{\text{eval}}\sqrt{G/k}\log(e/\varepsilon)\right). \quad (255)$$

When $\varepsilon = \Omega(\delta)$, the logarithm in (254) is simply $O(\log(e/\delta))$.

Proof. We prepare with a detuned winner mass m , chosen in two regimes:

$$\begin{aligned} m &= \begin{cases} \max\{3\delta/2, k/G\}, & \delta \leq 2p/3, \\ \max\{p - \varepsilon/2, k/G\}, & \delta > 2p/3, \end{cases} \\ m \leq p : & \begin{cases} 3\delta/2 \leq p \text{ and } k/G \leq p, & \delta \leq 2p/3, \\ p - \varepsilon/2 \leq p \text{ and } k/G \leq p, & \delta > 2p/3. \end{cases} \end{aligned} \quad (256)$$

Then $k/G \leq m \leq p$, $m - \delta \geq \delta/2$ in the first regime, and $m - \delta \geq \varepsilon/2$ in the second. Moreover $\sqrt{Gm/k} = O(1 + \sqrt{\delta G/k})$: this is immediate in the first regime, while $m = \Theta(\delta)$ in the second. The cap $m \leq p$ is essential: with $1 - p \geq c_0$, it keeps the zero-winner acceptance probability from adding a factor $(1 - m)^{-1/2}$ to the amplification cost.

Run the rejection construction of [proposition 6.8](#) with the label copied before amplification and the copy included in every reflection, and with detuned weights

$$a'_W = m/k, \quad a'_L = (1 - m)/(G - k), \quad a'_{\text{max}} = m/k.$$

Starting from the uniform label superposition, copy the label to an environment register, compute the membership bit, rotate an acceptance qubit with amplitude $\sqrt{a'_b/a'_{\text{max}}}$, and conditionally attach public purifications of ρ_W and ρ_L . A trial has acceptance probability

$$P_k = \frac{k}{Gm}, \quad P_0 = \frac{k(1 - m)}{m(G - k)} \quad (257)$$

on k -winner and zero-winner inputs, respectively. Both are at least a constant times $c_0 k/(Gm)$, because $m \leq p$ implies $1 - m \geq 1 - p \geq c_0$.

Apply fixed-point amplification simultaneously to the two possible acceptance angles [\[141\]](#). The fixed-point reflections act on every register the trial touches, including the environment copy of the label, which is uncomputed and recomputed inside each reflection. With

$$r = \frac{1}{8} \min\{\delta, \varepsilon\},$$

the acceptance-subspace residual can be made at most r^2 on both kinds of input using

$$O\left(\sqrt{Gm/k} c_0^{-1/2} \log(e/r)\right)$$

evaluator calls. The quadratic residual is necessary: overlap error of a purification becomes its square root in trace distance. Writing $q = k/G$, (257) gives $P_0/P_k = (1 - m)/(1 - q)$. Thus $m = q$ gives equal acceptance probabilities. When $q = o(m)$, however, an exact schedule tuned to the larger probability P_k under-rotates a zero-winner input and can leave $\Theta(m^2)$ junk probability, which exceeds the error budget for constant p . Fixed-point amplification avoids this mismatch.

Conditioned on acceptance, the output on a k -winner input is exactly (251) with winner mass m in place of p ; the zero-winner output is exactly $G^{-1} \sum_g |g\rangle\langle g| \otimes \rho_L$. Hence the respective errors are at most

$$(p - m) + r, \quad \eta_{\text{int}} + r.$$

If $\delta \leq 2p/3$, then $\varepsilon = p - \delta \geq \delta/2$ and $p - m \leq \varepsilon - \delta/2$, so both errors are below ε . If $\delta > 2p/3$, then $p - m \leq \varepsilon/2$ and $r \leq \varepsilon/8$, so again both errors are at most ε . Substituting the bound on m gives (254).

For (255), first decide whether there are zero or k winners by fixed-point search, reducing its error to $O(\varepsilon)$. If the decision is k , use the exactly- k circuit (AP3); if it is zero, prepare ρ_0 directly. The decision error contributes at most $O(\varepsilon)$ in trace distance, and the stated cost dominates both steps. Deciding first avoids circular use of the exactly- k promise. $\square$

The precision logarithm in (254) depends on $\min\{\delta, \varepsilon\}$, not on δ alone. In the second regime of (256), a fixed-point residual of order δ would not suffice when $\varepsilon \ll \delta$; the residual must be $O(\varepsilon)$.

Corollary 8.3 (Constant-visibility regime). *Let $f = \text{PARITY}_N$, let p be bounded away from zero and one, and suppose*

$$k = O(G/\log^2 G), \quad 4\eta_{\text{int}} \leq \varepsilon \leq (1-c)p$$

for a constant $c > 0$. Then, up to fixed-point precision logarithms,

$$Q_\varepsilon = \tilde{\Theta}\left(N\sqrt{G/k}\right).$$

By (255), the same scale holds for $\varepsilon < 4\eta_{\text{int}}$ when $\delta = \Omega(1)$. If ε exceeds the minimax radius of the target family, zero queries suffice; when the internal loser states agree, (172) bounds the public output's error by $\max\{p, k/G\}$.

Proof. Here $\delta = p - \varepsilon = \Omega(1)$ and $A = \Theta(N)$. The domain assumption makes (253) valid, so theorem 8.1 gives the lower bound. The upper bound is theorem 8.2. The zero-query statement follows by returning the public all-loser target and using (172). $\square$

Theorem 8.4 (Classical query complexity). *For $f = \text{PARITY}_N$, suppose $p \geq k/G$, $1-p = \Omega(1)$, and*

$$2\eta_{\text{int}} \leq \varepsilon, \quad Ck/G \leq \delta = p - \varepsilon \leq 1/2. \quad (258)$$

Here $C > 0$ is a sufficiently large universal constant. Then the randomized classical raw-query complexity is

$$R_\varepsilon = \Theta(N\delta G/k).$$

Proof. For the upper bound, inspect $t = \Theta(\delta G/k)$ uniformly random blocks without replacement and compute each parity exactly using N queries. Put $q = k/G$. The probability of finding at least one winner is

$$P_{\text{find}} = 1 - \frac{\binom{G-k}{t}}{\binom{G}{t}}.$$

The constant in t can be chosen so that $P_{\text{find}} \geq \delta$ throughout (258). This use of a linear budget relies on $\delta \leq 1/2$; as $\delta \rightarrow 1$, the correct sampling budget contains $(G/k) \log(1/(1-\delta))$.

Let E_{all} be the event that every sampled block is a winner. Its public probability is at most $q^t \leq q \leq \delta$. On E_{all} output a uniformly selected sampled winner. On $E_{\text{find}} \setminus E_{\text{all}}$, do the same with a public probability chosen so that the total probability of winner output is exactly δ ; this is possible because $\Pr(E_{\text{all}}) \leq \delta \leq \Pr(E_{\text{find}})$. In all other cases select uniformly among sampled losers. A loser is available whenever the latter rule is used. Exchangeability makes the selected winner uniform on S and the selected loser uniform on its complement. Attaching the corresponding public internal error state therefore prepares exactly the target shape with winner mass δ in place of p . Its trace error on a k -winner input is $p - \delta = \varepsilon$. On a zero-winner input the output is the uniform-label ρ_L state and has error $\eta_{\text{int}} \leq \varepsilon/2$. Thus $O(N\delta G/k)$ queries suffice.

For the lower bound, append exact N -query parity verification to a sampled output label and repeat the preparation $\Theta(1/\delta)$ times. This decides zero versus k winners with one-sided error and cost $O(\delta^{-1}(T + N))$. For the decision lower bound, use the k -copy embedding from theorem 8.1 and draw each original block uniformly conditioned on its parity. Every proper subset of a block is uniform, so a classical transcript is independent of the planted parity until all N positions of that block have been queried. A q -query tree completes at most q/N original blocks, and it hits the uniformly planted winner with probability at most $O(qk/(NG))$. Yao's principle therefore gives $\Omega(NG/k)$ decision queries. Consequently

$$\delta^{-1}(T + N) = \Omega(NG/k).$$

The lower restriction $\delta \geq Ck/G$ absorbs the additive N term. $\square$

We claim no lower bound when $\delta = o(k/G)$: there, a public output with uniform label may already meet the accuracy requirement. The upper restriction $\delta \leq 1/2$ in (258) is also needed: near one, the probability of finding a winner is no longer linear in the number of sampled blocks.

8.2 Composed small-success search

The verified-sample argument loses a factor $\sqrt{\delta}$ by repeating the preparation. For parity blocks, the polynomial method recovers it. We state the degree-division step as a lemma because section 8.3 also uses it.

Lemma 8.5 (Parity collapse). *Let a T -query quantum algorithm access $x \in \{\pm 1\}^{G \times N}$, and let $P(x)$ be the probability of a fixed final measurement event. For block parities $z_g = \prod_{j=1}^N x_{g,j}$, define*

$$q(z) = \mathbb{E}[P(x) \mid \prod_j x_{g,j} = z_g \text{ for every } g].$$

Here the conditional expectation is uniform over each parity fiber, independently across blocks. Then q is a real multilinear polynomial in $z \in \{\pm 1\}^G$ of degree at most $\lfloor 2T/N \rfloor$. If $0 \leq P \leq 1$, then $0 \leq q \leq 1$ on the Boolean cube.

Proof. The polynomial method writes P as a real multilinear polynomial of degree at most $2T$ [20]. Consider a monomial x_R . Conditional expectation factors over blocks. In one block, the average of a character on a parity coset is one when its share of R is empty, is z_g when that share is the full block, and is zero for every proper nonempty share. The last assertion follows by pairing inputs that differ in two coordinates, one inside and one outside the proper share. Thus a raw monomial with nonzero average becomes a monomial in z whose degree is the number of full blocks it contains, at most $|R|/N \leq 2T/N$. Averaging all monomials proves the claim. $\square$

Remark 8.6 (Computational check). We checked the character calculation in lemma 8.5 by exhaustive enumeration for $N = 2, 3, 4, 5$, and the full identity on random multilinear polynomials.

Theorem 8.7 (One-sided composed small-success bound). *Suppose a T -query quantum algorithm accepts with probability at most $\beta \leq \delta/2$ on every all-even input and at least δ on every input with exactly k odd-parity blocks. Then, for absolute constants $c_1, c_2 > 0$,*

$$T \geq c_1 N \sqrt{\delta G/k} - c_2 N. \quad (259)$$

Proof. Let $P(x)$ be the acceptance probability. By lemma 8.5, conditioning on all block parities gives a polynomial $q(z)$ of degree at most $D = \lfloor 2T/N \rfloor$, bounded between zero and one at every Boolean input. It is at most β at the all-even point and at least δ at every parity vector of odd-block weight k .

Average q over all block permutations. Minsky–Papert symmetrization [103] gives a univariate polynomial h of degree at most D such that

$$0 \leq h(w) \leq 1 \quad (w = 0, 1, \dots, G), \quad h(0) \leq \beta, \quad h(k) \geq \delta.$$

Fix a sufficiently small absolute $\gamma > 0$. First suppose $D \leq \gamma\sqrt{G}$. The Coppersmith–Rivlin inequality bounds a degree- D polynomial bounded at the $G+1$ equally spaced integer points by

$$\sup_{t \in [0, G]} |h(t)| \leq C(\gamma),$$

where $C(\gamma)$ is an absolute constant in this degree range [46, 86]. We use only the existence of the universal Coppersmith–Rivlin constants, which the original paper does not give numerically; this is the form quoted in Theorem 8 of Klauck–Špalek–de Wolf. Erdélyi proves a related explicit-constant inequality in a different, single-point-versus-supremum parametrization [59].

Markov's inequality, rescaled from $[-1, 1]$ to $[0, G]$, now gives

$$\sup_{[0, G]} |h'| \leq \frac{2D^2 C(\gamma)}{G}.$$

The mean value theorem on $[0, k]$ gives a point ξ with

$$h'(\xi) \geq \frac{h(k) - h(0)}{k} \geq \frac{\delta - \beta}{k} \geq \frac{\delta}{2k}.$$

Therefore $D = \Omega(\sqrt{\delta G/k})$. In the complementary case $D > \gamma\sqrt{G}$, the same conclusion follows directly from $\delta \leq 1 \leq k$. Finally, $D \leq 2T/N$, with an additive $O(N)$ allowance for integer rounding, proves (259). $\square$

To our knowledge, [theorem 8.7](#) is new as stated, but its ingredients are standard and experts could plausibly derive it. It is not a new general composition theorem.

Corollary 8.8 (Composed search). *If a quantum algorithm outputs an odd-parity block with probability at least δ whenever exactly k blocks are odd, then*

$$T = \Omega\left(\sqrt{\delta} N \sqrt{G/k}\right) - O(N).$$

This is tight up to constants and the additive N term.

Proof. Verify the output block by computing its parity exactly with N fresh queries. Acceptance is then zero on an all-even input and at least δ on an exactly- k input, so [theorem 8.7](#) applies to $T + N$ queries. Conversely, if $\delta \leq 4k/G$ a uniformly random label is odd with probability $k/G \geq \delta/4$ at zero queries. Otherwise $k/G < 1/4$, so the initial Grover angle $\theta = \arcsin \sqrt{k/G}$ is below $\pi/6$, and $t = \lceil c\sqrt{\delta G/k} \rceil \geq 1$ Grover iterations with $c = 1/6$ keep $(2t + 1)\theta \leq \pi/2$; each uses a $\Theta(N)$ -query parity marker, and together they produce a winner with probability $\sin^2((2t + 1)\theta) \geq \sin^2(2c\sqrt{\delta}) = \Theta(\delta)$. Direct exact verification costs another N queries. For one marked item, Zalka proved exact optimality [142]. The small-success scaling for k marked items follows from the angle bound in Theorem 2 of the published Dohotaru–Høyer paper (Theorem 9 of the arXiv version) plus the standard k -copy embedding. Carolan–Poremba Corollary 2.4 records the packaged inequality $\delta \leq 8(T + 1)^2 k/G$; our attribution uses the angle theorem and embedding, not their intermediate distance-based derivation [35, 55]. Zalka's result alone does not establish the k -marked statement. Aaronson's direct-product Theorem 4.6 instead concerns the find-all- k analogue [1]. $\square$

Corollary 8.9 (Full conversion curve). *Let $f = \text{PARITY}_N$, $p \geq k/G$, $1 - p = \Omega(1)$, $4\eta_{\text{int}} \leq \varepsilon < p$, and $\delta = p - \varepsilon \geq Ck/G$ for a sufficiently large constant C . Then*

$$Q_\varepsilon = \tilde{\Theta}\left(N \sqrt{(p - \varepsilon)G/k}\right). \quad (260)$$

The tilde hides only fixed-point precision logarithms: the hidden factor is $\log(e/\min\{\delta, \varepsilon\}) = \log(1/\varepsilon) + O(1)$ when $\varepsilon \ll \delta$, and can be unbounded relative to the main term for superpolynomially small ε . The lower bound is nonvacuous once $\delta G/k$ is a sufficiently large constant.

Proof. One preparation followed by label measurement and exact parity verification uses $T + N$ queries. It accepts with probability zero on a zero-winner input and at least $p - \varepsilon = \delta$ on a k -winner input. Applying [theorem 8.7](#) and absorbing its additive $O(N)$ term under the domain assumption gives the lower bound. The upper bound is [theorem 8.2](#); its additive one is absorbed on the same domain. $\square$

Unlike [theorem 8.1](#), this conclusion needs neither $k = O(G/\log^2 G)$ nor $G/k = \tilde{\Omega}(\delta^{-2})$. It is parity-specific: the gain comes from [lemma 8.5](#).

Corollary 8.10 (Mixed-output preparation profile). *Suppose a T -query circuit, on every exactly- k parity input, outputs a state within trace distance $\varepsilon \leq p/2$ of the winner-mass- p target. Then*

$$T = \Omega\left(\sqrt{p} N \sqrt{G/k}\right) - O(N). \quad (261)$$

Thus the first two lines of the preparation profile ([AP6](#)) are tight even for trace-close mixed outputs: the lower bounds are $\Omega(N\sqrt{G/k})$ in the subcritical phase and $\Omega(N(G/k)^{1/4})$ at criticality. In the supercritical phase the displayed lower bound is vacuous, so it does not show that the $O(N)$ upper bound is tight.

Proof. Run the circuit also on an all-even input, which is outside its preparation promise but on which the physical circuit remains defined. Measure a label and verify its parity exactly. Acceptance is zero on this input and is at least $p - \varepsilon \geq p/2$ on every exactly- k input. Apply [theorem 8.7](#) with $\delta = p/2$. The three phase statements follow by substituting the winner masses from [theorem 6.9](#): $p = \Theta(1)$, $p = \Theta(\sqrt{k/G})$, and $p = \Theta(k/G)$. $\square$

[Corollary 8.10](#) weakens the hypothesis of the preparation lower bounds in [section 6.9](#). The lower bound ([AP5](#)) was obtained by coherent extraction through ([AP4](#)), and therefore assumed access to a preparation unitary and its inverse. For parity blocks and $\varepsilon \leq p/2$, [corollary 8.10](#) replaces this access by the weaker requirement that the circuit output a trace-close mixed state. The upper bound ([AP3](#)) and the caveat on the supercritical phase in ([AP6](#)) are unchanged. The family of [section 7.7](#) has parity blocks and $k = 1$, so this shows which of its condensed central states reveal the hidden holonomy; the search-then-crossover approach ([246](#)) is unchanged.

Combining [corollary 8.9](#) and [theorem 8.4](#) gives, on their common domain,

model	query complexity	dependence on $\delta = p - \varepsilon$
quantum	$\tilde{\Theta}(N\sqrt{\delta G/k})$	square root
randomized classical	$\Theta(N\delta G/k)$	linear

Thus at every nontrivial accuracy, from $\delta = \Theta(k/G)$ to constant δ , the classical-to-quantum ratio is $\tilde{\Theta}(\sqrt{\delta G/k})$, as for search. It is $\tilde{\Theta}(1)$ near the public-output boundary $\delta = \Theta(k/G)$ and $\tilde{\Theta}(\sqrt{G/k})$ at constant δ .

Open Problem 8.11 (General-inner composition). Does every predicate f satisfy a composed small-success lower bound

$$\Omega\left(\sqrt{\delta} Q(f) \sqrt{G/k}\right)$$

for finding a winner with probability δ ? The adversary reduction in [theorem 8.1](#) gives only linear dependence on δ , while the parity proof uses the fact that every character on a proper nonempty part of a block averages to zero.

8.3 Uncoupled holonomy batches

The global trace row in [section 7.1](#) couples all components. For contrast, take G disjoint copies of the qutrit-holonomy component, each with its own public trace-one row, and no equality that joins two components. The path length and local costs are those of [equations \(179\) and \(183\)](#). Consequently the product has $L = 33GN$ constant-size PSD blocks, scalar row sparsity at most three, and scalar column sparsity at most two. If

$$h_g = \prod_{j=1}^N \sigma_{g,j}, \quad y_g = (1 + h_g)/2,$$

then [\(190\)](#) gives component optima $v_+ = 29/10$ and $v_- = 14/5$.

Theorem 8.12 (No amortization for vector output). *Returning estimates $\hat{v}_1, \dots, \hat{v}_G$ satisfying $|\hat{v}_g - v_{h_g}| \leq 1/25$ for every g , with worst-case joint success probability at least $2/3$, has quantum raw-query complexity*

$$\Theta(GN) = \Theta(L).$$

Moreover, there are constants $a, b > 0$ such that every algorithm with at most aGN queries has worst-case joint success probability at most 2^{-bG} .

Proof. Threshold every estimate at $57/20$ to recover all h_g . Their product is the parity of all GN raw signs, which has quantum query complexity $\Omega(GN)$; reading all signs is a matching upper bound. For the stronger claim, specialize Lee–Roland’s strong direct-product theorem to G copies of PARITY_N [91]. Its Theorem 1.1 with success parameter $3/4$ gives

$$Q_{1-(3/4)^{G/2}}(f^{(G)}) \geq \frac{G \ln(9/8)}{8000} Q_{1/4}(f).$$

Since $Q_{1/4}(\text{PARITY}_N) = \Theta(N)$, with fewer than a fixed constant times GN queries, simultaneous decoding succeeds with worst-case probability at most $(3/4)^{G/2} = 2^{-\Omega(G)}$. $\square$

Now average the G component objectives into one SDP. Since the components share no constraint, its optimal value is

$$V(\sigma) = \frac{1}{G} \sum_{g=1}^G v_{h_g} = \frac{14}{5} + \frac{1}{10G} \sum_{g=1}^G y_g. \quad (262)$$

Theorem 8.13 (Query complexity of the averaged value). *For $0 < \varepsilon \leq 1/100$,*

$$Q_\varepsilon(V) = \Theta(N \min\{G, 1/\varepsilon\}), \quad (263)$$

$$R_\varepsilon(V) = \Theta(N \min\{G, 1/\varepsilon^2\}). \quad (264)$$

Proof. An exact coherent query to y_g costs $\Theta(N)$ raw sign queries. Amplitude estimation estimates its mean to error 10ε using $O(1/\varepsilon)$ such calls [31]; when $1/\varepsilon > G$, compute all parities. This proves the quantum upper bound. Classically, sample $O(\min\{G, 1/\varepsilon^2\})$ groups without replacement, compute their parities, and use the empirical mean, scanning all groups when the minimum is G .

For every G , restricting to inputs in which all G groups carry the same N -bit string x makes V change by $1/10 > 2\varepsilon$ when the parity of x flips, so an ε -estimator computes PARITY_N and $T \geq N/2$; this settles $G < 6$, where $\min\{G, 1/\varepsilon\} = O(1)$. For $G \geq 6$, choose Hamming weights $\ell, \ell' \in [G/4, 3G/4]$ separated by $\Delta = \max\{1, \lceil 21\varepsilon G \rceil\}$. Their values in (262) differ by more than 2ε , so thresholding an estimator gives a bounded-error polynomial distinguishing the two weights. Nayak–Wu’s symmetric approximate-counting lower bound gives degree

$$\Omega\left(\frac{\sqrt{m(G-m)}}{\Delta}\right) = \Omega(\min\{G, 1/\varepsilon\}),$$

where m is the selected weight farther from $G/2$ [112]. Applying the already proved degree division in lemma 8.5 forces $\lfloor 2T/N \rfloor = \Omega(\min\{G, 1/\varepsilon\})$.

The same lower bound also follows from perfect negative-adversary composition: compose the partial symmetric outer gap-counting problem with G independent copies of PARITY_N and combine Nayak–Wu’s outer bound with the Belovs–Lee composition theorem [22]. Thus the parity-collapse argument here is a construction-specific proof, not a new composition theorem.

For the classical bound, we show that even an adaptive algorithm must read every sign of a group to learn its parity. Fix $h \in \{\pm 1\}^G$ and draw the raw signs of each group uniformly conditional on their product being h_g . While at least two signs of a group remain unseen, a

simulator answers each new query to that group with an independent fair sign. Only when the final unseen sign in a group is queried does the simulator query h_g and force the required product. This exactly reproduces the conditional distribution under arbitrary interleaving. Each parity-oracle query consumes N distinct raw queries, so a depth- q raw tree induces a parity-bit tree of depth at most $\lfloor q/N \rfloor$. The classical bounded-error complexity of estimating a G -bit mean to error 10ε is $\Theta(\min\{G, 1/\varepsilon^2\})$, by the standard Yao distribution on two nearby central Hamming weights. Multiplication by N proves (264). $\square$

The two bounds give three regimes:

accuracy	$Q_\varepsilon(V)$	$R_\varepsilon(V)$
$\varepsilon = \Omega(G^{-1/2})$	$\Theta(N/\varepsilon)$	$\Theta(N/\varepsilon^2)$
$G^{-1} = O(\varepsilon) = O(G^{-1/2})$	$\Theta(N/\varepsilon)$	$\Theta(NG)$
$\varepsilon = O(G^{-1})$	$\Theta(NG)$	$\Theta(NG)$

At $\varepsilon = \Theta(G^{-1/2})$, the costs are $\Theta(N\sqrt{G})$ and $\Theta(NG)$, exactly the quantum and classical scales of the coupled winner-take-all value bound (197). The quantum advantage therefore depends on how the accuracy scales with G ; it is not an exponential advantage.

Corollary 8.14 (Classical output of all Newton directions). *At the public point $X_{g,i} = I/3$ and path parameter $\mu = 1/(GP)$, the uncoupled batch has reduced root Hessian $(9/G)I$ and condition number one. Nevertheless, producing classical descriptions of all G root Newton directions to entrywise error less than $1/200$ costs $\Theta(GN)$ queries, and query budgets below a suitable constant times GN have $2^{-\Omega(G)}$ worst-case joint success probability.*

Proof. Using A_h from (201), direct elimination gives

$$\Delta Z_g = -\frac{1}{90}(H_{12} + H_{23} + h_g H_{13}).$$

Every full physical step remains positive definite, with minimum eigenvalue at least $14/45$. The hidden 13 entry has magnitude $1/90 > 1/200$, so its sign reveals h_g . Apply theorem 8.12. $\square$

These batch results concern raw queries to the signed copy rows. An oracle that supplies group parities, aggregate gradients, or component optima has already done the hard part: combining the raw signs into parities or aggregates of them. Theorem 8.12 concerns writing out all components classically, not preparing one normalized quantum state of the global direction. Such a state gives each group amplitude $G^{-1/2}$, so a preparation with constant error need not reveal any single entry, as remark 7.1 shows. Finally, the claims assume exact equalities; the path operator has small singular values, so they imply nothing about solutions with constant residuals.

8.4 Open questions

Three questions remain open. First, the decoder based on winner mass does not detect the internal-state discrepancy η_{int} . The minimax radius and query complexity of the conversion problem when η_{int} matters, especially when p is below the condensation threshold, are open. Second, open problem 8.11 asks for the $\sqrt{\delta}$ composition bound beyond parity. Third, our tight bounds exclude the range $\delta = o(k/G)$: there, public outputs with uniform label can already meet some accuracy requirements, and the exact boundary depends on the internal states, not only on the winner mass.

9 LP Loading and Recovery Lower Bounds

This section proves lower bounds on the number of raw coefficient queries needed to load or recover LP states. The main result ([theorem 9.7](#)) is a linear-size standard-form LP with bounded row and column degree whose reduced primal Newton matrix is a scalar identity at a public infeasible start and whose reduced log-barrier Hessian is a scalar identity along the entire central path. Nevertheless, preparing the state of its Newton direction in the original primal coordinates needs linearly many raw coefficient queries. The hard step is not the scalar solve but constructing the hidden affine offset of the feasible set and recovering the direction in the original coordinates.

All state, access, residual, and success conventions are those of [sections 2.2](#) and [2.3](#). Throughout this section $N \geq 2$. A residual without another qualifier is the global RHS-relative residual $\|Ax - b\|_2/\|b\|_2$, and an output state is unconditional. The same results hold for a heralded branch with input-independent constant success probability when repetitions are counted. A reduced-coordinate state is not an original-coordinate output, and an unobserved “good branch” does not satisfy an output requirement. We count all raw queries used for setup, preconditioning, right-hand-side formation, solution, and recovery.

The amplification techniques we use are known. Parity and the polynomial method are classical query-complexity tools [\[20\]](#); endpoint padding and biased clocks occur in HHL and Feynman–Kitaev constructions [\[19, 34, 76\]](#); and sparse QLS states carrying an endpoint bit occur in Wang–Zhang and Mori et al. [\[109, 133\]](#). Apers–Gribling Theorem 8.4 already gives a linear exponent for sparse-LP *value* estimation in balanced dimensions [\[10\]](#). None of the results in this section is a QLS lower bound at constant condition number with free matrix and right-hand-side preparation oracles; each is an end-to-end lower bound for LP access and output. What is new is the combination, in LP families, of regularity (boundedness, strict primal–dual feasibility, and a unique nondegenerate strictly complementary optimum), well-conditioned reduced geometry, original-coordinate output, sharp residual thresholds, and accounting of every raw query.

9.1 Parity amplification and feasible-state hardness

The hidden input is $\sigma_1, \dots, \sigma_N \in \{\pm 1\}$. Put

$$p_0 = 1, \quad p_i = \prod_{j=1}^i \sigma_j.$$

Represent a signed scalar at node j by a nonnegative pair (u_j, v_j) , with $d_j = u_j - v_j$ and $q_j = u_j + v_j$. A chain fixes

$$d_0 = 1, \quad d_i - \sigma_i d_{i-1} = 0 \quad (1 \leq i \leq N). \quad (265)$$

At node N , attach a complete binary copy tree with

$$M = 2^{\lceil \log_2(8(N+1)) \rceil}, \quad K = 2M - 2, \quad P = N + 1 + K, \quad (266)$$

and impose $d_j - d_{\pi(j)} = 0$ on every new node, where $\pi(j)$ is its parent. Thus every copy node has $d_j = p_N$, while row and column sparsity are at most four and three.

We use the following elementary decoder throughout.

Lemma 9.1 (Fixed-observable parity decoder). *Let O be an input-independent Hermitian observable with $\|O\|_2 \leq 1$. If a family of normalized ideal states obeys*

$$p_N \langle \psi_\sigma | O | \psi_\sigma \rangle \geq \delta > 0,$$

then the POVM $(I \pm O)/2$ guesses p_N with bias at least $\delta/2$. From any ρ_σ with $D_{\text{tr}}(\rho_\sigma, |\psi_\sigma\rangle\langle\psi_\sigma|) \leq \epsilon$, the bias is at least $(\delta - 2\epsilon)/2$. If this is positive by an absolute constant, preparing ρ_σ requires $\Omega(N)$ raw coefficient queries.

Proof. Trace duality changes the expectation of O by at most 2ϵ . The stated POVM has signed expectation $\text{Tr}(O\rho)$. Constant repetition turns any constant positive bias into bounded error. Every sparse row, column, position, or value query in the families below exposes at most one σ_i , even coherently, and is simulated by one call to the sign oracle. The parity lower bound completes the reduction. $\square$

First, minimize $\sum_j q_j$ subject to the chain and copy equations. Exact feasibility fixes every $d_j \in \{\pm 1\}$, and $q_j \geq 1$. Hence the unique optimum has $q_j = 1$ and value P . The nullspace has the public orthonormal basis

$$V_j = \frac{e_{u_j} + e_{v_j}}{\sqrt{2}}. \quad (267)$$

On the feasible affine space the barrier objective is

$$\Phi_\mu(q) = \sum_j \left[q_j - \mu \log \frac{q_j^2 - 1}{4} \right]. \quad (268)$$

Thus every central coordinate is $q(\mu) = \mu + \sqrt{\mu^2 + 1}$, and

$$V^\top \nabla^2 \Phi_\mu V = 2\mu[(q+1)^{-2} + (q-1)^{-2}]I. \quad (269)$$

At $\mu = 3/4$, $q = 2$, so every pair, ordered with the coordinate selected by the sign of d_j first (the *oriented pair*), is $(3/2, 1/2)$. If instead an exactly feasible point has objective gap at most one, writing $q_j = 1 + \delta_j$ gives $\sum_j \delta_j \leq 1$ and

$$\|x\|_2^2 = \sum_j \frac{q_j^2 + 1}{2} \leq P + \frac{3}{2}. \quad (270)$$

Since $K \geq 16(N+1) - 2$, more than 9/10 of this squared norm lies in the copy tree, and the first coordinate of an oriented copy pair has at least 9/10 of the pair's squared norm. The computational-basis decoder therefore succeeds with probability greater than 43/50; trace error 1/10 leaves success probability greater than 19/25. This proves an $\Omega(N)$ lower bound both for every such near-optimal primal state and for the exact $\mu = 3/4$ central state, even though $\kappa_{\text{red}} = 1$.

A capped variant makes every exactly feasible point hard, without an objective promise. Add $t_j \geq 0$ and

$$2u_j + 2v_j + 2t_j = 3 \quad (271)$$

at every node, and retain objective $\sum_j q_j$. Exact feasibility now gives

$$1 \leq q_j \leq \frac{3}{2}, \quad t_j = \frac{3}{2} - q_j. \quad (272)$$

The equality matrix has rank $2P$, and its public orthonormal null basis is

$$\bar{V}_j = \frac{e_{u_j} + e_{v_j} - 2e_{t_j}}{\sqrt{6}}. \quad (273)$$

The positive optimal columns form a basis; the opposite pair coordinate has dual slack two. Thus the LP is bounded and strictly primal-dual feasible, with a unique nondegenerate strictly complementary optimum. Its reduced central Hessian is again a scalar identity, because the reduced barrier is a sum of identical functions of q_j .

For $q \in [1, 3/2]$, one triple has squared norm

$$\bar{w}(q) = \frac{q^2 + 1}{2} + \left(\frac{3}{2} - q \right)^2 \in \left[\frac{5}{4}, \frac{13}{8} \right]. \quad (274)$$

The copy triples therefore carry more than 9/10 of the total mass, and the computational-basis decoder (+ on u_j , - on v_j , a random guess on t_j) succeeds on the copy triples with conditional probability at least 9/10. Consequently the amplitude state of *every* exactly feasible point has the same $\Omega(N)$ query lower bound at trace error 1/10. The proof also holds for every $x \geq 0$ with absolute global ℓ_1 residual $\|\bar{A}x - \bar{b}\|_1 \leq 1/50$: telescoping gives $p_N d_j \geq 49/50$ at every copy node, while the cap gives $q_j + t_j \leq 151/100$. The resulting copy mass is greater than 43/50, and the signed conditional expectation exceeds 63/100. Before trace error the total signed expectation is therefore

$$\frac{43}{50} \frac{63}{100} = \frac{2709}{5000}, \quad (275)$$

so trace error 1/10 leaves signed expectation at least

$$\frac{2709}{5000} - \frac{1}{5} = \frac{1709}{5000} > \frac{1}{3}.$$

9.2 Residual robustness and construction size

The ℓ_1 result above does not extend to a constant ℓ_2 residual on a single path of unit height $|d_j| = 1$. In the gauged variables $p_i d_i$, set

$$p_i d_i = 1 - \frac{i}{N}, \quad d_j = 0 \quad (j \text{ in the copy tree}), \quad (276)$$

and take $q_j = 1, t_j = 1/2$. All cap and copy equations are exact, the N chain residuals equal $1/N$, and the total residual is $N^{-1/2}$, yet every copy-tree difference d_j is zero. Using $1 - 2i/N$ instead puts the wrong value $d_j = -p_N$ on every copy node, at residual $2/\sqrt{N}$. This is the basic drift obstruction.

For power-of-two N , a bounded-height construction avoids it by running $R = N$ signed paths of length N in parallel. A binary fanout tree feeds the paths, and a binary tree with $L = 16N$ leaves is attached to every endpoint. The resulting tree has

$$P = (2N - 1) + N^2 + N(2L - 2) = 33N^2 - 1, \quad |S| = 16N^2 \quad (277)$$

vertices and designated output leaves S . At every vertex use $(u, v, h, t) \geq 0$. On $d = u - v$ impose $d_{\text{child}} - a_e d_{\text{parent}} = 0$ on every tree edge e , with $a_e = \sigma_i$ on the i -th edge of each signed path and $a_e = 1$ elsewhere; impose the same equations on h with all labels one; pin both root values to one; and impose $q + t - 2h = 0$. Minimize $\sum_j (2u_j + 2v_j + h_j + t_j)$.

Theorem 9.2 (Parallel-path lower bound for small residual). *This LP has $4P$ variables, $3P$ full-rank equalities, coefficients of magnitude at most two, and row and column sparsity at most four. It has the same regularity properties as the capped family and a scalar reduced central Hessian for every $\mu > 0$. If $x \geq 0$, $x \neq 0$, and*

$$\frac{\|Ax - b\|_2}{\|b\|_2} \leq \frac{1}{200}, \quad \|b\|_2 = \sqrt{2}, \quad (278)$$

then the computational-basis decoder on the output leaves S has ideal bias greater than 1/80. Consequently, preparing an amplitude state within trace distance 1/100 of any such point needs $\Omega(N) = \Omega(\sqrt{P})$ raw queries, even under an additional two-sided objective or centrality promise.

Proof. Let λ_j be the exact propagated tree label. Gauging removes all signs. Descendant counting in each binary tree and Cauchy–Schwarz on the N parallel paths give

$$\frac{\|\text{Diag}(\lambda)z - \mathbf{1}\|_2}{\sqrt{P}} \leq 12\|(z_r - 1, Dz)\|_2, \quad \frac{\|\text{Diag}(\lambda_S)z_S - \mathbf{1}\|_2}{\sqrt{|S|}} \leq 12\|(z_r - 1, Dz)\|_2. \quad (279)$$

The binary-level operator norms sum to $\sum_{\ell \geq 1} 2^{-\ell/2} = 1 + \sqrt{2} < 5/2$, while the path term is $\sqrt{N/R} = 1$. With $E = \|Ax - b\|_2 \leq \sqrt{2}/200$, (279) for d and h , plus the cap and nonnegativity, gives

$$\|x\|_2^2 < 11P, \quad \sum_{j \in S} \lambda_j d_j q_j > \frac{3}{5}|S|. \quad (280)$$

The two loss terms in the second inequality, divided by $|S|$, are at most

$$12E \left(1 + 2.18\sqrt{P/|S|}\right) < \frac{12\sqrt{2}}{200} \left(1 + 2.18\sqrt{33/16}\right) < \frac{2}{5}.$$

Hence the measurement bias is larger than

$$\frac{(3/5)16}{2 \cdot 11 \cdot 33} = \frac{8}{605} > \frac{1}{80}.$$

Trace error $1/100$ leaves positive constant bias, and lemma 9.1 applies. Full rank and regularity follow by pivoting successively on the pinned d , pinned h , and private t blocks. The public null vectors are the local vectors in (286); identical feasible intervals make the central curvature scalar. $\square$

Quadratic size is unavoidable when every constraint is a signed copy edge (proposition 9.3), but not for all bounded real coefficients.

Proposition 9.3 (Effective-resistance volume bound). *Let a connected graph have maximum degree Δ , a pinned root, and outputs S . Suppose every consistent edge constraint $d_v - a_{uv}(\sigma)d_u = 0$ has $a_{uv} \in \{\pm 1\}$ depending on at most k input bits, and the label propagated from the root to every output equals the parity p_N . Then there is a vector with root value one and wrong-parity value at every output whose residual is at most*

$$\frac{k\sqrt{2\Delta|V|}}{N}. \quad (281)$$

Thus excluding all wrong-output vectors below an absolute residual η requires $|V| > \eta^2 N^2 / (2\Delta k^2)$.

Proof. The product of edge labels on a root-to-output path is parity. Fourier degree is subadditive under products, so every such path has length at least N/k . Gauge the consistent labels away and consider a unit root-to- S flow. Removing circulations and decomposing into paths gives $\sum_e |f_e| \geq N/k$. Cauchy–Schwarz and Thomson’s principle yield

$$R_{\text{eff}}(r, S) \geq \frac{N^2}{k^2|E|}, \quad C_{\text{eff}}(r, S) \leq \frac{\Delta k^2|V|}{2N^2}.$$

The Dirichlet energy for voltage difference two is $4C_{\text{eff}}$; its harmonic minimizer lies in $[-1, 1]$. Taking the square root and undoing the gauge proves (281). $\square$

Edge coefficients of magnitude above one (gains) avoid this bound even on a single path. Replace the signed recurrence by $d_i - 2\sigma_i d_{i-1} = 0$, add a public reference $h_i - 2h_{i-1} = 0$, and retain $q_i + t_i - 2h_i = 0$. The exact scale is $h_i = |d_i| = 2^i$. With absolute residual $E = \|Ax - b\|_2$, rescaling by 2^{-i} gives

$$|2^{-i}p_i d_i - 1| \leq \left(1 + \frac{1}{\sqrt{3}}\right) E. \quad (282)$$

Put $K_0 = 1 + 1/\sqrt{3}$, and assume $E \leq 10^{-2}$. Applying the same estimate to the reference rows gives

$$p_i d_i \geq (1 - K_0 E)2^i, \quad 0 \leq h_i \leq (1 + K_0 E)2^i.$$

If $c_i = q_i + t_i - 2h_i$ is the cap residual, nonnegativity gives $q_i, t_i \leq 2h_i + |c_i|$. Therefore

$$\|x\|_2^2 \leq \sum_i [(q_i + t_i)^2 + h_i^2] \leq 12(1 + K_0 E)^2 4^N + 2E^2, \quad x_{\text{correct},N} \geq (1 - K_0 E)2^N. \quad (283)$$

The endpoint coordinate therefore carries a constant fraction of $\|x\|_2^2$ (ideal decoder bias greater than $3/100$ at $E = 10^{-2}$), giving a robust family of linear size. It has exponential range and poor conditioning, but shows that bounded local coefficients alone do not force quadratic size. The next construction replaces the long gain chain by a short gain prefix and a polynomial-height plateau.

9.3 The linear-size gain-plateau family

Set

$$T = \left\lceil \frac{1}{2} \log_2 N \right\rceil, \quad K = 16N, \quad P = 17N + 1, \quad H_i = 2^{\min\{i, T\}}, \quad H = 2^T \in [\sqrt{N}, 2\sqrt{N}]. \quad (284)$$

There are nodes $i = 0, \dots, N + K$, gains $g_i = H_i/H_{i-1} \in \{1, 2\}$, and labels $a_i = \sigma_i$ for $i \leq N$ and $a_i = 1$ afterward. Define

$$\tau_0 = 1, \quad \tau_i = \prod_{j=1}^i a_j, \quad R_\tau = \text{diag}(\tau_0, \dots, \tau_{P-1}).$$

At every node introduce $(u_i, v_i, h_i, t_i) \geq 0$, put $d_i = u_i - v_i$, $q_i = u_i + v_i$, and impose

$$\begin{aligned} d_0 &= 1, & d_i - g_i a_i d_{i-1} &= 0, \\ h_0 &= 1, & h_i - g_i h_{i-1} &= 0, \\ q_i + t_i - 2h_i &= 0. \end{aligned} \quad (285)$$

The base objective is $\sum_i (2u_i + 2v_i + h_i + t_i)$. Its public orthonormal null basis is

$$W_i = \sqrt{\frac{2}{3}} \left(\frac{1}{2} e_{u_i} + \frac{1}{2} e_{v_i} - e_{t_i} \right). \quad (286)$$

Theorem 9.4 (Linear robust primal-state lower bound). *The LP (285) has $4P$ variables, $3P$ full-rank equalities, coefficients of magnitude at most two, and row and column sparsity at most four and three. It is bounded and strictly primal-dual feasible, with a unique nondegenerate strictly complementary optimum. The optimum is $\Theta(N^{3/2})$, its largest coordinate is $\Theta(\sqrt{N})$, and its Euclidean norm is $\Theta(N)$.*

For every nonzero $x \geq 0$ with

$$\text{res}_p(x) \leq \frac{1}{100}, \quad (287)$$

the fixed plateau decoder has ideal bias greater than $1/30$. Preparing its original-primal state to trace error $1/100$ therefore needs $\Omega(N) = \Omega(P)$ raw queries. Moreover,

$$\kappa_{\text{red}}(\mu) < \frac{7}{6} \quad (0 < \mu \leq 1/16). \quad (288)$$

Proof. The pinned signed and unsigned bidiagonal blocks fix $d_i = H_i \tau_i$ and $h_i = H_i$; private t_i columns finish the rank proof. Exact feasibility is

$$H_i \leq q_i \leq 2H_i, \quad t_i = 2H_i - q_i. \quad (289)$$

The local objective is $q_i + 3H_i$, so the unique optimum has $q_i = t_i = H_i$. The sign-selected pair member together with h_i, t_i forms a triangular positive basis, and the opposite coordinate has

slack two. This proves all regularity claims. Since there are $\Theta(N)$ nodes of height $H = \Theta(\sqrt{N})$, the three scale estimates follow.

Write a central coordinate as $q_i = H_i + z_i$. Stationarity and curvature are

$$\frac{1}{\mu} = \frac{1}{2H_i + z_i} + \frac{1}{z_i} - \frac{1}{H_i - z_i}, \quad f_i'' = \mu [(2H_i + z_i)^{-2} + z_i^{-2} + (H_i - z_i)^{-2}]. \quad (290)$$

For $\mu \leq 1/16$, $15\mu/16 \leq z_i < \mu$, whence $1/\mu < f_i'' < 7/(6\mu)$. Changing to the orthonormal coordinates (286) multiplies every eigenvalue by $2/3$, proving (288).

For robustness put $E = \|Ax - b\|_2 \leq \sqrt{2}/100$. The weighted resistance is

$$\sum_{i \geq 1} H_i^{-2} < \frac{1}{3} + \frac{17N}{H^2} \leq \frac{52}{3}.$$

Weighted telescoping gives, simultaneously at every node,

$$\left| \frac{\tau_i d_i}{H_i} - 1 \right| < \frac{3}{40}, \quad \left| \frac{h_i}{H_i} - 1 \right| < \frac{3}{40}. \quad (291)$$

The cap then gives $q_i, t_i < 2.165H_i$, and

$$\|x\|_2^2 < 11 \sum_i H_i^2 < \frac{583}{3} NH^2. \quad (292)$$

On the $K = 16N$ output nodes, $p_N d_i q_i > (37H/40)^2$. The computational-basis decoder has bias

$$\frac{16N(37/40)^2 H^2}{2(583/3)NH^2} > \frac{1}{30}. \quad (293)$$

Trace error $1/100$ leaves bias greater than $7/300$, and lemma 9.1 completes the proof. $\square$

The conditioning bound (288) does not extend to large μ : for $\mu/H \rightarrow \infty$, local curvature scales as μ/H_i^2 , so without prefix rigidification (section 9.5.1) the condition ratio can reach $\Theta(H^2) = \Theta(N)$. The gain is also what lets this family escape the quadratic size bound for parallel-path bundles. If R internally disjoint length- N paths have reference height at most H , harmonic interpolation from the correct root value to a wrong endpoint value has squared residual at most $4RH^2/N$. Therefore exclusion below absolute residual η requires

$$RH^2 > \frac{\eta^2 N}{4}, \quad P = \Theta(RN) \implies PH^2 = \Omega(N^2). \quad (294)$$

Both the parallel-path and the gain-plateau constructions attain this bound. General mixture and cut theorems for gadget classes are given in section 10; (294) is the only path-bundle statement used here. Naive balanced-XOR or PCP-style encodings do not improve it: their truth-table points lie on the boundary, which obstructs strict feasibility, and homogenized shallow-circuit encodings still admit fractional points that drift across a small cut.

For one fixed input, theorem 9.4 gives a tight lower bound on the total query count:

$$Q_{\text{prep}} + \sum_{t=1}^R (Q_{\text{form},t} + Q_{\text{solve},t} + Q_{\text{recover},t}) + Q_{\text{out}} = \Omega(P). \quad (295)$$

Querying all N signs and caching their prefix products costs $O(P)$ and makes every later raw query free, so no $\Omega(RP)$ bound is possible for a fixed instance.

9.4 Reduced geometry versus the affine offset

In the gain–plateau family only the affine offset of the feasible set depends on σ . Every exactly feasible point has the form

$$x_\sigma(q)_i = \left(\frac{q_i + H_i \tau_i}{2}, \frac{q_i - H_i \tau_i}{2}, H_i, 2H_i - q_i \right), \quad (296)$$

and

$$\{x : A_\sigma x = b\} = x_\sigma^{\text{off}} + \text{range}(W), \quad (x_\sigma^{\text{off}})_i = H_i \left(\frac{1 + \tau_i}{2}, \frac{1 - \tau_i}{2}, 1, 1 \right). \quad (297)$$

Even the minimum-norm offset is hidden: minimizing the local squared norm in (296) gives $A_\sigma^\dagger b = x_\sigma(4H_i/3)$. By contrast, the full reduced barrier, all derivatives, central scalars, W , both projectors, and the diagonal reduced Hessian are independent of σ . For $\mu \leq 1/16$, its eigenvalues lie in $(2/(3\mu^2), 7/(9\mu^2))$ in the unscaled log-barrier convention of [definition 2.9](#); the μ -weighted Hessian $W^\top \nabla^2 \Phi_\mu W$ has eigenvalues in $(2/(3\mu), 7/(9\mu))$.

Corollary 9.5 (Affine-offset oracle dichotomy). *Give the algorithm free exact oracles for any input-independent reduced-geometry object, including all those listed above. For the gain–plateau family, loading any original-primal state satisfying (287) costs $\Omega(P)$, while the free reduced Hessian has condition number below $7/6$ for all $0 < \mu \leq 1/16$. For the parallel-path family, the analogous cost is $\Omega(\sqrt{P})$, while the reduced Hessian is an exact scalar identity for the entire central path. In both cases the hidden object is the affine offset, not the reduced solution.*

Proof. Every free oracle is the same for all σ and can be hardwired in the parity reduction. [Theorem 9.4](#) and [theorem 9.2](#) then apply unchanged. A free feasible offset, prefix table, or loading isometry would not be input-independent: one known output pair reveals p_N , and a normalized offset state carries a constant amount of output-pair mass. $\square$

9.5 Newton steps that land exactly on the central path

An *exact Newton secant* is a full Newton step from a public start that lands exactly on the central path. With the base objective, the gain–plateau family has reduced condition number below $7/6$ for $\mu \leq 1/16$, but its exact secant uses different complementarity parameters on the $O(\log N)$ gain nodes. A public reweighting of the objective removes this. At height H_i , replace the local objective by

$$(1 + \omega_i)(u_i + v_i) + h_i + t_i, \quad \omega_i = \frac{7}{36H_i}. \quad (298)$$

On the feasible line its nonconstant part is $\omega_i q_i$, so the unique optimum and all regularity properties are unchanged; the nonbasic reduced cost is $2\omega_i > 0$.

Set

$$\mu_1 = \frac{1}{16}, \quad x_i^1 = H_i \left(\frac{9}{8}, \frac{1}{8}, 1, \frac{3}{4} \right) \quad (299)$$

up to the hidden pair swap, and put $s_j^1 = \mu_1/x_j^1$. The reduced stationarity equation at $z_i = H_i/4$ is exact because

$$\frac{\omega_i}{\mu_1} = \frac{28}{9H_i} = \frac{1}{2H_i + H_i/4} + \frac{1}{H_i/4} - \frac{1}{H_i - H_i/4}. \quad (300)$$

Thus (x^1, y^1, s^1) is the exact global μ_1 -central point.

Use the public start

$$\begin{aligned} x_i^0 &= H_i(5/6, 5/6, 20/27, 5/9), \\ s_i^0 &= H_i^{-1}(10/27, 10/27, 5/12, 5/9), \quad y^0 = 0. \end{aligned} \quad (301)$$

Every coordinate product is

$$\mu_0 = \frac{25}{81}, \quad \hat{\mu} = \frac{25}{162} = \frac{\mu_0}{2}. \quad (302)$$

Every Newton right-hand side is public, because the pair differences at the start vanish. For either hidden pair orientation,

$$\frac{10}{27H_i} \frac{9H_i}{8} + \frac{5H_i}{6} \frac{1}{18H_i} = \frac{10}{27H_i} \frac{H_i}{8} + \frac{5H_i}{6} \frac{1}{2H_i} = \frac{25}{54}. \quad (303)$$

The h, t coordinates give the same $25/54$. Hence

$$S^0(x^1 - x^0) + X^0(s^1 - s^0) = \left(\frac{25}{54} - 2\frac{25}{81} \right) e = (\hat{\mu} - \mu_0)e. \quad (304)$$

Primal and dual endpoint feasibility provide the other two rows of the standard system (14); full row rank makes the correction unique. One full Newton step therefore reaches the exact global $\mu_1 = 1/16$ center. The endpoint parameter differs from $\hat{\mu}$ because the Newton equation omits $\Delta X \Delta s$.

Up to the hidden swap, the primal direction block is

$$\Delta x_i = H_i(7/24, -17/24, 7/27, 7/36), \quad \|\Delta x_i\|_2^2 = DH_i^2, \quad D = \frac{16139}{23328}. \quad (305)$$

The direction entry on the unselected pair coordinate has magnitude $17H_i/24$, against $7H_i/24$ on the selected one, so the squared gap is $5H_i^2/12$. Since $\sum_i H_i^2 < (17N + 4/3)H^2$, the fixed output-copy decoder has bias

$$\frac{16N(5/12)H^2}{2D(17N + 4/3)H^2} > \frac{1}{4}. \quad (306)$$

Thus the exact original-primal direction state at trace error $1/100$ needs $\Omega(P)$ raw queries.

In the unscaled log-barrier convention of [definition 2.9](#), the μ_1 -scaled endpoint reduced Hessian and the start reduced Newton matrix are

$$\mu_1 W^\top \nabla^2 F(x^1) W = \frac{182}{243} \text{diag}(H_i^{-2}), \quad W^\top (X^0)^{-1} S^0 W = \frac{22}{27} \text{diag}(H_i^{-2}). \quad (307)$$

Thus all but the $T = O(\log N)$ public gain modes form an exact scalar block. The public congruence $D_H = \text{diag}(H_i)$ makes both matrices scalar; as a standard-form diagonal change of variables it raises the coefficient scale to $\Theta(\sqrt{N})$. More generally, scaling by H_i^η gives

$$B_\eta = \Theta(H^\eta), \quad \kappa_\eta = H^{2(1-\eta)}, \quad B_\eta^2 \kappa_\eta = \Theta(N). \quad (308)$$

This tradeoff is specific to this diagonal family, not a general theorem.

Within this node gadget the T outlier modes are unavoidable.

Proposition 9.6 (Four-variable-node incompatibility). *Consider pairs (u_i, v_i) whose central endpoint values are $L_i > S_i > 0$, in an order fixed by the hidden input, at one common central parameter μ_1 , and suppose their differences are the prescribed heights $L_i - S_i = H_i$. Within the separable four-variable node gadget, the following three requirements are incompatible: a public start centered coordinatewise at one μ_0 ; one full standard Newton correction with a common target that works for both hidden orientations; and an $O(1)$ -conditioned endpoint Hessian in the raw orthonormal null basis. This statement does not cover added rows, weighted barriers, or other local gadgets.*

Proof. Write the two public start values as a_i, b_i , with slacks $\mu_0/a_i, \mu_0/b_i$. Equality of the first-coordinate secant cross terms for the two hidden orientations requires

$$\frac{\mu_0 L_i}{a_i} + \frac{a_i \mu_1}{L_i} = \frac{\mu_0 S_i}{a_i} + \frac{a_i \mu_1}{S_i}.$$

Since $L_i \neq S_i$, this gives $a_i^2 = \mu_0 L_i S_i / \mu_1$; the second coordinate similarly gives $b_i = a_i$. The common cross term is therefore

$$C_i = \sqrt{\mu_0 \mu_1} \frac{L_i + S_i}{\sqrt{L_i S_i}}.$$

One global Newton target makes C_i independent of i , hence the aspect ratio L_i/S_i is constant. The cap row $t_i = 2H_i - (L_i + S_i) > 0$ makes $L_i + S_i$ proportional to H_i and forces that constant aspect ratio to exceed three. Together with $L_i - S_i = H_i$, this makes both endpoint values proportional to H_i . The log-barrier curvature in the local orthonormal null direction consequently scales as H_i^{-2} , giving condition ratio $H^2 = \Theta(N)$. $\square$

The reweighted construction also shows that a small fixed relative residual of the full Newton system does not determine the normalized primal direction. On the K output nodes let $w = \sum_{i \in S} W_i$, and perturb the exact solution by

$$e_x = Mw, \quad e_y = 0, \quad e_s = -(X^0)^{-1} S^0 e_x. \quad (309)$$

The primal and complementarity residual perturbations vanish. On the plateau, direct substitution in (301) gives the sharp bound

$$\|(X^0)^{-1} S^0 w\|_2 \leq \frac{\|w\|_2}{H^2}$$

(the exact coefficient is below $0.856/H^2$), whereas the public right-hand side of the cap rows has norm $\Omega(H\sqrt{K})$. With $M = H^2$, the relative full-KKT residual is $O(H^{-1})$, and the normalized primal direction converges to the public, query-free state $|w/\|w\|\rangle$. The perturbed update has negative t -coordinates. Therefore the robust direction theorem below must keep the condition $x^0 + \Delta x \geq 0$; a full-KKT residual promise alone is not enough.

9.5.1 Prefix rigidification: reduced condition number one

To remove the T outlier modes, return to the constraints (285). At each of the T small-height nodes $i < T$, add the public row

$$q_i - \frac{5}{4}h_i = 0, \quad (310)$$

and use the uniform objective

$$\sum_i [(1 + \omega)(u_i + v_i) + h_i + t_i], \quad \omega = \frac{7}{36H}. \quad (311)$$

Write the augmented matrix and RHS as A'_σ, b' .

Theorem 9.7 (Hard original-primal Newton direction with reduced condition number one). *The prefix-rigidified family has $4P$ variables and $3P + T$ full-rank equalities. Coefficients have magnitude at most two, and row and column sparsity are at most four. It is bounded and strictly primal-dual feasible, with a unique nondegenerate strictly complementary optimum. In the raw orthonormal reduced coordinates,*

$$\kappa_{\text{red}}(\mu) = 1 \quad \text{for every } \mu > 0, \quad (312)$$

and the reduced Newton matrix at the public start (301) is also exactly condition one.

The unique standard Newton correction from that start with target $\hat{\mu} = 25/162$ reaches, after one full step, the exact global $\mu_1 = 1/16$ center (299). Its original-primal direction state has decoder bias greater than $1/4$, so trace error $1/100$ requires $\Omega(N) = \Omega(P)$ raw queries.

The same lower bound holds for a state of every nonzero $x \geq 0$ satisfying $\text{res}_p(x) \leq 1/100$, and for every nonzero direction $\tilde{\Delta}x$ whose update is nonnegative and obeys that residual, at state trace error $1/100$. In the latter case the fixed observable defined in the proof has bias greater than $1/54$.

Proof. For the row r_i in (310) and the old null vectors,

$$r_i^\top W_j = \sqrt{2/3} \delta_{ij}. \quad (313)$$

Hence the new rows are independent modulo the old row space and

$$\ker A'_\sigma = \text{span}\{W_i : i \geq T\}. \quad (314)$$

They fix $q_i = 5H_i/4$ on the prefix, leaving only equal-height H coordinates free. At the optimum all four prefix variables and three variables at every free node are positive, exactly $3P + T$ in total. A null vector supported on them must vanish because every nonzero free W_i touches that node's zero pair coordinate. The positive columns form a basis, and each nonbasic reduced cost is $2\omega > 0$. This proves rank and regularity.

All free nodes have the same height, objective slope, central scalar, and local curvature. Equation (314) therefore gives

$$W^\top \nabla^2 F(x(\mu)) W = \lambda(\mu) I_{P-T}.$$

At $\mu_1 = 1/16$, the free coordinate is $q = 5H/4$, while the fixed prefix coordinates already have the same local template. Positive slacks $s^1 = \mu_1(X^1)^{-1}e$ are orthogonal to the remaining kernel after subtracting the objective, so full row rank supplies the central multiplier. The secant identities (303)–(304) are unchanged, and the endpoint satisfies the new rows. This proves the exact Newton statement. At the start,

$$W^\top (X^0)^{-1} S^0 W = \frac{22}{27H^2} I_{P-T}. \quad (315)$$

The fraction-to-boundary maximum is exactly $\alpha_{\max} = 20/17 > 1$, so the full step preserves strict positivity; any safety factor at least $17/20$ permits it. The direction formula (305) survives and (306) proves the exact-state lower bound.

For a robust feasible state, deleting the public new residual coordinates gives the base residual bound with the same RHS norm $\|b'\|_2 = \sqrt{2}$; [theorem 9.4](#) applies. For a robust direction, set $\tilde{x} = x^0 + \tilde{\Delta}x \geq 0$. On every plateau node the same telescoping estimates give

$$\tau_i \tilde{d}_i > \frac{37}{40}H, \quad \frac{37}{40}H < \tilde{h}_i < \frac{43}{40}H, \quad \tilde{q}_i, \tilde{t}_i < \frac{433}{200}H. \quad (316)$$

Together with the public start, these imply

$$\|\tilde{\Delta}x\|_2^2 < 100NH^2, \quad \tilde{\Delta}h_i > \frac{199}{1080}H \quad (i \in S). \quad (317)$$

Let $|g_i\rangle = (|u_i\rangle - |v_i\rangle)/\sqrt{2}$, and use the public norm-one observable

$$O = \bigoplus_{i \in S} (|h_i\rangle\langle g_i| + |g_i\rangle\langle h_i|). \quad (318)$$

Equations (316)–(317) give signed expectation greater than

$$\frac{7363\sqrt{2}}{270000} > \frac{1}{27}, \quad (319)$$

so its associated binary POVM has bias greater than $1/54$. Trace error $1/100$ leaves positive constant bias. [Lemma 9.1](#) finishes the proof. $\square$

The condition-one claim concerns only the reduced primal matrix. At the same start, the raw OSS matrix $\mathcal{M}_0 = [-X^0(A')^\top \ S^0 W]$ and the augmented KKT matrix $\mathcal{K}_0$ with Δs eliminated satisfy

$$\kappa_2(\mathcal{M}_0) \geq \sqrt{681/44} H^2, \quad \kappa_2(\mathcal{K}_0) \geq \sqrt{1701/178} H^2, \quad (320)$$

so both are $\Omega(P)$. Exact block preconditioning can reduce the second condition to φ^2 , but its difference-row Schur block has the gauge form

$$G_\sigma^{(d)} = R_\tau G_+^{(d)} R_\tau, \quad (G_\sigma^{(d)})^{-1/2} = R_\tau (G_+^{(d)})^{-1/2} R_\tau. \quad (321)$$

The all-positive block is an irreducible tridiagonal Stieltjes matrix, whose inverse square root is entrywise strictly positive by its resolvent integral. Thus the sign of an entry of this preconditioner reveals a prefix product. Block encodings of the factor without entry access are analyzed in [section 12](#).

With the minimum-norm (Moore–Penrose) particular solution p_σ , all input dependence of the direction lies in p_σ :

$$\Delta x = p_\sigma + Wz, \quad p_\sigma = (A'_\sigma)^\dagger (b' - A'_\sigma x^0). \quad (322)$$

On every free node,

$$(p_d, p_q, p_h, p_t) = H(\tau_i, -4/27, 7/27, -2/27), \quad (323)$$

with pair coordinates recovered from p_d, p_q . The complete reduced system is public:

$$M = \frac{22}{27H^2} I, \quad z_i = -\frac{29}{108} \sqrt{\frac{3}{2}} H, \quad Mz = g. \quad (324)$$

Normalized at $\alpha_M = 22/(27H^2)$, it has $M/\alpha_M = I$, $\kappa = 1$, and

$$\alpha_M \frac{\|M^{-1}g\|_2}{\|g\|_2} = 1. \quad (325)$$

The states of g, z, Wz are public and free. For the fixed diagonal-pair observable

$$O_S = \sum_{i \in S} (|u_i\rangle\langle u_i| - |v_i\rangle\langle v_i|),$$

each output node contributes $u_i^2 - v_i^2 = p_d p_q = -(4/27)p_N H^2$. A free-node block has squared norm $(851/1458)H^2$, a prefix block has squared norm DH_i^2 with $D = 16139/23328 < 7/10$, and $\sum_i H_i^2 < (53/3)NH^2$. Hence

$$\left| \left\langle \frac{p_\sigma}{\|p_\sigma\|_2}, O_S \frac{p_\sigma}{\|p_\sigma\|_2} \right\rangle \right| > \frac{(64/27)NH^2}{(7/10)(53/3)NH^2} = \frac{1920}{10017} > \frac{1}{6}. \quad (326)$$

Its sign is $-p_N$, so preparing $|p_\sigma/\|p_\sigma\|\rangle$ to trace error $1/100$ costs $\Theta(P)$ raw queries. Because $p_\sigma \perp Wz$, their coherent combination succeeds with probability at least $1/2$, provided one has phase-referenced coherent preparation access to both normalized summands. A density-state output requirement alone does not provide that access, as [section 2.3](#) records. The split depends on the particular solution, since another one can exchange a null-space component with z ; only the end-to-end original-coordinate lower bound is independent of this choice.

For one fixed input, [theorem 9.7](#) gives the same tight total bound (295); the matching upper bound caches all signs in N queries. A bound that grows with the number of iterations holds only in a streaming model, in which round t releases a fresh independent sign block after output $t-1$ is committed. Before the next block is released, round t must return an unconditional state within trace distance $1/100$ of that round's hard original-coordinate Newton direction (or of a point under the robust output requirement in [theorem 9.7](#)). Let Q_t count every call to the

current block's coefficient oracle or inverse, all setup after that block is released, and all queries used to prepare classical or quantum advice consumed in that round. Conditioned on all past memory, the new block is independent, so [theorem 9.7](#) applies in every round and

$$\sum_{t=1}^R Q_t = \Omega(RN) = \Omega(RP). \quad (327)$$

This is a direct-sum bound over fresh inputs, not a per-iteration bound for one fixed LP.

9.6 Full-KKT and preconditioned formulations

The preceding results concern only the original-primal block of the direction. A related construction makes the complete primal–dual direction hard while keeping its unpreconditioned right-hand side public. Use the unrigidified gain–plateau constraints (285), the base objective $c_i = (2, 2, 1, 1)$, and the public start

$$x^0 = s^0 = e, \quad y^0 = 0, \quad \mu^0 = 1. \quad (328)$$

For a public centering target $\theta \in [0, 1]$, the Newton equations are

$$A_\sigma \Delta x = b - A_\sigma e, \quad A_\sigma^\top \Delta y + \Delta s = c - e, \quad \Delta x + \Delta s = (\theta - 1)e. \quad (329)$$

The RHS is input-independent: every hidden coefficient multiplies the zero pair difference at the start.

Let $R_{r,j} = \sum_{i=j}^{N+K} H_i^r$. Direct elimination and backward substitution give

$$\begin{aligned} \Delta x_{u_i} &= (4H_i - 8 + 3H_i \tau_i)/6, & \Delta x_{v_i} &= (4H_i - 8 - 3H_i \tau_i)/6, \\ \Delta x_{h_i} &= H_i - 1, & \Delta x_{t_i} &= (2H_i - 1)/3, \\ \alpha_j &= \tau_j R_{2,j}/(2H_j), & \beta_j &= [7R_{2,j} + (4 - 9\theta)R_{1,j}]/(3H_j), \\ \gamma_j &= (2H_j + 2)/3 - \theta. \end{aligned} \quad (330)$$

Here $(\alpha, \beta, \gamma) = \Delta y$ correspond to the signed, reference, and cap rows.

Theorem 9.8 (Public-RHS full-KKT direction). *Let $w_\sigma = (\Delta x, \Delta y, \Delta s)$ solve (329). Preparing its normalized state to trace distance $1/100$ takes at least $N/2$ raw coefficient queries. The same holds if an approximate vector obeys $\|\tilde{w} - w_\sigma\|_2 \leq \|w_\sigma\|_2/300$ and its state has trace error at most $1/300$.*

For $\theta = 1$, the two-block elimination has an explicit canonical sparse block encoding of normalization six and a public RHS state. The same $N/2$ lower bound holds for calls to that fully specified block encoding, not merely to its principal matrix block.

Proof. On the K output rows define $A_S = H^2 \sum_{\ell=1}^K \ell^2/4$. The height sums and (330) imply

$$\|\alpha\|_2^2 < 3A_S, \quad \|\beta\|_2^2 < \frac{484}{3}A_S, \quad \|w_\sigma\|_2^2 < 165A_S. \quad (331)$$

Moreover α_j has sign p_N and $\beta_j > 0$ on the output suffix, with $\beta_j/|\alpha_j| \geq 3$. The public norm-one observable that swaps the corresponding α_j, β_j coordinates therefore has parity-signed expectation

$$p_N \langle O_{\text{KKT}} \rangle > \frac{6}{165} = \frac{2}{55} > \frac{1}{30}. \quad (332)$$

Trace error $1/100$ changes this by at most $1/50$. The resulting expectation is a real polynomial of degree at most twice the query count; its nonzero parity Fourier coefficient forces degree at least N . Thus the query count is at least $N/2$. The triangle inequality for normalized rays gives the approximate-vector statement.

At $\theta = 1$, eliminate $\Delta s = -\Delta x$, put $y' = -\Delta y$, and write

$$\mathcal{K}_\sigma = \begin{bmatrix} I & A_\sigma^\top \\ A_\sigma & 0 \end{bmatrix}, \quad r_\sigma = \begin{pmatrix} e - c \\ b - A_\sigma e \end{pmatrix}. \quad (333)$$

The maximum absolute row and column sums are six. The standard magnitude-weighted row/column state-pair construction gives an exact normalization-six block encoding. Its support, magnitudes, failure arms, and left preparation are public; the right preparation uses one local sign phase, hence exactly one sign query per call, adjoint, or controlled call. Also $\|r_\sigma\|_2^2 = 3P + T + 1$, and its preparation is public. Removing Δs can only strengthen (332), so the same degree argument applies to this complete oracle implementation. $\square$

Ideal Schur preconditioning makes the matrix well conditioned but does not make the end-to-end raw-input task easy. Put

$$\mathcal{S}_\sigma = A_\sigma A_\sigma^\top, \quad Q_\sigma = \mathcal{S}_\sigma^{-1/2} A_\sigma, \quad P_\sigma = \text{diag}(I, \mathcal{S}_\sigma^{-1/2}).$$

Define

$$\bar{w}_\sigma = P_\sigma^{-1} \begin{pmatrix} \Delta x \\ y' \end{pmatrix}, \quad \bar{r}_\sigma = P_\sigma r_\sigma, \quad \bar{K}_\sigma = P_\sigma \mathcal{K}_\sigma P_\sigma.$$

Then

$$\bar{K}_\sigma = \begin{bmatrix} I & Q_\sigma^\top \\ Q_\sigma & 0 \end{bmatrix}, \quad \bar{K}_\sigma \bar{w}_\sigma = \bar{r}_\sigma, \quad (334)$$

and the primal block of $\bar{w}_\sigma$ is exactly Δx . Since $Q_\sigma Q_\sigma^\top = I$, the spectrum is 1 on the nullspace and

$$\frac{1 + \sqrt{5}}{2}, \quad \frac{1 - \sqrt{5}}{2} \quad (335)$$

on every coupled two-plane. Therefore

$$\|\bar{K}_\sigma\|_2 = \|\bar{K}_\sigma^{-1}\|_2 = \varphi, \quad \kappa_{\text{abs}}(\bar{K}_\sigma) = \varphi^2 = \frac{3 + \sqrt{5}}{2}. \quad (336)$$

This is the classical ideal-Schur spectrum of Murphy–Golub–Wathen [110].

Theorem 9.9 (Residual-robust Schur-preconditioned KKT state). *Suppose $\tilde{w} \neq 0$ obeys*

$$\|\bar{K}_\sigma \tilde{w} - \bar{r}_\sigma\|_2 \leq 10^{-3} \|\bar{r}_\sigma\|_2. \quad (337)$$

Preparing its normalized state to trace error $1/200$ takes at least $N/2$ total raw coefficient queries, counting preconditioner, transformed RHS, solve, and recovery access. The same bound holds for the prefix-rigidified family.

Proof. Let

$$O_x = \sum_{i \in S} (|D_i\rangle\langle h_i| + |h_i\rangle\langle D_i|), \quad |D_i\rangle = \frac{|u_i\rangle - |v_i\rangle}{\sqrt{2}}.$$

This norm-one, primal-supported observable has exact parity-signed expectation greater than $2/87$ on the preconditioned solution. Indeed,

$$\|\Delta x\|_2^2 \leq \frac{11}{3} P H^2, \quad p_N \frac{\langle \Delta x | O_x | \Delta x \rangle}{\|\Delta x\|_2^2} > \frac{1}{6}, \quad \|\bar{w}\|_2^2 < \frac{29}{4} \|\Delta x\|_2^2. \quad (338)$$

Equations (336)–(337) give $\|\tilde{w} - \bar{w}\|_2 \leq 10^{-3} \varphi^2 \|\bar{w}\|_2$. The total target-state distance is at most

$$\frac{1}{200} + \frac{2\varphi^2}{1000} < \frac{1}{87}, \quad (339)$$

which preserves the sign of the observable. The polynomial argument gives $N/2$.

For the prefix-rigidified transfer, use the same all-ones start. Every added row $q_i - \frac{5}{4}h_i = 0$ has the public primal residual $-3/4$, and the uniform objective is

$$c'_i = (1 + \omega, 1 + \omega, 1, 1), \quad \omega = \frac{7}{36H}.$$

The hidden difference coefficients still multiply $u_i - v_i = 0$, so the complete Newton RHS remains public. On every free node, null projection and exact feasibility give

$$\Delta q_i = \frac{4H - 6 - 2\omega}{3}, \quad \Delta h_i = H - 1, \quad \Delta u_i - \Delta v_i = H\tau_i.$$

The prefix variables are fixed by feasibility. These formulas yield

$$\|\Delta x\|_2^2 < 5 \sum_i H_i^2, \quad p_N \frac{\langle \Delta x | O_x | \Delta x \rangle}{\|\Delta x\|_2^2} > \frac{1}{8},$$

because the numerator is $K\sqrt{2}H(H-1)$, each prefix node contributes less than $5H_i^2$, and each free node less than $4H^2$. For the exact block preconditioner the transformed primal RHS is $e - c' = (-\omega, -\omega, 0, 0)$ per node, and

$$\frac{\|e - c'\|_2}{\|\Delta x\|_2} < \frac{21}{100}, \quad \|\bar{w}_\sigma\|_2^2 < \frac{5}{2}\|\Delta x\|_2^2.$$

Thus the parity-signed expectation on the complete preconditioned state exceeds $1/20$. Its condition remains φ^2 , while the solve and output errors in (339) change the expectation by less than $2/87 < 1/20$. The same $N/2$ conclusion follows. $\square$

No matrix-only tradeoff between normalization and condition number underlies this theorem. The row-space projector is public because $\ker A_\sigma$ is public. If Q_0 is a public row isometry with $Q_0^\top Q_0 = I - WW^\top$, then $O_\sigma = Q_\sigma Q_0^\top$ is orthogonal and

$$\text{diag}(I, O_\sigma)^\top \bar{K}_\sigma \text{diag}(I, O_\sigma) = \begin{bmatrix} I & Q_0^\top \\ Q_0 & 0 \end{bmatrix}, \quad (340)$$

which is wholly public. The input dependence has moved to the transformed RHS and the recovery map, so the formulation-independent statement is

$$Q_{\text{pre}} + Q_{\text{on}} \geq N/2. \quad (341)$$

It is tight: after N cached sign queries, later raw-query cost can vanish. Factor access is treated in section 12, not here.

9.7 Bounded data and the sharp residual scale

With all data bounded by absolute constants, the exact states remain hard, but residual robustness holds only at the sharp scale $1/\sqrt{N}$. Remove the gain prefix by setting every height and gain to one, retain a length- N signed chain followed by $K = 16N$ public copy edges, and put $P = 17N + 1$. Use the same four nonnegative variables at every node and the equations

$$\begin{aligned} d_0 &= 1, & d_i - a_i d_{i-1} &= 0, \\ h_0 &= 1, & h_i - h_{i-1} &= 0, \\ q_i + t_i - 2h_i &= 0, & a_i &= \begin{cases} \sigma_i & i \leq N, \\ 1 & i > N. \end{cases} \end{aligned} \quad (342)$$

Use the objective

$$c^\top x = \sum_i \left[\frac{43}{36}(u_i + v_i) + h_i + t_i \right]. \quad (343)$$

All matrix, RHS, objective, feasible-primal, and relevant slack coordinates are bounded by absolute constants; $\|b\|_2 = \sqrt{2}$. Exact feasibility is $d_i = \tau_i, h_i = 1, t_i = 2 - q_i, 1 \leq q_i \leq 2$, so the same basis (286) spans the nullspace.

Theorem 9.10 (Bounded exact-state separation). *The all-unit family is bounded and strictly primal-dual feasible, with a unique nondegenerate strictly complementary optimum. Its raw orthonormal reduced central Hessian has condition one for every $\mu > 0$, and its reduced Newton matrix at the public start below is $(22/27)I$.*

At $\mu_1 = 1/16$, one exact Newton correction from

$$x_i^0 = (5/6, 5/6, 20/27, 5/9), \quad s_i^0 = (10/27, 10/27, 5/12, 5/9), \quad y^0 = 0 \quad (344)$$

with $\mu_0 = 25/81$ and target $25/162$ lands at the exact center

$$x_i^1 = (9/8, 1/8, 1, 3/4), \quad s_i^1 = (1/18, 1/2, 1/16, 1/12), \quad (345)$$

up to the hidden pair swap. Preparing either the exact primal direction state or exact central-primal state to trace error $1/100$ requires $\Omega(N) = \Omega(P)$ raw queries.

Proof. On the feasible line the local objective is $3 + (7/36)q_i$, so the unique optimum has $q_i = 1$. The positive columns form a basis and the zero pair coordinate has reduced cost $7/18$. All central coordinates solve the same scalar equation

$$\frac{7}{36\mu} = \frac{1}{q+1} + \frac{1}{q-1} - \frac{1}{2-q},$$

so the reduced Hessian is scalar. At $q = 5/4, \mu = 1/16$, both sides are $28/9$, and direct evaluation gives the μ_1 -scaled reduced curvature $182/243$. Equations (303)–(304) with $H_i = 1$ prove the same exact Newton secant and $(22/27)I$ start matrix.

The primal direction is the constant block in (305). Its plateau decoder has bias

$$\frac{K(5/12)}{2DP} > \frac{1}{4}.$$

Each central block has squared norm $91/32$ and oriented pair gap $5/4$, so the central-state decoder has bias

$$\frac{K(5/4)}{2(91/32)P} = \frac{20K}{91P} > \frac{1}{5}.$$

Lemma 9.1 completes both lower bounds. □

Theorem 9.11 (Sharp all-unit residual threshold). *For every nonzero $x \geq 0$ satisfying the absolute residual bound*

$$\|A_\sigma x - b\|_2 \leq \frac{1}{16\sqrt{N}}, \quad (346)$$

the normalized primal state has constant correct-parity bias under the output decoder. Preparing it to trace error $1/100$ needs $\Omega(N) = \Omega(P)$ raw queries. In relative form, (346) is $1/(16\sqrt{2N})$; in particular it contains the $1/(100\sqrt{N})$ absolute residual tube.

The order is sharp: there are nonnegative points with the exact optimal objective and absolute residual $1/\sqrt{N}$, respectively $2/\sqrt{N}$, whose output-node differences d_i are all zero, respectively all have the wrong sign $-p_N$. After adding the public homogeneous rows $q_i - 5h_i/4 = 0$ on all non-output nodes, the wrong-sign point can also be chosen with exact dual feasibility and exact coordinatewise complementarity.

Proof. Let $E = \|Ax - b\|_2$, $z_i = \tau_i d_i$, and $\delta = \sqrt{P}E$. Telescoping the signed and reference recurrences gives

$$|z_i - 1| \leq \sqrt{P}E, \quad |h_i - 1| \leq \sqrt{P}E. \quad (347)$$

The factor $\sqrt{P}$ is exact by equality in Cauchy–Schwarz. Under (346), $P/N \leq 35/2$ gives $\delta < 4/15$, so on every output node $p_N d_i > 11/15$. The cap and nonnegativity give $q_i, t_i < 31/12$, $h_i < 19/15$, and $\|x\|_2^2 < 15P$. Thus the output decoder has bias

$$\frac{K(11/15)^2}{30P} > \frac{1}{70}. \quad (348)$$

Trace error $1/100$ leaves bias at least $1/70 - 1/100 = 3/700 > 0$.

For the zero-signal witness take

$$z_i = 1 - i/N \ (i \leq N), \quad z_i = 0 \ (i > N), \quad (349)$$

and for the wrong-parity witness replace this by

$$z_i = 1 - 2i/N \ (i \leq N), \quad z_i = -1 \ (i > N). \quad (350)$$

Set $d_i = \tau_i z_i$ and $h_i = q_i = t_i = 1$. All reference and cap rows are exact. The N signed residuals are respectively $1/N$ and $2/N$, proving the claimed norms. Both points are nonnegative and have the exact optimal objective.

For the stronger failure, add $q_i - 5h_i/4 = 0$ on all non-output nodes and average the exact centers of the N inputs obtained by flipping one sign. The average has the wrong common output template and the same $2/\sqrt{N}$ primal residual. Every coordinate is positive. Setting $\bar{s} = \mu_1 \bar{X}^{-1} e$ gives exact complementarity. The remaining nullspace is supported on output nodes, where $\bar{x}$ is itself an exact central template; hence $c - \bar{s} \perp \ker A$, which supplies an exact dual multiplier. $\square$

For the chain recurrences alone, the threshold is exactly $E < 1/\sqrt{P}$: taking $z_i = 1 - (i+1)/P$ makes all gauged residuals $-1/P$, reaches zero at the last node, and has norm $1/\sqrt{P}$. Inverting the robust scale over all hidden lengths L with $17L + 1 \leq P$ gives the worst-case query lower bound

$$Q(P, \epsilon) = \Omega(\min\{P, \epsilon^{-2}\}) \quad (351)$$

for absolute or RHS-relative residual, up to constants. The bound comes from choosing the hidden length for each accuracy; it does not show that one instance grows harder with accuracy.

At unit height, the public copy plateau is necessary for constant one-copy distinguishability. Flipping only σ_N swaps exactly $K + 1$ target blocks. For the central and direction states the normalized overlaps are

$$1 - \frac{32(K+1)}{91P}, \quad 1 - \frac{K+1}{DP}, \quad (352)$$

respectively. If $K = o(N)$, their trace distance is $o(1)$ even for opposite parities. Hence, with bounded data, the resource needed is $K = \Theta(N)$ copy nodes, not a dynamic range of $\Theta(\sqrt{N})$.

9.8 Dual homogeneity and the bounded full direction

On a unit-height chain the equality multipliers accumulate. One exact way to suppress this is to scale the dual variables by a positive constant, which is an elementary symmetry of the central and Newton equations, consistent with the general scale-invariance viewpoint in interior-point theory [129].

Lemma 9.12 (Dual homogeneity). *In the standard Newton system, leave (A, b, x^0) fixed and replace*

$$(c, y^0, s^0, \hat{\mu}) \longmapsto (\lambda c, \lambda y^0, \lambda s^0, \lambda \hat{\mu}), \quad \lambda > 0. \quad (353)$$

Then $(\Delta x, \Delta y, \Delta s)$ is replaced by $(\Delta x, \lambda \Delta y, \lambda \Delta s)$. The primal central trajectory is unchanged, with μ reparameterized as $\mu' = \lambda \mu$.

Proof. The primal equation is unchanged, while substitution multiplies the dual and complementarity equations by λ . Full row rank and positivity make the Newton solution unique. The same substitution in $Ax = b$, $A^\top y + s = c$, $Xs = \mu e$ proves the central-path statement. $\square$

For the full-state theorem use the all-unit signed chain and copy plateau, but pin each reference coordinate independently by $h_i = 1$. This changes the row representation, not the primal feasible set. Starting from (344), scale the objective, initial slacks, and complementarity data by

$$\lambda = \frac{1}{P}. \quad (354)$$

The exact primal correction remains the block (305). If the first coordinate is paired with the larger endpoint coordinate, the unscaled slack correction is

$$\Delta \bar{s}_i = (-17/54, 7/54, -17/48, -17/36), \quad (355)$$

and, with difference, height-pin, and cap multipliers,

$$\Delta y = \frac{1}{P}(\bar{\alpha}, \bar{\beta}, \bar{\gamma}), \quad \bar{\alpha}_j = \frac{2}{9}\tau_j(P - j), \quad \bar{\beta}_j = \frac{133}{48}, \quad \bar{\gamma}_j = \frac{11}{12}. \quad (356)$$

Theorem 9.13 (Bounded full-KKT direction). *Every coordinate of the complete $11P$ -dimensional direction $w = (\Delta x, \Delta y, \Delta s)$ in (354)–(356) is bounded by an absolute constant, and $\|w\|_2 = \Theta(\sqrt{P})$. Preparing its normalized state to trace error $1/100$ requires $\Omega(N) = \Omega(P)$ raw queries.*

The explicit canonical sparse block encoding of either the three-block Newton matrix or its two-block elimination has normalization five, a public RHS, and the same linear call lower bound. The reduced Newton condition remains one, but the central endpoint is at $\mu'_1 = 1/(16P)$, and both raw full-KKT matrices have condition $\Theta(P)$.

Proof. Equations (305), (355), and (356) give

$$\|\Delta x\|_\infty \leq 17/24, \quad \|\Delta y\|_\infty \leq 2/9, \quad \|\Delta s\|_\infty \leq 17/(36P).$$

Writing $D = 16139/23328$, direct summation gives

$$DP \leq \|w\|_2^2 < \left(D + \frac{1}{40}\right)P < \frac{3}{4}P. \quad (357)$$

Use the diagonal decoder convention $+1$ on every v coordinate and -1 on every u coordinate. The underlying u -minus- v squared-mass difference on each output node is $-p_N(5/12)$, so the decoder expectation contributes $+p_N(5/12)$ per node and has magnitude

$$\frac{(5/12)K}{(3/4)P} = \frac{5K}{9P} \geq \frac{32}{63} > \frac{1}{2}. \quad (358)$$

Trace error preserves constant bias. The polynomial method gives at least $N/2$ sign queries.

The maximum absolute row and column sums of both canonical matrices are five. The same magnitude-weighted state-pair construction used in theorem 9.8 therefore gives normalization five and one sign query per full oracle call. The RHS remains public because the start has zero pair difference. This proves the canonical-oracle claim.

Dual homogeneity gives $\mu'_1 = 1/(16P)$ and the reduced start matrix $(22/(27P))I$. If $z \in \ker A$ is unit, applying either raw KKT matrix to the corresponding nullspace test vector has norm at most $1/P$, while the largest singular value is bounded below by a constant. Conversely, the pinned signed incidence has smallest singular value $\Theta(P^{-1})$; bounded local changes reduce the remaining blocks to identities. The paired saddle eigenvalues then show that both conditions are $\Theta(P)$. $\square$

This comes at a cost: objective coefficients, positive slacks, reduced costs, and complementarity are $\Theta(P^{-1})$. The theorem does not give constant scale and constant full-system condition number at the same time. It also does not inherit the sharp relative-residual theorem, because independent height pins make $\|b\|_2 = \Theta(\sqrt{P})$.

9.9 Why a bounded full direction at constant dual scale remains open

The need for large multipliers or a small dual scale does not depend on signed pairs. At any positive start, eliminate Δs from the Newton equations and put

$$H = (X^0)^{-1}S^0, \quad g = (X^0)^{-1}r_c - r_d.$$

Then

$$H\Delta x - A^\top \Delta y = g, \quad \boxed{r_p^\top \Delta y = \Delta x^\top H\Delta x - g^\top \Delta x}. \quad (359)$$

This exact *work identity* holds for every row representation of the constraints. If $g = 0$, $H \succeq \rho I$, and $\|r_p\|_1 \leq R$, it gives

$$\|\Delta y\|_\infty \geq \frac{\rho \|\Delta x\|_2^2}{R}. \quad (360)$$

A constant-scale correction with $\Theta(P)$ squared norm and a sparse public residual r_p with $\|r_p\|_1 = O(1)$ therefore forces an $\Omega(P)$ multiplier, independently of redundant rows or branching.

There is also a two-input bound, the orthogonal-copy cut law, for constant-dimensional copy blocks. Let a flat output region S carry $z_v \in \mathbb{R}^k$ and orthogonal copy equations $z_v - U_e z_u = 0$. After public transports G_v , suppose stationarity is

$$\operatorname{div}_U(f)_v = D_v \Delta z_v + g_v, \quad D_v \succeq \rho I, \quad (361)$$

with the same numerical D_v, g_v for two opposite-parity inputs, and their transported codewords differ by a common w , $\|w\|_2 \geq 2a$. If a boundary row scale is at most B , summing the difference of (361) over S cancels every public source and internal force. Taking the inner product with w yields

$$\max_{p \in \{+, -\}} \|\Delta y_{\partial S}^p\|_{2,\infty} \geq \frac{\rho a |S|}{B |\partial S|}. \quad (362)$$

Thus swaps, simplex permutations, and public flat rotations do not prevent large multipliers from accumulating at a bounded interface.

The most tempting counterexample is a real quarter-turn $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, with $z_i = \sigma_i J z_{i-1}$ and closure $z_0 - J z_N = e_1$. For $4 \mid N$,

$$z_0(p) = \frac{1}{2}(1, p)^\top, \quad (363)$$

so all blocks are bounded and locally reveal parity after a public inverse rotation. Nevertheless the work identity gives the closure multiplier

$$e_1^\top \Delta y_{\text{close}} = \sum_{i=0}^N \|z_i\|_2^2 = \frac{N+1}{2}. \quad (364)$$

After gauging, the constraint is a twisted cycle whose singular values are

$$2 \left| \sin \frac{2\pi\ell \pm \pi/2}{2(N+1)} \right|, \quad \ell = 0, \dots, N, \quad (365)$$

so its least singular value is $2 \sin(\pi/[4(N+1)]) = \Theta(N^{-1})$. A positive/negative standard-form LP realization preserves bounded primal and slack directions and even allows a positive half-centering step, but (364) survives exactly. Circulations, public antisymmetric sources, and dense public residuals fail for the same work or two-parity cancellation reason unless they introduce linearly many independent hard interfaces.

Open Problem 9.14 (Bounded-scale full-KKT parity amplifier). Does a sparse standard-form LP family exist that simultaneously has:

- (i) linear dimension and bounded row and column degree;
- (ii) bounded coefficients, each depending on at most one input bit;
- (iii) public, simply preparable Newton right-hand sides;
- (iv) primal and slack starts bounded above and below by positive constants;
- (v) a coordinatewise bounded complete Newton direction; and
- (vi) a parity-hard primal component with constant mass in that complete direction state?

The work identity and the orthogonal-copy cut law rule out the natural signed, permutation, rotation, and flat-copy constructions. They do not cover arbitrary nonorthogonal mixed-mode constraints, input-dependent sources whose loading cost is counted separately, or output interfaces that quotient the equality multipliers.

The three main results of this section make different claims. The prefix-rigidified construction proves condition-one *reduced Newton geometry* together with a linear original-coordinate loading/recovery cost. The preconditioned full-KKT result proves a constant-condition residual theorem only when the queries that build its input-dependent access are counted. The bounded full direction obtained by dual homogeneity instead has canonical full-system condition number $\Theta(P)$.

10 Limits of Gadget Classes

[Section 9](#) made a local parity signal robust by replicating constraints, scaling rows, or increasing the height of the propagated signal. This section asks whether some other sparse LP gadget can avoid all three costs. For several broad, precisely stated classes the answer is no. The most general result is a convex-mixture theorem that assumes no graph structure. For signed-copy graphs and bounded-arity Walsh rows, electrical resistance gives a geometric refinement. Conservation of the dual multipliers shows that complete central and Newton outputs force a large row scale, inverse dual scale, multiplier size, or input incidence.

Throughout this section the hidden input is $\varsigma \in \{\pm 1\}^N$, with $N \geq 2$, and $p(\varsigma) = \prod_{i=1}^N \varsigma_i$. Input locality always refers to the raw sparse coefficient access of [definition 2.11](#). A residual written as $\|Ax - b\|_2$ is absolute; when a relative threshold is named, it is the global RHS-relative residual of [definition 2.26](#). The results are structural limits on output requirements. They become query lower bounds only after a separate reduction shows that the oracle can be simulated locally and that parity can be decoded.

10.1 Convex mixtures of neighboring witnesses

Consider a fixed-support family

$$\mathcal{F}_\varsigma = \{x \in \mathbb{R}_{\geq 0}^n : A_\varsigma x = b\}, \quad (366)$$

where b is public. Every row has at most s nonzeros, every coefficient has magnitude at most B , and each input-dependent position (r, j) is a function of one designated bit $\varsigma_{\ell(r, j)}$. Let M_{dep} be the number of these positions. A bit may label many positions, including several in one row. Fix a base input, let $\varsigma^{(i)}$ flip bit i , and suppose that each such neighbor has an exact witness

$$x^{(i)} \in \mathcal{F}_{\varsigma^{(i)}}, \quad \|x^{(i)}\|_\infty \leq H \quad (1 \leq i \leq N). \quad (367)$$

The magnitude bounds B, H are nonnegative. The residual estimates allow $s = 0$; the resource inequalities below that divide by s assume $s > 0$.

Theorem 10.1 (Single-flip convex mixture). *The average $\bar{x} = N^{-1} \sum_i x^{(i)}$ is nonnegative and satisfies*

$$\|A_\varsigma \bar{x} - b\|_2 \leq \frac{2BH\sqrt{sM_{\text{dep}}}}{N}. \quad (368)$$

If a public objective c has the same optimum v_ on every instance and the witnesses in (367) are optimal, then $c^\top \bar{x} = v_*$: the mixture is objective-exact, although it is generally infeasible for the base instance.*

Proof. Feasibility for the neighboring instances gives

$$N(A_\varsigma \bar{x} - b) = \sum_{i=1}^N (A_\varsigma - A_{\varsigma(i)})x^{(i)}.$$

Let D_r be the dependent positions in row r , and set $m_r = |D_r|$. In the r -th coordinate, position (r, j) contributes only to the summand $i = \ell(r, j)$, so this coordinate is a sum of at most m_r terms, each of magnitude at most $2BH$. Since $m_r \leq s$,

$$|N(A_\varsigma \bar{x} - b)_r|^2 \leq 4B^2H^2m_r^2 \leq 4sB^2H^2m_r.$$

Summing over rows and using $\sum_r m_r = M_{\text{dep}}$ proves (368). Convexity proves nonnegativity, and linearity proves the objective statement. $\square$

The theorem also covers a right-hand side whose entries each depend on one bit: apply it to $[A_\varsigma, -b_\varsigma](x, 1) = 0$. This uses row sparsity $s + 1$ and height $\max\{H, 1\}$, replaces B by a bound on every entry of $[A_\varsigma, -b_\varsigma]$, and counts the dependent RHS positions. Inequalities are covered after slack conversion only when the selected slack coordinates satisfy the same height bound.

The output hypothesis cannot be omitted. For an affine public score $\phi(x) = a^\top x - a_0$, suppose every selected witness for input ξ obeys $p(\xi)\phi(x^\xi) \geq \gamma > 0$. All single-flip neighbors have parity $-p(\varsigma)$, so $p(\varsigma)\phi(\bar{x}) \leq -\gamma$. Assume $b \neq 0$, as in definition 2.26, and call an output requirement *sound at threshold* $\eta \geq 0$ if it rejects every nonnegative, objective-exact point with the wrong output and relative residual at most η . A sound output requirement needs

$$M_{\text{dep}}B^2H^2 > \frac{\eta^2\|b\|_2^2}{4s}N^2. \quad (369)$$

In particular, for fixed $s, B, \eta, \|b\|_2$, linear incidence and bounded height are incompatible with soundness at a fixed threshold η (a fixed residual tube). Here and in later asymptotic statements about a fixed tube, $\eta > 0$.

The same conclusion holds for an output requirement on the normalized primal state $x/\|x\|_2$ under a concrete mixture-stability condition. Let $\mathcal{S}$ be $L \geq 1$ pairwise disjoint public output pairs (u_ℓ, v_ℓ) , and suppose

$$p(\xi)(u_\ell^\xi - v_\ell^\xi) \geq a > 0 \quad (\ell \in \mathcal{S}) \quad (370)$$

for every selected witness. Averaging gives $-p(\varsigma)(\bar{u}_\ell - \bar{v}_\ell) \geq a$. Nonnegativity gives $\bar{u}_\ell + \bar{v}_\ell \geq a$, hence

$$-p(\varsigma)(\bar{u}_\ell^2 - \bar{v}_\ell^2) \geq a^2.$$

For $Q_{\mathcal{S}} = \sum_{\ell \in \mathcal{S}} (|u_\ell\rangle\langle u_\ell| - |v_\ell\rangle\langle v_\ell|)$,

$$-p(\varsigma) \left\langle \frac{\bar{x}}{\|\bar{x}\|_2} \middle| Q_{\mathcal{S}} \middle| \frac{\bar{x}}{\|\bar{x}\|_2} \right\rangle \geq \frac{La^2}{nH^2}. \quad (371)$$

Measure a coordinate, output its pair sign on the selected pairs, and output an independent fair sign otherwise. The wrong-parity bias of this binary measurement is half the signed expectation

Table 3: Constructions attaining the one-bit-local mixture bound (369) up to constants, each by increasing one resource.

construction	M_{dep}	B	H
bounded-height parallel replication	$\Theta(N^2)$	$\Theta(1)$	$\Theta(1)$
direct row scaling	$\Theta(N)$	$\Theta(\sqrt{N})$	$\Theta(1)$
gain-plateau scaling	$\Theta(N)$	$\Theta(1)$	$\Theta(\sqrt{N})$

in (371). Thus $La^2 = \Omega(nH^2)$ gives a constant wrong-parity bias. Without an affine score, a common output template, the pair margin (370), or another explicit mixture-stability property, theorem 10.1 makes no claim about a normalized quadratic decoder.

Two refinements show the roles of sensitivity and uneven incidence. For a Boolean function f , fix a base input whose sensitive set S_f has size $k > 0$, and average only over its sensitive neighbors. If M_S counts the positions labelled by bits in S_f , the same proof gives

$$\|A_\zeta \bar{x}_S - b\|_2 \leq \frac{2BH\sqrt{sM_S}}{k}. \quad (372)$$

Under the common-objective and mixture-stable output hypotheses, soundness at threshold η therefore requires

$$M_S B^2 H^2 > \frac{\eta^2 \|b\|_2^2 k^2}{4s}.$$

More sharply, let M_i be the number of positions labelled by sensitive bit i . For weights $w_i \geq 0$ with $\sum_i w_i = 1$, set $x_w = \sum_i w_i x^{(i)}$. Cauchy-Schwarz in each row gives

$$\|A_\zeta x_w - b\|_2^2 \leq 4sB^2H^2 \sum_i M_i w_i^2. \quad (373)$$

When every $M_i > 0$, the right side is minimized by $w_i = M_i^{-1} / \sum_h M_h^{-1}$, which gives

$$\|A_\zeta x_w - b\|_2 \leq 2BH \sqrt{\frac{s}{\sum_i M_i^{-1}}}.$$

The corresponding soundness condition is

$$B^2 H^2 > \frac{\eta^2 \|b\|_2^2}{4s} \sum_i M_i^{-1}.$$

If some $M_i = 0$, that neighbor's witness is exactly feasible for the base matrix and already makes the output requirement unsound. These mixtures preserve every uniform affine or pair margin and the common objective. The resource that matters is therefore $\sum_i M_i^{-1}$, not merely the total incidence.

For a fixed relative tube and fixed $s, \eta, \|b\|_2$, Table 3 lists three constructions that attain (369) up to constants. Every row has $M_{\text{dep}} B^2 H^2 = \Theta(N^2)$. The parallel and gain-plateau families of theorems 9.2 and 9.4 attain the first and third rows. Scaling the propagation rows by $\sqrt{N}$ attains the second without changing the feasible set; the cost appears in the matrix normalization. This sharpness concerns only these scales, not the conditioning or QIPM cost of the formulations.

This is not an extension-complexity lower bound. Carr-Konjevod give an $O(N)$ -size dynamic-programming extended formulation for the parity polytope, and Ermel-Walter develop further flow formulations [36, 60]. Those formulations do not guarantee the correct parity output at every point of a fixed equality-residual tube. Sparse LP decoding also has classical fractional pseudocodewords and graph-cover descriptions [62, 87, 132]; these are conceptually related but do not give the quantitative bound (368). In particular, expander LP decoding may use an input-dependent objective, whereas exact objective preservation above requires one common objective and optimum.

10.2 Extensions: multi-bit coefficients, KKT systems, cones, and centrality

If coefficient position (r, j) may depend on a set $I_{rj} \subseteq [N]$ of input bits, define

$$d = \max_r \left| \bigcup_j I_{rj} \right|, \quad d_{\text{coeff}} = \max_{r,j} |I_{rj}| \leq d, \quad M = \sum_{(r,j)} |I_{rj}|,$$

and keep the row sparsity s , coefficient bound B , and witness height H .

Proposition 10.2 (Multi-bit coefficients). *The single-flip average obeys*

$$\|A_\zeta \bar{x} - b\|_2 \leq \frac{2BH\sqrt{d_{\text{coeff}}sM}}{N} \leq \frac{2BH\sqrt{dsM}}{N}. \quad (374)$$

For fixed d_{coeff}, s , the product bound $MB^2H^2 = \Omega(N^2)$ still holds under the objective and output assumptions above. The first bound recovers the one-bit constant when every coefficient depends on at most one bit, however many distinct bits appear in a row.

Proof. Write $f_{rji} = (A_{\zeta, rj} - A_{\zeta^{(i)}, rj})x_j^{(i)}$. It vanishes unless $i \in I_{rj}$, and $|f_{rji}| \leq 2BH$. Applying Cauchy–Schwarz first over the at most s dependent positions in a row, and then over each coefficient’s input set, gives

$$\left(\sum_j \sum_{i \in I_{rj}} f_{rji} \right)^2 \leq s \sum_j |I_{rj}| \sum_{i \in I_{rj}} f_{rji}^2 \leq 4sd_{\text{coeff}}B^2H^2 \sum_j |I_{rj}|.$$

Sum over rows and divide by N^2 . The row-union estimate follows from $d_{\text{coeff}} \leq d$. $\square$

A simultaneous primal–dual version also holds. Suppose A_ζ has maximum row and column sparsities s_r, s_c and M one-bit-dependent positions, and that the selected exact optimal triples (x^ξ, y^ξ, s^ξ) have all coordinates bounded in magnitude by H and a common optimal value v . Split $y = y^+ - y^-$, stack the primal feasibility and dual stationarity equations, and put

$$s_{\text{KKT}} = \max\{s_r, 2s_c + 1\}, \quad B_{\text{KKT}} = \max\{B, 1\}.$$

Each dependent position occurs once in the primal block and twice in the split dual block. [Theorem 10.1](#) therefore gives

$$\left\| \begin{bmatrix} A_\zeta \bar{x} - b \\ A_\zeta^\top \bar{y} + \bar{s} - c \end{bmatrix} \right\|_2 \leq \frac{2B_{\text{KKT}}H\sqrt{3s_{\text{KKT}}M}}{N}. \quad (375)$$

By linearity $c^\top \bar{x} = b^\top \bar{y} = v$, so the algebraic duality gap is exactly zero. This covers output requirements combining feasibility, stationarity, and gap requirements; the mixture does not preserve coordinatewise complementarity.

The mixture proof uses only convexity of the domain. Let $\mathcal{K}$ be a fixed convex set or cone, and suppose that for every selected input $\xi \in \{\zeta, \zeta^{(1)}, \dots, \zeta^{(N)}\}$ there is a witness $x^\xi \in \mathcal{K}$ with $A_\xi x^\xi = b$, $\|x^\xi\|_\infty \leq H$ in one fixed basis, and $c^\top x^\xi = v$ for one public objective and common value. If, in addition, a fixed parity-neutral linear map satisfies $Lx^\xi = p(\xi)y$ for a fixed nonzero y , then the half-base mixture

$$\hat{x} = \frac{1}{2}x^\zeta + \frac{1}{2N} \sum_i x^{\zeta^{(i)}}$$

belongs to $\mathcal{K}$, has the common objective, satisfies $L\hat{x} = 0$, and has half the residual in (374). For a sparse SDP this statement applies after fixing one svec identification, with all sparsity, locality, coefficient, and coordinate bounds measured in that basis. By itself it gives no conclusion about normalized states, rank, determinant neighborhoods, or complementarity.

A mixture of central points stays near centrality only under an explicit stability hypothesis. Keep the preceding row, column, one-bit-locality, and height bounds, and suppose the selected neighboring triples are exact central points for a common $\mu > 0$, so $x_j^{(i)} s_j^{(i)} = \mu$. Let $\bar{x}, \bar{s}$ be their averages. For a fixed coordinate j , write $t_i = x_j^{(i)}$. Symmetrizing over pairs i, k gives the exact identity

$$\frac{\bar{x}_j \bar{s}_j}{\mu} - 1 = \frac{1}{2N^2} \sum_{i,k=1}^N \frac{(t_i - t_k)^2}{t_i t_k}. \quad (376)$$

In particular, the excess is nonnegative and vanishes exactly when the neighboring j -th primal coordinates agree. If, for every coordinate j , $\max_i x_j^{(i)} / \min_i x_j^{(i)} \leq R$, where $R \geq 1$, then

$$\mu \leq \bar{x}_j \bar{s}_j \leq K(R)\mu, \quad K(R) = \frac{(R+1)^2}{4R} = 1 + \frac{(R-1)^2}{4R}. \quad (377)$$

Indeed, for $\alpha \leq t_i \leq \beta$, $t_i + \alpha\beta/t_i \leq \alpha + \beta$. Averaging and maximizing the product under this linear constraint gives $(N^{-1} \sum_i t_i)(N^{-1} \sum_i t_i^{-1}) \leq (\alpha + \beta)^2 / (4\alpha\beta)$, the scalar Kantorovich inequality [81].

Proposition 10.3 (Central-mixture soundness dichotomy). *Assume separately that the output is wrong for the base input uniformly over mixtures, in one of two forms. Write $Z^{(i)} = (x^{(i)}, y^{(i)}, s^{(i)})$. Either a fixed affine decoder Φ satisfies $p(\varsigma)\Phi(Z^{(i)}) \leq -\gamma < 0$ for every i , or the primal components have the public pair decoder of (371), with $-p(\varsigma)(u_\ell^{(i)} - v_\ell^{(i)}) \geq a$ for every i, ℓ and $La^2 = \Omega(nH^2)$. In the first case the decoded object is the stacked triple Z_w ; in the second it is the normalized primal state $|x_w\rangle = x_w / \|x_w\|_2$.*

Let w be any probability vector and set $Z_w = (x_w, y_w, s_w) = \sum_i w_i (x^{(i)}, y^{(i)}, s^{(i)})$. With M_i the one-bit incidence of bit i , and with $t_{ij} = x_j^{(i)}$, the stacked residual and the centrality defect obey

$$\left\| \begin{bmatrix} A_\varsigma x_w - b \\ A_\varsigma^\top y_w + s_w - c \end{bmatrix} \right\|_2^2 \leq 12s_{\text{KKT}} B_{\text{KKT}}^2 H^2 \sum_i M_i w_i^2, \quad (378)$$

$$\delta_j(w) := \frac{(x_w)_j (s_w)_j}{\mu} - 1 = \frac{1}{2} \sum_{i,k} w_i w_k \frac{(t_{ij} - t_{kj})^2}{t_{ij} t_{kj}}. \quad (379)$$

The weighted residual estimate admits the sharper constant

$$\left\| \begin{bmatrix} A_\varsigma x_w - b \\ A_\varsigma^\top y_w + s_w - c \end{bmatrix} \right\|_2^2 \leq 4(s_r + s_c) B^2 H^2 \sum_i M_i w_i^2. \quad (380)$$

Apply the weighted matrix-change estimate separately to the primal block and to the transposed coefficient changes in stationarity. The latter allows signed multipliers and has the same incidence profile, with column sparsity s_c . Adding the squared bounds proves (380); the earlier bound follows by increasing the constant. Thus the soundness inequalities below also hold with $12s_{\text{KKT}} B_{\text{KKT}}^2$ replaced by $4(s_r + s_c) B^2$. The algebraic objective difference, primal minus dual objective, remains exactly $n\mu$. This follows from feasibility, stationarity, and complementarity of each neighboring triple; the neighbors need not share their individual primal or dual objective values. An output requirement asking for absolute stacked residual at most ε and an ℓ_∞ neighborhood of width θ around the common μ is therefore sound only if, for every w ,

$$12s_{\text{KKT}} B_{\text{KKT}}^2 H^2 \sum_i M_i w_i^2 > \varepsilon^2 \quad \text{or} \quad \max_j \delta_j(w) > \theta. \quad (381)$$

Here $\varepsilon, \theta \geq 0$. For the point-centered convention, $\mu_w = x_w^\top s_w / n$ with $n > 0$, replace the second term by the exact width

$$\frac{\max_j |\delta_j - \bar{\delta}|}{1 + \bar{\delta}}, \quad \bar{\delta} = \frac{1}{n} \sum_j \delta_j.$$

For uniform weights, suppose for every coordinate j that

$$\frac{\max_i t_{ij}}{\min_i t_{ij}} \leq R.$$

Then the width is at most $K(R) - 1$ in either convention, so soundness requires the simpler dichotomy

$$\frac{12s_{\text{KKT}}B_{\text{KKT}}^2H^2M}{N^2} > \varepsilon^2 \quad \text{or} \quad \frac{(R-1)^2}{4R} > \theta. \quad (382)$$

Proof. The stacked mixture estimate is the single-flip argument applied to primal feasibility and stationarity. Pairwise symmetrization gives the displayed formula for $\delta_j(w)$, and linearity preserves the algebraic objective difference. The affine or pair-margin hypothesis makes the decoded object wrong for the base input. If neither strict inequality in (381) held, this mixture would satisfy every condition of the output requirement. The coordinatewise Kantorovich bound gives (382) for uniform weights. $\square$

The stability hypothesis is necessary. Fix $0 < \mu < 1$, let $b = \mu e, c = 0$, and let bit j choose $a_j \in \{\mu, 1\}$ in the diagonal LP $\text{Diag}(a)x = b$. Its exact μ -central point is

$$x_j = \mu/a_j, \quad s_j = a_j, \quad y_j = -1.$$

At the all- μ base input, each single-flip neighbor changes one coordinate from $(x, s) = (1, \mu)$ to $(\mu, 1)$. Thus every coordinate of the average has product

$$\bar{x}_j \bar{s}_j = \frac{N-1+\mu}{N} \frac{1+(N-1)\mu}{N},$$

which is $\Theta(1/N)$, rather than $\Theta(\mu)$, when $\mu \ll 1/N$, although the primal and stationarity residuals vanish as $O(N^{-1/2})$. Bounded coefficients and bounded central coordinates alone therefore do not keep central mixtures near the common μ . This particular uniform mixture has the same complementarity product in every coordinate, so its point-centered width is exactly zero. Its algebraic objective difference is still $N\mu$, whereas its complementarity sum is $N\bar{x}_j\bar{s}_j$; the two need not agree at an infeasible mixture.

A bounded-coordinate counterexample also exists for the point-centered convention. Append N public diagonal coordinates with coefficient μ and central coordinates $(x, s, y) = (1, \mu, -1)$. The moving coordinates share the product $q = (N-1+\mu)(1+(N-1)\mu)/N^2$, while the public coordinates have product μ . The point-centered mean is $(q+\mu)/2$, and every coordinate has relative deviation $(q-\mu)/(q+\mu)$. For $N \geq 8$ and $\mu = N^{-2}$, the exact values satisfy

$$\left\| \begin{bmatrix} A\bar{x} - b \\ A^\top \bar{y} + \bar{s} - c \end{bmatrix} \right\|_2^2 = \frac{(1+\mu^2)(1-\mu)^2}{N} \leq \frac{2}{N}, \quad \frac{q-\mu}{q+\mu} \geq \frac{1}{3}.$$

The unpadding example has the same residual and $\max_j \delta_j \geq N/8$ for $\mu = N^{-2}$, $N \geq 2$. Both constructions have row and column sparsity one, coefficient and coordinate magnitudes at most one, and exactly N dependent positions. Thus these bounds and a vanishing residual do not imply a point-centered width below any fixed $\theta < 1/3$.

Formal verification. The Lean development in `formal/QipmFormal/Mixture/` verifies the residual and centrality results of these two subsections from the actual finite matrices and neighboring witnesses. It covers weighted and sensitive-subset mixtures, the stronger coefficientwise multibit bound, RHS and slack augmentation, the split KKT construction, the direct KKT bound, harmonic incidence weights, affine and normalized pair decoders, the scalar variance and Kantorovich identities, both neighborhood conventions, and the explicit counterexamples. The soundness hypotheses are stated as conditions on candidate outputs; the proofs derive the residual, objective, and decoder properties of the mixture. The development uses Euclidean sums of squares and relies only on Lean’s standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The claim map and reproduction command are in `formal/MIXTURE.md`. The sharpness constructions in Table 3, the electrical bounds below, and the quantum query reductions are not covered by this verification. The noncommutative extension has a separate Lean development and claim map in `formal/SDP_MIXTURE.md`; see [section 14.2](#).

10.3 Electrical-resistance bounds

The graph argument constructs, within the base instance, a point that erases the outputs and a point that flips them, and shows the geometry behind the mixture bound: disjoint input cuts force a large effective resistance. Let $G = (V, E)$ be a connected oriented multigraph with root r , nonempty output set S , public heights $h_v > 0$ with $h_r = 1$, and propagation rows

$$\lambda_e(d_v - g_e a_e(\varsigma) d_u) = 0, \quad g_e = h_v/h_u, \quad (383)$$

where $e = (u, v)$, $\lambda_e \neq 0$ is a public scale, a_e is either a fixed sign or a fixed sign times one input bit, and the rows are cycle-consistent. Thus there are signs τ_v with $\tau_r = 1$ and $\tau_v = a_e \tau_u$. Assume separately that the designated outputs carry the parity:

$$\tau_s(\varsigma) = \epsilon_s p(\varsigma) \quad (s \in S), \quad \epsilon_s \in \{\pm 1\} \text{ public}. \quad (384)$$

Embed each d_v using the capped pair of [section 9.1](#): write $d_v = u_v - v_v$ and $q_v = u_v + v_v$, fix a reference variable $h_v^{\text{var}} = h_v$, impose $q_v + t_v - 2h_v^{\text{var}} = 0$, and use the local objective $2u_v + 2v_v + h_v^{\text{var}} + t_v$. Put

$$L = \max_{e=(u,v)} |\lambda_e| h_v \leq BH, \quad M = |E|,$$

where the inequality assumes $0 < |\lambda_e| \leq B$ and $h_v \leq H$.

Theorem 10.4 (Cut-resistance bound). *There is a nonnegative, objective-exact point that satisfies every root, reference, and cap row exactly, erases all output differences, and obeys*

$$\|Ax - b\|_2^2 \leq \frac{ML^2}{N^2} \leq \frac{MB^2H^2}{N^2}. \quad (385)$$

There is also a point with every output sign flipped, whose squared residual is at most four times this bound. If the two unit anchors are the only nonzero RHS entries, so $\|b\|_2 = \sqrt{2}$, any relative-residual output requirement with threshold η that rules out the erased point requires

$$MB^2H^2 > 2\eta^2 N^2. \quad (386)$$

If each edge label contains at most k raw signs, the residual bound (385) for the erased point acquires a factor k^2 , and soundness requires $Mk^2B^2H^2 = \Omega(N^2)$.

Proof. For bit i , let E_i be the edges whose label changes when bit i flips. By cycle consistency every cycle meets E_i evenly, so signed-graph switching identifies E_i with a cut separating the root from every output [75, 144]. One-bit locality makes the N cuts edge-disjoint.

Give edge $e = (u, v)$ conductance $c_e = (\lambda_e h_v)^2 \leq L^2$, and merge all outputs into one node. For any unit root-to-output flow f , one unit crosses every cut. Weighted Cauchy–Schwarz gives

$$\sum_{e \in E_i} \frac{f_e^2}{c_e} \geq \frac{1}{C_i}, \quad C_i = \sum_{e \in E_i} c_e.$$

Because the cuts are disjoint, Thomson’s principle and a second application of Cauchy–Schwarz give the weighted Nash–Williams bound

$$\mathcal{R}_{r,S} \geq \sum_i \frac{1}{C_i} \geq \frac{N^2}{\sum_i C_i} \geq \frac{N^2}{ML^2}. \quad (387)$$

We include this derivation to show where the oracle’s disjoint input cuts enter; the resistance inequality itself is classical [111].

Let $w \in [0, 1]^V$ be the harmonic voltage with $w_r = 1$ and $w_s = 0$ on every output. By Dirichlet’s principle and (387), its energy is at most ML^2/N^2 . Set

$$d_v = \tau_v h_v w_v, \quad q_v = h_v^{\text{var}} = t_v = h_v, \quad u_v = (h_v + d_v)/2, \quad v_v = (h_v - d_v)/2.$$

The maximum principle gives nonnegativity. Since every λ_e is nonzero and the graph is connected, every feasible point has $|d_v| = h_v$, hence $q_v = u_v + v_v \geq |d_v| = h_v$; the local objective reduces to $q_v + 3h_v$, so this point, which has $q_v = h_v$, is objective-exact. All nonpropagation rows hold exactly, while edge e has residual $\lambda_e \tau_v h_v (w_v - w_u)$, whose squared sum is the Dirichlet energy. Every output has $d_s = 0$. Replacing the output boundary value zero by -1 doubles every voltage difference and proves the statement about the sign-flipped point.

For k -local labels, an edge lies in at most k of the bit cuts. Hence $\mathcal{R}_{r,S} \geq k^{-1} \sum_i C_i^{-1}$, while $\sum_i C_i \leq kML^2$. This gives $\mathcal{R}_{r,S} \geq N^2/(k^2 ML^2)$ and the stated factor. $\square$

By [theorem 10.4](#), the gain–plateau family is optimal among cycle-consistent signed graphs: linear size and bounded row scales force $H = \Omega(\sqrt{N})$. The other two rows of [Table 3](#) attain the remaining extremes. Adaptivity does not rescue the pointwise output requirement, because the harmonic point exists at every requested stage. However, the theorem gives no unconditional lower bound on fresh queries per iteration: after all N fixed signs are cached, later raw-query cost may be zero under [definition 2.27](#).

The resistance argument extends to rows with several wires. Give wire v a public height $h_v \in (0, H]$ and intended value $h_v \chi_{J_v}(\varsigma)$, where $\chi_J = \prod_{i \in J} \varsigma_i$. Let each of m homogeneous rows have arity at most q :

$$\sum_{v \in V_a} \alpha_{av} \chi_{I_{av}}(\varsigma) d_v = 0, \quad |\alpha_{av}| \leq A, \quad |I_{av}| \leq k. \quad (388)$$

Roots have $J_v = \emptyset$, outputs have $J_v = [N]$, and every row holds on the intended wires for every input. We may assume that the comparison graph in the proof connects the roots to the outputs; otherwise the output character is unconstrained and can be erased with zero propagation residual. For the objective-exact conclusion, assume also that the intended full-height assignment is optimal for the capped LP. It suffices that every exactly feasible assignment has $|d_v| \geq h_v$ on every wire, since nonnegativity then forces $q_v \geq h_v$.

Theorem 10.5 (Bounded-arity Walsh-row bound). *The capped-pair embedding has a nonnegative, objective-exact point that satisfies the anchors, references, and caps, erases every output, and has*

$$\|r\|_2^2 \leq \frac{4k^2 q^3 m A^2 H^2}{N^2}. \quad (389)$$

Consequently, for fixed k, q , soundness at a constant absolute residual threshold requires $m A^2 H^2 = \Omega(N^2)$.

Proof. Substituting the intended wires into row a gives $\sum_v \beta_{av} \chi_{K_{av}} = 0$, where $\beta_{av} = \alpha_{av} h_v$ and $K_{av} = I_{av} \triangle J_v$. Since distinct Walsh characters are linearly independent, $\sum_{v \in G_{a,K}} \beta_{av} = 0$ within each equal-character group. Choose one pivot per group and join it to the other group members by comparison edges of conductance $q^2 \beta_{av}^2$. There are at most qm edges, each with conductance at most $q^2 A^2 H^2$. For arbitrary interpolation values w_v , a character group with pivot p has residual

$$\sum_{v \in G} \beta_{av} w_v = \sum_{v \in G \setminus \{p\}} \beta_{av} (w_v - w_p).$$

Applying Cauchy–Schwarz first over the at most q terms of a group and then over the at most q groups of a row gives

$$\|r\|_2^2 \leq \sum_e c_e (w_{h(e)} - w_{t(e)})^2.$$

The endpoints of a comparison edge satisfy $J_u \triangle J_v = I_{au} \triangle I_{av}$, so the edge crosses at most $2k$ of the vertex cuts $\{v : i \in J_v\}$. The bounded-overlap form of the Nash–Williams calculation proved above gives

$$\mathcal{R}_{\text{eff}} \geq \frac{N^2}{4k^2 q^3 m A^2 H^2}.$$

Take the harmonic interpolation equal to one at the roots and zero at the outputs. Its energy gives (389); the maximum principle and the capped-pair construction from the preceding proof give nonnegativity and exact auxiliary rows. The erased point has $q_v = h_v$ on every wire, hence the local objective of the intended assignment, which is optimal by hypothesis. $\square$

Formulations built from exact local XOR hulls have three further concrete defects. If a set U of unfixed variables meets every parity check in zero or at least two positions, then the point that sets $x_j = 1/2$ on U and keeps a satisfying Boolean codeword elsewhere lies in every local parity polytope. Indeed, for a check meeting U in $r \geq 2$ variables, this point is the uniform average of the 2^{r-1} assignments with the required residual parity. This stopping-set pseudocodeword can make an output exactly unbiased. If the checks instead peel along a dependency chain of length D , linearly interpolating the gauged wire error from zero to one violates one opposing facet at each gate by $1/D$, for total Euclidean violation $1/\sqrt{D}$. Finally, every satisfying truth-table assignment is a vertex of its exact gate hull, so some standard-form slacks vanish; a Boolean circuit forced by its inputs is not strictly primal feasible. Thickening the hulls restores positive slacks only by admitting fractional drift. These facts explain the LP-decoding analogy but are not a universal lower bound for all inequality extended formulations.

10.4 Multiplier conservation

Residual robustness constrains primal witnesses. Complete central and Newton outputs face a second constraint: conservation of the dual multipliers. In a signed-copy graph $G = (V \cup T, E)$, vertices in V carry variables with $d_v = x_v^+ - x_v^-$ and $q_v = x_v^+ + x_v^-$, and vertices in T are fixed terminals used for anchor rows. A variable–variable edge $e = (u, v)$ represents $d_v - a_e d_u = 0$. A terminal edge is oriented from its terminal to its variable endpoint, with the fixed terminal term moved to the RHS. Let B_a and B be the signed and the ordinary oriented incidence matrices, with all terminal columns omitted. For $S \subseteq V$, the boundary ∂S consists of the edges to $V \setminus S$ and all terminal edges incident to S . Each normalized edge equation has a multiplier y_e ; an actual LP row may be multiplied by a public nonzero scale $r_e \neq 0$, in which case its scale-invariant row force is $f_e = r_e y_e$. Assume exact primal–dual centrality, $d_v \neq 0$, a pair-symmetric objective, and pair-symmetric columns for all equality constraints outside the graph.

Theorem 10.6 (Gauge flow and isoperimetric law). *Let $\tau_v = \text{sgn}(d_v)$. After the edge gauge $z_e = \tau_{h(e)}y_e$, signed stationarity becomes the positive-divergence law*

$$B^\top z = g, \quad g_v = \frac{2\mu|d_v|}{q_v^2 - d_v^2} > 0. \quad (390)$$

For every vertex set S with nonempty boundary,

$$\sum_{e \in \partial S} |y_e| \geq G_S, \quad \|y_{\partial S}\|_\infty \geq \frac{G_S}{|\partial S|}, \quad \|y_{\partial S}\|_2 \geq \frac{G_S}{\sqrt{|\partial S|}}, \quad G_S = \sum_{v \in S} g_v. \quad (391)$$

If $S \neq \emptyset$ but $\partial S = \emptyset$, no such positive central point exists. If $|d_v| \geq a > 0$ and $q_v \leq Q$ on S , then

$$\|y_{\partial S}\|_\infty \geq \frac{2\mu a}{Q^2 - a^2} \frac{|S|}{|\partial S|}, \quad \|y_{\partial S}\|_2 \geq \frac{2\mu a}{Q^2 - a^2} \frac{|S|}{\sqrt{|\partial S|}}. \quad (392)$$

For a row written as $r_e(d_v - a_e d_u) = 0$, all statements hold for the row force $f_e = r_e y_e$; hence, if $|r_e| \leq B_{\text{row}}$, the lower bounds apply to $B_{\text{row}}\|y\|_\infty$, not to the multiplier alone, which depends on the row normalization.

Proof. Subtract the two stationarity equations of each pair:

$$2(B_a^\top y)_v + s_v^+ - s_v^- = 0.$$

Since $x_v^\pm = (q_v \pm d_v)/2$ and $s_v^\pm = \mu/x_v^\pm$, this becomes $(B_a^\top y)_v = 2\mu d_v/(q_v^2 - d_v^2)$. Feasibility implies $a_e = \tau_u \tau_v$ on variable-variable edges; terminal rows have the same single variable coefficient in B_a and B . Switching the signed incidence matrix therefore gives (390). Summing over S cancels the internal edges; the triangle inequality and Cauchy-Schwarz give (391). Positivity gives $q_v > |d_v|$, and the margin hypotheses bound every g_v below by $2\mu a/(Q^2 - a^2)$. A diagonal row scaling enters stationarity only through $r_e y_e$, which proves the final statement. $\square$

Pair symmetry is not needed to compare two inputs if the pair-antisymmetric source h_v below is the same vector, in the original pair coordinates, for both. Let two exact central points, denoted $+$ and $-$, have row forces $f_e^\pm = r_e y_e^\pm$, and suppose their subtracted stationarity equations are

$$(B_{a^\pm}^\top f^\pm)_v = h_v + \frac{2\mu_\pm d_v^\pm}{(q_v^\pm)^2 - (d_v^\pm)^2}.$$

If $d_v^+ > 0 > d_v^-$ on S , then switching the two inputs, adding their equations, and summing over S cancels both the common source and every internal edge. Hence

$$\sum_{e \in \partial S} (|f_e^+| + |f_e^-|) \geq \sum_{v \in S} (g_v^+ + g_v^-), \quad g_v^\pm = \frac{2\mu_\pm |d_v^\pm|}{(q_v^\pm)^2 - (d_v^\pm)^2}. \quad (393)$$

If $\mu_\pm \geq \mu_0$, $|d_v^\pm| \geq a$, $q_v^\pm \leq Q$, and $|r_e| \leq B_{\text{row}}$, this gives

$$\max_{p \in \{+, -\}} \|y_{\partial S}^p\|_\infty \geq \frac{2\mu_0 a}{Q^2 - a^2} \frac{|S|}{B_{\text{row}} |\partial S|}. \quad (394)$$

Thus a public asymmetric objective can cancel the demand g_v for one sign pattern of d on S , but not for both when ∂S is small. Coefficients of another public asymmetric constraint do not suffice unless their contribution through the multipliers is also the same vector h for both inputs.

The identity for the work $b_a^\top y_a$ does not need a graph. Let $Cd = b_a$ be an arbitrary pair-antisymmetric equality block, with columns C , $-C$, and let all other local data be pair-symmetric.

Theorem 10.7 (Central and Newton work identities). *At an exact central point,*

$$\boxed{b_a^\top y_a = \sum_v \frac{2\mu d_v^2}{q_v^2 - d_v^2} \geq 0.} \quad (395)$$

The work is strictly positive exactly when $d \neq 0$. At a pair-symmetric Newton start with pair-symmetric dual and complementarity right-hand sides, pair-symmetric other equality columns, $x_v^{0,+} = x_v^{0,-} = \xi_v$, and $s_v^{0,+} = s_v^{0,-} = \zeta_v$, the antisymmetric block satisfies

$$C_\zeta \Delta d = r_a, \quad C_\zeta^\top \alpha = D \Delta d, \quad D = \text{Diag}\left(\frac{\zeta_v}{2\xi_v}\right), \quad \boxed{r_a^\top \alpha = \Delta d^\top D \Delta d.} \quad (396)$$

Both pairings are invariant under $(C, b_a, y_a) \mapsto (UC, Ub_a, U^{-\top} y_a)$, and likewise for (C_ζ, r_a, α) . If $D \succeq \rho I$, $|\Delta d_v| \geq a$ on L coordinates, and $\|r_a\|_1 \leq R$, then $\|\alpha\|_\infty \geq \rho a^2 L/R$. Likewise, if $|d_v| \geq a$ and $q_v \leq Q$ on L coordinates, then

$$\|y_a\|_\infty \geq \frac{2\mu a^2 L}{(Q^2 - a^2)\|b_a\|_1}, \quad \|y_a\|_2 \geq \frac{2\mu a^2 L}{(Q^2 - a^2)\|b_a\|_2}. \quad (397)$$

A zero denominator with nonzero work means that no such point exists.

Proof. Pair subtraction gives $C^\top y_a = 2\mu d/(q^2 - d^2)$ coordinatewise. Taking the inner product with d and using $Cd = b_a$ proves (395). At the Newton start, subtracting the dual and complementarity equations and eliminating the slack difference gives $C_\zeta^\top \alpha = D \Delta d$; primal antisymmetric feasibility gives $C_\zeta \Delta d = r_a$. Taking inner products proves (396), and Hölder's inequality gives the norm bounds. The contragredient transformations preserve both scalar pairings. $\square$

For the bounded secant start of [section 9.8](#), $D = (2/9)I$, $\Delta d_v = \tau_v$, and the antisymmetric RHS at the root is a unit vector. Equation (396) therefore gives the root multiplier $2P/9$. Scaling all dual data by λ changes this to $2\lambda P/9$; bounded full Newton directions require $\lambda = O(1/P)$, exactly the inverse dual scale used in [theorem 9.13](#). [Section 9.9](#) states the corresponding unrestricted Newton virtual-work identity. We use the paired forms here because they expose the gadget resources; [section 14](#) places them in the general involution/symmetry framework.

The cut laws also hold directly for Newton directions. Under a pair-symmetric start, pair-symmetric Newton right-hand sides, and homogeneous signed-copy primal equations, switching by $\text{sgn}(\Delta d_v)$ gives

$$B^\top z = g^N, \quad g_v^N = \frac{\zeta_v}{2\xi_v} |\Delta d_v|. \quad (398)$$

The proof is the Newton half of [theorem 10.7](#) followed by the cut summation in [theorem 10.6](#). In particular, if $\zeta_v/\xi_v \geq r$ and $|\Delta d_v| \geq a$ on S , then

$$\|\Delta y_{\partial S}\|_\infty \geq \frac{ra}{2} \frac{|S|}{|\partial S|}.$$

Now assume that terminal values, labels, and terminal Newton residuals are public and fixed, and fix a base input such that it and all its single-flip neighbors have exact central points with nonzero d_v . Let O be a fixed set of K output vertices on which $\text{sgn } d_v$ flips under every single-bit flip, and define

$$S_j = \{v : \text{sgn } d_v(\zeta^{(j)}) = -\text{sgn } d_v(\zeta)\}.$$

Let M_j count the signed rows that change under flip j , and put $M_{\text{dep}} = \sum_j M_j$. Feasibility on an internal edge gives $a_e = \text{sgn}(d_u) \text{sgn}(d_v)$, so its label changes exactly when one endpoint is in S_j . Public terminal signs act as fixed outside vertices, so the same boundary count includes terminal

edges. Hence $|\partial S_j| \leq M_j$, with equality when rows rather than expanded coefficient positions are counted, and $O \subseteq S_j$. The base multiplier must satisfy all N cut inequalities simultaneously. Assume explicitly that the base central point has $|d_v| \geq a = \Omega(1)$ and $q_v \leq Q = O(1)$ on O . Applying this central margin only on O and summing over j proves

$$M_{\text{dep}} B_{\text{row}} \|y\|_{\infty} \geq \frac{2\mu a}{Q^2 - a^2} K N. \quad (399)$$

For Newton directions, suppose instead that the base and every single-flip system satisfy the hypotheses of (398), define S_j from the signs of Δd , and assume that every output direction flips with $|\Delta d_v| \geq a = \Omega(1)$ on O . The same argument gives

$$M_{\text{dep}} B_{\text{row}} \|\Delta y\|_{\infty} \geq \rho a K N, \quad \rho = \min_{v \in O} \frac{s_v^0}{2x_v^0}. \quad (400)$$

Thus $M_{\text{dep}} = \Theta(N)$, $K = \Theta(N)$, bounded row scale, and bounded multipliers, together with the explicit constant margin $a = \Omega(1)$ and central range $Q = O(1)$, force $\mu = O(1/N)$ in the central system and $\rho = O(1/N)$ in the Newton system. Equation (400) is the general conservation law behind the multiplier lower bound beyond signed pairs in section 9.9. Every signed-copy gadget must make the input incidence, row scale, multiplier size, or inverse dual scale large. Branching and redundant copy rows only move the cost into the relevant boundary count.

Pair-asymmetric local data do not provide a cheap universal cancellation. Suppose a prescribed direct source $h_v(\varsigma)$ changes central stationarity to

$$(B_a^{\top} f)_v = \frac{2\mu d_v}{q_v^2 - d_v^2} + h_v(\varsigma), \quad f_e = r_e y_e.$$

On a fixed set O of K output vertices, assume $\text{sgn}(d_v) = \kappa_v p(\varsigma)$, $|d_v| \geq a$, and $q_v \leq Q$ for every input. With the uniform Fourier convention $\widehat{h}_v([N]) = 2^{-N} \sum_{\varsigma} p(\varsigma) h_v(\varsigma)$, put $\varepsilon_O = K^{-1} \sum_{v \in O} |\widehat{h}_v([N])|$.

Theorem 10.8 (Fourier obstruction to direct cancellation). *For some input ς^* ,*

$$\sum_{e \in \partial O} |r_e y_e(\varsigma^*)| \geq K \left(\frac{2\mu a}{Q^2 - a^2} - \varepsilon_O \right). \quad (401)$$

In particular, bounded-bit juntas that depend on strictly fewer than N input bits, and all sources of Fourier degree less than N , have zero full-parity coefficient and cannot cancel the demand for all inputs. The bound is tight: the direct source $h_v = -\kappa_v p(\varsigma) g_0$ cancels a constant demand g_0 exactly.

Proof. After switching, sum the stationarity equations over O . The direct contribution is $H(\varsigma) = \sum_{v \in O} \kappa_v p(\varsigma) h_v(\varsigma)$, whose average over inputs is at least $-K\varepsilon_O$. Some ς^* has $H(\varsigma^*)$ at least this average. The positive central demand is at least $2\mu a K / (Q^2 - a^2)$; internal edge forces cancel, and the triangle inequality proves (401). $\square$

There is an exact Newton analogue. At a pair-symmetric start put $x_v^{0,\pm} = \xi_v$, $s_v^{0,\pm} = \zeta_v$, and allow asymmetric dual and complementarity right-hand sides. Pair subtraction gives

$$(B_a^{\top} \Delta f)_v = \frac{\zeta_v}{2\xi_v} \Delta d_v + h_v^N(\varsigma), \quad h_v^N = \frac{r_{d,v}^+ - r_{d,v}^-}{2} - \frac{r_{c,v}^+ - r_{c,v}^-}{2\xi_v}.$$

If $\text{sgn}(\Delta d_v) = \kappa_v p(\varsigma)$, $|\Delta d_v| \geq a$, and $\zeta_v / (2\xi_v) \geq \rho$ on O , then the same proof gives some input ς^* with

$$\sum_{e \in \partial O} |r_e \Delta y_e(\varsigma^*)| \geq K(\rho a - \varepsilon_O^N), \quad \varepsilon_O^N = K^{-1} \sum_{v \in O} |\widehat{h}_v^N([N])|. \quad (402)$$

This Fourier coefficient must be computed after division by the start coordinates: input dependence in $1/\xi_v$ can create full parity from raw data that are otherwise local. The word *direct* also matters. An unknown multiplier of another antisymmetric constraint may depend on all input bits even when that row is local, so [theorem 10.8](#) does not assume that such a multiplier has low Fourier degree.

Switching and conservation of incidence flows are classical [\[75, 144\]](#); the work identities are special cases of standard KKT virtual-work and Schur-energy identities. Electrical flows on interior-point central paths are also established algorithmic tools [\[98\]](#). The only contribution claimed here is that the local input cuts, the parity-sized output region, multiplier conservation, row scaling, and incidence accounting must all be satisfied at once. Effective resistance has a separate established role in quantum graph algorithms [\[9, 23, 79\]](#); those works do not imply the LP residual or multiplier bounds of this section.

10.5 Relative-value estimation and long shortcut rows

The path family also gives a simple lower bound for value outputs. Put $p_0 = 1$ and $p_i = \prod_{k=1}^i s_k$. On a single signed path, minimize

$$\frac{1}{N+1} \sum_{i=0}^N (u_i + v_i) + \frac{1}{4}(u_N - v_N) \quad (403)$$

subject to $d_0 = 1$ and $d_i - \varsigma_i d_{i-1} = 0$.

Theorem 10.9 (Relative-value gap). *In raw sparse coefficient access, estimating the optimum of (403) to target-relative error $\epsilon < 1/4$ requires $\Omega(N)$ queries. The LP has a finite, attained optimum, is strictly primal-dual feasible, and has a unique, nondegenerate, strictly complementary optimal solution. Its objective norm is bounded by an absolute constant, and its primal log-barrier Hessian, restricted to the feasible affine space, has condition number one along the entire central path.*

Proof. Exact feasibility fixes $d_i = p_i$ and $q_i = u_i + v_i \geq 1$. On this affine space the objective is $(N+1)^{-1} \sum_i q_i + p_N/4$, so the unique optimum has every $q_i = 1$ and

$$\text{OPT} = 1 + \frac{1}{4}p(\varsigma) \in \left\{ \frac{3}{4}, \frac{5}{4} \right\}. \quad (404)$$

The two target-relative error intervals are disjoint for every $\epsilon < 1/4$; hence such value estimation requires $\Omega(N)$ raw queries by the parity lower bound. Even an exactly feasible point with objective at most $3\text{OPT}/2$ distinguishes the two cases by a threshold between $9/8$ and $5/4$. Each sparse coefficient query is simulated by one raw sign query. Also $\|c\|_2^2 = 2/(N+1) + 1/8 = O(1)$.

The equality matrix is full row rank, and its public orthonormal null basis consists of $(e_{u_i} + e_{v_i})/\sqrt{2}$. Taking every $q_i = 2$ proves strict primal feasibility. Taking $s_{u_i} = s_{v_i} = 1/(N+1)$ gives a positive slack vector for which $c - s$ is the endpoint difference term; it is orthogonal to the nullspace and hence lies in $\text{range}(A^\top)$, proving strict dual feasibility. At the optimum, one positive coordinate from each pair gives a triangular basis, and every opposite coordinate has positive reduced cost $2/(N+1)$. This proves nondegeneracy and strict complementarity.

Finally, on the feasible slice the barrier objective is

$$\sum_{i=0}^N \left[\frac{q_i}{N+1} - \mu \log \frac{q_i^2 - 1}{4} \right] + \frac{1}{4}p_N.$$

Every central coordinate solves the same scalar equation $(N+1)^{-1} = 2\mu q/(q^2 - 1)$, so the Hessian in the displayed orthonormal null basis is a positive scalar multiple of the identity. $\square$

This is a value reduction with the normalization guarantees stated above, not a query lower bound for generic LPs.

One tempting repair is a long two-coordinate shortcut row. Suppose $0 \leq i < j \leq N$ and a homogeneous row $a(\varsigma)d_i + b(\varsigma)d_j = 0$ holds on every exact signed path, with both $a(\varsigma)$ and $b(\varsigma)$ nonzero for every input covered by the conclusion. Substitution then gives

$$-\frac{a(\varsigma)}{b(\varsigma)} = \frac{p_j}{p_i} = \prod_{k=i+1}^j \varsigma_k. \quad (405)$$

Thus the row itself carries the parity of bits $i+1, \dots, j$, invariant under common row rescaling, and generally costs $\Omega(j-i)$ raw queries to simulate. Otherwise, replicated-path and expander constructions are covered by the mixture, resistance, and Walsh-row theorems above.

10.6 Scope of the conclusions

The parallel-path and gain-plateau constructions of [section 9](#) are optimal in their stated bounded-height and linear-size classes, respectively. The $\Theta(N^{-1/2})$ residual threshold of the all-unit family in [theorem 9.11](#) is the matching threshold without amplification, and dual homogeneity attains the scaling $\rho = \Theta(1/N)$ that [\(400\)](#) forces. None of these conclusions says that parity needs a quadratic-size LP, that every bounded-range LP fails, or that an adaptive QIPM pays a fresh linear query cost at every iteration.

A bounded-range construction with linear incidence must violate at least one hypothesis used here. The escape routes are:

- (i) use an output promise that is nonlinear or otherwise not mixture-stable, and prove its robustness directly;
- (ii) require only central or Newton outputs, and allow the neighboring central coordinates to leave every fixed Kantorovich neighborhood [\(377\)](#);
- (iii) use nonlocal input coefficients and charge their full raw-query cost;
- (iv) give up the common objective or common optimum, so neighbor mixtures need not be objective-exact; or
- (v) impose integrality or another nonconvex promise, which leaves linear programming.

The second route is not automatic: [\(381\)](#)–[\(382\)](#) show exactly how much central instability it requires.

Two problems remain open within the present scope. First, can an arbitrary sparse bounded-range equality LP with a nonlinear normalized-state decoder be robust in a fixed global residual tube without nonlocal coefficient access? Second, can the bounded-scale full-KKT problem of [open problem 9.14](#) be solved by a nonorthogonal mixed-mode gadget when all input-dependent sources and recovery maps are charged? Together with the orthogonal-copy cut law in [section 9.9](#), the theorems here rule out the natural mixture-stable, signed-copy, constant-arity Walsh, permutation, rotation, and flat-copy approaches; they do not settle either question.

11 SDP Hardness Families

This section proves query lower bounds for SDP families built directly as SDPs, rather than by embedding the LP families of [section 9](#). In each, the requested output reveals the parity of hidden signs. The bounds concern three kinds of output. For the optimal value, the first family has a constant gap between the parities. For a loaded state, the second (Schur) family has a public Schur-complement solve, but loading the requested state in the original coordinates is

expensive. For block encodings, preparing the Newton-direction or complete KKT state needs linearly many calls even to an exact constant-normalization block encoding of a canonical KKT matrix, and first constructing a tangent-projector encoding needs linearly many raw queries. All claims use the access models and output requirements of [sections 2.2](#) and [2.3](#).

We also determine which families are SOCP-representable. Projection onto X_0 identifies the feasible set of the first, trace-normalized family with the density slice of $\mathbb{S}_+^3$. In the Schur family, 3×3 blocks leave only a free $\mathbb{S}_+^2$ and are SOCP-representable; 4×4 blocks are the smallest whose free Schur complement is $\mathbb{S}_+^3$, which has no finite second-order-cone lift.

11.1 An intrinsic-spectrum trace-normalized family

Let $n_{\text{bit}} \geq 1$, let

$$K = 16n_{\text{bit}}, \quad P = 2K + n_{\text{bit}} = 33n_{\text{bit}}, \quad h = \prod_{j=1}^{n_{\text{bit}}} \varsigma_j, \quad (406)$$

and put $D_s = \text{diag}(s, 1, 1)$. There are P blocks $X_0, \dots, X_{P-1} \in \mathbb{S}_+^3$. Consecutive blocks obey

$$X_i = G_i X_{i-1} G_i^\top. \quad (407)$$

Along the path of blocks, the first $K-1$ and last K edges have $G_i = I$; the intervening n_{bit} edges have $G_i = D_{\varsigma_j}$ in input order. Add $\text{tr } X_0 = 1$. With $R_0 = I$ and $R_i = G_i \cdots G_1$, elimination shows that the feasible set is exactly

$$X_i = R_i Z R_i^\top, \quad Z \succeq 0, \quad \text{tr } Z = 1. \quad (408)$$

In the Frobenius-isometric coordinates

$$\text{svec}(X) = (X_{11}, \sqrt{2}X_{12}, \sqrt{2}X_{13}, X_{22}, \sqrt{2}X_{23}, X_{33}), \quad (409)$$

congruence by D_s acts as $\text{diag}(1, s, s, 1, 1, 1)$. Each copy row has two nonzeros, the trace row has three, and each scalar column occurs in at most two rows. Each input sign changes only the two predecessor coefficients in the 12 and 13 rows. Ordering the copy rows along the path shows that they are linearly independent; their kernel is the six-dimensional root family in [\(408\)](#), and the trace row removes one further dimension.

For $a < b$, write $H_{ab} = e_a e_b^\top + e_b e_a^\top$, and set

$$u = \frac{1}{10}, \quad q = \frac{K}{P} = \frac{16}{33}, \quad \gamma = \frac{u}{q} = \frac{33}{160}. \quad (410)$$

Use the public costs

$$C_i = \begin{cases} 3I + uH_{23} + \gamma H_{12}, & 0 \leq i < K, \\ 3I + uH_{23}, & K \leq i < K + n_{\text{bit}}, \\ 3I + uH_{23} + \gamma H_{13}, & K + n_{\text{bit}} \leq i < P, \end{cases} \quad (411)$$

and consider

$$\min \left\{ \frac{1}{P} \sum_{i=0}^{P-1} \langle C_i, X_i \rangle : (407), \text{tr } X_0 = 1, X_i \succeq 0 \right\}. \quad (412)$$

Because $\sqrt{u^2 + \gamma^2} < 1/4$, all local costs have spectrum in $(11/4, 13/4)$. The point $X_i = I/3$ is strictly primal feasible, and zero equality multipliers with slacks C_i/P prove strict dual feasibility.

Theorem 11.1 (Intrinsic qutrit parity). *The program (412) has optimal values*

$$v_+ = \frac{29}{10}, \quad v_- = \frac{14}{5}. \quad (413)$$

Consequently, estimating the optimal value to additive error $1/25$ or target-relative error $1/100$ requires $\Omega(P)$ raw coefficient queries. More precisely, the query complexities for errors ϵ and η are

$$Q_{\text{add}}(\epsilon) = \begin{cases} \lceil n_{\text{bit}}/2 \rceil, & \epsilon < 1/20, \\ 0, & \epsilon \geq 1/20, \end{cases} \quad Q_{\text{rel}}(\eta) = \begin{cases} \lceil n_{\text{bit}}/2 \rceil, & \eta < 1/57, \\ 0, & \eta \geq 1/57. \end{cases} \quad (414)$$

The feasible spectrahedron has no lift over any finite product of second-order cones.

Proof. The H_{23} term is invariant under every R_i . The anchor γH_{12} on the first K blocks remains in the root frame, and the anchor γH_{13} on the last K blocks acquires the final sign h . Since $q\gamma = u$, pulling the objective through (408) gives

$$\overline{C}_h = 3I + uA_h, \quad A_h = H_{12} + H_{23} + hH_{13}. \quad (415)$$

For $h = +1$, A_h has eigenvalues $2, -1, -1$; for $h = -1$, its signed cycle product is negative and its eigenvalues are $1, 1, -2$. Hence

$$\text{spec}(\overline{C}_+) = \left\{ \frac{16}{5}, \frac{29}{10}, \frac{29}{10} \right\}, \quad \text{spec}(\overline{C}_-) = \left\{ \frac{31}{10}, \frac{31}{10}, \frac{14}{5} \right\}. \quad (416)$$

Rayleigh–Ritz on $Z \succeq 0$, $\text{tr } Z = 1$, proves (413). The common trace nine rules out an orthogonal congruence followed by a nontrivial positive rescaling. Adding a multiple of I does not help: $\overline{C}_+$ has a simple largest and a repeated smallest eigenvalue, and $\overline{C}_-$ the reverse. The gap is therefore intrinsic to the optimization problem.

The additive intervals are disjoint exactly when $\epsilon < 1/20$ and meet at $57/20$. The relative intervals meet exactly at

$$\eta_* = \frac{v_+ - v_-}{v_+ + v_-} = \frac{1}{57}, \quad v_+(1 - \eta_*) = v_-(1 + \eta_*) = \frac{812}{285}. \quad (417)$$

Below either threshold, comparison with a public separating point computes parity with bounded error. One coefficient query uses at most one sign query, and the approximate-degree bound recalled in section 2.4 gives the bounded-error lower bound $\lceil n_{\text{bit}}/2 \rceil$. Conversely, apply Deutsch’s one-query circuit to each pair and query an unpaired last sign; this computes h in exactly that many queries. At and above the thresholds the displayed intersection points are public zero-query answers.

Projection onto X_0 identifies the feasible set with $\mathcal{D}_3 = \{Z \succeq 0 : \text{tr } Z = 1\}$. If it had a finite SOC lift, replace each affine lift equation $Ay = b$ by $Ay = tb$ and impose $t \geq 0$. For $t > 0$ the projection is $t\mathcal{D}_3$. At $t = 0$, a homogeneous lift direction could be added at arbitrary scale to a fixed feasible lift; compactness of $\mathcal{D}_3$ forces its matrix projection to be zero. The homogenized projection is exactly $\mathbb{S}_+^3$, contradicting Fawzi’s theorem [61]. $\square$

With $r_+ = (1, 1, 1)^\top / \sqrt{3}$ and $r_- = (1, -1, 1)^\top / \sqrt{3}$, the strictly complementary reduced optimal pairs are

$$Z_+^* = \frac{1}{2}(I - r_+ r_+^\top), \quad Z_-^* = r_- r_-^\top, \quad \overline{S}_+^* = \frac{3}{10} r_+ r_+^\top, \quad \overline{S}_-^* = \frac{3}{10}(I - r_- r_-^\top). \quad (418)$$

Lifting by $X_i^* = R_i Z_i^* R_i^\top$ and $S_i^* = P^{-1} R_i \overline{S}_i^* R_i^\top$ gives a strictly complementary product-cone optimum. Stationarity follows because the remaining coefficient tuple is orthogonal to every trace-zero feasible tangent and hence lies in the equality adjoint’s range.

11.1.1 A public, condition-one Newton step

Use

$$\Phi_\mu(X) = \frac{1}{P} \sum_i \langle C_i, X_i \rangle - \mu \sum_i \log \det X_i, \quad X_i^0 = I/3, \quad \mu = 1/P. \quad (419)$$

At this point, in the coordinates (409),

$$g_i = (C_i - 3I)/P, \quad \nabla^2 \Phi_\mu(X^0) = (9/P)I. \quad (420)$$

Thus both right-hand sides of

$$\tilde{A}_\zeta \Delta x = 0, \quad (9/P) \Delta x - \tilde{A}_\zeta^\top \Delta y = -g \quad (421)$$

are public. For trace-zero Y , define the isometry $\mathcal{W}_\zeta(Y) = P^{-1/2}(R_0 Y R_0^\top, \dots, R_{P-1} Y R_{P-1}^\top)$. Then

$$\mathcal{W}_\zeta^* \nabla^2 \Phi_\mu(X^0) \mathcal{W}_\zeta = (9/P)I_5. \quad (422)$$

In the unnormalized root coordinate Z , the Hessian is $9I_5$ and

$$\Delta Z_h = -\frac{u}{9} A_h. \quad (423)$$

The Newton decrement and the standard full-step bound are

$$\lambda = \frac{1}{\sqrt{150}}, \quad \lambda_{\text{next}} \leq \frac{1}{(\sqrt{150} - 1)^2} < \frac{1}{126}. \quad (424)$$

Indeed $\lambda^2 = \langle u A_h, (9I)^{-1} u A_h \rangle = 1/150$. The post-step root eigenvalues are $(14/45, 31/90, 31/90)$ for $h = +1$ and $(29/90, 29/90, 16/45)$ for $h = -1$, so the undamped step remains positive in every block. The bound on λ_{next} is the standard self-concordance bound, not a statement about any primal-dual neighborhood defined in this paper.

Theorem 11.2 (Newton and complete-KKT state hardness). *At (419), preparing the normalized root or global primal Newton-direction state to trace distance $1/100$ requires $\Omega(P)$ raw queries. The same bound holds for the exact feasible lift of every reduced solution Y satisfying*

$$\frac{\|9Y + u A_h\|_F}{\|u A_h\|_F} \leq \frac{1}{10}. \quad (425)$$

It also holds for the normalized complete primal-and-multiplier state of (421), with the displayed unit row scaling and multiplier convention; for this state, a public parity observable has expectation of magnitude at least $1489/2097 > 7/10$.

Proof. Up to global sign,

$$|\delta z_h\rangle = \frac{|12\rangle + |23\rangle + h|13\rangle}{\sqrt{3}}. \quad (426)$$

The public contraction $T = |12\rangle\langle 13| + |13\rangle\langle 12|$ has expectation $2h/3$. At block i , both swapped coordinates acquire the same prefix sign, so the same expectation holds for $T_{\text{glob}} = \bigoplus_i T$. The POVM $(I \pm T)/2$ succeeds with probability $5/6$, and trace error $1/100$ leaves constant bias.

For the residual claim, (425) is exactly $\|Y - \Delta Z_h\|_F \leq \eta \|\Delta Z_h\|_F$ with $\eta \leq 1/10$. The projective sine angle from the exact ray is at most η . Direct minimization gives the sharp bound

$$h\langle T \rangle \geq m(\eta) := \frac{2}{3} - \frac{\eta^2}{3} - \frac{2\sqrt{2}}{3} \eta \sqrt{1 - \eta^2}, \quad 0 \leq \eta \leq \frac{1}{3}. \quad (427)$$

To verify it, at sine angle s write the only useful tangent coefficient as $x \in [-1, 1]$. The expectation is $\frac{2}{3}(1 - s^2) - s^2 + \frac{2\sqrt{2}}{3}s\sqrt{1 - s^2}x + \frac{4}{3}s^2x^2$. For $s \leq 1/3$ its unconstrained minimizer is below -1 ,

so $x = -1$, and the result decreases with s . At $\eta = 1/10$ it is $199/300 - \sqrt{198}/150 > 17/30$. Also $\lambda_{\min}(I/3 + Y) > 3/10$.

For the complete vector, put

$$\alpha = 16/33, \quad c = \sqrt{2}\gamma/P, \quad d = \sqrt{2}u/9, \quad v_e = \begin{cases} c(1 - \alpha)e, & e \leq K, \\ c\alpha(P - e), & K < e < P. \end{cases} \quad (428)$$

If τ_i is the prefix sign, the three primal chains are $-d\tau_i, -dh\tau_i, -d$ in coordinates 12, 13, 23. Switching signed path incidence to ordinary incidence shows that the only nonzero copy multipliers are $-\tau_e v_e$ on 12 and $h\tau_e v_{P-e}$ on 13. The public swap of coordinates 12 and 13 within the primal and within the multiplier part then has expectation $2h(D - C)/(3D + 2V)$, where

$$\begin{aligned} D &= \frac{11n_{\text{bit}}}{1350}, \\ V &= \frac{289n_{\text{bit}}}{4950} + \frac{17}{158400n_{\text{bit}}}, \\ C &= \frac{577n_{\text{bit}}}{9900} + \frac{1}{9900n_{\text{bit}}}. \end{aligned} \quad (429)$$

Its sign is $-h$. The leading ratio is $1489/2097$; the ratio of positive $1/n_{\text{bit}}$ corrections is $16/17 > 1489/2097$, so finite size only increases the magnitude. Reversing outcome labels decodes parity. Each such preparation would therefore compute parity and costs $\Omega(n_{\text{bit}}) = \Omega(P)$ queries. $\square$

The complete-state bound is coordinate-specific: rescaling equality rows changes multiplier amplitudes. Also $DA_-D = -A_+$ for $D = \text{diag}(1, -1, 1)$, so the two direction rays differ by an input-dependent orthogonal change of coordinates (gauge). The value gap is gauge-invariant; the state bound asks for output in fixed public coordinates, consistent with [section 6](#).

11.1.2 Ambient robustness and access objects

For arbitrary $X_i \succeq 0$, define

$$e_0 = \text{tr } X_0 - 1, \quad E_i = X_i - G_i X_{i-1} G_i^\top, \quad e_0^2 + \sum_{i=1}^{P-1} \|E_i\|_F^2 \leq \epsilon^2. \quad (430)$$

Gauging to $Z_i = R_i^\top X_i R_i$ and telescoping gives

$$|P^{-1} \sum_i \text{tr } X_i - 1| < \sqrt{P}\epsilon, \quad \left| P^{-1} \sum_i \langle C_i, X_i \rangle - \langle \bar{C}_h, Z_0 \rangle \right| < \frac{13}{4} \sqrt{P}\epsilon. \quad (431)$$

The second bound writes the difference as suffix costs against $Z_i - Z_{i-1}$ and uses $\|C_i\|_{\text{op}} < 13/4$. If the objective is at most $v_h + 1/100$ and $\epsilon \leq 1/(400\sqrt{P})$, define

$$\rho(X) := \frac{\bigoplus_i X_i}{\sum_i \text{tr } X_i}. \quad (432)$$

The trace bounds in (431) make this density operator well defined. The public contraction

$$M = \frac{1}{2/5} \left(\bigoplus_i C_i - \frac{57}{20} I \right) \quad (433)$$

satisfies $h \text{tr}(M\rho(X)) > 1/16$. Indeed, the normalized cost is $> 23/8$ for $h = +1$ and $< 113/40$ for $h = -1$. Thus any procedure that, for every input, returns a density operator $\tilde{\rho}$ satisfying

$D_{\text{tr}}(\tilde{\rho}, \rho(X)) \leq 1/100$ for some PSD tuple X meeting the displayed residual and objective conditions requires $\Omega(P)$ raw queries, even if it chooses the tuple adversarially. The residual order $P^{-1/2}$ is necessary: averaging the n_{bit} neighboring negative-parity optimizers against a fixed positive-parity input gives the wrong value and residual

$$\frac{4}{3\sqrt{n_{\text{bit}}}} = \Theta(P^{-1/2}). \quad (434)$$

For block encodings, use the following state-pair construction. If a real matrix B has public support and magnitudes and maximum absolute row and column sums at most a , give each supported pair (r, c) its own token. Row and column states with amplitudes $\sqrt{|B_{rc}|/a}$ and $\text{sgn}(B_{rc})\sqrt{|B_{rc}|/a}$, completed on disjoint failure tokens, are orthonormal and have overlap B_{rc}/a . Unitary completions therefore give an exact normalization- a block encoding.

Put $y' = -\Delta y$ in (421) and apply this to

$$M_{\zeta} = \begin{pmatrix} (9/P)I & \tilde{A}_{\zeta}^{\top} \\ \tilde{A}_{\zeta} & 0 \end{pmatrix}. \quad (435)$$

Its maximum absolute row and column sums equal three; the trace row attains three and a primal row has sum at most $2 + 9/P < 3$. One query phases the four symmetric hidden KKT tokens. This gives a fully specified exact normalization-three block encoding. Substituting it into the decoders gives

$$q_{\text{BE}} \geq \left\lceil \frac{n_{\text{bit}}}{2} \right\rceil \quad (436)$$

for preparing the root, global, or complete-KKT state. The bound is relative to this completion; an input-dependent junk part may contain the parity, and the queries used to build it must be counted.

Let Π_{ζ} project onto $\ker \tilde{A}_{\zeta}$. Projecting (421) and calculating norms gives

$$\|\Pi_{\zeta}g\|_2^2 = \frac{3}{50P}, \quad \|g\|_2^2 = \frac{41}{400P}, \quad \|\Pi_{\zeta}|\hat{g}\rangle\|_2^2 = \frac{24}{41}. \quad (437)$$

After one use of a normalization-one encoding of Π_{ζ} with operator error $1/32$, the signed expectation is at least $16/41 - \sqrt{24/41}/16 - 1/1024 > 0.341$. Hence its first construction costs at least $\lceil n_{\text{bit}}/2 \rceil$ raw queries. Conversely, applying all prefix signs to the 12, 13 chains constructs the tangent isometry; conjugating the public trace-zero root projector gives an exact encoding with $2n_{\text{bit}}$ queries and no QRAM. Constructing it therefore costs $\Theta(n_{\text{bit}})$ queries; a supplied projector encoding already combines all signs.

The condition number one in (422) is for the reduced Hessian, not the unscaled saddle matrix M_{ζ} . Switching removes all path signs without changing singular values. On each off-diagonal coordinate, the path-difference operator has singular values $2\sin(j\pi/(2P))$, $1 \leq j < P$, so $s_{\min}^+(\tilde{A}_{\zeta}) \leq \pi/P$.

The trace-coupled diagonal subsystem gives the lower bound. Let $x = (x^{(1)}, x^{(2)}, x^{(3)})$ be orthogonal to its kernel, and write $m_j = P^{-1}\mathbf{1}^{\top}x^{(j)}$. Orthogonality to every constant triple whose coefficients sum to zero implies $m_1 = m_2 = m_3 =: m$. Put $x^{(j)} = m\mathbf{1} + z^{(j)}$, with each $z^{(j)}$ mean zero, and define

$$E^2 = \sum_{j=1}^3 \|Bz^{(j)}\|_2^2, \quad t = \sum_{j=1}^3 x_0^{(j)}, \quad (438)$$

where B is ordinary path incidence. The path Poincaré inequality and telescoping give

$$\sum_j \|z^{(j)}\|_2^2 \leq \frac{P^2}{4} E^2, \quad |z_0^{(j)}| \leq \sqrt{P} \|Bz^{(j)}\|_2. \quad (439)$$

Since $3m = t - \sum_j z_0^{(j)}$, it follows that $\|x\|_2^2 \leq (9/4)P^2(t^2 + E^2)$. The last parenthesis is exactly the squared norm of the diagonal subsystem applied to x . Combining diagonal and off-diagonal coordinates, and using the maximum absolute row and column sums three and two, yields

$$\frac{2}{3P} \leq s_{\min}^+(\tilde{A}_\zeta) \leq \frac{\pi}{P}, \quad s_{\max}(\tilde{A}_\zeta) \leq \sqrt{6}. \quad (440)$$

The kernel gives the saddle eigenvalue $9/P$, while every positive equality singular value s gives eigenvalues $\{9/P \pm \sqrt{(9/P)^2 + 4s^2}\}/2$. The two-sided bounds in (440) therefore prove

$$\kappa_2(M_\zeta) = \Theta(P). \quad (441)$$

Nevertheless, the off-diagonal sparsity graph of M_ζ is a forest: each coordinate forms a subdivided path and the trace multiplier joins three distinct diagonal paths at one new vertex. Thus its treewidth is one.

11.2 Non-SOCP Schur-block families

Fix $k \geq 2$, let $K = 16n_{\text{bit}}$ and $L = 17n_{\text{bit}} + 1$, and set $a_i = \varsigma_i$ for $i \leq n_{\text{bit}}$ and $a_i = 1$ thereafter, with $\tau_i = \prod_{j=1}^i a_j$. For $X_i \in \mathbb{S}_+^{k+1}$, put $F_j = (e_j e_{k+1}^\top + e_{k+1} e_j^\top)/\sqrt{2}$ and $E = e_{k+1} e_{k+1}^\top$. Impose

$$\begin{aligned} \langle F_1, X_0 \rangle &= \sqrt{2}, & \langle F_1, X_i \rangle - a_i \langle F_1, X_{i-1} \rangle &= 0, \\ \langle F_j, X_0 \rangle &= 0, & \langle F_j, X_i \rangle - \langle F_j, X_{i-1} \rangle &= 0 \quad (2 \leq j \leq k), \\ \langle E, X_0 \rangle &= 1, & \langle E, X_i \rangle - \langle E, X_{i-1} \rangle &= 0 \end{aligned} \quad (442)$$

for $1 \leq i < L$. Every row and scalar column has at most two nonzeros. Every feasible block is

$$X_i = \begin{pmatrix} A_i & \tau_i e_1 \\ \tau_i e_1^\top & 1 \end{pmatrix}, \quad Z_i = A_i - e_1 e_1^\top \succeq 0. \quad (443)$$

Put $\mu_0 = 1/L$ and use $C_0 = \mu_0 \text{diag}(I_k, 2)$ on every block.

Theorem 11.3 (Exact Schur path and SOCP boundary). *The unique, strictly complementary optimum has $Z_i = 0$. With $t = \mu/\mu_0$, the complete central path is*

$$X_{\tau_i}^{(k)}(t) = \begin{pmatrix} tI_k + e_1 e_1^\top & \tau_i e_1 \\ \tau_i e_1^\top & 1 \end{pmatrix}, \quad S_i(\mu) = \begin{pmatrix} \mu_0 I_k & -\mu_0 \tau_i e_1 \\ -\mu_0 \tau_i e_1^\top & \mu + \mu_0 \end{pmatrix}. \quad (444)$$

In every public Frobenius-orthonormal tangent basis, $H_{\text{red}}(\mu) = (\mu_0/\mu)^2 I = t^{-2} I$. All primal, slack, and multiplier coordinates are bounded for $0 < \mu \leq \mu_0$. The feasible block slice has a finite SOCP lift exactly when $k \leq 2$.

Proof. Schur complementation gives (443), $\det X_i = \det Z_i$, and $\langle C_0, X_i \rangle = \mu_0(\text{tr } Z_i + 3)$. Thus $Z_i = 0$ is the unique optimum, and $\mu_0 \text{tr } Z_i - \mu \log \det Z_i$ is uniquely minimized at tI_k , proving the primal formula. Direct inversion gives the slack. With $S = C - \mathcal{A}^* y$, the nonzero chain multipliers are

$$\alpha_i = \sqrt{2}\mu_0(L - i)\tau_i, \quad \beta_i = (\mu_0 - \mu)(L - i). \quad (445)$$

Their divergences reproduce $C - S$. On $0 < t \leq 1$, primal entries are at most two, slack entries at most $2/L$, and $\|\alpha\|_\infty \leq \sqrt{2}$, $\|\beta\|_\infty \leq 1$. At $t = 0$ the rank-one primal and rank- k displayed slack are strictly complementary. For dual uniqueness, put $w_i = (\tau_i e_1, 1)^\top$, so $X_i^{\text{opt}} = w_i w_i^\top$. Zero gap for any dual optimum gives $S_i w_i = 0$. Stationarity in the unconstrained upper-left coordinates fixes $(S_i)_{1:k, 1:k} = \mu_0 I_k$; annihilating w_i then forces the last column and row to be $(-\mu_0 \tau_i e_1, \mu_0)^\top$, exactly the $t = 0$ limit in (444). Finally, full row rank of $\mathcal{A}$ makes $\mathcal{A}^*$ injective, so the equality multipliers are unique as well.

Every feasible tangent is $\Delta X_i = \begin{pmatrix} H_i & 0 \\ 0 & 0 \end{pmatrix}$. Using the inverse of (444), whose upper-left block is $t^{-1}I_k$ because $\tau_i^2 = 1$, the reduced log-det Hessian is

$$\sum_i \text{tr}(X_i^{-1} \Delta X_i X_i^{-1} \Delta Y_i) = \frac{\mu_0^2}{\mu^2} \sum_i \langle H_i, K_i \rangle,$$

so the reduced Hessian is a multiple of the identity. The two local centers do not commute: $[X_+^{(k)}(t), X_-^{(k)}(t)] = 2t(e_{k+1}e_1^\top - e_1e_{k+1}^\top)$. Finally, (443) is an affine bijection from $\mathbb{S}_+^k$. $\mathbb{S}_+^2$ is a Lorentz cone, while $\mathbb{S}_+^3$ has no finite SOC lift; larger k contain it as a face. This proves the SOCP characterization. $\square$

An input-dependent congruence removes the noncommutativity: in the Z_i coordinates the problem is public. The product-cone representation matters: one unrestricted $(k+1)L$ -dimensional PSD block with off-block entries forced to zero needs quadratically many scalar equalities.

11.2.1 Tight loading bounds and the accuracy–query tradeoff

For $k = 3$ and $0 < t \leq 1$, define

$$\rho_\varsigma(t) = \frac{\bigoplus_i X_{\tau_i}^{(3)}(t)}{L(3t+2)}. \quad (446)$$

On the suffix $T = \{n_{\text{bit}} + 1, \dots, n_{\text{bit}} + K\}$, every $\tau_i = h$. Let O equal $e_1e_4^\top + e_4e_1^\top$ on these blocks and zero elsewhere. Then

$$h \text{tr}(O\rho_\varsigma(t)) = \frac{2K}{L(3t+2)} \geq \frac{32n_{\text{bit}}}{5(17n_{\text{bit}}+1)} > \frac{1}{3}. \quad (447)$$

Thus preparing this density operator to trace distance $1/100$ costs $\Omega(L)$ queries.

For the complete central triple $g_\varsigma(t) = (\text{svec } X, y, \text{svec } S)$,

$$\|g_\varsigma(t)\|_2^2 = L(3t^2 + 2t + 4) + [2 + (1-t)^2] \frac{(L+1)(2L+1)}{6L} + \frac{5 + (1+t)^2}{L} < 11L. \quad (448)$$

In each suffix block, swapping the primal coordinates with values 1 and $\sqrt{2}\tau_i$ gives a public contraction W with $h\langle W \rangle = 2\sqrt{2}K/\|g_\varsigma(t)\|_2^2 > 1/5$. Hence preparing the normalized complete-amplitude state also costs $\Omega(L)$ queries for every $0 < t \leq 1$.

We now construct a canonical normalization-two block encoding. The equality matrix has public unit-magnitude entries and maximum absolute row and column sums two. Give each row and column two public slots. A degree-two row or column prepares the equal superposition of its two entry tokens; a degree-one row or column sends the unused slot to its own failure label, and a degree-zero free column goes entirely to its failure label. Distinct rows, columns, and failure arms use disjoint labels. Public two-slot Hadamards, reversible token maps, and a fixed lexicographic bijection on invalid addresses give fixed unitary completions L_A and $R_A^{(0)}$, with every hidden sign set to $+1$ in $R_A^{(0)}$ while retaining the public predecessor minus signs. A phase unitary S_ς recognizes the predecessor token in signed transition i and phases it by ς_i , while acting as the identity on all other entry, failure, and invalid labels. Extending the sign oracle by the public dummy value $\varsigma_0 = 1$ makes this one coherent raw query. Therefore

$$U_A = L_A^\dagger S_\varsigma R_A^{(0)}, \quad \langle 0, r | U_A | 0, c \rangle = \frac{(\mathcal{A}_\varsigma)_{rc}}{2}. \quad (449)$$

After padding the row and column address spaces to a common public dimension, this is an exact, fully specified normalization-two encoding, and every ordinary, adjoint, or controlled call uses one raw query. Thus $q_{\text{BE}} \geq n_{\text{bit}}/2$ for either output. As for (436), an input-dependent

completion, junk action, prefix table, or preprocessing stage is allowed only if all queries used to build it are counted.

For arbitrary $X_i \succeq 0$, let $d_i = \sqrt{2}(X_i)_{14}$ and suppose

$$\|\mathcal{A}_\zeta(X) - b\|_2 \leq \epsilon, \quad \sum_i \langle C_0, X_i \rangle \leq 6, \quad \epsilon \leq \frac{1}{4\sqrt{L}}. \quad (450)$$

Define the trace-normalized density output of such a point by

$$\rho(X) := \frac{\bigoplus_i X_i}{\sum_i \text{tr } X_i}. \quad (451)$$

It is well-defined because the root residual forces a nonzero off-diagonal entry and a PSD matrix with such an entry has positive trace. If $r_0 = d_0 - \sqrt{2}$ and $r_i = d_i - a_i d_{i-1}$, switching the recurrence gives $\tau_i d_i = \sqrt{2} + \sum_{j \leq i} \tau_j r_j$. Summing over T and applying Cauchy–Schwarz gives $h \sum_{i \in T} d_i > K$. Positive semidefiniteness and the objective bound give $\sum_i \text{tr } X_i \leq 6L$. Since $\text{tr}(O_i X_i) = \sqrt{2} d_i$ on the suffix,

$$h \text{tr}(O\rho(X)) = \frac{\sqrt{2}h \sum_{i \in T} d_i}{\sum_i \text{tr } X_i} > \frac{\sqrt{2}K}{6L} > \frac{1}{5}. \quad (452)$$

Therefore preparing any $\tilde{\rho}$ with $D_{\text{tr}}(\tilde{\rho}, \rho(X)) \leq 1/100$ for some blockwise PSD point meeting (450) costs $\Omega(L)$ raw queries. The point may depend adversarially on the input, and neither centrality nor matrix-norm proximity is required. Conversely, the residual order $L^{-1/2}$ is necessary: averaging the exact centers obtained by flipping each sign once gives the wrong suffix parity and

$$\|\mathcal{A}_\zeta(\bar{X}) - b\|_2 = 2\sqrt{2/n_{\text{bit}}} = \Theta(L^{-1/2}). \quad (453)$$

For precision parameters $\epsilon, \eta > 0$, let $Q_{\text{abs}}(P, \epsilon)$ be the worst-case raw-query cost, including setup, of first preparing an unconditional density operator $\tilde{\rho}$ within trace distance $1/100$ of $\rho(X)$ for some blockwise PSD X with at most P blocks, absolute equality residual at most ϵ , and the objective bound of (450). Define $Q_{\text{rel}}(P, \eta)$ identically with $\|\mathcal{A}_\zeta(X) - b\|_2 / \|b\|_2 \leq \eta$. For a heralded procedure, its success probability and all repetition or amplification costs are counted as in [definition 2.25](#); reusable preprocessing is counted as in [definition 2.27](#).

Theorem 11.4 (Accuracy–query tradeoff). *For at most P blocks and the output requirement above,*

$$Q_{\text{abs}}(P, \epsilon) = \Omega(\min\{P, \epsilon^{-2}\}), \quad Q_{\text{rel}}(P, \eta) = \Omega(\min\{P, \eta^{-2}\}) \quad (454)$$

for $0 < \epsilon, \eta \leq 1/8$. The relative residual is the global RHS-relative residual of [definition 2.26](#). Public padding extends the bounds to exactly P blocks, and they are tight for the family used in the proof.

Proof. For absolute error choose $n_{\text{bit}} = \lfloor \min\{(P-1)/17, 1/(2048\epsilon^2)\} \rfloor$. Apart from a constant-size range, this is $\Theta(\min\{P, \epsilon^{-2}\})$ and ensures the last inequality in (450). The decoder computes parity from every valid output. Since $\|b\|_2 = \sqrt{3}$, replacing 2048 by 6144 proves the relative claim.

For exact P , add $P-L$ public 4×4 dummy blocks fixed by ten one-coordinate rows to $P^{-1}I_4$, and set the common cost scale and parameter to $1/P$. The cap becomes $6L/P + 5(P-L)/P^2$, so the total trace is at most $6L + 5$ and the bias on the original blocks remains above $1/5$. Dummy blocks have no tangent coordinates, have bounded multipliers and a strictly complementary optimum, and do not change the active non-SOCP slice. Their RHS makes $\|b_{\text{pad}}\|_2^2 < 4$, so 8192 replaces 6144 for exact- P relative padding. Reading all active signs gives an exact valid output in $O(n_{\text{bit}})$ queries, while (453) matches the residual exponent. $\square$

The hidden coordinate in block i is the prefix product τ_i , whose query complexity is $\Theta(\min\{i, n_{\text{bit}}\})$. A classical scan streams all blocks with exactly n_{bit} queries and constant extra workspace. Quantumly, use

$$X_\tau^{(k)}(t) = t \sum_{j=1}^k e_j e_j^\top + w_\tau w_\tau^\top, \quad w_\tau = \tau e_1 + e_{k+1}, \quad (455)$$

to prepare a public local purification. A coherent prefix-phase loop queries and unqueries each sign, using at most $2n_{\text{bit}}$ queries, logarithmic workspace, and no QRAM. Public coordinate predicates extend it to the complete-triple state. Both first-preparation complexities are therefore $\Theta(L)$.

With $q_k = (k+1)(k+2)/2$, a path decomposition whose bags contain two adjacent block cliques and their $k+1$ transition multipliers proves

$$\text{tw}(K_{\text{KKT}}) \leq 2q_k + k. \quad (456)$$

Thus the central solve in free coordinates uses no input queries, while loading the center in the original coordinates uses $\Theta(L)$. This complements bounded-treewidth results on chordal conversion [146]. The bound is for first preparation: cached prefixes, with their queries counted, can make later centers cheap. It gives no bound for outputting the reduced Z or the value, which are public.

11.3 Abstract tree-holonomy loading

Theorem 11.5 (Tree-holonomy loading). *Let $\mathcal{T} = (V, E)$ be a public rooted tree with edge signs ς_e and transported labels $\tau_v = \prod_{e \in \text{path}(r, v)} \varsigma_e$. For public constant-dimensional $B_+, B_- \succeq 0$ of common trace T_0 , put*

$$\rho_\varsigma = \frac{1}{|V|T_0} \bigoplus_{v \in V} B_{\tau_v}, \quad d = \frac{1}{2} \|B_+/T_0 - B_-/T_0\|_1. \quad (457)$$

If public constant-size circuits purify the local states $B_\pm/T_0$, a purification of (457) needs at most $4|E|$ sign queries and logarithmic workspace, without QRAM. If an unknown path of length n_{bit} is followed by $K = \Theta(n_{\text{bit}})$ public descendants, $|V| = \Theta(n_{\text{bit}})$, and $d > 0$, preparation to trace distance below a constant depending only on d and $K/|V|$ needs at least $n_{\text{bit}}/2$ queries on the first preparation.

Proof. In a public depth-first order, an edge is on the root-to- v path exactly when v lies in the child subtree's preorder interval. Prepare a uniform entangled vertex address, query every edge, toggle a label on that interval when the sign is negative, and unquery. Select B_{τ_v}/T_0 , then reverse the loop to erase the label. The two compute-uncompute loops use four queries per edge and logarithmic workspace.

For the lower bound, project onto the K descendants. This has public probability $\alpha = K/|V|$. Conditioned on it, the local state is B_h/T_0 , whose Helstrom success is $(1+d)/2$. Guessing randomly outside the descendants gives success probability $1/2 + \alpha d/2$. Preparation error below $\alpha d/4$ leaves constant advantage, so parity costs $n_{\text{bit}}/2$ queries. A classical tree scan matches the linear order and may restrict itself to the union of requested root paths. $\square$

11.4 Matrix-valued group holonomy

Let $U : G \rightarrow O(d)$ be an orthogonal representation of a group G , let $g_1, \dots, g_{n_{\text{bit}}}$ be the hidden symbols, put $k = 3d$, and define $W = U_{g_{n_{\text{bit}}}} \cdots U_{g_1}$. Use the construction of [section 11.1](#) with

$P = 33n_{\text{bit}}$ and hidden transports $\mathcal{D}_g = \text{diag}(U_g, I_d, I_d)$. Replace $3I$ in (411) by kI_k and replace scalar H_{ab} by $H_{ab}(I_d)$. Elimination gives

$$\overline{C}_W = kI_k + uA_W, \quad A_W = \begin{pmatrix} 0 & I_d & W^\top \\ I_d & 0 & I_d \\ W & I_d & 0 \end{pmatrix}. \quad (458)$$

Conjugating W by Q conjugates A_W by $\text{diag}(Q, Q, Q)$.

Theorem 11.6 (Holonomy spectrum).

$$\text{spec}(A_W) = \bigcup_{e^{i\theta} \in \text{spec}(W)} \left\{ 2 \cos \frac{\theta + 2\pi\ell}{3} : \ell = 0, 1, 2 \right\}. \quad (459)$$

Thus the optimal value $v(W) = k + u\lambda_{\min}(A_W)$ is a class function of W . For the natural permutation representation of S_3 , the identity, transposition, and three-cycle values are

$$\frac{89}{10}, \quad \frac{44}{5}, \quad 9 + \frac{1}{5} \cos \frac{8\pi}{9}. \quad (460)$$

Proof. On a W -eigenvector with eigenvalue $e^{i\theta}$, the complexified operator is $\begin{pmatrix} 0 & 1 & e^{-i\theta} \\ 1 & 0 & 1 \\ e^{i\theta} & 1 & 0 \end{pmatrix}$, whose characteristic polynomial is $\lambda^3 - 3\lambda - 2\cos\theta$. Substituting $\lambda = 2\cos\phi$ proves the spectrum. The natural S_3 eigenvalue lists are $(1, 1, 1)$, $(1, 1, -1)$, and $(1, e^{2\pi i/3}, e^{-2\pi i/3})$, with smallest A_W eigenvalues $-1, -2, 2\cos(8\pi/9)$. Identity and three-cycles agree in the abelianization but have different values, so the unrestricted value depends on nonabelian information. $\square$

For signed-permutation representations, each svec copy row still has two unit-magnitude nonzeros, though the hidden symbol may move the predecessor entry. The natural S_3 example uses 9×9 blocks and public costs with spectra in $(9 - 1/4, 9 + 1/4)$, and is strictly primal-dual feasible.

Theorem 11.7 (S_3 value and Newton hardness). *Restrict every symbol to $\{e, t\}$, where t is a fixed transposition. Estimating the optimal value to additive error $1/25$ or target-relative error $1/200$ takes $\Omega(P)$ raw queries. At $X_i = I_9/9$, $\mu = 1/P$, the ambient and reduced primal Hessians have condition number one, the KKT right-hand side is public, and*

$$\Delta Z_W = -\frac{u}{81}A_W, \quad \lambda = \frac{\sqrt{2}}{30}, \quad \lambda_{\min}(I_9/9 + \Delta Z_W) \geq \frac{44}{405}. \quad (461)$$

Preparing the normalized root or global primal direction, or the normalized complete primal-plus-multiplier KKT state in the stated row scaling, to trace distance $1/100$ requires $\Omega(P)$ raw queries. The public observable for the complete state has expectation of magnitude greater than $33/100$.

Proof. The restricted word is identity or t according to parity. The first two values in (460) differ by $1/10$ and are separated at $177/20$; their relative intervals are disjoint below $1/177$. A sparse position query is simulated by a constant number of bit queries, so parity hardness implies value hardness.

At the public start the gradient is $(C_i - 9I)/P$ and the ambient Hessian is $(81/P)I$. Orthogonal transport makes the normalized tangent map isometric, so the Hessian in the unnormalized root coordinate is $81I$. Since $\text{tr } A_W = 0$, the direction is as displayed. Also $\|A_W\|_F^2 = 18$ and $\|A_W\|_{\text{op}} \leq 2$, so $\lambda^2 = 2u^2/9 = 1/450$ and $1/9 - 2u/81 = 44/405$. The next decrement is $2/(30 - \sqrt{2})^2 < 1/400$.

For identity and a transposition, the normalized A_W overlap is $7/9$. The root-direction states have trace distance $4\sqrt{2}/9$, so their fixed Helstrom measurement succeeds with probability

$\frac{1}{2}(1 + 4\sqrt{2}/9) > 0.81$. Let J swap corresponding Frobenius-isometric svec coordinates in the $(1, 2)$ and $(1, 3)$ blocks of the partition in (458), and act as zero on every other coordinate. It has expectation $2/3$ for identity and $2/9$ for a transposition; centering as $T_G = (9J - 4I)/13$ gives $2h/13$ on every block X_i and hence on the global direction.

For the complete state, a public basis makes $U_t = \text{diag}(-1, 1, 1)$. Two modes are public and one is the signed qutrit mode. With $d_0 = \sqrt{2}u/81$, its primal contribution is $D = Pd_0^2 = 11n_{\text{bit}}/109350$ and its multiplier tent has the V, C in (429). Swapping 12, 13 only in the signed mode gives

$$|\langle T_{\text{sign}} \rangle| = \frac{2239504n_{\text{bit}}^2 + 3888}{6759216n_{\text{bit}}^2 + 12393} > \frac{33}{100}, \quad (462)$$

by cross multiplication. These fixed decoders retain positive bias at trace error $1/100$, and parity proves all state lower bounds. $\square$

For the canonical KKT block encoding, each block has 45 scalar coordinates. A copy-row state uses amplitude $1/3$ on two tokens and a trace-row state uses amplitude $1/3$ on nine. The column preparation routes predecessor tokens through the svec involution induced by t , which one symbol query implements by toggling its public orbit bit. This exactly encodes the equality dilation with normalization nine. Adding $(81/P)$ times the public primal-sector projector P_x by a two-term LCU gives

$$\alpha_M = 9 + \frac{81}{P} < 12, \quad (463)$$

and $q_{\text{BE}} \geq \lceil n_{\text{bit}}/2 \rceil$ for the three outputs above. The completion is fixed; the caveat after (436) applies.

The tangent projector again has signal probability $24/41$. Combining it with the global decoder gives $48/533$. At operator error $1/64$, the signed expectation remains $48/533 - 2\sqrt{24/41}/64 - 1/4096 > 0.065$. The first construction of a projector encoding therefore costs $\Omega(n_{\text{bit}})$ queries. Accumulating each prefix svec involution and conjugating the public trace-zero projector gives an exact $2n_{\text{bit}}$ query construction, so the bound is tight. The scalar KKT graph is a forest: a layer permutation makes disjoint paths, and the trace multiplier joins distinct diagonal paths without a cycle.

If setup uses S queries and later iteration t uses q_t , then

$$S + \sum_t q_t \geq n_{\text{bit}}/2, \quad (464)$$

whether a position query XORs its answer into an arbitrary target register or writes it into a blank one. Indeed, the answer is one of two public indices selected by one hidden sign, so after a public linear change of the target register one call to O_ζ (or its controlled version) simulates the query, and bounded-error parity needs $n_{\text{bit}}/2$ calls. A linear-query setup may store the word or its prefixes and make later iterations cheap.

The unrestricted objective detects nonabelian information, but every proved $\Omega(P)$ query theorem above restricts inputs to the abelian subgroup $\{e, t\} \cong \mathbb{Z}_2$. These are parity lower bounds, not bounds for a promise on ordered words whose letters do not commute.

By construction, the optimal value is invariant under a simultaneous orthogonal gauge. The direction decoders instead use the fixed natural permutation basis and the root frame. A common public gauge only conjugates both public measurements; an input-dependent output gauge would have to be learned or supplied and lies outside the stated output requirement.

Open Problem 11.8 (Intrinsically nonabelian word hardness). Prove a lower bound on local-symbol queries for distinguishing conjugacy classes under a promise on words whose letters do not commute, and carry it over to the SDP through (458), without restricting the alphabet to one cyclic subgroup.

11.5 Relation to prior work and summary

Signed switching and cycle products are classical graph invariants [80]. Group synchronization frames the same construction as gauge transport and holonomy [64], and orthogonal synchronization is routinely relaxed to SDPs [96]. General group oracle classification and symmetric-oracle query problems concern different promises [45, 145]. Fawzi’s theorem gives the exact SOC-lift boundary, not a coefficient-query lower bound.

Our first examples, which use sparse 2×2 blocks, reduce to paired LPs in exact scalar form; their shared-trace variant has a nonpolyhedral feasible set and hence no LP lift. The first noncommuting Schur example, with 3×3 blocks, is still SOCP-representable, so we state the Schur results for 4×4 blocks. The family of [section 11.1](#) uses two asymmetric anchors: its two effective costs, and hence its central matrices, have different spectra and are not related by any orthogonal similarity.

A separate free- $\mathbb{S}_+^3$ construction, whose central Hessian depends on the parity, gives a lower bound for building a Hessian block encoding. For

$$\overline{C}_h = I + a(H_{12} + hH_{13}), \quad a = 8/33, \quad \mathcal{L}_h(Y) = \overline{C}_h Y \overline{C}_h, \quad (465)$$

the reduced Hessian has condition number below five, but building its block encoding with public constant normalization to unscaled error $1/8$ costs at least $n_{\text{bit}}/2$ raw queries. Indeed $\text{svec}(\mathcal{L}_h(E_{33})) = (a^2, 0, \sqrt{2}ha, 0, 0, 1)^\top$; swapping its 13, 33 coordinates has signed expectation $2\sqrt{2}a/(1+a^2)^2 > 0.61$, and error $1/8$ leaves constant bias. This concerns the first construction: a supplied Hessian oracle reveals parity in one call. In this free- $\mathbb{S}_+^3$ family—unlike the two-anchor family—the two Hessians are related by an input-dependent orthogonal gauge.

Linear lower bounds for sparse optimal values already follow from LP constructions, including Theorem 8.4 of Apers–Gribling [10], and general SDP-value lower bounds predate these families [130]. The sparse QLS lower bound of Mori et al. also propagates parity [109]. Our contribution is to prove the value, state, and block-encoding lower bounds together in SDP families built directly as optimization problems; parity, its propagation, and the linear exponent are not new.

The qutrit component used in [section 7](#) is exactly (406) with costs (411) and $N = n_{\text{bit}}$. Its spectra are (416) and its value gap is $1/10$. Nothing here implies that the public-start Newton state of that section reveals the winner; trace allocation is governed by [section 6](#). As in the LP families of [section 9](#), reduced conditioning, KKT sparsity, data access, and requested output coordinates are separate resources. On these parity promises the only quantum improvement is Deutsch’s factor of two; classical reading takes n_{bit} signs, so there is no asymptotic quantum speedup.

12 Preconditioning and Interface Obstructions

“Precondition it” and “keep the solution as a quantum state” are incomplete complexity arguments. Preconditioning can lower the condition number while raising factor normalization, the cost of preparing the transformed right-hand side, or recovery sensitivity. A normalized Newton state may not suffice to build the coordinatewise diagonal that the next Newton system needs. This section quantifies these tradeoffs in the access models of [section 2.2](#). Each conclusion bounds a product (such as normalization times condition number) or the total cost of all phases, never one spectral quantity in isolation.

The analytic condition-versus-normalization statements, including the minimax bound for fixed preconditioners, are model free. Their query-counting corollaries, the parity reductions, and the lazy-oracle results use the raw sparse coefficient model of [definition 2.11](#). The state-interface results count calls to the state-preparation access of [definition 2.14](#), not raw coefficient queries. The lazy-oracle results explain why the caching ceiling of [section 9.3](#) rules out an unconditional per-iteration raw-query lower bound on a fixed instance: once the hidden input is cached, later iterations may need no raw queries.

12.1 Factor normalization and whitening

We first study the signed block $d_i - g_i a_i d_{i-1} = 0$ of the LP in [theorem 9.4](#). Let $N \geq 2$, $M = 17N$, and

$$T = \lceil \frac{1}{2} \log_2 N \rceil, \quad H = 2^T, \quad H_i = 2^{\min\{i, T\}}, \quad g_i = H_i / H_{i-1},$$

and take $a_i = \varsigma_i$ for $i \leq N$ and $a_i = 1$ afterward. Here ς_i is the hidden sign denoted σ_i in [theorem 9.4](#), and H is the plateau height; it is unrelated to the normal matrix H_μ used later. On $d = (d_0, \dots, d_M)^\top$, define the pinned lower-bidiagonal factor

$$(B_\varsigma d)_0 = d_0, \quad (B_\varsigma d)_i = d_i - g_i a_i d_{i-1}. \quad (466)$$

Put $\tau_0 = 1$, $\tau_i = \prod_{j=1}^i a_j$, and $D_\tau = \text{diag}(\tau_0, \dots, \tau_M)$. Let B_+ be (466) with every sign positive and set $S_+ = B_+ B_+^\top$. Then

$$B_\varsigma = D_\tau B_+ D_\tau, \quad S_\varsigma := B_\varsigma B_\varsigma^\top = D_\tau S_+ D_\tau. \quad (467)$$

Thus the singular values do not depend on the signs, but applying the gauge D_τ requires the prefix products τ_i . This switching identity and the corresponding inverse formulas are standard signed-graph facts [3].

Lemma 12.1 (Plateau singular scale). *If $L = M - T + 1$ is the number of plateau nodes, then*

$$\sigma_{\min}(B_\varsigma) \leq 2 \sin \frac{\pi}{2(L+1)} \leq \frac{\pi}{L+1} < \frac{\pi}{16N}, \quad \|B_\varsigma^{-1}\|_2 = \Theta(N). \quad (468)$$

Proof. Take a Dirichlet sine vector supported on nodes $T, \dots, M$, with virtual zeros at $T-1$ and $M+1$. On this support B_+ is the first-difference operator. Adding the omitted final difference can only increase its energy, and the full Dirichlet path quotient is $4 \sin^2(\pi/(2(L+1)))$. Equation (467) gives the signed case, while $\sin x \leq x$ and $L+1 > 16N$ give the two remaining inequalities.

For $i \geq j$, direct forward substitution gives

$$(B_+^{-1})_{ij} = H_i / H_j. \quad (469)$$

The column $B_+^{-1} e_0 = (H_i)_i$ has norm $\Theta(N)$, whereas summing the squares in (469) gives $\|B_+^{-1}\|_F = O(N)$. Orthogonal equivalence in (467) proves the final claim. $\square$

Theorem 12.2 (Sharp plateau whitening tradeoff). *Let P_ς be any square factor such that, for $K \geq 1$,*

$$\frac{1}{K} I \preceq P_\varsigma S_\varsigma P_\varsigma^* \preceq I. \quad (470)$$

Then

$$\|P_\varsigma\|_2 \sqrt{K} \geq \frac{1}{\sigma_{\min}(B_\varsigma)} > \frac{16N}{\pi}. \quad (471)$$

Consequently every exact $(\alpha_P, a_{\text{anc}}, 0)$ block encoding of P_ς obeys

$$\boxed{\alpha_P \sqrt{K} > \frac{16N}{\pi}}. \quad (472)$$

For $1 \leq a \leq \Theta(N)$, the regularized family

$$P_\varsigma^{(a)} = D_\tau (S_+ + a^{-2} I)^{-1/2} D_\tau \quad (473)$$

has $\|P_\varsigma^{(a)}\|_2 = \Theta(a)$ and, after a constant input-independent normalization of the largest eigenvalue,

$$K = \Theta((N/a)^2), \quad \|P_\varsigma^{(a)}\|_2 \sqrt{K} = \Theta(N). \quad (474)$$

Thus (471) is attained in order throughout this range, not only at its endpoints.

Proof. Equation (470) implies $\sigma_{\min}(P_\zeta B_\zeta) \geq K^{-1/2}$. On a right singular vector of B_ζ for its least singular value,

$$\sigma_{\min}(P_\zeta B_\zeta) \leq \|P_\zeta\|_2 \sigma_{\min}(B_\zeta).$$

This and lemma 12.1 prove (471); the normalization of an exact block encoding is at least the encoded norm.

For tightness, the eigenvalues of the congruence in (473) are

$$\frac{\lambda}{\lambda + a^{-2}}, \quad \lambda \in \sigma(S_+).$$

The bidiagonal factor has bounded norm, a diagonal entry of one, and inverse norm $\Theta(N)$ by lemma 12.1. Hence $\lambda_{\max}(S_+) = \Theta(1)$ and $\lambda_{\min}(S_+) = \Theta(N^{-2})$. It follows that the factor norm is $\Theta(a)$ and that the ratio of the displayed largest and smallest eigenvalues is $\Theta((N/a)^2)$ in the stated range. This proves (474). $\square$

The theorem specializes a known result, not a new general obstruction to quantum preconditioning: Theorem 3 and Corollary 4 of Nie, Lai, and An already prove this limitation for arbitrary preconditioners supplied as a separate block encoding [114]. New here are the constant $16/\pi$, the tight interpolation (473)–(474), and the raw-query theorem below, which counts preprocessing. Directly encoding a classically formed or algebraically regrouped product is a different access model, as both that work and proposition 5.8 emphasize.

The same argument covers approximate whitening and inversion. If the factor is supplied through an exact $(\alpha_P, a_{\text{anc}}, 0)$ block encoding, then, for fixed $\delta < 1$,

$$\|P_\zeta S_\zeta P_\zeta^* - I\|_2 \leq \delta \implies \alpha_P \geq \frac{\sqrt{1-\delta}}{\sigma_{\min}(B_\zeta)}, \quad (475)$$

$$\|P_\zeta B_\zeta - I\|_2 \leq \delta \implies \alpha_P \geq \frac{1-\delta}{\sigma_{\min}(B_\zeta)}. \quad (476)$$

Indeed, the first inequality gives $\sigma_{\min}(P_\zeta B_\zeta) \geq \sqrt{1-\delta}$, while the second gives $\sigma_{\min}(P_\zeta B_\zeta) \geq 1-\delta$; the same singular-vector argument then applies. Hence constant-error whitening or inversion by an exactly block-encoded factor has $\alpha_P = \Omega(N)$. More generally, if $\beta = \|P_\zeta S_\zeta P_\zeta^*\|_2$ and the congruence has condition number at most K , the scale-invariant bound is

$$\|P_\zeta\|_2 \sqrt{K/\beta} \geq \sigma_{\min}(B_\zeta)^{-1}. \quad (477)$$

The division by β is essential, since scalar rescaling of P_ζ would defeat an unnormalized bound. For a left factor normalized by $\sigma(P_\zeta B_\zeta) \subseteq [1/K, 1]$ and supplied through an exact $(\alpha_P, a_{\text{anc}}, 0)$ block encoding, the same argument gives

$$\alpha_P K \geq \|P_\zeta\|_2 K > \frac{16N}{\pi}. \quad (478)$$

The factor norm ignores the raw sign queries needed to build the factor. Define

$$F_\zeta = \text{diag}(B_\zeta^{-1}, B_+^{-1}), \quad G_\zeta = \text{diag}(B_\zeta, B_+), \quad |r\rangle = \frac{|0, e_0\rangle + |1, e_0\rangle}{\sqrt{2}}. \quad (479)$$

By (469),

$$F_\zeta |r\rangle = \frac{1}{\sqrt{2}} ((H_i \tau_i)_i \oplus (H_i)_i), \quad \|F_\zeta r\|_2 = \left(\sum_i H_i^2 \right)^{1/2} = \Theta(N). \quad (480)$$

On the $16N$ nodes $i > N$, where τ_i is the full sign parity, pair corresponding signed and unsigned coordinates and measure their side qubit in the X basis. The parity-signed expectation is

$$\frac{16NH^2}{\sum_i H_i^2} > \frac{16N}{17N + 4/3} > \frac{9}{10}. \quad (481)$$

Theorem 12.3 (Reusable inverse access: query cost versus normalization). *Suppose an offline stage makes s raw sign queries and produces a reusable exact $(\alpha, a, 0)$ block encoding of $\tilde{F}_\zeta$, and that one call to it, its inverse, or its controlled version makes at most q further raw queries. If, for fixed $\delta \leq 1/20$,*

$$\|\tilde{F}_\zeta G_\zeta - I\|_2 \leq \delta, \quad (482)$$

then

$$\boxed{s + O(q\alpha/N) = \Omega(N)}, \quad \boxed{Ns + q\alpha = \Omega(N^2)}. \quad (483)$$

The constants hidden in O and Ω are absolute.

Proof. Since $G_\zeta F_\zeta = I$,

$$\|(\tilde{F}_\zeta - F_\zeta)r\|_2 \leq \delta \|F_\zeta r\|_2.$$

Thus the normalized approximate-inverse state retains a constant parity bias under the decoder in (481), and $\|\tilde{F}_\zeta r\|_2 \geq (1 - \delta)\Theta(N)$. Fixed-point amplitude amplification prepares it with constant success using $O(\alpha/N)$ block-encoding calls. Including setup, the resulting parity algorithm uses $s + O(q\alpha/N)$ raw queries. Quantum parity needs $\Omega(N)$ queries [20], proving the first inequality; multiplication by N gives the second. Equation (476) also gives $\alpha = \Omega(N)$, so the amplification count is never subconstant. $\square$

Three remarks limit these statements. First, a directly supplied preconditioned product can be the identity: $P_\zeta = B_\zeta^{-1}$ gives $P_\zeta B_\zeta = I$, and exact Schur whitening gives $P_\zeta S_\zeta P_\zeta^* = I$. Hence the normalization and condition number of the product alone imply no lower bound. Second, applying D_τ is not free: any implementation with operator-norm error below $1/3$ needs $\Omega(N)$ raw queries (apply it to $(|0\rangle + |M\rangle)/\sqrt{2}$ and measure in the $(|0\rangle \pm |M\rangle)/\sqrt{2}$ basis to recover the endpoint parity). Third, a factor-independent bound on the total cost needs a specific hard right-hand side and required output. Consider the plateau solve $G_\zeta x = r$ from (479), and require an output state within trace distance $1/20$ of $F_\zeta|r\rangle/\|F_\zeta r\|_2$. For any preconditioned implementation of this task, compose its output with the fixed X -basis decoder of (481). The reduction gives

$$q_{\text{set}} + q_{\text{rhs}} + q_{\text{rec}} + mq_{\text{BE}} = \Omega(N), \quad (484)$$

where the solve phase makes raw queries only inside its m solver calls, each costing q_{BE} , and q_{set} , q_{rhs} , and q_{rec} count the raw queries for setup, transformed right-hand-side preparation, and recovery. Indeed, the composition is a bounded-error parity algorithm. Without this right-hand side and output we claim no such bound: a sign-independent solution need not reveal the signs. An orthogonal rotation of the factor can move the parity cost between phases. The statement concerns the pinned plateau block, not every preconditioner of the full KKT system.

12.2 Congruence condition versus physical-state recovery

A second tradeoff arises even when the preconditioner only scales the active and inactive subspaces by two scalars. Let the fixed Euclidean space split orthogonally as $\mathbb{R}^r = U \oplus K$, with projectors Q and R , and suppose

$$H_\mu = \mu^{-1}C_\mu + \mu D_\mu, \quad 0 < \mu \leq \mu_0. \quad (485)$$

Assume $C_\mu = QC_\mu Q$ and $C_\mu|_U \rightarrow C \succ 0$; $D_\mu \succeq 0$ is uniformly bounded; $D_{KK,\mu} \rightarrow D_{KK} \succ 0$; and all other blocks converge. For a fixed nonzero $f \in U$, put

$$p = C^{-1}f, \quad g = D_{KK}^{-1}D_{KU}p, \quad (486)$$

and assume the coupling is nontrivial, $g \neq 0$. These are asymptotics as $\mu \downarrow 0$ for one fixed strictly complementary instance, not uniform claims in the dimension.

For $0 \leq \gamma \leq 1$, define the power family

$$T_{\mu,\gamma} = \mu^{\gamma/2}Q + \mu^{-\gamma/2}R, \quad G_{\mu,\gamma} = T_{\mu,\gamma}H_\mu T_{\mu,\gamma}, \quad \eta_\mu = H_\mu^{-1}f, \quad z_{\mu,\gamma} = T_{\mu,\gamma}^{-1}\eta_\mu. \quad (487)$$

Because $f \in U$, $T_{\mu,\gamma}f = \mu^{\gamma/2}f$, so the normalized transformed right-hand side is exactly the original right-hand-side state. The tradeoff below is therefore between condition number and recovery, not right-hand-side loading. The normalized contraction that recovers the physical (unscaled) coordinates is

$$R_{\mu,\gamma} = T_{\mu,\gamma}/\|T_{\mu,\gamma}\|_2 = \mu^\gamma Q + R, \quad \Phi_{\mu,\gamma}(w) = \frac{R_{\mu,\gamma}w}{\|R_{\mu,\gamma}w\|_2}. \quad (488)$$

For a unit vector w , define the local sensitivity of the recovered state by

$$\chi_{\mu,\gamma}(w) = \|D\Phi_{\mu,\gamma}(w)|_{w^\perp}\|_{2 \rightarrow 2}. \quad (489)$$

Theorem 12.4 (Power congruence tradeoff). *Under the preceding assumptions, for each fixed $\gamma \in [0, 1]$ as $\mu \downarrow 0$,*

$$\kappa_2(G_{\mu,\gamma}) = \Theta(\mu^{-2(1-\gamma)}), \quad (490)$$

$$\eta_\mu = \mu \begin{pmatrix} p \\ -g \end{pmatrix} + o(\mu), \quad (491)$$

$$z_{\mu,\gamma} = \begin{pmatrix} \mu^{1-\gamma/2}(p + o(1)) \\ -\mu^{1+\gamma/2}(g + o(1)) \end{pmatrix}. \quad (492)$$

Write $\hat{z} = z/\|z\|_2$. Normalized recovery is exact, and direct postselection succeeds with probability P_{rec} :

$$\Phi_{\mu,\gamma}(\hat{z}) = \frac{\eta_\mu}{\|\eta_\mu\|_2}, \quad P_{\text{rec}} = \|R_{\mu,\gamma}\hat{z}\|_2^2 = \Theta(\mu^{2\gamma}), \quad (493)$$

while

$$\chi_{\mu,\gamma}(\hat{z}) = \Theta(\mu^{-\gamma}). \quad (494)$$

Consequently

$$\boxed{\sqrt{\kappa_2(G_{\mu,\gamma})} \chi_{\mu,\gamma}(\hat{z}) = \Theta(1/\mu)}, \quad \sqrt{\kappa_2(G_{\mu,\gamma})} P_{\text{rec}}^{-1/2} = \Theta(1/\mu). \quad (495)$$

Proof. Set $a_\mu = \mu^{-(1-\gamma)}$ and $b_\mu = \mu^{1-\gamma} = a_\mu^{-1}$. In $U \oplus K$ blocks,

$$G_{\mu,\gamma} = \begin{pmatrix} a_\mu C_\mu + \mu^{1+\gamma} D_{UU,\mu} & \mu D_{UK,\mu} \\ \mu D_{KU,\mu} & b_\mu D_{KK,\mu} \end{pmatrix}. \quad (496)$$

The upper-left block A_μ is $\Theta(a_\mu)I_U$, and its Schur complement is

$$b_\mu D_{KK,\mu} - \mu^2 D_{KU,\mu} A_\mu^{-1} D_{UK,\mu} = \Theta(b_\mu)I_K.$$

The elimination factor has off-diagonal norm $O(\mu b_\mu) = o(1)$. A uniformly conditioned near-identity congruence therefore reduces (496) to blocks of sizes $\Theta(a_\mu)$ and $\Theta(b_\mu)$, proving (490).

The K block of $H_\mu \eta_\mu = f$ gives exactly

$$\eta_{K,\mu} = -D_{KK,\mu}^{-1} D_{KU,\mu} \eta_{U,\mu}.$$

Substitution in the U block gives

$$[\mu^{-1}C_\mu + \mu E_\mu] \eta_{U,\mu} = f, \quad E_\mu = D_{UU,\mu} - D_{UK,\mu} D_{KK,\mu}^{-1} D_{KU,\mu},$$

where E_μ is bounded. Hence $\eta_{U,\mu} = \mu(C_\mu + \mu^2 E_\mu)^{-1} f = \mu(p + o(1))$, which proves (491); applying T^{-1} proves (492).

Because $Rz = \mu^{\gamma/2} \eta$, normalized recovery is exact. The two expansions give

$$\|R\hat{z}\|_2 = \mu^\gamma \frac{(\|p\|_2^2 + \|g\|_2^2 + o(1))^{1/2}}{(\|p\|_2^2 + \mu^{2\gamma} \|g\|_2^2 + o(1))^{1/2}} = \Theta(\mu^\gamma),$$

proving (493). Differentiation gives, with $\hat{\eta} = \Phi(w)$,

$$\chi_{\mu,\gamma}(w) = \frac{\|(I - \hat{\eta}\hat{\eta}^\top)R_{\mu,\gamma}(I - ww^\top)\|_2}{\|R_{\mu,\gamma}w\|_2}. \quad (497)$$

This is $O(\mu^{-\gamma})$ at $w = \hat{z}$. For $\gamma > 0$, project $g/\|g\|$ onto $\hat{z}^\perp$. The resulting unit tangent is $g/\|g\| + o(1)$ in K , while the recovered state tends to $(p, -g)/(\|p\|^2 + \|g\|^2)^{1/2}$. Since $p \neq 0$, the numerator in (497) is bounded below by $\|p\|/(\|p\|^2 + \|g\|^2)^{1/2} > 0$. This proves the matching lower bound. For $\gamma = 0$, $R = I$ and $\chi = 1$. Multiplying the estimates proves (495). $\square$

One LP with two nonzeros per row satisfies the hypotheses of theorem 12.4 for every γ . Consider

$$\min_{x \geq 0} x_2 + x_3, \quad x_1 + x_2 = 1, \quad x_2 - x_3 = 0, \quad (498)$$

so $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$, $b = e_1$, and $c = (0, 1, 1)^\top$. Its unique optimum is $(1, 0, 0)$. Along the central path $x_2 = x_3 = t_\mu$, $x_1 = 1 - t_\mu$, with $t_\mu = \mu + O(\mu^2)$, and

$$H_\mu = A(XS^{-1})A^\top = \frac{(1 - t_\mu)^2}{\mu} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{t_\mu^2}{\mu} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}. \quad (499)$$

Indeed, dual feasibility and complementarity give equal inactive slacks $q_\mu = 1 + \mu/[2(1 - t_\mu)]$ and $t_\mu q_\mu = \mu$, which yields the expansion. For $U = \text{span}\{e_1\}$ and $K = \text{span}\{e_2\}$, the limiting blocks in (485) are $C = e_1 e_1^\top$ and $D = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. Thus $p = 1$, $g = 1/2$, and $H_\mu^{-1}b = \mu(1, -1/2)^\top + o(\mu)$. This fixed LP therefore satisfies all conclusions of theorem 12.4. At $\gamma = 1$, the condition number of its scaled matrix has a finite positive limit and its recovery amplitude $P_{\text{rec}}^{1/2}$ is asymptotic to $\sqrt{5/4}\mu$.

Among all scalar block-diagonal SPD congruences (positive t_U, t_K), exactly those whose ratio t_U/t_K lies, up to constants, between μ and 1 attain the optimal order $\Theta(1/\mu)$ of $\sqrt{\kappa_2}\chi$; these include the power family. Let

$$T_\mu = t_U(\mu)Q + t_K(\mu)R, \quad r_\mu = t_U/t_K, \quad M_\mu = \max\{r_\mu/\mu, \mu/r_\mu\}, \quad K_\mu = \max\{r_\mu, r_\mu^{-1}\}. \quad (500)$$

Put

$$\tilde{G}_\mu = T_\mu H_\mu T_\mu, \quad \tilde{z}_\mu = T_\mu^{-1} \eta_\mu, \quad \tilde{R}_\mu = T_\mu / \|T_\mu\|_2, \quad \tilde{\Phi}_\mu(w) = \frac{\tilde{R}_\mu w}{\|\tilde{R}_\mu w\|_2}. \quad (501)$$

Define $\tilde{P}_{\text{rec}}$ by applying the squared norm in (493) to $\tilde{R}_\mu$ and $\tilde{z}_\mu$, and define $\tilde{\chi}_\mu$ by (489) with $\tilde{\Phi}_\mu$.

Theorem 12.5 (General scalar-block tradeoff). *Uniformly over all positive scalar functions t_U, t_K ,*

$$\sqrt{\kappa_2(\tilde{G}_\mu)} = \Theta(M_\mu), \quad \tilde{P}_{\text{rec}}^{-1/2} = \Theta(K_\mu), \quad \tilde{\chi}_\mu = \Theta(K_\mu). \quad (502)$$

Consequently

$$\sqrt{\kappa_2(\tilde{G}_\mu)} \tilde{\chi}_\mu = \Theta(M_\mu K_\mu) \geq c/\mu. \quad (503)$$

The same holds with $\tilde{P}_{\text{rec}}^{-1/2}$ in place of $\tilde{\chi}_\mu$. Equality in order, $M_\mu K_\mu = \Theta(1/\mu)$, holds if and only if

$$r_\mu = \Omega(\mu), \quad r_\mu = O(1). \quad (504)$$

More explicitly,

$$M_\mu K_\mu = \begin{cases} \mu/r_\mu^2, & r_\mu < \mu, \\ 1/\mu, & \mu \leq r_\mu \leq 1, \\ r_\mu^2/\mu, & r_\mu > 1. \end{cases} \quad (505)$$

Proof. The assumptions imply uniform spectral equivalence

$$c_0(\mu^{-1}Q + \mu R) \preceq H_\mu \preceq C_0(\mu^{-1}Q + \mu R). \quad (506)$$

For the lower bound, write a vector as $u + k$. Fixed $c, d, M > 0$ give

$$(u + k)^\top H_\mu (u + k) \geq (c/\mu - 2M^2\mu/d)\|u\|^2 + (d\mu/2)\|k\|^2,$$

after Young's inequality; for small μ this proves the lower half of (506), and boundedness proves the upper half. Congruence by T_μ shows that the two spectral scales are t_U^2/μ and μt_K^2 , proving the condition estimate in (502).

Equation (491) gives $\|\eta_{U,\mu}\| = \Theta(\mu)$ and $\|\eta_{K,\mu}\| = \Theta(\mu)$. Therefore

$$\|\tilde{R}_\mu \hat{z}_\mu\| = \frac{\|\eta_\mu\|}{\max\{t_U, t_K\} \sqrt{\|\eta_{U,\mu}\|^2/t_U^2 + \|\eta_{K,\mu}\|^2/t_K^2}} = \Theta(K_\mu^{-1}).$$

This proves the recovery-probability claim. In the plane spanned by the normalized U and K components, differentiating the output angle of $\text{diag}(t_U, t_K)$ gives tangent gain

$$\frac{r_\mu^{-1}\|\eta_{U,\mu}\|^2 + r_\mu\|\eta_{K,\mu}\|^2}{\|\eta_{U,\mu}\|^2 + \|\eta_{K,\mu}\|^2} = \Theta(K_\mu).$$

This is a lower bound on $\tilde{\chi}_\mu$; differentiation as in (497) gives the matching upper bound. The elementary three-case calculation (505) proves the product and the equality class. $\square$

This is not a lower bound for all preconditioners. It does not cover arbitrary left preconditioners, input-dependent transformations mixing U and K , direct preparation of the physical state, variable-time recovery, or algorithms that request only an observable. Nor does it count constructing Q or the block-encoding normalization. If $g = 0$, the recovery conclusion can fail. The state-preserving left equilibration of theorem 5.7 is not a congruence followed by recovery and avoids this tradeoff. The theorem is thus a sharp bound for one specified family, realized by (498).

12.3 The parity preconditioner dichotomy

The parity LP of section 9.1 shows a different limit. Its reduced primal Hessian has condition number one, but its equality normal matrix is a signed grounded-tree Laplacian. No fixed preconditioner serves all sign patterns, whereas a perfect (exactly whitening) data-dependent sparse factor is locally trivial, one sign per row. The raw-query cost of the global sign information then appears in applying the factor, preparing the transformed right-hand side, or recovery.

The minimax bound for fixed preconditioners already holds on a path. Let

$$B = \begin{pmatrix} 1 & & & \\ -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix}, \quad C = BB^\top, \quad R = \text{diag}(1, -1, 1, -1, \dots) \quad (507)$$

where B is $m \times m$ and $m \geq 1$. Both C and RCR arise from valid sign inputs.

Theorem 12.6 (Minimax fixed-preconditioner bound). *For SPD matrices write $\kappa = \lambda_{\max}/\lambda_{\min}$. For every input-independent SPD matrix M ,*

$$\max \left\{ \kappa(M^{-1/2}CM^{-1/2}), \kappa(M^{-1/2}RCRM^{-1/2}) \right\} \geq \frac{4m^2 - 1}{3} \geq m^2. \quad (508)$$

The choice $M = I$ gives condition number at most $4m^2$ for either signing, so it is minimax optimal in order. Dense M is allowed. More generally, the bound holds for every fixed invertible factor P , with the two matrices $P^\top CP$ and $P^\top RCP$. The bound $(4m^2 - 1)/3$ comes from test vectors and is not claimed to be the exact minimax value.

Proof. For SPD A_0, A_1, M , generalized Rayleigh quotients give

$$\kappa(A_0^{-1/2}A_1A_0^{-1/2}) \leq \kappa(M^{-1/2}A_0M^{-1/2})\kappa(M^{-1/2}A_1M^{-1/2}). \quad (509)$$

Indeed, bound the numerator and denominator of $x^\top A_1 x / (x^\top A_0 x)$ by the extremal eigenvalues in M coordinates; the independent scales cancel in the quotient of extrema.

For $r = (m, m-1, \dots, 1)^\top$,

$$x^\top Cx = \sum_{j=0}^{m-2} (x_j - x_{j+1})^2 + x_{m-1}^2, \quad r^\top Cr = m.$$

Summing the alternating witness exactly gives

$$r^\top RCRr = \sum_{k=1}^m (2k-1)^2 = \frac{m(4m^2 - 1)}{3}.$$

Put $t = (4m^2 - 1)/3$. The generalized quotient is t at r and $1/t$ at Rr , so $\kappa(C^{-1/2}RCRC^{-1/2}) \geq t^2$. Apply (509) with $A_0 = C$ and $A_1 = RCR$; one of its two factors must be at least t . The same argument in coordinates $P^{-1}r$ and $P^{-1}Rr$ proves the assertion for any invertible P , without a polar-decomposition assumption.

For the upper bound, write $d_i = x_i - x_{i+1}$ for $i < m-1$ and $d_{m-1} = x_{m-1}$. Then $x_i = \sum_{j=i}^{m-1} d_j$ and $x^\top Cx = \sum_j d_j^2$. Cauchy–Schwarz gives $\|x\|_2^2 \leq m^2 x^\top Cx$, while $(a-b)^2 \leq 2a^2 + 2b^2$ gives $x^\top Cx \leq 4\|x\|_2^2$. Thus $\kappa(C) \leq 4m^2$. Since R is orthogonal, the same bound holds for RCR ; more generally every signed path is an orthogonal diagonal conjugate of C . $\square$

Theorem 12.6 is formalized in Lean, including the incidence matrices, their positive definiteness, the extremal-eigenvalue condition numbers, inequality (509), the exact test-vector ratio, the upper bound for $M = I$ under every edge signing, and the statement for an arbitrary factor P , with the SPD case $P = M^{-1/2}$. The data-dependent factor and the quantum query results that follow are not formalized. The repository file `formal/PRECONDITIONER.md` maps these claims to the Lean proofs and lists the verification commands.

For data-dependent preconditioning, let D_ζ be the square grounded signed incidence matrix of the chain-plus-copy tree of [section 9.1](#): its root row is $e_0^\top$ and the row of a nonroot node j is $e_j^\top - \tau_j e_{\pi(j)}^\top$, where τ_j is its edge sign. With pair variables the equality matrix is

$$A_\zeta = D_\zeta[I, -I], \quad C_\zeta = D_\zeta D_\zeta^\top. \quad (510)$$

Every row of D_ζ has at most two nonzeros, each determined by one local sign, yet

$$D_\zeta^{-1} C_\zeta D_\zeta^{-\top} = I, \quad D_\zeta^{-1} e_0 = (p_j)_j, \quad (511)$$

where p_j is the root-to- j sign product. Set $R_\zeta = \text{diag}(p_j)$, and let D_+ be the unsigned grounded incidence matrix on the same rooted tree. The switching identity is $D_\zeta = R_\zeta D_+ R_\zeta$. Thus a

sparse perfect preconditioner is easy to construct, but applying its inverse exposes all prefix products. This is the classical requirement that a useful preconditioner be cheap to apply, in support-theory language [27].

The right-hand side e_0 in (511) occurs in an actual Newton system. At the infeasible starting point $x = s = \mathbf{1}$, $y = 0$, $\mu = 1$ for the pair LP with $c = \mathbf{1}$ and $b = e_0$, the point is centered and dual feasible, and $A_\zeta \mathbf{1} = 0$. With the residual convention of (13), any centering target $0 < \sigma_c \leq 1$ gives

$$A_\zeta \Delta x = e_0, \quad A_\zeta^\top \Delta y + \Delta s = 0, \quad \Delta x + \Delta s = (\sigma_c - 1)\mathbf{1}, \quad A_\zeta A_\zeta^\top \Delta y = e_0. \quad (512)$$

Since $A_\zeta A_\zeta^\top = 2D_\zeta D_\zeta^\top$, the signed incidence congruence transforms this exact Newton right-hand side into $\hat{D}_\zeta^{-1}e_0$, up to a public scalar.

For the state lower bound, we amplify the parity signal. Take $N \geq 2$ and attach two isomorphic complete binary copy trees, one to the root and one to the endpoint of the hidden spine with N edges. Each has M leaves and $J = 2M - 2$ new nodes, where

$$M = 2^{\lceil \log_2(16(N+1)) \rceil}. \quad (513)$$

Here M and $m = N + 1$ below are local to this amplified-tree construction and are unrelated to the plateau length M and path dimension m used earlier. All new edges have positive sign; write $\hat{D}_\zeta$ for this grounded incidence factor and define

$$\hat{A}_\zeta = \hat{D}_\zeta[I, -I], \quad \hat{C}_\zeta = \hat{D}_\zeta \hat{D}_\zeta^\top. \quad (514)$$

At the same infeasible-start point $x = s = \mathbf{1}$, $y = 0$, $\mu = 1$, now for the amplified pair LP with $c = \mathbf{1}$ and $b = e_0$, the Newton equations are

$$\hat{A}_\zeta \Delta x = e_0, \quad \hat{A}_\zeta^\top \Delta y + \Delta s = 0, \quad \Delta x + \Delta s = (\sigma_c - 1)\mathbf{1}, \quad \hat{A}_\zeta \hat{A}_\zeta^\top \Delta y = e_0. \quad (515)$$

Again the signed incidence congruence maps this Newton right-hand side to $\hat{D}_\zeta^{-1}e_0$, up to a public scalar.

Away from the start, the normal matrix still has an exact factor with at most two nonzeros per row. At any positive interior point let $\Theta = XS^{-1} = \text{diag}(\Theta_+, \Theta_-)$, where $\Theta_\pm = \text{diag}(\theta_j^\pm)_j$, and set

$$\Omega = \text{diag}(\omega_j)_j, \quad \omega_j = \theta_j^+ + \theta_j^- > 0. \quad (516)$$

Then

$$\hat{A}_\zeta \Theta \hat{A}_\zeta^\top = \hat{D}_\zeta \Omega \hat{D}_\zeta^\top = L_\zeta L_\zeta^\top, \quad L_\zeta := \hat{D}_\zeta \Omega^{1/2}. \quad (517)$$

Below, g is the right-hand side of the corresponding eliminated normal equation. In particular, $4/5 \leq x_i, s_i \leq 6/5$ implies $4/3 \leq \omega_j \leq 3$, hence $\Gamma := \max_j \omega_j / \min_j \omega_j \leq 9/4 < 4$.

Theorem 12.7 (Transformed Newton right-hand-side hardness for one factor). *Every unconditional preparation (definition 2.25) with only raw coefficient access that, on each sign input, outputs a state within trace distance $1/20$ of*

$$|z_\zeta\rangle = \frac{\hat{D}_\zeta^{-1}e_0}{\|\hat{D}_\zeta^{-1}e_0\|_2} \quad (518)$$

makes $\Omega(N)$ raw coefficient queries.

The conclusion remains true for the weighted nearby factor L_ζ in (517) and eliminated RHS g whenever

$$\frac{\max_j \omega_j}{\min_j \omega_j} \leq 4, \quad \|g - e_0\|_1 \leq \frac{1}{100}; \quad (519)$$

the target is the normalized state proportional to $\Omega^{-1/2} \hat{D}_\zeta^{-1}g$. If g or Ω is input-dependent, the count includes all raw queries used to construct and access it. Thus the theorem applies at every nearby central iterate satisfying (519). It does not cover free side oracles whose numerical values themselves encode the parity.

Proof. In the exact case, the prefix vector is +1 on every node of the root copy tree and endpoint parity p_N on every node of the endpoint copy tree. Pair corresponding nodes and measure their side qubit in the X basis. If $m = N + 1$, then $J \geq 31m$ and the probability of a paired node is

$$\frac{2J}{m + 2J} > \frac{62}{63}. \quad (520)$$

Conditional on that event the measurement returns p_N exactly; outputting a fair bit otherwise succeeds with probability greater than $1/2 + 31/63 = 125/126$. Trace error $1/20$ leaves success above $2/3$, so the fixed decoder computes parity and requires $\Omega(N)$ queries.

For (519), first take $g = e_0$ and put $\Gamma = \max \omega / \min \omega \leq 4$. Paired nodes have squared mass at least

$$\alpha \geq \frac{2J}{2J + \Gamma m} \geq \frac{31}{33}.$$

The amplitudes of a corresponding pair are a and $p_N b$ with $a/b \in [1/2, 2]$. The X measurement is correct with probability $1/2 + ab/(a^2 + b^2) \geq 9/10$. The full ideal success is therefore at least $1/2 + (2/5)(31/33) = 289/330$.

For $\delta = g - e_0$, every coordinate of $\widehat{D}_\zeta^{-1}\delta$ is a signed ancestor sum, so, if P is the number of tree nodes,

$$\|\widehat{D}_\zeta^{-1}\delta\|_2 \leq \sqrt{P}\|\delta\|_1, \quad \frac{\|\Omega^{-1/2}\widehat{D}_\zeta^{-1}\delta\|_2}{\|\Omega^{-1/2}\widehat{D}_\zeta^{-1}e_0\|_2} \leq \sqrt{\Gamma}\|\delta\|_1 \leq \frac{1}{50}.$$

The corresponding normalized pure states are at Euclidean distance at most $1/25$. Including trace error $1/20$ changes outcome probabilities by at most $9/100$; subtracting this from $289/330$ leaves success above $2/3$. Parity again proves the claim. $\square$

The proof needs the ℓ_1 hypothesis. Define $\tilde{d}$ by $\tilde{d}_j = p_j(1 - j/N)$ on the signed spine, one on the root copy tree, and zero on the endpoint copy tree. Then

$$\|\widehat{D}_\zeta \tilde{d} - e_0\|_2 = N^{-1/2}, \quad (521)$$

although the amplified endpoint signal has vanished and $\tilde{d}$ stays at constant relative distance from the prefix vector. Thus even a vanishing ℓ_2 feasibility residual does not imply the state closeness used above. This shows only that this proof fails, not that every ℓ_2 -robust reduction fails.

The hardness of the transformed right-hand side also depends on the factor. On the amplified tree, let $\widehat{R}_\zeta = \text{diag}(p_j)$ be the prefix gauge and let $\widehat{D}_+$ be the unsigned grounded incidence matrix. Then $\widehat{D}_\zeta = \widehat{R}_\zeta \widehat{D}_+ \widehat{R}_\zeta$, and the equally perfect factor

$$\widehat{P}_\zeta = \widehat{D}_+^{-1} \widehat{R}_\zeta = \widehat{R}_\zeta \widehat{D}_\zeta^{-1} \quad (522)$$

satisfies

$$\widehat{P}_\zeta \widehat{C}_\zeta \widehat{P}_\zeta^\top = I, \quad \widehat{P}_\zeta e_0 = \widehat{D}_+^{-1} e_0 = \mathbf{1}. \quad (523)$$

The transformed right-hand side is now input independent. Recovery uses $\widehat{P}_\zeta^\top = \widehat{R}_\zeta \widehat{D}_+^{-\top}$, so the parity cost moves to applying the factor or recovering the original coordinates. Because an orthogonal change of factor can move the cost this way, [theorem 12.7](#) is stated for one factor.

A statement valid for every factor must count the whole Newton step. Specialize (515) to $\sigma_c = 1$ and take the full step. With $p = \widehat{D}_\zeta^{-1}e_0$, the unique minimum-norm direction and its update are

$$\Delta x = \frac{1}{2}(p, -p), \quad \Delta s = -\Delta x, \quad x^+ = \mathbf{1} + \Delta x. \quad (524)$$

Every pair of x^+ is $(3/2, 1/2)$ or its swap, and $X^+ s^+ = (3/4)\mathbf{1}$; this is the exact feasible center at $\mu = 3/4$. The endpoint copy tree occupies more than $15/32$ of all pairs. Conditional on measuring such a pair, the computational-basis coordinate identifies p_N with probability $9/10$. Returning a fair bit otherwise gives ideal success greater than $1/2 + (2/5)(15/32) = 11/16$, and trace error $1/100$ leaves success above $2/3$.

Theorem 12.8 (Preconditioner-oracle accounting). *Let P_ς be any possibly data-dependent invertible factor for the normal equation in (515), with no sparsity or locality assumption, and let an end-to-end Newton procedure output a state within trace distance $1/100$ of the normalized x^+ in (524). Partition all raw coefficient queries into setup, transformed right-hand-side preparation, iterative solve, and recovery of the original coordinates with the update. Then*

$$\boxed{Q_{\text{set}} + Q_{\text{rhs}} + Q_{\text{solve}} + Q_{\text{rec}} = \Omega(N)}. \quad (525)$$

If the preconditioned system has condition number at most K and the solve phase's only raw queries occur inside at most $T(K, \epsilon)$ calls to its preconditioned oracles, each costing at most q_{on} raw queries, then

$$Q_{\text{set}} + Q_{\text{rhs}} + T(K, \epsilon)q_{\text{on}} + Q_{\text{rec}} = \Omega(N). \quad (526)$$

In particular, for $T(K, \epsilon) = \tilde{O}(K \log(1/\epsilon))$ and hidden polylogarithmic factor $L(N, \epsilon)$,

$$Q + O(Kq_{\text{on}}L(N, \epsilon)) = \Omega(N), \quad Q := Q_{\text{set}} + Q_{\text{rhs}} + Q_{\text{rec}}. \quad (527)$$

Proof. All four phases together form one coefficient-query algorithm. Compose its output with the fixed decoder following (524); this computes parity of the N hidden spine signs with bounded error. Each LP coefficient query is simulated by $O(1)$ sign queries, proving (525). Under the oracle-only solve hypothesis, $Q_{\text{solve}} \leq T(K, \epsilon)q_{\text{on}}$, which gives (526), and substitution of the stated solver call bound gives (527). $\square$

A hybrid solver that also queries raw data directly is still covered by (525); such queries are counted in Q_{solve} .

The bound (525) is tight and is not a quantum–classical separation. A classical algorithm reads the N signs, computes every prefix, and stores the gauge in $\Theta(N)$ queries and work. Ladner–Fischer prefix scan reduces parallel depth to $O(\log N)$ with $O(N)$ bounded-fan-in work [89]. After caching, no further hidden-sign queries are needed, but the unsigned triangular operations are not free: applying $\hat{D}_+^{-1}$ or $\hat{D}_+^{-\top}$ still takes $\Theta(N)$ work and $O(\log N)$ parallel depth by prefix or subtree aggregation. Multiplication by the cached diagonal gauge is local. Quantum parity changes only the constant. The three results of this subsection are lemmas and counterexamples: [theorem 12.6](#) is a finite-dimensional minimax lower bound, [theorem 12.7](#) locates the cost for one canonical factor, and [theorem 12.8](#) shows that another factor can move this cost between phases but not remove it on this instance. Together they are not a general lower bound for preconditioning. Recent preconditioned QIPM analyses that assume prepared transformed data are consistent with this; implementing those oracles from raw LP access is a separate, counted step [139].

12.4 Newton-state output is not an iterate oracle

We now change oracle models. Let $v \in \mathbb{R}_{>0}^n$ have known norm $r_v = \|v\|_2$, and grant controlled state-preparation access

$$U_v|0\rangle = |\hat{v}\rangle = \frac{1}{r_v} \sum_{j=1}^n v_j |j\rangle, \quad (528)$$

together with $U_v^\dagger$ and a fixed real phase, as in [definition 2.14](#). This is stronger than receiving copies of the state but differs from coordinate-value, raw-coefficient, QRAM, or compiled diagonal access; these models are not generally ordered by strength. The next results count calls to U_v and $U_v^\dagger$, not raw LP queries.

Theorem 12.9 (State-to-diagonal normalization–query product). *Suppose a controlled circuit W_v , using q calls to $U_v, U_v^\dagger$, is an $(\alpha, a, 1/8)$ block encoding of $\text{diag}(v)$. If the reported normalization satisfies a known input-uniform bound $\alpha \leq A$, where $A \geq 1$, then in the worst case*

$$\boxed{qA = \Omega(\sqrt{n})}. \quad (529)$$

The product is tight in order on the hard near-uniform family of the proof.

Proof. For $n \geq 10$, take the positive equal-norm pair

$$v^{(0)} = (1, \dots, 1), \quad v^{(1)} = (2, \beta, \dots, \beta), \quad \beta = \sqrt{\frac{n-4}{n-1}}. \quad (530)$$

Both norms are $\sqrt{n}$, and

$$\frac{1}{\sqrt{n}} \leq \|v^{(0)}/\sqrt{n} - v^{(1)}/\sqrt{n}\|_2 \leq \sqrt{2/n}. \quad (531)$$

Choose unitary extensions U_0, U_1 related by the minimum planar rotation between the prepared states. Then $\|U_0 - U_1\|_2 = \Theta(n^{-1/2})$, so the hybrid argument requires $\Omega(\sqrt{n})$ oracle calls for constant-bias distinction.

On the other hand, $O(A)$ controlled calls to W_v estimate $\text{Re}\langle 0^a, 1 | W_v | 0^a, 1 \rangle$ to additive error $1/(16A)$. Multiplication by the reported α contributes error at most $1/16$, and block-encoding error contributes at most $1/8$. The first diagonal entry is therefore known within $1/4$, distinguishing one from two. The composed distinguisher uses $O(qA)$ original oracle calls, proving (529).

Rattew and Rebentrost give an exact constant-query block encoding of $\text{diag}(v/r_v)$ from U_v [121]. As an encoding of $\text{diag}(v)$ it has normalization $r_v = \sqrt{n}$ on (530), so it attains the product in order. This worst-case tightness is not a lower bound for every vector and structured preparation circuit. $\square$

With copy-only access, distinguishing the pair in (530) requires $\Omega(n)$ copies: the one-copy overlap is $1 - \Theta(1/n)$, so only $k = \Omega(n)$ copies have constant trace distance. Thus the inverse access in definition 2.14 changes the exponent but does not remove the state-to-diagonal cost.

Near a strictly complementary endpoint the obstruction grows as μ decreases. Let $n = 2m$, $m \geq 5$, $0 < \mu \leq 1$, and define

$$\beta = \sqrt{\frac{m-4}{m-1}}, \quad v = (2, \beta \mathbf{1}_{m-1}), \quad r = (m-1)(1-\beta) = \frac{3}{1+\beta}, \quad z = 2(r, \mathbf{1}_{m-1}). \quad (532)$$

For the one-sparse LP

$$A = [I_m \ 0], \quad b = \mathbf{1}_m, \quad c = (z, \mathbf{1}_m), \quad (533)$$

take the common $x = (\mathbf{1}_m, \mu \mathbf{1}_m)$ and

$$\begin{aligned} s^{(0)} &= (\mu \mathbf{1}_m, \mathbf{1}_m), & y^{(0)} &= z - \mu \mathbf{1}_m, \\ s^{(1)} &= (\mu v, \mathbf{1}_m), & y^{(1)} &= z - \mu v. \end{aligned} \quad (534)$$

Both triples are feasible. Since $\|v\| = \|\mathbf{1}_m\|$ and $z^\top(v - \mathbf{1}_m) = 0$, the corresponding slack norms and dual norms agree.

Theorem 12.10 (State-access lower bound near a strictly complementary optimum). *In controlled state-preparation access to the slack and dual states in (534), distinguishing*

$$\|Xs - \bar{\mu} \mathbf{1}\|_2 \leq \bar{\mu}/4 \quad \text{from} \quad \|Xs - \bar{\mu} \mathbf{1}\|_2 \geq 3\bar{\mu}/4 \quad (535)$$

requires

$$\boxed{\Omega(\sqrt{n}/\mu)} \quad (536)$$

total controlled preparation and inverse calls. With copy-only access, the bound is $\Omega(n/\mu^2)$. The result holds although both triples approach the same strictly complementary optimum. The fixed LP data, the common x , all vector norms, positivity, and the real phase conventions are free.

Proof. The zero triple is exactly centered with $\bar{\mu}_0 = \mu$. For the other, $\bar{\mu}_1 < \mu$ while the first complementarity product is 2μ , so

$$\|Xs^{(1)} - \bar{\mu}_1 \mathbf{1}\|_2 > \mu > \bar{\mu}_1. \quad (537)$$

The normalized slack-state distance is $\Theta(\mu/\sqrt{n})$. The duals differ by the same unnormalized vector $\mu(v - \mathbf{1}_m)$ and have common norm $\Omega(\sqrt{m})$, so their normalized-state distance is also $O(\mu/\sqrt{n})$. Choose close minimum-rotation extensions for both oracles. A hybrid over queries to either oracle gives (536). For independent copies, fidelity multiplicativity gives the square of this dependence. Finally both triples converge to $((\mathbf{1}_m, 0), z, (0, \mathbf{1}_m))$, proving strict complementarity. $\square$

A similar pair arises when the objective of a sparse LP is available only as a state. Put $d = v - \mathbf{1}_m$, $z_0 = 2(r, \mathbf{1}_{m-1})$, and $z_\mu = z_0 - (\mu/2)d$. Then

$$z_0^\top d = 0, \quad \mathbf{1}^\top d = 1 - r, \quad 1 \leq \|d\|_2^2 \leq 2. \quad (538)$$

Theorem 12.11 (State-access lower bound for an objective given as a state). *Use the common one-sparse $A = [I_m \ 0]$, $b = \mathbf{1}_m$, and objectives*

$$c^{(0)} = (z_\mu, \mathbf{1}_m), \quad c^{(1)} = (z_\mu + \mu d, \mathbf{1}_m). \quad (539)$$

Let the only input-dependent access be a controlled garbage-free preparation of the normalized objective state and its inverse; the common norm, A , and b are free. Distinguishing the inputs, deciding which one is centered at the common pair $x = (\mathbf{1}_m, \mu \mathbf{1}_m)$, $y = z_\mu - \mu \mathbf{1}_m$, or estimating the optimum to additive error $\mu/8$, needs $\Omega(\sqrt{n}/\mu)$ controlled oracle calls. Copy-only access needs $\Omega(n/\mu^2)$ copies.

Moreover, multiply both objectives symmetrically by n , take $b = \mathbf{1}_m/6$, and use

$$\tilde{c}^{(0)} = n(z_0 - \frac{\mu}{2}d, \mathbf{1}_m), \quad \tilde{c}^{(1)} = n(z_0 + \frac{\mu}{2}d, \mathbf{1}_m).$$

The normalized objective oracles do not change, the value gap obeys

$$|\text{OPT}_1 - \text{OPT}_0| = \frac{n\mu}{6}(r - 1) \geq \frac{n\mu}{12}, \quad (540)$$

and additive value accuracy $n\mu/48$ has the same query lower bound. This is a fixed fraction of the centered total duality gap $n\mu$; here $n = \nu$ is the canonical LP barrier parameter of [section 2.1](#).

Proof. The objective norms agree because $2\mu z_\mu^\top d + \mu^2 \|d\|^2 = 0$, and their normalized-state distance is $\Theta(\mu/\sqrt{n})$. Minimum-rotation objective-state oracles therefore require $\Omega(\sqrt{n}/\mu)$ calls to distinguish; fidelity multiplicativity gives $\Omega(n/\mu^2)$ copies. The displayed common pair is centered for input zero and violates (535) for input one. Every feasible point has first block $\mathbf{1}_m$ and its positive-cost second block vanishes at optimum, so the original value gap is $\mu|\mathbf{1}^\top d| \geq \mu/2$. Additive error $\mu/8$ distinguishes the inputs.

For the scaled claim, the common point

$$\tilde{x} = (\mathbf{1}_m/6, (\mu/n)\mathbf{1}_m), \quad \tilde{y} = n(z_0 - \frac{\mu}{2}d) - 6\mu\mathbf{1}_m$$

has every complementarity product equal to μ for input zero; input one remains positive because $n(1 - \beta) \leq 5 < 6$ and lies outside every fixed N_2 neighborhood of radius below one. Its centered total duality gap is $n\mu$. Feasibility fixes the first primal block to $\mathbf{1}_m/6$, which gives (540); accuracy $n\mu/48$ distinguishes the inputs. The inverse- μ factor is therefore not caused by demanding accuracy finer than the centered duality-gap scale. $\square$

Theorem 12.11 depends on the access model. A binary coefficient-value oracle reveals the known exceptional coordinate, whose two values differ by $\Theta(\mu)$, in one query at sufficient precision: oracles writing different bit strings into a value register are not close unitaries merely because the encoded numbers are close. Likewise, the naturally normalized Newton diagonals of the two inputs differ by a constant in their first entry, so a free Newton block encoding distinguishes them in $O(1)$ calls. The $\Omega(\sqrt{n}/\mu)$ bound therefore holds only when the objective and updates are available as states alone, not for exact coefficient-value or precompiled Newton access.

State access also obstructs the fraction-to-boundary rule.

Corollary 12.12 (Fraction-to-boundary). *For $n \geq 4$, let $x = \mathbf{1}_n$ and compare the equal-norm directions*

$$\delta^{(0)} = -\frac{1}{2}\mathbf{1}_n, \quad \delta^{(1)} = (-1, -\gamma, \dots, -\gamma), \quad \gamma = \frac{1}{2}\sqrt{\frac{n-4}{n-1}}.$$

Their largest steps keeping x nonnegative are two and one, while their normalized states are $\Theta(n^{-1/2})$ apart. A step rule with only direction-state access that returns at most one on the second input and at least $3/2$ on the first needs $\Omega(\sqrt{n})$ controlled state-preparation calls.

Proof. Choose the two preparation unitaries as minimum rotations. Any valid output distinguishes them with constant bias, while a Q -query hybrid has distance $O(Q/\sqrt{n})$. Thus $Q = \Omega(\sqrt{n})$. $\square$

Skipping the diagonal block encoding and solving the following well-conditioned diagonal system directly from the state is also costly. For $n \geq 4$, fix $0 < \eta \leq 1/4$ and define

$$v^{(0)} = \mathbf{1}_n, \quad v^{(1)} = (1 + \eta, \beta_\eta, \dots, \beta_\eta), \quad \beta_\eta = \sqrt{1 - \frac{2\eta + \eta^2}{n-1}}. \quad (541)$$

Both norms equal $\sqrt{n}$, and their normalized distance d_η satisfies

$$\frac{\eta}{\sqrt{n}} \leq d_\eta \leq \frac{2\eta}{\sqrt{n}}. \quad (542)$$

Let $D_b = \text{diag}(v^{(b)})$, $r = (e_1 + e_2)/\sqrt{2}$, and $|z_b\rangle = D_b^{-1}r/\|D_b^{-1}r\|_2$. These systems are one-sparse and $\kappa(D_b) < 2$.

Theorem 12.13 (State to next solve). *Give an algorithm controlled access to the minimum-rotation preparations $U_b, U_b^\dagger$ of the vectors in (541). If its output ρ_b satisfies*

$$D_{\text{tr}}(\rho_b, |z_b\rangle\langle z_b|) \leq \eta/12 \quad (543)$$

for both inputs, then it uses $\Omega(\sqrt{n})$ oracle calls. With copy-only access to the iterate state, it needs $\Omega(n)$ copies. In particular, $\eta = 1/4$ gives a constant-accuracy lower bound for a system with condition number below two.

Proof. For $b = 0$, measuring coordinate one in $|z_b\rangle$ has probability $1/2$. For $b = 1$, put $q = \beta_\eta/(1 + \eta)$. The probability is $q^2/(1 + q^2)$, and

$$\frac{1}{2} - \frac{q^2}{1 + q^2} \geq \frac{1}{2} - \frac{1}{1 + (1 + \eta)^2} \geq \frac{\eta}{3}. \quad (544)$$

Contractivity and (543) therefore make the two algorithm outputs at least $\eta/6$ apart in trace distance. A Q -query hybrid is at most $Qd_\eta \leq 2Q\eta/\sqrt{n}$ apart, whence $Q \geq \sqrt{n}/12$. For k copies the input trace distance is at most $\sqrt{k}d_\eta = O(\sqrt{k}\eta/\sqrt{n})$, forcing $k = \Omega(n)$. $\square$

The Hermitian dilation $\begin{pmatrix} 0 & D_b \\ D_b & 0 \end{pmatrix}$ has the same sparsity and condition number, so this is also a standard Hermitian QLS instance. The theorem does not contradict QLS algorithms polylogarithmic in the dimension: they assume a matrix oracle, which here must be built from the iterate state or avoided.

Per-round costs add up only if each round brings a fresh, independent oracle. At round $t = 1, \dots, T$, after earlier outputs are committed, an adversary chooses a fresh bit b_t independent of the current workspace and supplies only a new minimum-rotation pair $U_{t,b_t}, U_{t,b_t}^\dagger$ for (541); future oracles are unavailable. Before the next round, the algorithm must either output a state satisfying (543) or, after S_t setup queries, provide a reusable controlled $(\alpha_t, a, \eta/8)$ block encoding of D_{b_t} with $\alpha_t \leq A_t$ (we call such an algorithm a compiler). One invocation of this encoding may use q_t further state-oracle calls.

Theorem 12.14 (Online fresh-oracle direct sum). *In the preceding interface, the worst-case numbers Q_t of state-oracle calls in round t of an algorithm that outputs next-solution states satisfy*

$$\sum_{t=1}^T Q_t = \Omega(T\sqrt{n}) \quad (545)$$

and, for algorithms that output reusable diagonal block encodings,

$$\boxed{\sum_{t=1}^T (\eta S_t + q_t A_t) = \Omega(T\sqrt{n})} \quad (546)$$

For a fixed short-step parameter η , the latter is equivalent to $\sum_t (S_t + q_t A_t) = \Omega(T\sqrt{n})$.

Proof. Condition on the complete workspace and transcript at the start of round t . Freshness makes that side information independent of b_t , so it is a fixed ancilla in theorem 12.13, which gives $Q_t = \Omega(\sqrt{n})$. For a compiler, run setup once and estimate the first encoded entry to $O(\eta/A_t)$ using $O(A_t/\eta)$ invocations. This distinguisher makes $S_t + O(q_t A_t/\eta)$ state-oracle calls, but the close input oracles require $\Omega(\sqrt{n}/\eta)$ calls. Hence $\eta S_t + q_t A_t = \Omega(\sqrt{n})$ in each conditioned round. Summation proves both claims. $\square$

Every part of the access model in theorem 12.14 is needed:

- The counted operations are controlled calls to the current U_{t,b_t} and its inverse; we give no gate-complexity lower bound.
- The common norm $\sqrt{n}$, the two vector formulas, r , and the round number are free; only b_t is hidden.
- No coordinate-value, QRAM, objective-coefficient, or compiled matrix oracle is available. Each could reveal the exceptional coordinate in one query.
- The action of each oracle on all inputs, not only on $|0\rangle$, is fixed by the model: the adversary may choose the close minimum-rotation completions. Specifying only $U|0\rangle$ would not define a query lower bound.
- One-time setup and reusable advice are counted in S_t , and the cost of each invocation in q_t . Advice consumed by calls must be prepared in sufficient quantity and cannot be cloned for free.
- Each round's output is fixed before the next fresh oracle arrives; this ordering lets the per-round conditional bounds be added.

A heralded compiler, which succeeds with probability bounded below, obeys the same bounds after counting repetitions. An unheralded constant failure probability is not covered when $\eta \rightarrow 0$, because it can exceed the $\Theta(\eta)$ separation of the target states.

The hard vector sequence stays positive and moves by short relative steps: $|v_i^{(b_{t+1})} - v_i^{(b_t)}| \leq \eta v_i^{(b_t)}$. Thus positivity and small relative changes do not weaken the online theorem, but neither do they make it a per-iteration QIPM lower bound on a fixed instance:

1. *Fresh oracles are a stronger assumption than a fixed instance.* Every QIPM iterate is determined by one static input and the prior computation; no independent oracle normally arrives after each output.
2. *Worst-case hardness for one vector does not automatically add up over rounds.* If every iterate depends on one hidden bit, learning it once makes every later diagonal easy.
3. *Circuits must be counted in underlying queries.* A circuit for $U_{x_{t+1}}$ may call U_{x_t} , which contains the prior history. Treating each circuit as one unit-cost call hides this nesting; counting both as independent primitives counts it twice.
4. *What an oracle reveals away from $|0\rangle$ depends on the model.* The hybrid argument chooses close extensions, whereas a concrete solver circuit may reveal coordinate or trajectory data on other inputs, so its entire oracle action must be fixed.
5. *Coherent access to all rounds is a different problem.* If a single multiplexed query reaches every time-indexed oracle, the round-by-round proof fails and a separate coherent direct-sum theorem is needed.
6. *The QIPM equations may permit a shortcut.* We have not embedded the diagonal systems above as consecutive Newton systems of one fixed sparse LP; cancellation, scale-free formulations, or history-state methods might avoid building each diagonal separately.
7. *The required output matters.* An algorithm that outputs a reusable diagonal or a Newton state in every round meets the theorem's output requirement; one that proves its steps analytically and outputs only one final observable need not.

Thus an $\Omega(T\sqrt{n})$ bound for a fixed LP would require T independent hard pieces in one sparse instance, a trajectory that reveals them one by one, and a direct-sum theorem for the static input oracle. We prove none of these.

Finally, coherent updates accumulate errors. If

$$\|\tilde{v}\rangle - |v/r_v\rangle\|_2 \leq \xi, \quad |\tilde{r}_v - r_v| \leq \delta_r, \quad (547)$$

then lifting $|\tilde{v}\rangle$ and $\tilde{r}_v$ to a block encoding of $\text{diag}(v)$ gives absolute operator error at most $\delta_r + \tilde{r}_v \xi$. For coherent LCU updates $X_{t+1} = X_t + h_t \Delta_t$, let $E_t := \|\tilde{X}_t - X_t\|_2$ be the accumulated absolute operator error before update t . If the step lengths and LCU arithmetic are exact and the only new error is the lifted direction error, the operator-norm triangle inequality gives

$$E_{t+1} \leq E_t + |h_t|(\delta_{r,t} + \tilde{r}_t \xi_t), \quad E_T \leq E_0 + \sum_{t < T} |h_t|(\delta_{r,t} + \tilde{r}_t \xi_t). \quad (548)$$

Errors from rounding, imperfect LCU coefficients, or postselection are separate additive terms. Late-path centrality therefore requires bounding the cumulative weighted error, not only each preparation's error. Literal circuit reuse also nests costs. Let C_t be the cost of the current iterate-preparation circuit, let Q_t be the number of its calls made by the next direction circuit, and let G_t be all nonrecursive work in that update. With literal inlining, additive uniform gate/query costs, and no sharing or memoization, the exact accounting is

$$C_{t+1} = G_t + (1 + Q_t)C_t. \quad (549)$$

This identity describes literal composition; it is not a lower bound for algorithms with direct structured or history-state oracles.

12.5 Lazy central oracles and the caching ceiling

Changing every diagonal weight does not force every weight to be rewritten, as the following fixed sparse LP shows. For K disjoint blocks of length L , let $d_{a,i} = u_{a,i} - v_{a,i}$ and impose

$$d_{a,0} = 1, \quad d_{a,i} - \varsigma_{a,i} d_{a,i-1} = 0 \quad (1 \leq i \leq L), \quad (550)$$

with all $u, v \geq 0$. Minimize

$$\sum_{a=1}^K \left[\sum_{i=0}^L (u_{a,i} + v_{a,i}) + \alpha(u_{a,L} - v_{a,L}) \right], \quad 0 < \alpha < 1. \quad (551)$$

Rows and columns have constant sparsity and the instance size is $\Theta(KL)$. With

$$p_{a,0} = 1, \quad p_{a,i} = \prod_{j=1}^i \varsigma_{a,j}, \quad q(\mu) = \mu + \sqrt{\mu^2 + 1}, \quad (552)$$

the primal central path is exactly

$$(u_{a,i}(\mu), v_{a,i}(\mu)) = \left(\frac{q(\mu) + p_{a,i}}{2}, \frac{q(\mu) - p_{a,i}}{2} \right). \quad (553)$$

Indeed feasibility fixes the difference $p_{a,i}$; differentiating the one-dimensional pair barrier gives the common sum $q(\mu)$. In the fixed orthonormal null basis $(e_{u_{a,i}} + e_{v_{a,i}})/\sqrt{2}$, the reduced log-barrier Hessian $W^\top X^{-2}W$ is the scalar

$$2[(q(\mu) + 1)^{-2} + (q(\mu) - 1)^{-2}]I,$$

so it has condition number one along the whole path. The only changing numerical information is the public scalar $q(\mu)$; all hidden information is in the static prefix-product table.

Give reversible raw access to the bits $b_{a,i}$, where $\varsigma_{a,i} = (-1)^{b_{a,i}}$. An endpoint central lookup at block a returns (553) for $i = L$, so it reveals $P_a = p_{a,L}$, even if each returned coordinate has error below $1/2$.

Theorem 12.15 (Lazy endpoint lookup). *One exact coherent endpoint lookup, or one whose decoded sign is correct with probability at least $2/3$, needs $\Omega(L)$ raw queries in the worst case, even when the block index is in superposition. The bound is tight.*

Proof. Fix the block register to any basis value a and fix all other blocks. Decoding the returned pair computes parity of the remaining L bits, which has bounded-error quantum query complexity $\Omega(L)$. Conversely, the exact quantum parity algorithm computes the selected block's parity in $O(L)$ queries while carrying a coherently as a spectator; reversible arithmetic then combines it with $q(\mu)$. The same circuit works in superposition over a . $\square$

Call a resource an *exact reusable endpoint oracle* if, once constructed, it lets a fixed processor obtain all K endpoint values without further raw queries, and it survives, or otherwise supports, those K invocations.

Theorem 12.16 (Reusable endpoint preprocessing). *Constructing an exact reusable endpoint oracle needs at least*

$$\left\lceil \frac{KL}{2} \right\rceil \quad (554)$$

raw queries. A bounded-error resource also needs $\Omega(KL)$ queries when all K successive answers are jointly correct with probability at least $2/3$ on every input.

Proof. After preprocessing, invoke the resource on every block and multiply the decoded signs. The result is

$$\prod_{a=1}^K P_a = (-1)^{\bigoplus_{a=1}^K \bigoplus_{i=1}^L b_{a,i}},$$

the parity of all KL raw bits. Exact quantum parity requires $\lceil KL/2 \rceil$ queries, and bounded-error parity requires $\Omega(KL)$. The proof allows entangled program states and implicit formulas; what matters is reusability, not an explicit classical table. $\square$

Theorem 12.17 (Explicit-coordinate trajectory service). *Fix arbitrary parameters $\mu_1, \dots, \mu_K > 0$. A service that at stage a returns the classical endpoint pair of block a at μ_a , with both coordinate errors below $1/2$ and all answers jointly correct with probability at least $2/3$ on every input, uses*

$$\boxed{\Omega(KL)} \tag{555}$$

raw queries in total over preprocessing and all stages, hence $\Omega(L)$ amortized per stage.

Proof. Each returned pair reveals the independent P_a . Multiplying the K answers computes parity of all KL input bits, so bounded-error quantum parity proves (555). $\square$

Theorem 12.17 is a lower bound for a service that must return explicit coordinates, not for every QIPM. Quantum cell-probe lower bounds likewise require a specified storage-and-probe model and do not apply automatically to an analytic central path [124]. We prove no unconditional lower bound on per-iteration refresh cost, for five separate reasons:

1. A fixed LP has no fresh per-iteration information. Once all KL bits have been processed, unrestricted persistent memory can amortize that setup over arbitrarily many iterations.
2. The path parameter may be low dimensional. In (553) every changing coordinate is generated by one scalar $q(\mu)$, so a stored structural oracle needs only $O(1)$ extra arithmetic when μ changes.
3. Cell-probe lower bounds need a specified query sequence. An IPM follows an analytic trajectory chosen by the algorithm, not an adversarial sequence of cell updates and requests; theorem 12.17 applies only once the distinct endpoint values are required outputs.
4. Coherent lookup defeats a naive sum over blocks. One circuit with $O(L)$ queries evaluates the selected block in superposition; a bound K times larger requires classical outputs or a suitable direct-product theorem.
5. A normalized state is weaker than a data structure. It need not support reusable random access to every coordinate; assuming that it does silently adds the cost of tomography or QRAM construction.

This is the caching ceiling of (32) and of the total query count following theorem 9.4: caching all signs can reduce later raw-query costs to zero. A true per-iteration refresh lower bound would need a memory limit, mandatory coordinate monitoring, independent invocations that cannot share work, or a fixed-LP trajectory proved online-hard. Sparsity and changing diagonal weights alone provide none of these.

12.6 Synthesis

In each of the five settings, the irreducible cost is a product or a total over all phases. For a separately block-encoded exact factor, $\alpha_P \sqrt{K} = \Omega(N)$, and with reusable compilation, $Ns + q\alpha = \Omega(N^2)$. Scalar congruence obeys $\sqrt{\kappa} \chi \geq c/\mu$, with equality in order exactly when $r_\mu = \Omega(\mu)$ and

$r_\mu = O(1)$. On the parity instance, the total cost of setup, right-hand-side preparation, solve, and recovery remains $\Omega(N)$ even when a perfect factor makes one phase trivial. Converting a prepared state into a diagonal block encoding obeys $qA = \Omega(\sqrt{n})$, and these costs add over rounds only with fresh oracles. Lazy evaluation shows why static preprocessing can defeat any unqualified per-iteration lower bound.

These are obstructions within specified access models, not a verdict against preconditioning or state reuse. The positive modules in [section 13](#) count costs the same way: they assemble favorable operators directly and state which projectors or cached structures are counted and which output they produce.

13 Positive Algorithmic Modules

The lower bounds in [sections 9](#) and [12](#) do not rule out useful quantum subroutines, but they determine which costs a positive result must include. This section proves such results: a pathwise bound on how often a reused Newton direction must be recomputed; a tail solver that needs no supplied partition, with exact repair of errors inside the optimal faces; a dual method with implicit primal and slack coordinates and compressed output; and a block-angular solver that returns the border variables explicitly. Some combine into conditional theorems for whole trajectories. None is an unconditional end-to-end speedup for arbitrary sparse LPs.

Every result below obeys five accounting rules for positive QIPM claims, which also serve as a checklist for other such claims.

1. A small spectral condition number does not make loading, normalization, RHS preparation, or output recovery free. Every block encoding and every output compiler (which produces the required output) must be expanded into the access model of [sections 2.2](#) and [2.3](#).
2. A strict-complementarity construction applies only after μ falls below its stated, data-dependent threshold. The separation margin and active singular-value gaps are parameters, not consequences of sparsity.
3. Compressed dual output is useful only when the retained dimension satisfies $m = o(n)$, and only against a classical algorithm costed in the same model.
4. A coherent primal-coordinate oracle counts as real output. When its values are well spread, value-to-amplitude conversion has constant overhead, so the oracle is at least as informative as primal-state preparation.
5. A dual-only certificate can fall outside a lower bound for primal output. Claiming a contradiction with such a bound requires an explicit low-query reduction to its output requirement.

These rules apply the cost accounting of [section 12](#); every cost below includes operator normalization, RHS amplification, state-to-residual conversion, tomography, and static input construction.

13.1 Projective lazy refresh and its limits

Consider a realized sequence of nonsingular real systems, $t = 0, \dots, T$, with $f_t \neq 0$:

$$H_t z_t = f_t, \quad q_t = \|z_t\|, \quad \psi_t = z_t/q_t, \quad M_t = \frac{q_t}{\|f_t\|} H_t, \quad (556)$$

and define

$$\delta_t = \|M_{t+1}(\psi_{t+1} - \psi_t)\|, \quad V_{\text{proj}} = \sum_{t < T} \delta_t. \quad (557)$$

We call V_{proj} the projective variation. Here z_t is expressed in the fixed basis in which the method stores it (for the OSS systems below, the fixed V spanning $\ker A$), so δ_t and V_{proj} depend on

that basis. Refresh policy: fix $\eta > 0$ and $\epsilon \geq 0$. At the initial checkpoint $s = 0$ and at every later refresh, store a classical vector $\widehat{z}_s$ satisfying $\|H_s \widehat{z}_s - f_s\|/\|f_s\| \leq \epsilon$. At each later time t , let s be the most recent checkpoint and choose the scalar a minimizing $\|H_t(a\widehat{z}_s) - f_t\|$. For $v = H_t \widehat{z}_s \neq 0$ this scalar is $\langle v, f_t \rangle / \|v\|^2$; for $v = 0$ choose zero. Reuse the rescaled vector if this minimum relative residual is at most η ; otherwise refresh, that is, solve the system at t and make t the new checkpoint. The minimization and threshold comparison are exact. Write R for the total number of checkpoint solves, including the initial one.

Theorem 13.1 (Pathwise residual-certified refresh). *Suppose $G > 0$ and, for every test at time t with most recent checkpoint s , including a test that triggers a refresh,*

$$\|M_t M_j^{-1}\| \leq G \quad (s \leq j \leq t). \quad (558)$$

We call $\|M_t M_j^{-1}\|$ the cross-gain and G its bound. If $\epsilon \leq \eta/(2G)$, then the number R of checkpoint solves satisfies

$$\boxed{R \leq 1 + \frac{2GV_{\text{proj}}}{\eta}}. \quad (559)$$

The statement is pathwise: the systems may depend adaptively on earlier inexact steps.

Proof. Put $u_s = \widehat{z}_s/q_s$ and $e_s = M_s(u_s - \psi_s) = (H_s \widehat{z}_s - f_s)/\|f_s\|$, so $\|e_s\| \leq \epsilon$. The nonzero right-hand sides imply $q_s > 0$. For a test at t , the least-squares scalar is no worse than $a = q_t/q_s$. The candidate's normalized residual vector is

$$M_t(u_s - \psi_t) = M_t M_s^{-1} e_s - \sum_{j=s}^{t-1} M_t M_{j+1}^{-1} M_{j+1}(\psi_{j+1} - \psi_j).$$

The triangle inequality and (558) give

$$\frac{\|H_t(a\widehat{z}_s - z_t)\|}{\|f_t\|} \leq G \left(\epsilon + \sum_{j=s}^{t-1} \delta_j \right).$$

Consequently, a test fails only after the projective variation since the last checkpoint exceeds $\eta/(2G)$. The intervals between consecutive checkpoints are disjoint and contained in $\{0, \dots, T-1\}$, which proves (559), including when no test fails. $\square$

Formal verification and scope. The Lean development in `formal/QipmFormal/Refresh/` verifies the normalization identities, the scalar least-squares minimization, the residual transport identity, and the bound for the sequential reuse-or-refresh procedure, including the initial solve, equality at the acceptance threshold, and the final interval with no further refresh. The map from claims to Lean statements and the reproduction command are in `formal/REFRESH.md`. The proofs use no axioms other than `propext`, `Classical.choice`, and `Quot.sound`. The development does not establish the gain and checkpoint-accuracy hypotheses for an IPM trajectory, nor verify the tail, OSS, cycling, or quantum-cost results below. A residual test computed with numerical error needs a guarantee on rejection as well as on acceptance: rejecting a vector that meets the threshold can add a checkpoint solve that is not paid for by accumulated variation of at least $\eta/(2G)$, so the counting argument no longer applies.

For $H_t = AD_t A^\top$, the test uses $A^\top \widehat{z}_s$, the current diagonal, and A ; it costs $O(\text{nnz}(A))$ classical arithmetic without forming H_t . This cost is paid at every iteration; the theorem amortizes QLS and readout costs, not all outer-loop work. This residual-tested classical cache differs from coherent eigenpath traversal [26] and from classical Krylov recycling or inverse maintenance [93, 120]. Classical warm starts and reoptimization reuse information across perturbed problem instances [68, 140], not a direction along one fixed central tail.

Under strict complementarity, the quantities in theorem 13.1 stay bounded on the exact central tail, even when the unscaled normal matrix becomes ill conditioned.

Theorem 13.2 (Directed strict-complementarity tail). *Let Q project onto U , let $Q^\perp = I - Q$, and suppose*

$$H_\mu = \mu^{-1}C_\mu + \mu D_\mu, \quad C_\mu = QC_\mu Q, \quad (560)$$

where, uniformly on $0 \leq \mu \leq \mu_0$,

$$c_-Q \preceq C_\mu \preceq c_+Q, \quad D_\mu \succeq 0, \quad \|D_\mu\| \leq d_+, \quad Q^\perp D_\mu Q^\perp \succeq d_-Q^\perp.$$

Assume C_μ, D_μ are C^1 with bounded derivatives. For nonzero $f \in U$, set $z_\mu = H_\mu^{-1}f$, $\psi_\mu = z_\mu/\|z_\mu\|$, and $N_\mu = (\|z_\mu\|/\|f\|)H_\mu$. There are finite constants B, L, G depending only on the displayed bounds and μ_0 such that

$$\sup_\mu \|N_\mu\| \leq B, \quad \sup_{0 < \mu \leq \nu \leq \mu_0} \|N_\mu N_\nu^{-1}\| \leq G, \quad \sup_\mu \|\psi'_\mu\| \leq L. \quad (561)$$

Hence, on every decreasing discretization of the tail, the refresh policy above with $\epsilon \leq \eta/(2G)$ satisfies

$$V_{\text{proj}} \leq BL\mu_0, \quad R \leq 1 + \frac{2GBL\mu_0}{\eta}, \quad (562)$$

independently of the final value of μ and the number of steps.

Proof. Put $\mathcal{A}_\mu = C_\mu + \mu^2 D_\mu$ and $w_\mu = \mathcal{A}_\mu^{-1}f$. Then $z_\mu = \mu w_\mu$ and $N_\mu = a_\mu \mathcal{A}_\mu$, where $a_\mu = \|w_\mu\|/\|f\|$. In $U \oplus U^\perp$ blocks, Schur elimination gives

$$E_\mu = D_{UU} - D_{U\perp} D_{\perp\perp}^{-1} D_{\perp U} \succeq 0, \quad S_\mu = C_{UU} + \mu^2 E_\mu, \\ w_U = S_\mu^{-1}f, \quad w_\perp = -D_{\perp\perp}^{-1} D_{\perp U} w_U.$$

The hypotheses bound these factors and their derivatives and bound $\|w_\mu\|$ above and away from zero. Thus a_μ and the normalized ray are C^1 at zero, proving the first and third bounds.

The only unbounded block of $\mathcal{A}_\nu^{-1}$ is $\nu^{-2} D_{\perp\perp, \nu}^{-1}$ in its lower-right block. Every block of $\mathcal{A}_\mu$ that can multiply it contains μ^2 , so the product contains the factor μ^2/ν^2 . For $\mu \leq \nu$ this factor is at most one, so $\|\mathcal{A}_\mu \mathcal{A}_\nu^{-1}\| = O(1)$. The bounded ratio a_μ/a_ν proves the second (directed) bound in (561). Finally the fundamental theorem of calculus gives

$$\sum_t \|N_{\mu_{t+1}}(\psi_{\mu_{t+1}} - \psi_{\mu_t})\| \leq B \int_0^{\mu_0} \|\psi'_\mu\| d\mu \leq BL\mu_0,$$

and [theorem 13.1](#) gives the refresh count. $\square$

The restriction $\mu \leq \nu$ in (561) is essential. For $C = \text{diag}(1, 0)$, $D = \text{diag}(0, 1)$, and $f = e_1$, the ray and forward gain are constant, whereas the reverse gain is $(\nu/\mu)^2$. On an exact LP central path, $C_\mu = A_B \text{diag}(x_B^2) A_B^\top$, $D_\mu = A_N \text{diag}(s_N^{-2}) A_N^\top$, and $f = b \in \text{range}(A_B)$. Analyticity of the central path up to its limit gives the required regularity even for degenerate optimal faces [73, 108]. The constants depend on the instance and need not be bounded by sparsity or dimension alone.

At checkpoint s , let $\Gamma_{f,s}$ be the RHS-state amplification factor, $\mathcal{K}_s = \alpha_s \|H_s^{-1}\|$ the effective QLS parameter, and $\rho_s = \alpha_s \|z_s\|/\|f_s\|$ the factor converting normalized-state error into relative residual. Suppose these are bounded by $\Gamma_f, \mathcal{K}, \rho$ on the tail. A refresh needs residual accuracy $O(\eta/G)$ by [theorem 13.1](#); hence its QLS precision contributes $\log(G/\eta)$, and coherent tomography, linear in the inverse accuracy, uses $\tilde{O}(mG\rho/\eta)$ controlled preparations. The refreshes cost

$$\tilde{O}\left[\left(1 + \frac{GBL\mu_0}{\eta}\right) \Gamma_f \mathcal{K} \frac{mG\rho}{\eta} \log \frac{G}{\eta}\right], \quad (563)$$

queries, with RHS preparation and amplification counted explicitly. Tomography from state copies instead depends quadratically on the inverse of the required normalized-state accuracy $O(\eta/(G\rho))$ [131]. The $T = O(\sqrt{n} \log(\mu_0/\epsilon_{\text{opt}}))$ ordinary sparse updates and residual tests are additional costs.

Corollary 13.3 (Exact-central feasible OSS). *With a fixed full-rank V spanning $\ker A$, let*

$$\mathcal{M}_\mu = [-X_\mu A^\top \quad S_\mu V], \quad f_\mu = -\tau \mu e. \quad (564)$$

On the exact central path, its normalized solution ray has finite projective variation on the tail and bounded directed cross-gain. Every coefficient vector, reconstructed as $\Delta x = V\lambda$, $\Delta s = -A^\top \Delta y$, preserves both feasibility equations exactly. Consequently residual-certified reuse with checkpoint accuracy $\epsilon \leq \eta/(2G)$ has a refresh count on the exact-central tail that is independent of the final precision.

Proof. At the center,

$$\mathcal{M}_\mu^\top \mathcal{M}_\mu = \text{diag}(AX_\mu^2 A^\top, V^\top S_\mu^2 V),$$

so $\|\mathcal{M}_\mu^{-1}\| = O(1/\mu)$. Analyticity gives $\|\mathcal{M}_\mu - \mathcal{M}_\nu\| = O(|\mu - \nu|)$, and therefore $\|\mathcal{M}_\mu \mathcal{M}_\nu^{-1}\| = O(1)$ for $\mu \leq \nu$. Differentiating centrality shows that the OSS solution divided by μ is an analytic nonzero vector. Thus the normalized exact-central OSS ray has finite variation. The reconstruction identities prove exact feasibility, and [theorem 13.1](#) gives the refresh conclusion. $\square$

Finite variation does not extend to every standard inexact trajectory.

Proposition 13.4 (Cycling and proximal no-go). *There is a row-3-sparse LP and a feasible three-phase trajectory in a fixed $N_2(\theta)$ neighborhood, satisfying the standard relative OSS residual condition with a fixed tolerance, for which*

$$\sum_{t < T} \delta_t = \Omega\left(\log \frac{\mu_0}{\epsilon_{\text{opt}}}\right). \quad (565)$$

Moreover, no Tikhonov (ridge) rule that relies only on that neighborhood and residual condition can guarantee all of finite projective variation, an admissible OSS residual, and bounded reconstructed steps. A ridge scale that keeps the residual admissible does not improve the worst-case conditioning.

Proof. Use

$$\min \sum_{j=4}^n x_j \quad \text{subject to} \quad x_1 + x_2 + x_3 = 1, \quad x \geq 0.$$

Set $\tau = a_0/\sqrt{n}$, $\mu_t = (1 - \tau)^t \mu_0$, and use

$$u_0 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad u_1 = \frac{1}{\sqrt{2}}(0, 1, -1), \quad u_2 = \frac{1}{\sqrt{2}}(-1, 0, 1).$$

Cycle

$$x_{1:3}^{(t)} = e/3 + d u_{t \bmod 3}, \quad x_{4:n}^{(t)} = \mu_t e, \quad y^{(t)} = -3\mu_t, \quad s_{1:3}^{(t)} = 3\mu_t e, \quad s_{4:n}^{(t)} = e.$$

Then $\|X_t s_t - \mu_t e\| = 3d\mu_t$, and the feasible step to the next phase has OSS residual $3d\mu_t(u_{j+1} - \tau u_j)$. Taking $d = \Theta(1/\sqrt{n})$ meets the usual relative residual tolerance $\eta_n = \Theta(1/\sqrt{n})$. With the fixed sparse basis $V = [e_1 - e_3, e_2 - e_3, e_4, \dots, e_n]$, the exact OSS coefficient rays converge cyclically to the three distinct lines $(1, -1)$, $(0, 1)$, and $(-1, 0)$. The normalized matrix M_{t+1} of the next system, as in [\(556\)](#), acts on their differences at constant scale, so $\delta_t = \Omega(d)$ eventually. Since $T = \Theta(\sqrt{n} \log(\mu_0/\epsilon_{\text{opt}}))$, this proves [\(565\)](#).

Unlike the cycling part, which takes $d = \Theta(1/\sqrt{n})$ and $\eta_n = \Theta(1/\sqrt{n})$, the proximal part fixes constants $d > 0$ and $0 < \theta < 1$ such that

$$3d \leq \theta, \quad \frac{c_1 d}{\sqrt{a_0^2 + 9d^2}} > \eta, \quad (566)$$

where $c_1 > 0$ is the constant separation of the three projective lines. For the ridge selector

$$z_\lambda = \arg \min_z \{\|Mz - f\|^2 + \lambda\|z - c\|^2\},$$

let $z_* = M^{-1}f$, v be a right singular vector of singular value σ , and $e = c - z_*$. Direct diagonalization gives the exact identities

$$\begin{aligned} \langle v, z_\lambda - c \rangle &= -\frac{\sigma^2}{\sigma^2 + \lambda} \langle v, e \rangle, \\ \left| \left\langle \frac{Mv}{\sigma}, Mz_\lambda - f \right\rangle \right| &= \frac{\lambda\sigma}{\sigma^2 + \lambda} |\langle v, e \rangle|. \end{aligned}$$

On the cycling face $\sigma = \Theta(\mu)$. A ray-locked sequence of bounded steps and finite projective variation is Cauchy, so on at least one of every three phases it stays a fixed distance from that phase's exact face target. If the ridge moves a nonvanishing fraction toward those targets infinitely often, its projective variation diverges. Otherwise its residual is at least $c_1 d \mu_t$ on an infinite separated subsequence. Since $\|f_t\| = \mu_t \sqrt{a_0^2 + 9d^2}$, (566) makes this exceed the admissible $\eta\|f_t\|$.

Conversely the universal bound

$$\|Mz_\lambda - f\| \leq \frac{\sqrt{\lambda}}{2} \|e\| \quad (567)$$

is sharp, since $\sigma\lambda/(\sigma^2 + \lambda)$ is maximized at $\sigma = \sqrt{\lambda}$. With $\|f\| = \Theta(\mu)$ and $\|e\| = \Theta(1)$, the largest worst-case admissible ridge parameter is $\lambda = O(\eta^2 \mu^2)$. The augmented system $[M^\top \sqrt{\lambda} I]^\top$ then retains condition $\Theta(1/\mu)$; its normal matrix retains $\Theta(1/\mu^2)$. Thus regularization cannot guarantee finite projective variation, and at any admissible ridge parameter it does not improve the worst-case conditioning. $\square$

Residual-certified reuse still gives a conditional QIPM. If the feasible short-step theorem accepts every coefficient vector satisfying $\|\mathcal{M}_t \hat{z}_t - f_t\| \leq \eta\|f_t\|$, then exact OSS reconstruction preserves feasibility, and the published neighborhood, gap, and iteration guarantees apply unchanged [107]. Under finite realized V_{proj} , the directed cross-gain bound, and checkpoint accuracy $\epsilon \leq \eta/(2G)$, theorem 13.1 also applies. Assume block encodings of $A^\top/\alpha_A$ and V/α_V , coherent value access to the current x_t, s_t , and known bounds $R_x \geq \|x_t\|_\infty$ and $R_s \geq \|s_t\|_\infty$. Given $F_{\infty,t} \geq \|f_t\|_\infty$, define

$$\alpha_{\mathcal{M},t} = R_x \alpha_A + R_s \alpha_V, \quad \mathcal{K}_t = \alpha_{\mathcal{M},t} \|\mathcal{M}_t^{-1}\|, \quad \Gamma_{f,t} = \frac{\sqrt{n} F_{\infty,t}}{\|f_t\|},$$

and the conversion factor $\rho_t = \alpha_{\mathcal{M},t} \|z_t\|/\|f_t\|$. With dense output, the total cost is

$$C_{\text{iter}} + O(T \text{nnz } \mathcal{M}) + \sum_{s \in \mathcal{R}} \tilde{O}\left(\Gamma_{f,s} \mathcal{K}_s \frac{n \rho_s}{\epsilon}\right), \quad (568)$$

plus $O(nR)$ checkpoint writes. Here C_{iter} is the full cost of maintaining the assumed coherent iterate oracles; rebuilding dense memories for x_t, s_t in every round gives $C_{\text{iter}} = \Theta(nT B_{\text{bit}})$, and the term vanishes only if the outer implementation already provides coherent reads without such rebuilds. A quantum residual test costs $\tilde{O}((\Lambda_{t,s} + \Gamma_{f,t})/\eta)$, where $\Lambda_{t,s} = \alpha_{\mathcal{M},t} \|\hat{z}_s\|/\|\mathcal{M}_t \hat{z}_s\|$; on the cycling witness $\Lambda_{t,s} = \Omega(1/\mu_t)$. Generic quantum verification is therefore not free [126]. Proximal QLS warm starts help for an individual certified system, but they do not remove the cost of verification along the sequence [85]. Iterative-refinement tomography can improve a single high-accuracy solve, but it does not remove the per-refresh and output factors in (568) [105].

13.2 Partition-free equilibration and exact face repair

The stable solver in [section 5.4](#) assumes a supplied active range; on a strict-complementarity tail, it can be computed from the current coordinates. Suppose

$$\underline{x} \leq x_i \leq \bar{x} \ (i \in B), \quad \underline{s} \leq s_i \leq \bar{s} \ (i \in N), \quad (569)$$

and $(1 - \theta)\mu \leq x_i s_i \leq (1 + \theta)\mu$.

Lemma 13.5 (Constant-margin tail selector). *If*

$$\mu \leq \frac{1}{2(1 + \theta)} \min\{\underline{x}^2, \underline{s}^2\}, \quad (570)$$

then $E = \text{diag}(\mathbf{1}\{x_i \geq s_i\})$ equals P_B , and every comparison has absolute margin at least $\frac{1}{2} \min\{\underline{x}, \underline{s}\}$.

Proof. For $i \in B$, $s_i \leq (1 + \theta)\mu/\underline{x} \leq \underline{x}/2 \leq x_i/2$. The symmetric calculation gives $x_i \leq s_i/2$ for $i \in N$. $\square$

This is the classical tail rule of Mehrotra–Ye, evaluated coherently [37, 102]. It requires oracles for current coordinate values and enough bits to resolve the promised margin; it does not convert amplitude states into coordinate values.

Theorem 13.6 (Constructive stable active-range equilibration). *Under [lemma 13.5](#), define $F = I - E$,*

$$\begin{aligned} C_\mu &= \mu A E X S^{-1} E A^\top, & D_\mu &= \mu^{-1} A F X S^{-1} F A^\top, \\ Q_U &= \Pi_{\text{range}(AE)}, & \widehat{M}_\mu &= C_\mu + D_\mu - (1 - \mu^2) Q_U D_\mu. \end{aligned} \quad (571)$$

Then $H_\mu = A X S^{-1} A^\top = \mu^{-1} C_\mu + \mu D_\mu$, and direct block encodings have normalizations

$$\alpha_C \leq \alpha_A^2 \frac{\bar{x}^2}{1 - \theta}, \quad \alpha_D \leq \alpha_A^2 \frac{1 + \theta}{\underline{s}^2}. \quad (572)$$

If $\sigma_B = \sigma_{\min}^+(A_B) > 0$, QSVT constructs Q_U to error δ in

$$O\left(\frac{\alpha_A}{\sigma_B} \log \frac{1}{\delta}\right) \quad (573)$$

queries. If the tail block bounds of [theorem 5.7](#) hold with fixed constants, $\widehat{M}_\mu$ has μ -independent normalization and inverse norm. For $f \in \text{range}(AE)$,

$$\widehat{M}_\mu^{-1} f = \mu^{-1} H_\mu^{-1} f \quad (574)$$

for the exact projector, and constant projector error gives constant solution-state error. With RHS factor Γ_f , state preparation costs

$$\tilde{O}\left(\Gamma_f \Gamma_M \|\widehat{M}_\mu^{-1}\| \frac{\alpha_A}{\sigma_B}\right), \quad \Gamma_M = \alpha_C + (2 - \mu^2) \alpha_D, \quad (575)$$

with no polynomial $1/\mu$ factor.

Proof. The selector splits the columns exactly. On B , $\mu x_i/s_i = \mu x_i^2/(x_i s_i) \leq \bar{x}^2/(1 - \theta)$; on N , $\mu^{-1} x_i/s_i = (x_i s_i)/(\mu s_i^2) \leq (1 + \theta)/\underline{s}^2$. This proves (572). QSVT thresholding of AE/α_A at its smallest nonzero singular value gives (573) [65]. With $P_\mu = \mu Q_U + \mu^{-1}(I - Q_U)$, multiplication gives $\widehat{M}_\mu = P_\mu H_\mu$. Its $U \oplus U^\perp$ diagonal blocks are uniformly invertible and the upper off-diagonal block is $O(\mu^2)$; the Schur complement bounds its inverse. Since $P_\mu f = \mu f$, the solution identity follows. Direct LCU gives Γ_M , and a Neumann bound makes constant projector error sufficient. Multiplying projector, QLS, RHS, and direct encoding factors yields (575). $\square$

Theorem 13.6 removes the supplied-partition assumption left open in [section 5.4](#). Its scope is narrower than that of the published infeasible-QIPM preconditioner that estimates the partition [\[139\]](#): it gives constant tail conditioning, by direct assembly, only for right-hand sides in the active range, and it is not a complete infeasible-QIPM theorem. The input-dependent gap and output costs remain in the bound, consistent with classical active-partition preconditioning [\[39\]](#).

The next theorem computes exactly the OSS-residual components that cause errors inside the primal and dual optimal faces (the face leakage). Let the columns of Z and Y be orthonormal bases of $\ker A_B$ and $\ker A_B^\top$, and define

$$\begin{aligned} G_p &= Z^\top X_B^{-1} S_B Z, & g_p(r) &= Z^\top X_B^{-1} r_B, \\ G_d &= Y^\top A_N S_N^{-1} X_N A_N^\top Y, & g_d(r) &= Y^\top A_N S_N^{-1} r_N. \end{aligned} \quad (576)$$

Write $\sigma_N = \sigma_{\min}(A_N^\top|_{\ker A_B^\top})$ when $\ker A_B^\top \neq \{0\}$.

Theorem 13.7 (Face leakage, repair, and restricted stability). *For any feasible OSS direction error with residual r , the primal and dual face errors are*

$$p_F = Z G_p^{-1} g_p(r), \quad q_F = -Y G_d^{-1} g_d(r). \quad (577)$$

Under [\(569\)](#), the primal map has sharp two-sided $\Theta(1/\mu)$ amplification when $\ker A_B \neq \{0\}$, and the dual map has the same amplification when $\ker A_B^\top \neq \{0\}$; otherwise the corresponding map is zero. The correction

$$\widehat{\Delta x}_B^+ = \widehat{\Delta x}_B - p_F, \quad \widehat{\Delta y}^+ = \widehat{\Delta y} + Y G_d^{-1} g_d(r), \quad \widehat{\Delta s}^+ = \widehat{\Delta s} - A^\top Y G_d^{-1} g_d(r) \quad (578)$$

preserves primal and dual feasibility and makes $g_p(r^+) = g_d(r^+) = 0$ exactly. It also obeys $\|r^+\| \leq C_{\text{rep}} \|r\|$, and the inverse of the OSS map restricted by these two equations is uniformly bounded as $\mu \downarrow 0$. The total residual-induced face displacement is finite under the worst-case sharp condition

$$\sum_t \left(\frac{\|g_p(r_t)\|}{\mu_t} + \frac{\|g_d(r_t)\|}{\mu_t} \right) < \infty. \quad (579)$$

Proof. Writing the direction error as (e_x, e_y, e_s) gives $Ae_x = 0$, $e_s = -A^\top e_y$, and $Se_x - XA^\top e_y = r$. Multiply the B rows by $Z^\top X_B^{-1}$; the dual term vanishes and $Z^\top X_B^{-1} S_B(e_x)_B = g_p(r)$. This is the normal equation for the $X_B^{-1} S_B$ -weighted projection of $(e_x)_B$ onto $\ker A_B$. Eliminating e_x gives $AXS^{-1}A^\top e_y = -AS^{-1}r$; left-multiplying by $Y^\top$ and using $Y^\top A_B = 0$ gives $Y^\top (A_N X_N S_N^{-1} A_N^\top) e_y = -g_d(r)$, the normal equation for the H_μ -weighted projection of e_y onto $\ker A_B^\top$, whose coordinate is $-G_d^{-1} g_d(r)$ because $Y^\top H_\mu Y = G_d$. These are [\(577\)](#).

Complementarity and the tail margins imply

$$c_p \mu I \preceq G_p \preceq C_p \mu I, \quad c_d \mu I \preceq G_d \preceq C_d \mu I,$$

where, when $\ker A_B^\top \neq \{0\}$, the dual constants also contain $\sigma_N^2 > 0$. This proves both sides of the $1/\mu$ amplification on each nonzero face. The primal correction lies in $\ker A$ and is supported on B ; the dual correction lies in $\ker A_B^\top$. They therefore preserve feasibility and affect disjoint projected residual components, giving $g_p(r^+) = g_d(r^+) = 0$. The displayed spectral bounds and the bounded tail coordinates give $\|r^+\| \leq C_{\text{rep}} \|r\|$.

At $\mu = 0$, the kernel of the limiting OSS map consists precisely of primal motions in $\ker A_B$ and dual motions in $\ker A_B^\top$. The two corrected equations remove those components. Compactness of the unit sphere on the remaining complement and continuity give a uniform smallest restricted singular value. Finally each step can realize face displacement comparable to $\|g_p\|/\mu + \|g_d\|/\mu$; aligned successive residuals attain that order. Summability is therefore sufficient and worst-case sharp. $\square$

Condition (579) controls the residual-induced motion inside the optimal faces, not the realized projective variation. A local analysis near a strictly complementary solution also needs the transverse-forcing condition, which bounds the total change of r_t/μ_t :

$$W_r = \sum_t \left\| \frac{r_{t+1}}{\mu_{t+1}} - \frac{r_t}{\mu_t} \right\| < \infty. \quad (580)$$

Define the total face leakage

$$\mathcal{L}_\infty = \sum_t (\|p_F(r_t)\| + \|q_F(r_t)\|). \quad (581)$$

Even together with the tail bound $\sum_{k \geq t} (\|p_F(r_k)\| + \|q_F(r_k)\|) = O(\mu_t)$, the sums $\mathcal{L}_\infty$ and W_r do not bound the realized V_{proj} . Take the row-3-sparse family of [proposition 13.4](#) with face displacement $d_t = \varrho_t \bmod 3\mu_t$ and $\varrho = (1, 3, 1)$. It has convergent $\mathcal{L}_\infty$ and W_r , satisfies that tail bound, and stays inside the neighborhood and residual condition, yet its realized V_{proj} grows linearly in the number of steps: an oscillating relative face displacement d_t/μ_t moves the normalized ray by $\Theta(1)$ per step, which the absolute sums $\mathcal{L}_\infty$ and W_r do not detect. With the constant ratio $\varrho = (1, 1, 1)$, the same family has bounded V_{proj} when δ_t is measured in an orthonormal basis of $\ker A$, but not in the skew stored basis $V = [e_1 - e_3, e_2 - e_3, e_4, \dots, e_n]$ of [proposition 13.4](#): there the three cycling directions have coefficient norms $1, 1/\sqrt{2}, 1/\sqrt{2}$, so $\|z_t\|$ depends on the phase, $\delta_t = \Theta(1)$ on two of every three steps, and V_{proj} still grows linearly. Neither $o(\mu_t)$ leakage nor convergence of d_t/μ_t suffices either: $d_t = \varrho_t \bmod 3\mu_t/\sqrt{t}$ gives convergent sums and an $o(\mu_t)$ tail while V_{proj} diverges logarithmically. Finite realized V_{proj} therefore remains a separate trajectory hypothesis of [theorem 13.1](#); we do not show that reuse is impossible when it fails. For repaired directions, r_t denotes the residual after correction and retesting.

The repair also has a basis-free form. With $T = E\sqrt{XS^{-1}}E$, $B_r = E(XS)^{-1/2}E$, $K_F = AT$, $Q = \Pi_{\ker(AE)^\top}$, $D_N = FXS^{-1}F$, and $H_N = AD_NA^\top$,

$$p_F(r) = T\Pi_{\ker K_F}B_r r, \quad q_F(r) = -(QH_NQ)^+QAFS^{-1}Fr. \quad (582)$$

Indeed the weighted kernel projector equals $W^{1/2}Z(Z^\top WZ)^{-1}Z^\top W^{1/2}$ on B , with $W = X_B^{-1}S_B$; sandwiching it yields the first expression. The second is the inverse of the H_N form on $\ker A_B^\top$. Given known positive lower bounds on the gaps σ_B and σ_N , QSVT implements the projectors with

$$\chi_p = O\left(\frac{\alpha_A \bar{x}}{\sigma_B \underline{x}} \sqrt{\frac{1+\theta}{1-\theta}}\right), \quad \chi_d = O\left(\frac{\alpha_A^2}{\sigma_N^2} \frac{1+\theta \bar{s}^2}{1-\theta \underline{s}^2}\right), \quad (583)$$

both independent of μ . With

$$R_T = \frac{\bar{x}}{\sqrt{(1-\theta)\mu}}, \quad R_B = \frac{1}{\sqrt{(1-\theta)\mu}}, \quad \alpha_p = R_T R_B = \Theta(1/\mu),$$

the primal correction map has RHS factor $\Gamma_p = \alpha_p \|Er\|/\|p_F(r)\|$. If $c_N(r) = QAFS^{-1}Fr$, the dual factor is

$$\Gamma_d(r) = \frac{\alpha_A \underline{s}^{-1} \|Fr\|}{\|c_N(r)\|}.$$

With $\Gamma_{r,B}, \Gamma_{r,N}$ the factors for preparing the selected residual components, preparing the normalized correction states conservatively costs

$$Q_p = \tilde{O}(\Gamma_{r,B} \Gamma_p \chi_p), \quad Q_d = \tilde{O}(\Gamma_{r,N} \Gamma_d \chi_p \chi_d), \quad (584)$$

queries, with norm recovery and the RHS-specific success amplitude of the inverse counted separately. The repair is thus an operator with its own cost, not a free correction.

Replacing the fixed face radius in the row-3-sparse example by $d_t > 0$ gives exactly

$$r_t = 3\mu_t(d_{t+1}u_{j+1} - \tau d_t u_j),$$

and, for adjacent radii within a fixed factor, its leakage is $\Theta(d_t + d_{t+1})$. This identity shows that (579) is necessary in the worst case. It does not give the threshold for projective variation claimed in a preliminary version of this analysis: exact solves show that when $d_t \gg \mu_t$, the phase-dependent transverse coefficient changes by $\Theta(\mu_t/d_t)$, while the normalized OSS operator amplifies that component by $\Theta(d_t/\mu_t)$, so $\delta_t = \Theta(1)$ rather than $\Theta(d_t + d_{t+1})$.

On a geometric schedule with $d_t = \Theta(\mu_t^\gamma)$ and $0 < \gamma < 1$, both $\mathcal{L}_\infty$ and W_r converge, but realized V_{proj} grows linearly with the number of steps. This family violates the tail requirement $\sum_{k \geq t} (\|p_F(r_k)\| + \|q_F(r_k)\|) = O(\mu_t)$, but the oscillating-displacement family above satisfies it and still has divergent projective variation. This corrects the claim of that preliminary version about this family. Residuals of order $O(\mu^2)$ at every step suffice for (579), but this is not the sharp leakage condition.

Separating the eigenvalue clusters cannot be skipped. Without the selector, using $T_{\text{all}} = (X^{-1}S)^{-1/2}$ and $K_{\text{all}} = AT_{\text{all}}$, the weighted kernel formula reconstructs the entire Newton error, not merely face leakage. For

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}, \quad x = (1/2, 1/2, \mu, \mu), \quad s = (2\mu, 2\mu, 1, 1),$$

$\kappa(K_{\text{all}}) = 1/(2\mu)$ and $\kappa(AXS^{-1}A^\top) = 1/(4\mu^2)$. A Tikhonov kernel filter $\lambda(K_{\text{all}}^\top K_{\text{all}} + \lambda I)^{-1}$ accurate to $\delta < 1/2$ forces $\lambda \leq 2\delta\mu/(1 - \delta)$ and hence condition $\Omega(1/(\delta\mu^2))$. Separating the eigenvalue clusters, by the coordinate selector or an equivalent spectral cutoff, is therefore necessary.

Finally, if a repair must be written out classically within the pointwise budget $O(\tau\mu_t)$, an arbitrary constant-norm correction in dimension d needs absolute tomography error $O(\tau\mu_t)$. Generic coherent tomography costs

$$\tilde{\Theta}\left(\frac{d}{\tau\mu_t}\right) \tag{585}$$

calls to the state-preparation unitary. This limits generic extraction; it is not an LP-specific output lower bound. A concise formula for the correction, sparse output, or the compressed representation below can avoid it.

13.3 Implicit dual coordinates and compressed output

The state-to-diagonal obstruction of section 12.4 does not apply when the changing diagonal is computed from a smaller explicit iterate instead of from a state-preparation oracle. Let the columns of A have at most d_c nonzeros, and let the method store only the dual iterate $y_t \in \mathbb{R}^m$. A static block encoding of A/α_A costs C_A per use, a coherent read of one retained-iterate entry costs $C_{y,\text{read}}$, and B_{bit} is the common arithmetic word width. The one-time cost $C_{\text{build}}(A)$ of building this access is counted separately. Coherent RAM may make $C_{y,\text{read}}$ polylogarithmic; otherwise the actual cost of a multiplexed read must be substituted below.

Theorem 13.8 (Implicit slack oracle and affine closure). *Given static coherent sparse-column access, coherent value access to c_i , and coherent reads of y_t , a reversible circuit computes*

$$s_i(y_t) = c_i - a_i^\top y_t \tag{586}$$

to absolute error ϵ_s using d_c column accesses and

$$\tilde{O}(d_c B_{\text{bit}} + d_c C_{y,\text{read}} + \log(1/\epsilon_s))$$

gates. If $s_i \geq \underline{s}_t > 0$, the circuit gives a unit-normalized encoding of $\underline{s}_t S_t^{-1}$ with operator error

$$O\left(\frac{\epsilon_s}{\underline{s}_t} + \epsilon_{\text{arith}}\right), \quad (587)$$

and therefore an encoding of $AS_t^{-2}A^\top$ with normalization $\alpha_A^2/\underline{s}_t^2$. It suffices to make the error in (587) at most

$$O\left(\frac{\eta}{J_t \rho_t \mathcal{K}_{M,t}}\right),$$

with ρ_t as in (594); $1/\mathcal{K}_{M,t}$ converts operator error into relative vector error, and $1/(J_t \rho_t)$ is the vector accuracy target of [theorem 13.9](#). On active coordinates this follows, for example, from absolute slack error $O(\eta \mu_t / (J_t \rho_t \mathcal{K}_{M,t}))$. Thus finite-precision error is counted within the residual tolerance of the transformed system. Updating $y_{t+1} = y_t + \lambda_t \Delta y_t$ updates every implicit slack exactly, at a cost independent of the number of past updates and without rewriting n coordinates.

Proof. Query c_i and the support and values of a_i , then read the matching entries of y_t and reversibly accumulate (586). Approximate $\underline{s}_t/z$ on the promised interval and use it as a controlled-rotation amplitude; uncomputation yields the diagonal encoding. Product encodings give the normal matrix. The affine identity $c_i - a_i^\top y_{t+1} = s_i(y_t) - \lambda_t a_i^\top \Delta y_t$ proves exact closure. No state-preparation oracle for s is queried, so the premise of the state-to-diagonal obstruction does not hold. $\square$

The dual method stores no primal iterate, so its selector uses the implicit primal coordinate that exact centrality assigns to the current slack,

$$x_i^b = \frac{\mu_t}{s_i(y_t)}, \quad E_t = \text{diag}\left(\mathbf{1}\{x_i^b \geq s_i(y_t)\}\right) = \text{diag}\left(\mathbf{1}\{s_i(y_t)^2 \leq \mu_t\}\right). \quad (588)$$

This is [lemma 13.5](#) with exact complementarity ($\theta = 0$), applied to $(x^b, s(y_t))$; we assume the tail margins for this implicit pair. When $\mu_t \leq \frac{1}{2} \min\{\underline{x}^2, \underline{s}^2\}$, an active coordinate has $x_i^b \geq 2s_i$ and an inactive coordinate has $s_i \geq 2x_i^b$. Consequently the comparison of x^b with s has the constant margin of that lemma; equivalently, the directly queryable squared test is at least $\mu_t/2$ from its threshold. It is resolved with $O(\log(1/\mu_t))$ bits and needs no stored x_t .

We next stabilize the dual Newton equation without a supplied partition:

$$G_t \Delta y_t = g_t, \quad G_t = \mu_t AS_t^{-2} A^\top, \quad g_t = b - \mu_t AS_t^{-1} e. \quad (589)$$

Using (588), define $C_t = \mu_t^2 AES_t^{-2} EA^\top$, $D_t = AFS_t^{-2} FA^\top$, $Q_t = \Pi_{\text{range}(AE)}$, and

$$M_t = C_t + D_t - (1 - \mu_t^2) Q_t D_t = P_t G_t, \quad P_t = \mu_t Q_t + \mu_t^{-1} (I - Q_t). \quad (590)$$

With direct component normalizations $\alpha_{C,t}$ and $\alpha_{D,t}$,

$$\alpha_{M,t} \leq \alpha_{C,t} + (2 - \mu_t^2) \alpha_{D,t}, \quad \mathcal{K}_{M,t} = \alpha_{M,t} \|M_t^{-1}\|. \quad (591)$$

The assembled block encoding of M_t is an $(\alpha_{M,t}, \cdot, \epsilon_{M,t})$ encoding in the sense of [definition 2.13](#), with precision fixed so that $\epsilon_{M,t} \leq \alpha_{M,t}$. Given the tail gaps assumed above, both component normalizations and $\mathcal{K}_{M,t}$ are independent of μ_t . Since $b \in \text{range}(A_B)$, the transformed RHS can be assembled without a block encoding of the poorly normalized P_t :

$$\begin{aligned} \tilde{g}_t &= P_t g_t = \mu_t b - \mu_t^2 AES_t^{-1} E e - \mu_t^2 Q_t AFS_t^{-1} F e \\ &\quad - (I - Q_t) AFS_t^{-1} F e. \end{aligned} \quad (592)$$

Let $\Gamma_{g,t}$ be the actual LCU normalization of this sum divided by $\|\tilde{g}_t\|$, so that cancellation is counted. The system actually solved is

$$M_t \Delta y_t = \tilde{g}_t. \quad (593)$$

Theorem 13.9 (Conditional compressed dual QIPM). *Assume the implicit-access model of theorem 13.8, the strict-complementarity margins and known gaps used above, and the inexact dual-barrier theorem of [138] (their Theorem 3.3). With $B_t = S_t^{-1}A^\top$ and $\delta_t^N = \|B_t\Delta y_t\| \leq 1/2$, that theorem accepts any step $\hat{z}_t$ satisfying $\|B_t(\hat{z}_t - \Delta y_t)\| \leq \eta\delta_t^N$ for a fixed $\eta \leq 0.1$, and it needs $T = O(\sqrt{n} \log(n\mu_0/\epsilon_{\text{opt}}))$ iterations. Assume throughout the tail that $0 < \mu_t \leq 1$ and range $Q_t \neq \{0\}$. For $g_t \neq 0$ set*

$$J_t = \frac{\|P_t^{-1}\| \|P_t g_t\|}{\|g_t\|}, \quad \rho_t = \frac{\alpha_{M,t} \|\Delta y_t\|}{\|\tilde{g}_t\|}. \quad (594)$$

Solving (593) and coherently reconstructing its m -vector direction $\hat{z}_t$ with

$$\|\hat{z}_t - \Delta y_t\| \leq \frac{\eta}{2J_t\rho_t} \|\Delta y_t\|$$

produces

$$(y_T, \mu_T, O_{s,T}, O_{x,T}), \quad x_i = \frac{\mu_T}{s_i} + \frac{\mu_T}{s_i^2} a_i^\top \Delta y_T,$$

with the positivity, neighborhood, and convergence guarantees of the outer method. The T outer updates use solves $t < T$; one additional terminal solve at (y_T, μ_T) supplies the displayed Δy_T and final coordinate oracle. The query count is

$$Q_{\text{path}} = \tilde{O} \left(\sum_{t \leq T} \frac{mJ_t\rho_t}{\eta} \Gamma_{g,t} \mathcal{K}_{M,t} \chi_{Q,t} \right), \quad (595)$$

where $\chi_{Q,t} = O((\alpha_A/\sigma_B) \log(1/\delta_Q))$. The data-loading and gate cost is

$$C_{\text{build}}(A) + \tilde{O} \sum_{t \leq T} \left[\frac{mJ_t\rho_t}{\eta} \Gamma_{g,t} \mathcal{K}_{M,t} \chi_{Q,t} (C_A + d_c(B_{\text{bit}} + C_{y,\text{read}})) + mB_{\text{bit}} \right]. \quad (596)$$

There is no term of order nT for writing iterates. If every displayed factor and η^{-1} are $O(1)$, then $Q_{\text{path}} = \tilde{O}(Tm)$ on the tail.

Proof. Let $e_t = \hat{z}_t - \Delta y_t$ with $\|e_t\| \leq (\eta/(2J_t\rho_t)) \|\Delta y_t\|$. Since $G_t = P_t^{-1}M_t$ and $\|M_t\| \leq \alpha_{M,t} + \epsilon_{M,t} \leq 2\alpha_{M,t}$ by the definition of a block encoding and the precision convention $\epsilon_{M,t} \leq \alpha_{M,t}$ fixed after (591), $\|G_t\| \leq 2\|P_t^{-1}\| \alpha_{M,t}$, while (594) gives $J_t\rho_t = \alpha_{M,t} \|P_t^{-1}\| \|\Delta y_t\| / \|g_t\|$. Using $g_t = G_t\Delta y_t$ and $G_t^2 \preceq \|G_t\|G_t$,

$$J_t\rho_t \geq \frac{\|G_t\| \|\Delta y_t\|}{2\|G_t\Delta y_t\|} \geq \frac{\|G_t\|^{1/2} \|\Delta y_t\|}{2\|\Delta y_t\|_{G_t}}.$$

With $G_t = \mu_t B_t^\top B_t$ we have $\|B_t e_t\| = \mu_t^{-1/2} \|e_t\|_{G_t} \leq \mu_t^{-1/2} \|G_t\|^{1/2} \|e_t\|$ and $\delta_t^N = \|B_t\Delta y_t\| = \mu_t^{-1/2} \|\Delta y_t\|_{G_t}$, so $\|B_t e_t\| \leq \eta\delta_t^N$. Thus the reconstructed step meets the weighted acceptance condition of the outer theorem.

The operator-error budget of theorem 13.8 supplies this vector accuracy; its factor $1/\rho_t$ enters only logarithmic precision terms. Normalized-state and norm errors $O(\eta/(J_t\rho_t))$ suffice; coherent tomography then uses $\tilde{O}(mJ_t\rho_t/\eta)$ controlled preparations. Multiplication by RHS, QLS, and projector costs gives (595). The slack and selector queries depend only on the current explicit y_t , so each accepted direction costs $\tilde{O}(mB_{\text{bit}})$ to store and never materializes an n -vector. Expanding one state preparation yields (596). Since every step meets the accuracy required by the outer theorem, that theorem applies. $\square$

A well-conditioned M_t is not enough; J_t must also be bounded. Under the assumptions of [theorem 13.9](#) and $g_t \neq 0$, if $u_t = \|Q_t g_t\|$ and $v_t = \|(I - Q_t)g_t\|$, direct substitution gives

$$J_t = \frac{\sqrt{u_t^2 + \mu_t^{-4} v_t^2}}{\sqrt{u_t^2 + v_t^2}}. \quad (597)$$

Thus $v_t \leq C\mu_t^2 u_t$ implies $J_t \leq \sqrt{1 + C^2}$, while purely inactive forcing ($u_t = 0$) gives $J_t = \mu_t^{-2}$. The favorable case therefore requires bounded J_t (stable forcing), $\Gamma_{g,t}$ (RHS cancellation), and ρ_t (output conversion). Full face-repair tomography still obeys [\(585\)](#); classically, the same uniformly conditioned sparse repair can be computed by Krylov iteration with only logarithmic precision dependence.

The matching classical baseline applies the same stable operator without forming it, at $\Theta(nd_c)$ arithmetic per multiplication and

$$\tilde{O}\left(nd_c \log \frac{J_t}{\eta}\right) \quad (598)$$

per direction under uniform conditioning. Ignoring the difference between query and gate units, the compressed quantum route can beat this baseline only if

$$\frac{mJ_t \rho_t \Gamma_{g,t} \mathcal{K}_{M,t} \chi_{Q,t}}{\eta} = o(nd_c). \quad (599)$$

Randomized column sketches may improve the classical side further. Also, $C_{\text{build}}(A) = \Omega(nd_c B_{\text{bit}})$ for an explicit sparse input file, so no sublinear end-to-end wall-clock claim follows unless a reusable oracle or concise input circuit is supplied.

The condition-one prefix family of [section 9](#) neither contradicts nor confirms [theorem 13.9](#). Its public start is primal-dual infeasible, the selector's tail begins only at $\mu = O(1/P)$, and its retained dimension is $m = \Theta(P)$, so generic tomography already costs $\tilde{\Omega}(P)$.

Proposition 13.10 (Compressed-coordinate compiler bound). *On the prefix family, suppose a preprocessing algorithm (a compiler) uses S raw queries to the LP coefficients and returns data from which a coherent coordinate oracle can be implemented for either the exact central primal point or a nonzero $x \geq 0$ satisfying the robust residual condition $\text{res}_p(x) \leq 1/100$ of [theorem 9.7](#). If one oracle use costs q further raw queries, then*

$$S + q = \Omega(P). \quad (600)$$

The same conclusion holds for a direction oracle whose update is nonnegative and satisfies that residual condition. In particular, the affine compressed output above, including its Newton correction, has $q = O(1)$; whenever it meets this robust approximate-feasibility condition, $S = \Omega(P)$.

Proof. Here the plateau height is $H = \Theta(\sqrt{P})$. Both the central vector and every output admitted by the robust condition have $\Theta(P)$ plateau coordinates with $\|x\| = \Theta(H\sqrt{P})$ and $\|x\|_\infty = \Theta(H)$. Query the value oracle in a uniform superposition and rotate by $x_j/(CH)$. The success probability is constant, so constant-round amplification prepares the hard primal state with $S + O(q)$ raw queries. The central and robust primal-state clauses of [theorem 9.7](#) give $\Omega(P)$, proving the claim. For a direction oracle, add the public start before applying the same argument. $\square$

The hardness on this family comes from input loading or from tomography in the retained dimension, not from the equilibrated condition number. A dual-only certificate without a primal coordinate oracle is not covered by that reduction.

13.4 Checkpoint spans and the limits of classical sketching

Lazy refresh also gives a compact representation of the dual iterate. If the checkpoint directions are $\hat{z}_{s_1}, \dots, \hat{z}_{s_R}$, then scalar reuse of the latest checkpoint direction gives the exact affine representation

$$y_t = y_0 + \sum_{r \leq R_t} \theta_{r,t} \hat{z}_{s_r}, \quad R_t \leq R. \quad (601)$$

Theorem 13.11 (Checkpoint-span trajectory oracle). *Under the hypotheses of [theorem 13.1](#), if the stored vectors have B_{bit} bits per entry and columns of A have sparsity d_c , a query to one coordinate of y_t uses $R_t + 1$ coherent reads and a query to one slack coordinate uses $O(d_c R_t)$ reads. Including reversible arithmetic, a y_t query costs $\tilde{O}(R_t B_{\text{bit}})$ gates after the public y_0 term is supplied, and a slack query costs $\tilde{O}(d_c (R_t + 1) B_{\text{bit}})$ gates. All vector loads and scalar updates over the path cost*

$$\tilde{O}\left(m B_{\text{bit}} \left(1 + \frac{GV_{\text{proj}}}{\eta}\right) + T B_{\text{bit}}\right). \quad (602)$$

Proof. Collect consecutive scalar steps using one checkpoint to obtain (601). A coordinate query reads that coordinate from each stored vector and evaluates the affine sum. A slack query repeats this for the at most d_c entries of its column. Each checkpoint loads one m -vector and each ordinary iteration changes one scalar. Sum these costs and apply [theorem 13.1](#). $\square$

This representation assumes that accepted steps meet the outer method's residual condition. A fixed refresh schedule on the exact tail can avoid per-iteration residual tests; a version that refreshes on residual tests must pay for them at every iteration.

As an explicit tail example, let B be the oriented incidence matrix of a fixed-degree balanced bipartite expander on m vertices. For each vertex i and copy $k \in [K]$, introduce u_{ik}, v_{ik} with respective columns $K^{-1/2}e_i$ and $-K^{-1/2}e_i$ and costs $a_{ik} \in [1, 2]$:

$$\begin{aligned} \min \quad & \sum_{i,k} a_{ik}(u_{ik} + v_{ik}) \\ \text{subject to} \quad & Bx + K^{-1/2} \sum_{k=1}^K (u_{\cdot k} - v_{\cdot k}) = B\mathbf{1}, \quad x, u, v \geq 0. \end{aligned} \quad (603)$$

Here $n = |E(B)| + 2mK = \Theta(mK)$; K is the copy multiplicity, unrelated to the QLS parameters $\mathcal{K}$ above. On a sufficiently small tail, $|y_i|/\sqrt{K} \leq 1/2$. The quantum and classical algorithms get the same query access (coherent and random, respectively) to (a_{ik}) and to the formula-defined sparse columns. If the costs arrive as an explicit file, loading already costs $\Omega(nB_{\text{bit}})$. In this family the direct normalization is constant, and the condition number and projective constants are independent of K , but it is not an example of quantum advantage:

Proposition 13.12 (Bounded-average sketch obstruction). *For the scaled-copy family, the inactive contribution at vertex i is*

$$D_{ii}(y) = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{(a_{ik} - y_i/\sqrt{K})^2} + \frac{1}{(a_{ik} + y_i/\sqrt{K})^2} \right], \quad (604)$$

where all summands lie in a fixed bounded interval on the promised tail. Sampling $O(\varepsilon^{-2} \log(m/\delta))$ terms independently for every vertex produces $\|\hat{D} - D\| \leq \varepsilon$ with probability at least $1 - \delta$, in $O(m\varepsilon^{-2} \log(m/\delta))$ classical queries, independently of K .

Proof. Scalar Hoeffding at each vertex and a union bound give the diagonal operator bound. Uniform invertibility of the equilibrated matrix and the resolvent identity convert it to $O(\varepsilon)$ relative direction error. $\square$

The column scale $K^{-1/2}$ is essential. The inactive block has row sparsity $2K$, column sparsity one, and entries $K^{-1/2}$, so its standard sparse-access normalization is $O(1)$. Making one hidden column contribute $\Theta(1)$ requires magnitude $\Theta(1)$, which raises the sparse-access normalizations of A_N and of the Gram matrix H_N (a product of two such factors) to

$$\alpha_{A_N}^{\text{sparse}} = \Theta(\sqrt{K}), \quad \alpha_{H_N}^{\text{sparse}} = \Theta(K). \quad (605)$$

Since H_N remains diagonal with $\|H_N\| = \Theta(1)$, a direct diagonal encoding can keep constant normalization, but assembling its value oracle from raw column access requires locating the exceptional columns.

At constant IPM accuracy, this $\tilde{O}(m)$ classical sketch matches the unavoidable readout of m coordinates at a checkpoint. More generally, well-spread bounded contributions admit classical leverage or matrix-concentration sampling. If a rare column is made influential, a standard sparse block encoding acquires a large normalization, and locating the exceptional columns separately is only Grover preprocessing. The open problem is a structured access model that gives a quantum operator with constant normalization without also giving a classical sampling distribution with bounded leverage. We know no explicit sparse LP family with both properties.

13.5 A block-angular border module

Block-angular problems give the arrowhead Newton system

$$\begin{pmatrix} D_1 & & & B_1 \\ & \ddots & & \vdots \\ & & D_N & B_N \\ B_1^\top & \cdots & B_N^\top & -D_0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} f \\ g \end{pmatrix}, \quad (606)$$

where $D_i \succ 0$, $D_0 \succeq 0$, local blocks have size at most p , and the border has size k . Classical block-angular IPMs use the same Schur-complement decomposition [25, 71, 82].

Theorem 13.13 (Explicit border and coherent local directions). *Let*

$$S = D_0 + \sum_{i=1}^N B_i^\top D_i^{-1} B_i \succ 0, \quad h = \sum_{i=1}^N B_i^\top D_i^{-1} f_i - g, \quad v = S^{-1}h.$$

Suppose one coherent block query returns (D_i, B_i, f_i) and local factorization costs $\text{poly}(p, k)$. Put $M = Np + k$, $\alpha_z = \max_i \|D_i^{-1/2} f_i\|_\infty$, and $\chi_h = \alpha_z / \|v\|_S$ for $v \neq 0$, and suppose a known upper bound $\bar{\chi}_h \geq \chi_h$ is supplied. There is a bounded-error algorithm using

$$\tilde{O}\left(\sqrt{Mk} \left(\frac{1}{\epsilon} + \frac{\bar{\chi}_h}{\eta}\right)\right) \quad (607)$$

block queries that returns an explicit $\tilde{v}$ satisfying

$$\frac{\|\tilde{v} - v\|_S}{\|v\|_S} \leq \frac{\epsilon + \eta}{1 - \epsilon}. \quad (608)$$

It also implements, with one additional block query, the coherent local oracle $|i, 0\rangle \mapsto |i, D_i^{-1}(f_i - B_i \tilde{v})\rangle$. No tomography of the Np local coordinates and no condition-number factor is required. Without a known relative bound, one must instead request absolute inverse-energy error δ_h , replacing the second term by α_z / δ_h , or pay for adaptive estimation of $\|v\|_S$.

Proof. Stack $C_i = D_i^{-1/2} B_i$ and $D_0^{1/2}$ into C , so $S = C^\top C$, and stack $z_i = D_i^{-1/2} f_i$. The tall-matrix spectral approximation and inverse-local-norm mean-estimation primitives used by

Apers–Gribbling [10] return explicit $\tilde{S}, \tilde{h}$ with $(1 - \epsilon)S \preceq \tilde{S} \preceq (1 + \epsilon)S$ and $\|\tilde{h} - h\|_{S^{-1}} \leq \eta\|v\|_S$ at the cost in (607). Cancellation with g is exactly the $\bar{\chi}_h$ factor. With $\tilde{v} = \tilde{S}^{-1}\tilde{h}$,

$$\tilde{v} - v = \tilde{S}^{-1}[\tilde{h} - h + (S - \tilde{S})v].$$

The spectral sandwich bounds the first operator by $(1 - \epsilon)^{-1}$ in S norm and the bracket by $(\epsilon + \eta)\|v\|_S$, proving (608). Block elimination gives each local formula. $\square$

The $\sqrt{N}$ dependence is optimal for block queries. Set the border data $D_0 = 0$ and $g = 0$. For scalar blocks take a known base scenario with $D_{\text{base}} = B_{\text{base}} = f_{\text{base}} = 1$ and hidden $z_i \in \{0, 1\}$ of Hamming weight zero or one with $D_i = 1$, $B_i = z_i$, and $f_i = 0$. Then $S = 1 + \sum_i z_i$, $h = 1$, and the border value is 1 or $1/2$, while the augmented system remains uniformly conditioned. Computing the border value to constant additive accuracy therefore computes the OR of the z_i , giving $\Omega(\sqrt{N})$ quantum and $\Omega(N)$ randomized block-query lower bounds.

An exact LP realizes this lower bound at the public center $x = s = e$, $y = 0$ with one dense equality row: hidden coefficient pairs are (δ, δ) or $(1, 2\delta - 1)$ with $\delta = 1/(2N)$. Its scalar normal matrix has condition number one, and its centering multiplier differs by a constant between marked and unmarked instances. This lower bound is for column or coefficient queries, not for LPs with constant row sparsity. Indeed, for normalized, uniformly conditioned bounded-degree matrices, polynomial approximation of $1/x$ implies that inverse entries decay exponentially with graph distance, so a bounded local perturbation far from a fixed border cannot change the solution at the border by a constant amount. A growing condition number, RHS norm, output support, or graph degree can avoid this obstruction.

The module is not yet an end-to-end QIPM. First, the product log barrier still has parameter $\Theta(Np)$: partial minimization preserves this parameter and can produce $-N \log x$ (parameter N) even when the projected set admits $-\log x$ (parameter one). Second, static blocks can be cached classically. Third, later iterations need a nonrecursive representation of the updated local variables. The concrete open problem is a projected barrier of parameter $\tilde{O}(k + \text{poly}(p))$ whose local center and derivatives can be recomputed from one scenario block and the explicit border. Only with such a barrier would [theorem 13.13](#) yield an $\tilde{O}(\sqrt{N} \text{poly}(p, k))$ QIPM with border output, against the $\Omega(N)$ classical block-query lower bound.

13.6 Supporting exact crossover

For a tall integer LP $Az \geq b$ in dimension d , with coefficient magnitude at most 2^ℓ , row sparsity r , and a unique nondegenerate strictly complementary optimum, let

$$D = (\sqrt{r} 2^\ell)^d, \quad M = \max\{1, \sqrt{r} 2^\ell \sqrt{d} (\sqrt{r} 2^\ell)^{d-1}\}.$$

A feasible point of objective error

$$E = \frac{1}{8D^2M} \tag{609}$$

has true row slacks at most $1/(8D)$ on active rows and at least $7/(8D)$ on inactive rows. Evaluate each slack with error at most $1/(8D)$ and mark the row when the computed value is at most $1/(2D)$; these bounds make the classification exact. Its reversible arithmetic uses $O(d(\ell + \log d))$ -bit precision (and the corresponding polynomial gate cost). Thus $\log(1/E) = O(d(\ell + \log d))$, independent of the number n of rows. Coherent enumeration identifies all binding rows in $\tilde{O}(\sqrt{nd})$ row queries; exact rational solves for the primal and dual variables and a Grover feasibility check then return an exact KKT certificate. For this sparse-certificate procedure the enumeration cost is tight, as d independent search blocks show.

This procedure is only a supporting crossover step for the Apers–Gribbling QIPM for tall LPs with a row oracle. It assumes the structural promise and coherent row access, and loading an explicit input can remove any sublinear wall-clock time. More importantly, Objois–Vladu

give an exact quantum Clarkson algorithm with $\tilde{O}(\sqrt{n} d^3)$ row queries under a comparable nondegeneracy promise (a unique optimal basis of d constraints) but without the integrality, bit-size, or strict-complementarity structure, and Dadush–Végh–Zambelli give an exact primal–dual oracle framework [50, 117]. The procedure is therefore not the best known exact LP result.

13.7 Synthesis

The positive results give three ways around the obstructions of sections 5, 9 and 12: amortize classical readout over the projective variation, compute the changing diagonals from a smaller affine iterate, or request an explicit low-dimensional border instead of the full Newton vector. Each route has a condition that cannot be dropped: lazy reuse, as bounded here, needs finite realized V_{proj} and a fixed residual tolerance; partition-free equilibration needs the tail margins, known active gaps, stable forcing, and bounded RHS and output factors; compressed output needs $m = o(n)$ and pays for its coordinate compiler; block aggregation needs a projected barrier with a small parameter and a nonrecursive representation of the local states.

Simply combining the modules is not enough. A standard feasible inexact neighborhood permits face cycling with logarithmically divergent projective variation. Proximal regularization reduces motion only by increasing the residual, and at any admissible scale it does not improve conditioning. Face repair without the selector recreates the ill-conditioned Newton inverse. Generic full-vector tomography brings back a cost proportional to the inverse precision. When a tall family’s contributions are well-spread averages, classical sketches are equally cheap; when rare columns are influential, the cost moves into the normalization or a Grover search.

This section thus proves reusable subroutines and conditional end-to-end theorems with compressed output, not an unconditional quantum advantage. An advantage still requires one explicit problem family and access model with all of the following: stable direct normalization and inverse, stable forcing and RHS preparation, finite projective variation or a barrier with a small parameter, cheap static construction, a useful compressed output, and a classical lower bound that holds against randomized sketching and preprocessing. No module above supplies any of these items without an explicitly stated assumption.

14 Structural Identities and Obstruction Tools

This section collects structural facts that the preceding lower bounds use and that apply beyond any single hard family: a symmetry identity (section 14.1), a bound on averaged SDP centers of neighboring inputs (section 14.2), a rank bound for SDP Newton directions (section 14.3), and what sparse support, bounded treewidth, and hardware locality do and do not imply. Most ingredients are classical; we state in each case which part is new.

14.1 Symmetry as virtual work

Work in a finite-dimensional Euclidean space. Let R be an orthogonal involution, let Ω be an open convex set with $R\Omega = \Omega$, and let $F : \Omega \rightarrow \mathbb{R}$ be differentiable, convex, and R -invariant. Assume also that $R^\top c = c$ and use the stationarity convention

$$A^\top y = c + \mu \nabla F(x), \quad \mu > 0. \quad (610)$$

Thus the equality term in the Lagrangian is $-y^\top (Ax - b)$. Put

$$x_- := \frac{I - R}{2}x, \quad d := x - Rx = 2x_-, \quad r_- := Ax_-. \quad (611)$$

We call r_- the source and $r_-^\top y$ the work of the antisymmetric part. Defining the source as Ax_- , rather than as a selected block of b , keeps the following statement valid when equality rows are mixed.

Theorem 14.1 (Involution work identity). *Under the preceding assumptions,*

$$\begin{aligned}
r_-^\top y &= \mu \langle x_-, \nabla F(x) \rangle \\
&= \frac{\mu}{4} \langle x - Rx, \nabla F(x) - \nabla F(Rx) \rangle \\
&= \frac{\mu}{2} D_F(Rx, x) = \frac{\mu}{2} D_F(x, Rx) \geq 0,
\end{aligned} \tag{612}$$

where $D_F(u, v) = F(u) - F(v) - \langle \nabla F(v), u - v \rangle$. If $F \in C^2(\Omega)$ and

$$\overline{H}_x := \int_0^1 \nabla^2 F(Rx + td) dt, \tag{613}$$

then, exactly,

$$r_-^\top y = \frac{\mu}{4} d^\top \overline{H}_x d = \mu x_-^\top \overline{H}_x x_-. \tag{614}$$

For a standard self-concordant barrier, set $x_+ = (x + Rx)/2$ and $\delta = \|x_-\|_{x_+} < 1$. Integrating the Hessian comparison along the chord gives the outer bounds, which are sharper than a worst-point comparison; the middle bound is a stronger lower bound:

$$\mu \left(1 - \delta + \frac{\delta^2}{3} \right) \delta^2 \leq \frac{\mu \delta^2}{1 + \delta} \leq r_-^\top y \leq \frac{\mu \delta^2}{1 - \delta}. \tag{615}$$

Proof. The invariant objective does no antisymmetric work:

$$\langle x_-, c \rangle = \frac{1}{2} \langle x, c - R^\top c \rangle = 0.$$

Pairing (610) with x_- gives the first equality in (612). Differentiating $F(Rx) = F(x)$ gives $\nabla F(Rx) = R \nabla F(x)$. Since $Rd = -d$,

$$\langle d, \nabla F(Rx) \rangle = \langle Rd, \nabla F(x) \rangle = -\langle d, \nabla F(x) \rangle,$$

which proves the second equality. Gradient monotonicity proves nonnegativity. Invariance also gives

$$D_F(Rx, x) = D_F(x, Rx) = \langle \nabla F(x), x - Rx \rangle,$$

so the Bregman equalities follow. The fundamental theorem of calculus on the chord gives (614).

For the last claim, parameterize the chord as $x_+ + ux_-$, $-1 \leq u \leq 1$. Self-concordant Hessian comparison bounds the curvature in direction x_- along the chord by

$$(1 - |u|\delta)^2 \delta^2 \leq x_-^\top \nabla^2 F(x_+ + ux_-) x_- \leq (1 + |u|\delta)^2 \delta^2.$$

The relevant chord averages are

$$\int_0^1 (1 - u\delta)^2 du = 1 - \delta + \delta^2/3, \quad \int_0^1 (1 - u\delta)^{-2} du = (1 - \delta)^{-1},$$

so these constants are exact integrals of the comparison bounds and are generally not attained. For the stronger lower bound, put

$$\phi(u) = x_-^\top \nabla^2 F(x_+ + ux_-) x_-.$$

Self-concordance gives $|(\phi^{-1/2})'| \leq 1$ and $\phi(0) = \delta^2$, hence $\phi(u) \geq \delta^2/(1 + u\delta)^2$ for $0 \leq u \leq 1$. Invariance makes ϕ even and $r_-^\top y = \mu \int_0^1 \phi(u) du$, which proves the middle lower bound. It strictly dominates the chord-comparison lower constant for $0 < \delta < 1$, since

$$\frac{1}{1 + \delta} - \left(1 - \delta + \frac{\delta^2}{3} \right) = \frac{\delta^2(2 - \delta)}{3(1 + \delta)} > 0.$$

□

The identity does not assume that A has full row rank. If $\tilde{y}$ is another multiplier, then $A^\top(\tilde{y} - y) = 0$ and hence $(Ax_-)^\top(\tilde{y} - y) = 0$. Moreover, under any invertible row change

$$A' = UA, \quad y' = U^{-\top}y, \quad r'_- = Ur_-, \quad (r'_-)^\top y' = r_-^\top y. \quad (616)$$

Only the source-multiplier pairing is invariant; an ill-conditioned U can shrink a multiplier norm while enlarging the source or the input normalization. If the rows split as $A_+R = A_+$ and $A_-R = -A_-$ and $Ax = b = (b_+, b_-)$, then $r_- = (0, b_-)$ and the pairing is the familiar $b_-^\top y_-$. Arbitrary row mixing preserves (616), but not this coordinate form.

Two special cases give the constants used in earlier sections. For paired positive variables $x = (u, v)$, $R(u, v) = (v, u)$, and $F = -\sum_j (\log u_j + \log v_j)$, put $d = u - v$ and $q = u + v$. Then

$$r_-^\top y = \frac{\mu}{2} \sum_j \frac{d_j^2}{u_j v_j} = \sum_j \frac{2\mu d_j^2}{q_j^2 - d_j^2}. \quad (617)$$

This is exactly (395); theorem 10.7 proves this paired-coordinate case, not a separate general identity.

For the log-determinant barrier on $\mathbb{S}_{++}^r$, let $\mathcal{R}(X) = QXQ$ for a symmetric orthogonal Q , and set $X_- = (X - QXQ)/2$. With the Frobenius pairing,

$$(\mathcal{A}X_-)^\top y = \frac{\mu}{2} [\text{tr}(QXQX^{-1}) - r] \quad (618)$$

$$= \mu \int_0^1 \|Z_t^{-1/2} X_- Z_t^{-1/2}\|_F^2 dt, \quad Z_t = QXQ + t(X - QXQ). \quad (619)$$

The bracket in the first line is half the symmetrized Stein loss, so the work is $\mu/4$ times that loss; it is not the Jensen–Bregman LogDet divergence (S-divergence). The second line is the exact integrated-Hessian formula and does not assume that X and QXQ commute. If $\ell I \preceq X \preceq UI$, the work lies between $\mu\|X_-\|_F^2/U^2$ and $\mu\|X_-\|_F^2/\ell^2$. Here U is a spectral endpoint, unrelated to the row-change matrix in (616).

The same argument applies to a Newton step from a symmetric start $x^0 = Rx^0$. If $G_0 \succ 0$ commutes with R and

$$A\Delta x = r_p, \quad G_0\Delta x - A^\top\Delta y = h_+, \quad Rh_+ = h_+, \quad (620)$$

then, for $\Delta x_- = (I - R)\Delta x/2$,

$$\boxed{(\mathcal{A}\Delta x_-)^\top \Delta y = \Delta x_-^\top G_0 \Delta x_- \geq 0.} \quad (621)$$

This recovers (396). A nonzero antisymmetric right-hand-side component h_- adds $-\langle \Delta x_-, h_- \rangle$, so the pairing can become negative.

More generally, let a finite or compact group $\mathcal{G}$ act orthogonally, and assume $g\Omega = \Omega$ for every $g \in \mathcal{G}$. Let

$$P_0 = \int_{\mathcal{G}} g \, d\nu(g), \quad x_0 = P_0 x, \quad x_\perp = (I - P_0)x \quad (622)$$

be the Reynolds projection. The orbit is compact, and Carathéodory's theorem shows that its convex hull is a compact subset of the convex domain; thus $x_0 \in \Omega$ and the segment from x_0 to x stays in Ω .

Theorem 14.2 (Compact-group work). *If F and c are $\mathcal{G}$ -invariant and (610) holds, then*

$$(\mathcal{A}x_\perp)^\top y = \mu \langle x_\perp, \nabla F(x) - \nabla F(x_0) \rangle \geq 0. \quad (623)$$

When the Hessian exists, the complete chain is

$$\begin{aligned}
(Ax_\perp)^\top y &= \mu \langle x_\perp, \nabla F(x) - \nabla F(x_0) \rangle \\
&= \mu \int_0^1 \langle x_\perp, \nabla^2 F(x_0 + tx_\perp) x_\perp \rangle dt \\
&\geq 0.
\end{aligned} \tag{624}$$

At a group-fixed Newton start, suppose $G_0 \succeq 0$ commutes with the action and

$$A\Delta x = r_p, \quad G_0\Delta x - A^\top \Delta y = h_0, \quad gh_0 = h_0 \quad (g \in \mathcal{G}). \tag{625}$$

Define $\Delta x_\perp = (I - P_0)\Delta x$. Then

$$(A\Delta x_\perp)^\top \Delta y = \Delta x_\perp^\top G_0 \Delta x_\perp \geq 0. \tag{626}$$

Proof. Both c and $\nabla F(x_0)$ are fixed by the group and hence orthogonal to $x_\perp$. Pair stationarity with $x_\perp$, then apply gradient monotonicity between x_0 and x . Integration gives (624). The Newton identity follows by pairing the stationarity equation with the nontrivial-representation component; G_0 preserves the orthogonal representation decomposition. $\square$

For coordinate permutations, P_0 replaces each orbit by its average; for the logarithmic barrier the work on an orbit O is $\mu(\bar{x}_O \sum_{i \in O} x_i^{-1} - |O|)$. The simplex gives the same formula with $\bar{x} = 1/n$. For the standard $SO(k)$ action and the unit-ball barrier, it is $2\mu\|z\|^2/(1 - \|z\|^2)$. These examples explain the permutation and rotation gadgets excluded in [section 10.6](#).

Remark 14.3 (Scope and provenance). The sign reverses under the opposite multiplier convention. An asymmetric objective or Newton right-hand side contributes a linear term of uncontrolled sign. Convexity without strict curvature gives only nonnegativity. The scalar identity extends to a nonorthogonal linear involution by using $\nabla F(Rx) = R^{-\top} \nabla F(x)$, but the Euclidean orthogonal decomposition does not. Finally, arbitrary edge transports (such as the hidden signs or group elements on edges in [section 11](#)) need not come from a single public, that is input-independent, symmetry action on all edges at once, and then there may be no public Reynolds projection. Even when they come from a common representation, the Reynolds identity gives no conservation law across a cut; the signed and gain incidence and holonomy analysis of [section 11](#) is a separate argument.

The divergence in (612) is the classical symmetrized Bregman divergence [32, 115]; Reynolds reduction and symmetric criticality are likewise classical [119]. Our contribution is narrower: the source (611), whose pairing with the multiplier is invariant under row mixing, is a template for obstruction arguments against sparse QIPM gadgets.

14.2 Central mixtures in semidefinite programming

This subsection extends the scalar Kantorovich argument of [section 10.2](#) to SDPs without commutativity assumptions, by bounding the KKT residual and centrality error of an average of neighboring centers for the base input.

Consider SDPs with common C, b and input-dependent measurement maps $\mathcal{A}_\xi$, with matrix order $r \geq 1$ and $N \geq 1$ neighboring inputs. Fix one Frobenius-isometric $\text{svec} : \mathbb{S}^r \rightarrow \mathbb{R}^{r(r+1)/2}$. All sparsity and coefficient bounds below refer to the resulting matrix A_ξ , not to the unvectorized data. At a base input ς , suppose every one-bit neighbor $\varsigma^{(i)}$ has an exact center at the same $\mu > 0$:

$$\mathcal{A}_{\varsigma^{(i)}}(X_i) = b, \quad \mathcal{A}_{\varsigma^{(i)}}^*(y_i) + S_i = C, \quad S_i = \mu X_i^{-1}, \quad X_i \succ 0. \tag{627}$$

For weights $w_i \geq 0$, $\sum_i w_i = 1$, write

$$\begin{aligned}
X_w &= \sum_i w_i X_i, \quad y_w = \sum_i w_i y_i, \quad S_w = \sum_i w_i S_i, \\
D_i &= X_i - X_w, \quad Z_w = X_w^{1/2} (S_w / \mu) X_w^{1/2}.
\end{aligned} \tag{628}$$

Lemma 14.4 (Matrix multiplicative variance). *The following identity holds:*

$$\boxed{Z_w - I = X_w^{-1/2} \left(\sum_i w_i D_i X_i^{-1} D_i \right) X_w^{-1/2}.} \quad (629)$$

Consequently $Z_w \succeq I$, with equality exactly when all positively weighted X_i equal X_w .

Proof. Expanding the middle term and using $\sum_i w_i X_i = X_w$ gives

$$\sum_i w_i D_i X_i^{-1} D_i = X_w \left(\sum_i w_i X_i^{-1} \right) X_w - X_w.$$

Congruence by $X_w^{-1/2}$ proves the identity. Every summand $w_i D_i X_i^{-1} D_i$ is positive semidefinite, and a positive weighted sum of them vanishes only when each corresponding D_i vanishes. $\square$

Identity (629) is an elementary exact expansion of the gap between the arithmetic and harmonic means; we do not claim it as a new matrix inequality.

Theorem 14.5 (Noncommutative central sandwich). *If $0 < m \leq M$, $mI \preceq X_i \preceq MI$ for every i , and $R = M/m$, then*

$$\boxed{I \preceq Z_w \preceq K(R)I, \quad K(R) = \frac{(R+1)^2}{4R}.} \quad (630)$$

The constant is sharp, already for an equal scalar mixture of m and M . The matrices X_i need not commute.

Here and below $R = M/m$ is a spectral ratio, unrelated to the involution R in [section 14.1](#).

Proof. The lower bound is [lemma 14.4](#). Functional calculus, applied to one X_i at a time, gives

$$X_i + mMX_i^{-1} \preceq (m+M)I. \quad (631)$$

Averaging and congruencing by $X_w^{1/2}$ yields

$$Z_w \preceq \frac{(m+M)X_w - X_w^2}{mM}.$$

Every eigenvalue λ of X_w is in $[m, M]$, and

$$\frac{\lambda(m+M-\lambda)}{mM} \leq \frac{(m+M)^2}{4mM} = K(R).$$

Only X_i and X_w individually entered functional calculus, so no commutation was assumed. $\square$

The variance identity also gives dimension-free bounds that refine the spectral-ratio bound. If only $X_i \succeq mI$ with $m > 0$ is assumed, then

$$\|Z_w - I\|_F \leq \frac{1}{m^2} \sum_i w_i \|D_i\|_F^2, \quad (632)$$

$$\|Z_w - I\|_{\text{op}} \leq \frac{1}{m^2} \sum_i w_i \|D_i\|_{\text{op}}^2. \quad (633)$$

Indeed, the two inverse factors come separately from the outer congruence and X_i^{-1} in (629). Multiplying by μ gives the corresponding complementarity-residual bounds. They remain valid after the standard recentering: if

$$\mu_w = \frac{\langle X_w, S_w \rangle}{r} = \mu\alpha, \quad \alpha = \frac{\text{tr } Z_w}{r} \geq 1, \quad (634)$$

then

$$\mu_w^{-1} X_w^{1/2} S_w X_w^{1/2} - I = \frac{(Z_w - I) - [\text{tr}(Z_w - I)/r]I}{\alpha}.$$

Removing the trace component is an orthogonal projection in Frobenius norm, and it does not increase the operator norm because $Z_w - I \succeq 0$. We call $\|Z_w - I\|_{\text{op}}$ the common- μ operator width and $\|\mu_w^{-1} X_w^{1/2} S_w X_w^{1/2} - I\|_{\text{op}}$ the point-centered operator width; Frobenius widths use $\|\cdot\|_F$.

The following objective identity does not require the average to be feasible. Every point in (627) has algebraic gap $r\mu$, so linearity gives

$$\boxed{\langle C, X_w \rangle - b^\top y_w = r\mu.} \quad (635)$$

By contrast, $\langle X_w, S_w \rangle - [\langle C, X_w \rangle - b^\top y_w] = \mu \text{tr}(Z_w - I)$ is nonnegative. Under the common sandwich $0 < m \leq M$ and $mI \preceq X_i \preceq MI$, it is at most $r[K(M/m) - 1]\mu$. This difference is an arithmetic-harmonic Jensen gap at a point that is generally infeasible for the base input, not a consequence of weak duality.

We next bound the KKT residual of the average for the base input. Let s_r, s_c bound the row and column sizes of one fixed support containing the supports of all A_ξ . Let M_{dep} count the input-dependent positions; each is a function of one designated bit $\varsigma_{\ell(r,c)}$ (a bit may control several positions) and has magnitude at most $B \geq 0$. Assume that every coordinate of $\text{svec}(X_i), y_i, \text{svec}(S_i)$ has magnitude at most $H \geq 0$. Set

$$s_{\text{KKT}} = \max\{s_r, s_c + 1\}, \quad B_{\text{KKT}} = \max\{B, 1\}. \quad (636)$$

For the uniform average of all N one-bit neighbors, the incidence count of [theorem 10.1](#), applied to the primal and dual blocks, gives

$$\left\| \begin{bmatrix} \mathcal{A}_\varsigma(X_w) - b \\ \text{svec}(\mathcal{A}_\varsigma^*(y_w) + S_w - C) \end{bmatrix} \right\|_2 \leq \frac{2B_{\text{KKT}}H\sqrt{2s_{\text{KKT}}M_{\text{dep}}}}{N}. \quad (637)$$

Keeping the dual multipliers signed gives the stronger weighted estimate

$$\left\| \begin{bmatrix} \mathcal{A}_\varsigma(X_w) - b \\ \text{svec}(\mathcal{A}_\varsigma^*(y_w) + S_w - C) \end{bmatrix} \right\|_2^2 \leq 4(s_r + s_c)B^2H^2 \sum_i M_i w_i^2, \quad (638)$$

where M_i counts dependent coefficient positions assigned to bit i . For uniform weights this is $4(s_r + s_c)B^2H^2M_{\text{dep}}/N^2$, which implies (637). The PSD variables need no coordinatewise sign split: convexity preserves the cone, and the identity block on S is input independent. If a coefficient depends on several bits, the degree factor of [proposition 10.2](#) enters, as in [section 10.2](#).

Proposition 14.6 (SDP mixture phase boundary). *Suppose $0 < m \leq M$, every neighboring center obeys the common sandwich $mI \preceq X_i \preceq MI$, and $R = M/m$. Assume the one-bit-local dependence stated before (637), and that all N bits are sensitive. Assume also that the selected neighboring centers share a mixture-stable output (one preserved under averaging) that is wrong for the base input. This output may be given either by a fixed affine score with a uniform signed margin, or by a fixed symmetric contraction O such that*

$$f(\xi)\langle O, X^\xi \rangle \geq \gamma \text{tr } X^\xi \quad (639)$$

for an output $f \in \{-1, +1\}$ and a positive margin $0 < \gamma \leq 1$. The contraction condition means $-I \preceq O \preceq I$; the binary measurement has effects $(I + O)/2$ and $(I - O)/2$ on $X/\text{tr } X$. Suppose that, for the base input, the correct output is required from every positive-definite triple with stacked primal and dual residual (the left-hand side of (637)) at most $\varepsilon \geq 0$, algebraic gap $r\mu$, and common- μ or point-centered operator width at most θ . This requirement can be met only if

$$\boxed{\frac{8s_{\text{KKT}}B_{\text{KKT}}^2H^2M_{\text{dep}}}{N^2} > \varepsilon^2 \quad \text{or} \quad \frac{(R-1)^2}{4R} > \theta.} \quad (640)$$

For a Frobenius neighborhood of width η_F , the second alternative is

$$\sqrt{r} \frac{(R-1)^2}{4R} > \eta_F, \quad (641)$$

where the variance bound (632) may replace the left-hand side when it is smaller. The residual alternative can be sharpened to $4(s_r + s_c)B^2H^2M_{\text{dep}}/N^2 > \varepsilon^2$. Under point centering the algebraic gap remains $r\mu$, not $r\mu_w$.

Proof. Equation (637) bounds the residual, (635) gives the algebraic gap, and theorem 14.5 gives operator width at most $K(R) - 1 = (R-1)^2/(4R)$ under either centering convention. The affine score is preserved by averaging. Under (639), trace weighting gives

$$-f(\varsigma) \text{tr} \left(O \frac{X_w}{\text{tr } X_w} \right) \geq \gamma,$$

so the normalized density matrix has constant bias for the wrong sign. If neither strict inequality in (640) held, the average would meet all tolerances while giving the wrong output. Finally, $\|Z_w - I\|_F \leq \sqrt{r}\|Z_w - I\|_{\text{op}}$ gives (641); the same projection argument used after (633) covers point centering. $\square$

Since

$$K(R) - 1 = \sinh^2 \left(\frac{\log R}{2} \right), \quad (642)$$

under a common sandwich $0 < m \leq M$ and $mI \preceq X_i \preceq MI$, with $R = M/m \geq 1$, the common- μ Frobenius width of every mixture satisfying this sandwich is at most $\eta_F \geq 0$ if and only if

$$\log R \leq 2 \text{arsinh}(\sqrt{\eta_F} r^{-1/4}) = O(r^{-1/4}). \quad (643)$$

Indeed, the equal mixture $X_1 = mI$, $X_2 = MI$ attains $Z_w = K(R)I$ in every dimension. The same ratio criterion is sufficient for point centering, but this example has zero point-centered width and does not show necessity there. A nonasymptotic consequence of the criterion is $\log R \leq 2\sqrt{\eta_F} r^{-1/4}$. The actual Frobenius variance can be small even when the global spectral ratio is large.

The common sandwich cannot be dropped. Individual condition numbers cannot replace its ratio: the scalars $X_1 = 1$, $X_2 = 9$ each have condition number one, but their equal mixture has $\|Z_w - I\|_{\text{op}} = 16/9$.

Remark 14.7 (Seven scope limits). The common sandwich $mI \preceq X_i \preceq MI$ is a strong assumption. Operator-norm control loses a generic factor $\sqrt{r}$ in Frobenius norm. The factors m^{-2} in the additive variance bounds are a real limitation. Every sparsity and height parameter depends on the chosen Frobenius-isometric svec. Output stability is a separate hypothesis, and (639) says nothing about normalized svec amplitudes. The result only asserts that a bad mixed point exists; it is not a query lower bound without a reduction. Finally, (635) is an algebraic identity at an infeasible point.

The arithmetic–harmonic operator inequality and its Kantorovich reverse are classical [8, 88, 95, 100]. The closest related idea is warm starting for conic IPMs [42], which moves one old solution to a centered start for a perturbed problem. Here, instead, centers are averaged across instances and combined with a count of input-dependent coefficients, simultaneous primal and dual residuals, and a wrong output. The scalar LP case is [proposition 10.3](#); the present result is its operator version for SDPs.

Formal verification. The Lean development in `formal/QipmFormal/SDPMixture/` verifies the finite matrix-mixture argument in this subsection. It includes the variance identity and equality case, the sharp noncommutative sandwich, operator and Frobenius variance bounds, trace recentering, and the algebraic-gap discrepancy. A constructed Frobenius-isometric symmetric vectorization connects actual SDP measurement maps and their adjoints to weighted, uniform, and multibit residual bounds. The soundness results combine these properties with explicit output requirements (an affine score or a trace-normalized observable) under both centering conventions. The scalar ratio criterion, endpoint sharpness, and the distinction between the centering conventions are also checked. The formalized claims, their qualifications, and the reproduction command are listed in `formal/SDP_MIXTURE.md`. The only axioms used are `propext`, `Classical.choice`, and `Quot.sound`. The development does not assert that neighboring centers exist for arbitrary inputs and does not formalize an IPM trajectory or a quantum query reduction.

14.3 Rank required for SDP complementarity progress

A low-rank SDP optimum does not imply that an accurate Newton direction has low rank. We state the bound in Nesterov–Todd (NT) coordinates. At an exact center $XS = \mu I$, let W be the NT scaling, so $WSW = X$, put $P = W^{-1/2}$, and define

$$U = P\Delta XP^\top, \quad V = P^{-\top}\Delta SP^{-1}. \quad (644)$$

Here U is the NT-scaled primal direction, unrelated to the matrices denoted U above. We write the symmetric scaled complementarity equation as

$$\sqrt{\mu}(U + V) = -(1 - \sigma)\mu I + \tilde{R}. \quad (645)$$

This equation defines $\tilde{R}$, so no residual convention needs to be chosen.

Theorem 14.8 (Sharp rank–progress barrier). *Let $0 \leq \sigma < 1$. For $1 \leq p < \infty$, if*

$$\|\tilde{R}\|_{S_p} \leq \theta(1 - \sigma)\mu n^{1/p}, \quad 0 \leq \theta < 1, \quad (646)$$

then

$$\boxed{\text{rank}(\Delta X) + \text{rank}(\Delta S) \geq \lceil (1 - \theta^p)n \rceil}. \quad (647)$$

For spectral error,

$$\|\tilde{R}\|_{\text{op}} < (1 - \sigma)\mu \implies \text{rank}(\Delta X) + \text{rank}(\Delta S) \geq n. \quad (648)$$

The result is invariant under primal congruence and holds blockwise on a product PSD cone.

Proof. Congruence preserves rank, so, with $k = \text{rank}(\Delta X) + \text{rank}(\Delta S)$,

$$\text{rank}(U + V) \leq \text{rank}(U) + \text{rank}(V) = k.$$

Equation (645) makes $\tilde{R}$ the error of approximating $(1 - \sigma)\mu I$ by a rank-at-most- k matrix. Eckart–Young–Mirsky gives

$$\|\tilde{R}\|_{S_p} \geq (1 - \sigma)\mu(n - k)_+^{1/p}.$$

Combining this with (646) proves (647). If $k < n$, a kernel vector of $U + V$ sees residual exactly $(1 - \sigma)\mu$, proving (648). $\square$

The bound is sharp as a matrix statement: take $U = -(1 - \sigma)\sqrt{\mu} \text{diag}(I_k, 0)$ and $V = 0$. The residual is precisely $(1 - \sigma)\mu \text{diag}(0, I_{n-k})$. The approximation theorem used in the proof is classical [58, 104]; what is new here is its interpretation as a rank requirement for complementarity progress.

14.3.1 A path example and the choice of norm

Let L be the Laplacian of the n -vertex path and consider

$$\min \langle L, X \rangle \quad \text{subject to} \quad \text{tr } X = 1, \quad X \succeq 0. \quad (649)$$

For $u = \mathbf{1}/\sqrt{n}$, the unique optimum is $X^* = uu^\top$, while the dual optimum has slack $S^* = L$; hence strict complementarity holds. Its exact central path is

$$y(t) = -t, \quad S(t) = L + tI, \quad Z(t) = \text{tr}(S(t)^{-1}), \quad \mu(t) = Z(t)^{-1}, \quad X(t) = \mu(t)S(t)^{-1}. \quad (650)$$

With $T(t) = \text{tr}(S(t)^{-2})$, differentiation gives

$$\dot{S} = I, \quad \dot{\mu} = \mu^2 T, \quad \dot{X} = \dot{\mu}S^{-1} - \mu S^{-2}. \quad (651)$$

If $0 = \lambda_0 < \lambda_1 \leq \dots$ are the eigenvalues of L , the eigenvalues of $\dot{X}$ are

$$\frac{\mu}{(t + \lambda_i)^2} \left[\frac{T}{Z}(t + \lambda_i) - 1 \right]. \quad (652)$$

The zero-mode entry is negative. Every other entry is positive whenever $t \leq \lambda_1/(2\sqrt{n})$, because $T/Z > 1/(t + \lambda_1)$ follows from $\lambda_1^2 + \lambda_1 t - (n - 1)t^2 > 0$. Thus both tangent matrices are full rank arbitrarily close to a rank-one optimum. The exact Newton direction targeting $\sigma\mu$ is the scalar multiple

$$(\Delta X, \Delta S, \Delta y) = -\frac{(1 - \sigma)\mu}{\dot{\mu}}(\dot{X}, \dot{S}, \dot{y}). \quad (653)$$

Full algebraic rank alone says little about approximation, so consider the NT-scaled spectrum. Put $d_i = t + \lambda_i$ and $\beta = T/Z$. Since $W = \sqrt{\mu}S^{-1}$ on (650), the NT-scaled tangent eigenvalues are

$$u_i = \sqrt{\mu}(\beta - d_i^{-1}), \quad v_i = \sqrt{\mu}d_i^{-1}, \quad u_i + v_i = \sqrt{\mu}\beta. \quad (654)$$

If $t \leq \lambda_1/(4\sqrt{n})$, then $\beta d_i \geq 3$ for $i \geq 1$:

$$\beta \lambda_1 \geq \frac{\lambda_1/t}{1 + (n - 1)t/\lambda_1} > 3.$$

Hence

$$\frac{2}{3}\sqrt{\mu}\beta < u_i < \sqrt{\mu}\beta \quad (i \geq 1), \quad (655)$$

and every rank- $k < n - 1$ approximation obeys

$$\|U - U_k\|_F \geq \frac{2}{3}\sqrt{\mu}\beta\sqrt{n - 1 - k}. \quad (656)$$

Using the path trace identities below to control the exceptional zero mode, any fixed relative Frobenius target error below $2/3$ therefore requires rank $\Omega(n)$ in the NT metric. More generally, for any fixed target error below one, shrinking the constant in $t \leq \lambda_1/(4\sqrt{n})$ makes all $n - 1$ nonexceptional singular values an arbitrarily large fixed fraction of $\sqrt{\mu}\beta$ and again forces rank $\Omega(n)$.

The choice of coordinates matters. For the path,

$$\lambda_k = 4 \sin^2 \frac{\pi k}{2n}, \quad \text{tr } L^+ = \frac{n^2 - 1}{6}, \quad \text{tr}(L^+)^2 = \frac{2n^4 + 5n^2 - 7}{180}. \quad (657)$$

As $t \downarrow 0$, the raw tangent spectrum tends to $\{-\text{tr } L^+, \lambda_1^{-1}, \dots, \lambda_{n-1}^{-1}\}$, whose stable rank tends to $7/5$ and whose best rank-one relative Frobenius error tends to $\sqrt{2/7}$. At $t = \lambda_1/(2\sqrt{n})$, by contrast, the $n - 1$ nonzero-mode NT singular values are asymptotically flat. This regime occurs late on the path and is ill conditioned:

$$t = \frac{\pi^2}{2n^{5/2}}(1 + o(1)), \quad \text{cond}(L + tI) = \frac{8}{\pi^2}n^{5/2}(1 + o(1)). \quad (658)$$

For smaller t only the lower bound $\Omega(n^{5/2})$ is uniform.

14.3.2 Ways around the rank barrier

The rank theorem rules out reading out the NT direction for a single PSD block as a low-rank factorization (low-rank *factor tomography*). It does not prove an $\Omega(n^2)$ storage bound, and the path example shows three ways around it.

First, chordal conversion replaces the single block by edge blocks

$$Y_i = \begin{pmatrix} x_{ii} & x_{i,i+1} \\ x_{i,i+1} & x_{i+1,i+1} \end{pmatrix} \succeq 0 \quad (1 \leq i < n), \quad (659)$$

with overlap equations $(Y_i)_{22} = (Y_{i+1})_{11}$ and trace equation

$$(Y_1)_{11} + \sum_{i=1}^{n-1} (Y_i)_{22} = 1. \quad (660)$$

The objective is the sum of inner products with $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Restriction of a global PSD matrix is feasible in this formulation. Conversely, consistent PSD edge matrices have a PSD completion because the path is chordal: construct Gram vectors successively, adding one orthogonal coordinate per new vertex. Objective and trace use only specified entries, so the reformulation is exact. Its barrier Hessian has constant-size blocks; elimination gives a tridiagonal normal matrix with one dense border. A tridiagonal factorization and one scalar Schur complement use $O(n)$ memory and arithmetic per Newton solve. The conservative short-step total is $O(n^{3/2} \log(n/\epsilon))$ arithmetic and $O(n)$ memory. This example illustrates the formulation dependence emphasized in the chordal-conversion literature [146].

Second, the tangent itself has the implicit resolvent representation (651). More generally, accurate low-rank SDP methods may use low-rank endpoint structure to precondition or reformulate a dense direction rather than factorizing that direction [21, 44]. These approaches are consistent with theorem 14.8.

Third, a matrix of high approximate rank can still have an easily prepared amplitude encoding. Writing $a = \sqrt{\mu}\beta$, the path estimates give

$$\|U/a - (I - uu^\top)\|_{\text{op}} = O(n^{-1/2}), \quad \|U/a - (I - uu^\top)\|_F = O(1). \quad (661)$$

After normalization, $\text{vec}(U)$ is $O(n^{-1/2})$ -close to

$$\frac{|I\rangle - |uu^\top\rangle}{\sqrt{n-1}}. \quad (662)$$

For $u = \mathbf{1}/\sqrt{n}$, $|I\rangle/\sqrt{n}$ is a maximally entangled index state and $|uu^\top\rangle$ is a product of uniform states. A two-term LCU prepares (662) with constant success probability and $\tilde{O}(\log n)$ gates. The theorem therefore makes no state-preparation lower-bound claim.

Analyticity of the central path at the boundary under strict complementarity is classical [74].

14.4 Sparse support and sparse-system embeddings

The next results concern standard-form complementarity equations, not arbitrary optimization algorithms. At an exact LP center $XSe = \mu e$, let $0 \leq \sigma < 1$. An inexact centering direction satisfies

$$S\Delta x + X\Delta s = -(1 - \sigma)\mu e + r, \quad \|r\|_2 \leq \eta\mu. \quad (663)$$

Let $\text{psupp}(\Delta x, \Delta s)$ be the set of indices i with $\Delta x_i \neq 0$ or $\Delta s_i \neq 0$.

Theorem 14.9 (Complementarity support barrier). *For $0 \leq \sigma < 1$, every direction satisfying (663) obeys*

$$|\text{psupp}(\Delta x, \Delta s)| \geq n - \frac{\eta^2}{(1 - \sigma)^2}. \quad (664)$$

For a short step $1 - \sigma = a/\sqrt{n}$ with $\eta < a$, this is $[1 - (\eta/a)^2]n$. If the direction is reconstructed from a k -sparse latent vector z as $(\Delta x, \Delta s) = Bz$, and each column of B is nonzero in at most g complementarity pairs $(\Delta x_i, \Delta s_i)$ (its pair support), then

$$gk \geq n - \frac{\eta^2}{(1 - \sigma)^2}. \quad (665)$$

Proof. Let Z contain the indices at which both update coordinates vanish. For each $i \in Z$, (663) forces $r_i = (1 - \sigma)\mu$. Hence

$$|Z|(1 - \sigma)^2\mu^2 \leq \|r\|_2^2 \leq \eta^2\mu^2,$$

which proves (664). The support of Bz is contained in the union of the pair supports of at most k columns, of total size at most gk , proving (665). $\square$

The next LP has sparse data and a sparse optimum, but its central direction has full support and is incompressible. For $n \geq 3$, consider

$$\min x_2 \quad \text{subject to} \quad x_1 + x_2 = 1, \quad x_{i-1} - x_i = 0 \quad (3 \leq i \leq n), \quad x \geq 0. \quad (666)$$

Its equality matrix has maximum row and column sparsity two (the endpoint columns have only one nonzero), and its unique optimum is e_1 . The feasible segment is $x(t) = (1 - t, t, \dots, t)$, and, for $0 < t < (n - 1)/n$, the central parameter is

$$\mu(t) = \frac{t(1 - t)}{n - 1 - nt}. \quad (667)$$

Indeed, eliminating the equalities gives the one-variable barrier $t - \mu[\log(1 - t) + (n - 1)\log t]$; its stationarity equation is the displayed formula. At $t = 1/2$, every nonzero central tangent is proportional to $(-1, 1, \dots, 1)$. Its normalized best k -sparse approximation error is exactly $\sqrt{1 - k/n}$; a sparse optimum therefore does not make the central direction compressible.

Moreover, every positive unit vector is a central-path tangent of some bound-constrained QP with a one-sparse optimum. Given any $u \in \mathbb{R}_{++}^n$ with $\|u\|_2 = 1$, choose $\mu_0 > 0$ so that $u_i^{-2} > 4\mu_0$ and set

$$c_1 = -\sqrt{u_1^{-2} - 4\mu_0}, \quad c_i = \sqrt{u_i^{-2} - 4\mu_0} \quad (i \geq 2). \quad (668)$$

For the bound-constrained QP

$$\min_{x \geq 0} \frac{1}{2}\|x\|_2^2 + c^\top x, \quad (669)$$

the unique optimum is the one-sparse vector $(-c_1)e_1$, with strict complementarity. Central stationarity gives $s = x + c$ and

$$x_i(\mu) = \frac{-c_i + \sqrt{c_i^2 + 4\mu}}{2}, \quad x'_i(\mu) = \frac{1}{\sqrt{c_i^2 + 4\mu}}, \quad (670)$$

so $x'(\mu_0) = u$. The row-scaled complementarity Jacobian is $\text{diag}(2x(\mu_0) + c) = \text{diag}(1/u_i)$, which is one-sparse and has bounded condition number for near-uniform u . Reading out a unit vector u to ℓ_2 error ϵ from a unitary preparing it needs $\tilde{\Omega}(n/\epsilon)$ queries, already for a hard family of near-uniform vectors [131]; by this construction, the same bound applies to reading out the central-path tangent of (669) from its state-preparation unitary. It applies only to algorithms that access the solution through that unitary; the coefficients (668) themselves reveal u .

Sections 9.5.1 and 9.7 give lower bounds in the original coordinates under reduced condition number one, and section 14.5 states the limitation on QRAM-write claims.

14.4.1 A query-preserving OSS progress step

We next ask whether standard sparse-QLS lower bounds apply to the linear system of a valid QIPM Newton step. They do: we embed a sparse linear system into a single OSS step. Let $H \in \mathbb{R}^{n \times n}$ be real symmetric, invertible, with $\|H\|_2 = 1$. Assume sparse oracle access to rows and columns: for each row and column the oracle lists at most d candidate positions (slots), which may hold zeros; we call d the declared slot bound. Let $\|q\|_2 = 1$. Fix $\theta \in (0, 1)$, choose

$$0 < \delta \leq \min\{\theta, 1/2\}, \quad 0 \leq \alpha \leq \delta/4, \quad a = \delta q/\sqrt{2}, \quad D = \text{diag}(a), \quad \sigma = 1 - \alpha/\sqrt{n}. \quad (671)$$

Consider

$$\min c^\top x \quad \text{subject to} \quad Ax = 0, \quad x \geq 0, \quad A = \begin{bmatrix} H & -H \end{bmatrix}, \quad c = (e + a, e - a). \quad (672)$$

At $x^0 = e_{2n}, y^0 = 0, s^0 = c$, the complementarity average is one and the N_2 residual norm is δ , while the N_∞ residual is at most $\delta/\sqrt{2}$. Thus the point belongs to both radius- θ neighborhoods.

Theorem 14.10 (Feasible OSS embedding). *With null-space basis $V = (I, I)^\top$ and $r = 1 - \sigma$, the feasible OSS step is*

$$\underbrace{\begin{bmatrix} -H & I + D \\ H & I - D \end{bmatrix}}_{M_{\text{OSS}}} \begin{pmatrix} \Delta y \\ \lambda \end{pmatrix} = \begin{pmatrix} -re - a \\ -re + a \end{pmatrix}. \quad (673)$$

It has row sparsity at most $d + 1$, column sparsity at most $2d$, and $\text{cond}(M_{\text{OSS}}) = \Theta_\delta(\text{cond}(H))$. Its exact solution is

$$\lambda = -re, \quad \Delta y = \sigma H^{-1}a. \quad (674)$$

The normalized solution has probability at least $2/3$ in the Δy block. The physical step lands at

$$x^1 = \sigma e, \quad s^1 = (e + ra, e - ra), \quad \mu^1 = \sigma, \quad (675)$$

with centrality-residual norm $\sigma r \delta$. Every sparse query to the LP or OSS data is simulated with $O(1)$ queries to H and evaluations of the known q .

Proof. Primal and dual feasibility at the start are immediate, and $\|X^0 s^0 - e\|_2 = \|(a, -a)\|_2 = \delta$. Substitution of $\Delta x = V\lambda$ and $\Delta s = -A^\top \Delta y$ into the centering equation gives (673). The orthogonal row transform

$$Q = 2^{-1/2} \begin{bmatrix} -I & I \\ I & I \end{bmatrix}$$

gives

$$QM_{\text{OSS}} = \sqrt{2} \begin{bmatrix} H & -D \\ 0 & I \end{bmatrix}. \quad (676)$$

The triangular factor has constant norm and inverse $\begin{bmatrix} H^{-1} & H^{-1}D \\ 0 & I \end{bmatrix}$; since $\|D\| \leq \delta/\sqrt{2}$, its inverse norm is $\Theta_\delta(\|H^{-1}\|)$.

Applying Q to the right-hand side gives $H\Delta y - D\lambda = a$ and $\lambda = -re$. Since $De = a$, this proves (674). Also $\|\Delta y\|_2^2 \geq \sigma^2\delta^2/2$, whereas $\|\lambda\|_2^2 = \alpha^2$, so the block probability is at least

$$\frac{\sigma^2\delta^2/2}{\sigma^2\delta^2/2 + \alpha^2} > \frac{2}{3}.$$

Direct substitution proves (675) and gives $X^1s^1 - \mu^1e = \sigma r(a, -a)$. All remaining entries and sparse positions are fixed, proving the oracle simulation. $\square$

Combined with a symmetric dilation, the theorem shows that the $\Omega(\kappa\sqrt{d})$ fixed-error and $\Omega(\kappa \log(1/\epsilon))$ constant-sparsity QLS lower bounds of Mori et al. apply to a strictly feasible short progress step [109]. The theorem embeds known QLS lower bounds; it is not a new QLS technique or an end-to-end LP lower bound. Indeed, the physical step $(\Delta x, \Delta s)$ above is explicit; only the block Δy is hard, and the statement does not cover algorithms that avoid preparing the OSS solution state accurately. A symmetric dilation of a nonsymmetric source matrix requires bidirectional row- and column-slot access. The clock family of Mori et al. has known successor and predecessor slot locations and reversible hidden updates, so both directions are simulated with constant query overhead.

14.4.2 Bounded treewidth

The $\log(1/\epsilon)$ precision lower bound still holds for some KKT matrices of treewidth at most two. The hard matrix of Mori et al. has the form

$$A = I - \rho B, \tag{677}$$

where B is a clock permutation and every row and column has one identity and one permutation position. Form its symmetric dilation

$$K_A = \begin{bmatrix} 0 & A^\top \\ A & 0 \end{bmatrix}, \quad \hat{b} = \begin{pmatrix} 0 \\ b \end{pmatrix}. \tag{678}$$

Theorem 14.11 (Treewidth-two KKT precision). *There are real symmetric, indefinite, two-sparse equality-KKT matrices K_A whose support graphs have treewidth at most two and for which preparing an ϵ -approximation to the normalized solution requires*

$$\Omega(\text{cond}(K_A) \log(1/\epsilon)) \tag{679}$$

queries in the parameter range of the underlying QLS theorem.

Proof. One has $K_A^{-1}\hat{b} = (A^{-1}b, 0)^\top$, and the singular values of K_A are those of A , each repeated. The identity positions of A give one perfect matching in the bipartite support graph of K_A , while the clock-permutation positions give a second. Their union is a disjoint union of paths and even cycles, so its treewidth is at most two. Successor and predecessor clock queries simulate every sparse query with constant overhead. The solution state and the condition number are preserved, so the QLS precision lower bound applies to K_A . Finally, K_A is the equality-KKT matrix of $\min 0$ subject to $Ax = b$. $\square$

The two-sparse clock dilation of Wang and Zhang also has low-treewidth support, but it proves a different, query-depth lower bound and does not give the $\log(1/\epsilon)$ precision factor [133]. Theorem 14.11 does not say that bounded-treewidth optimization is hard in general. Classically the opposite holds: given an elimination ordering of width τ , Cholesky factorization solves an N -dimensional SPD Newton system with $O(N\tau^2)$ scalar work and $O(N\tau)$ storage. A dense classical output already costs $\Omega(N)$, so any full-output speedup measured in the dimension

alone is at most $O(\tau^2)$. Nearly linear-time IPMs for bounded treewidth strengthen this classical baseline [56, 72].

Nevertheless, approximating the value of an LP can require many queries even when its constraint-intersection graph has a prescribed exact treewidth. For Boolean matrices $Z^i \in \{0, 1\}^{c \times b}$, start from the pre-dual program

$$\begin{aligned} & \max_{w, v^1, \dots, v^a} \sum_{k=1}^c w_k \\ \text{subject to } & w - \sum_{i=1}^a Z^i v^i \leq 0, \quad e^\top v^i \leq 1 \quad (1 \leq i \leq a), \\ & w, v^i \geq 0. \end{aligned} \tag{680}$$

Its value is

$$\sum_{i=1}^a \max_{j \in [b]} \sum_{k=1}^c Z_{ijk}. \tag{681}$$

For each i , add a nonnegative zero-objective variable z_i with coefficient $+1$ in each of the c top inequalities (the first block in (680)) and in the i th bottom inequality $e^\top v^i \leq 1$.

Theorem 14.12 (LP value hardness at exact treewidth). *For integers $a, b, c \geq 1$, the augmented LP has $p = c + a(b + 1)$ variables, $q = c + a$ constraints, and constraint-intersection treewidth exactly $\tau = c$. When $c \leq a$, constant-additive approximation of its scalar value requires*

$$\Omega(a\sqrt{b}c) = \Omega(\tau\sqrt{pq}) \tag{682}$$

fixed-position quantum coefficient queries. The corresponding randomized bound is $\Omega(abc) = \Omega(p\tau)$.

Proof. For fixed v^i , each top constraint is tight at an optimum, so summing the coordinates of w separates the a simplex problems in v^i and gives (681). Setting $z = 0$ extends every original feasible point. Conversely, deleting positive z from an augmented point only relaxes its inequalities, so the value is unchanged.

Let C be the c top-constraint vertices. The column of z_i makes $C \cup \{i\}$ a clique, and no variable column contains two bottom vertices. Thus the intersection graph is exactly $K_c \vee I_a$, with bags $C \cup \{i\}$, and has treewidth c . The dimensions follow by counting w, v, z . The published Boolean reduction for (681) gives $\Omega(a\sqrt{b}c)$ quantum and $\Omega(abc)$ randomized fixed-position queries. Since

$$a\sqrt{b}c = \tau\sqrt{(q - \tau)(p - q)}, \quad abc = \tau(p - q),$$

and $c \leq a$ makes $q - \tau = \Theta(q)$ and $p - q = \Theta(p)$, the displayed bounds follow. $\square$

This theorem restates an existing reduction [10] in terms of treewidth; it is not a new adversary argument. It applies to individual coefficient queries; a whole column reveals $\Theta(\tau)$ coefficients, and a whole top row can reveal far more. Treewidth also appears in earlier QLS lower-bound work [84]. Of the treewidth results here, only theorem 14.11 is new.

14.5 Geometric locality of central-point access

The sparse-query model ignores where the input is stored. On hardware with bounded-range gates, input placement can itself force a depth lower bound. Let hidden signs $\varsigma_1, \dots, \varsigma_N \in \{\pm 1\}$ and introduce $d_i = u_i - v_i$ with $u_i, v_i \geq 0$. For fixed $\alpha \in (0, 1)$, consider

$$\begin{aligned} & \min \sum_{i=0}^N (u_i + v_i) + \alpha(u_N - v_N), \\ \text{subject to } & d_0 = 1, \\ & d_i - \varsigma_i d_{i-1} = 0 \quad (1 \leq i \leq N). \end{aligned} \tag{683}$$

Every row has at most four nonzeros, every column at most two, and the constraint-intersection graph is a path. Put $p_i = \prod_{j=1}^i \varsigma_j$ and $q_i = u_i + v_i$. Feasibility gives $d_i = p_i$. Taking every $q_i = 2$ is strictly primal feasible; with the convention $A^\top y + s = c$, the choice $y = 0$ gives $s = c > 0$ and strict dual feasibility. The barrier restricted to the feasible affine space is

$$\sum_{i=0}^N q_i + \alpha p_N - \mu \sum_{i=0}^N \log \frac{q_i^2 - 1}{4}. \quad (684)$$

Thus every central coordinate equals

$$q(\mu) = \mu + \sqrt{\mu^2 + 1}. \quad (685)$$

At $\mu = 3/4$, $q = 2$ and $(u_i, v_i) = (3/2, 1/2)$ or $(1/2, 3/2)$ according to p_i . Therefore, reading the final cell (u_N, v_N) with coordinate error below $1/2$ reveals the parity p_N .

Reduced conditioning does not reflect the cost of reading this parity. The public vectors $V_i = (e_{u_i} + e_{v_i})/\sqrt{2}$ form an orthonormal null-space basis, and for the full logarithmic-barrier Hessian $H_x = X^{-2}$,

$$V^\top H_x V = 2[(q+1)^{-2} + (q-1)^{-2}]I. \quad (686)$$

It has condition number one for every $\mu > 0$.

Place the input register ς_i at a site v_i of a hardware graph Γ (here v_i is a site, not the LP variable). A depth- T , range- r circuit has T layers of gates, each of hardware diameter at most r . Its fixed initial workspace may contain arbitrary input-independent entanglement. A measurement at site o is locally parity-readable if it returns p_N with probability at least $2/3$ for every input.

Theorem 14.13 (Central-oracle light cone). *Let $R_\Gamma(o; v) = \max_{1 \leq i \leq N} \text{dist}_\Gamma(o, v_i)$ and let $b = \max_{w \in \Gamma} |\{i : v_i = w\}|$ be the maximum site occupancy. Every such circuit that makes the final central cell locally parity-readable at o satisfies*

$$T \geq \left\lceil \frac{R_\Gamma(o; v)}{r} \right\rceil. \quad (687)$$

Consequently, when at most b input registers share a site, a path requires $T \geq (\lceil N/b \rceil - 1)/(2r)$ for every choice of output site, which is $\Omega(N/(br))$ once $N \geq 2b$, and a d -dimensional nearest-neighbor grid requires $(2rT + 1)^d \geq \lceil N/b \rceil$, hence $T \geq ((N/b)^{1/d} - 1)/(2r)$, which is $\Omega((N/b)^{1/d}/r)$ once $N \geq 2^d b$. Pairwise-distinct placement is the case $b = 1$.

Proof. Pull a local output observable backwards through the circuit. Its support can grow by hardware distance at most r per layer, so after T layers it is contained in $B_\Gamma(o, rT)$. If some v_j lies outside this ball, two inputs differing only at j have identical reduced input states on the backward light cone. Their workspace state is also identical, including all of its input-independent entanglement. The local output distributions are therefore identical, although flipping ς_j flips p_N ; one distribution cannot return both answers with probability $2/3$.

Thus the ball $B_\Gamma(o, rT)$ contains every occupied site, proving (687). Under occupancy at most b , at least $\lceil N/b \rceil$ distinct sites are occupied. On a path a radius- rT ball has at most $2rT + 1$ sites, and on the grid at most $(2rT + 1)^d$, which proves the two consequences. $\square$

The proof is the standard backward-light-cone argument [17]; our only addition is to apply it to the strictly feasible central point with reduced condition number one constructed above. The theorem gives no lower bound in the all-to-all query model or with free QRAM access; for a homogeneous self-dual embedding with a universal start; when the answer may be left in a global observable whose measurement is free; or when unboundedly many input registers may share one site. The same lower bound holds classically, so it is an obstruction to a claimed quantum depth

advantage, not a quantum–classical separation. Input-dependent initial entanglement is excluded because it can already contain the answer, which only moves the construction cost into advice.

The theorem does not imply that each IPM iteration must write $\Omega(N)$ QRAM entries. A lazy oracle, which stores the input once and computes each updated weight on demand from the queried index and a scalar path parameter, avoids such writes. An $\Omega(N)$ per-iteration write bound would need a model in which the updated array must be stored explicitly, a cell-probe model, or a lower bound for the composed evaluation oracle. Numerical change in all coordinates is not enough.

14.6 Synthesis

These tools bound different resources and should not be merged into one informal claim that sparsity fails. Symmetry forces a nonnegative work term that is invariant under mixing of equality rows. For sparse, one-bit-local input dependence, a correct output rule needs a residual tolerance below a sparsity bound or a width tolerance below a spectral-spread bound. Complementarity progress requires large pair support or large NT rank. Sparse-QLS lower bounds apply to a valid OSS step and even to a treewidth-two KKT matrix, although bounded treewidth also admits strong classical algorithms. Physical locality forces a communication cost only when input and output are stored locally with bounded occupancy per site.

Each tool can be avoided in a specific way. A nonlinear output may not be mixture-stable; central matrices may have a large common spectral spread; a nonlocal reconstruction map may keep a latent state sparse; chordal conversion or an implicit resolvent may avoid matrix-factor tomography; an algorithm may bypass the OSS; and a global or lazy oracle may avoid storing the data locally. The results above show which assumption such an algorithm must break. They do not give a universal quantum query or runtime lower bound.

15 Corrections to Prior Work and Recorded Dead Ends

This section collects corrections and negative results that mark the limits of the positive theorems. First, we correct a norm estimate in a publicly posted complexity derivation for the quantum central-path method (QCPM). Next, we show that a sparse barrier Hessian (the metric) can have a dense inverse, and that a feasible central start in the original form can reveal the input parity. We then list natural shortcuts that are false. None of these statements is a general argument against Hamiltonian optimization, preconditioning, or state reuse.

15.1 Correction to the QCPM simulator norm

We consider arXiv:2311.03977v2 (16 October 2024), the quantum central-path method (QCPM) of Augustino, Leng, Nannicini, Terlaky, and Wu [15]. All formulas in this subsection refer to that version. Let $d = m + n + 2$ be the dimension of its self-dual embedding, and use its notation

$$s(z) = Mz + q, \quad F(z) = z \odot s(z), \quad f_\mu(z) = \frac{1}{2} \|F(z) - \mu e\|_2^2. \quad (688)$$

The central-path Hamiltonian is $H(\mu) = -(h^2/2)\nabla^2 + f_\mu$, and Algorithm 1 sets

$$h(t) = a\mu(t)^2, \quad a = \frac{\gamma^2}{2R_1 C_{d,\delta}}, \quad (689)$$

where $C_{d,\delta} = \sqrt{d}/2 + (3/4)\log(2/\delta)$ is the concentration factor displayed in Algorithm 1. We assume $d \geq 1$, $a, \eta > 0$, and $0 < \varepsilon < 1$, where η multiplies ∂_t in (692) below. The spatial domain Ω_D is fixed and compact and contains e ; statements involving $z(\mu)$ also require that point to

belong to Ω_D . The real-space simulation theorem invoked there measures a time-dependent potential V on a fixed domain Ω_D by

$$\|V\|_{\infty,1} = \int \sup_{z \in \Omega_D} |V(z, \tau)| d\tau. \quad (690)$$

The QCPM proof truncates to a fixed box containing the central-path neighborhood and replaces the supremum in (690) by an assumed bound $\sup_{\Omega_D} f_\mu \leq K\gamma^2\mu^2$. That bound is false.

Proposition 15.1 (Global-supremum counterexample). *If Ω_D contains the initial center $e = z(1)$, then, for every $0 < \mu \leq 1$,*

$$\sup_{z \in \Omega_D} f_\mu(z) \geq f_\mu(e) = \frac{d}{2}(1 - \mu)^2. \quad (691)$$

Consequently, at $\mu = \varepsilon \leq 1/2$, the asserted bound would require $K \geq d/(8\gamma^2\varepsilon^2)$.

Proof. The self-dual embedding is initialized so that $s(e) = e$. Thus $F(e) = e$, and (688) gives

$$f_\mu(e) = \frac{1}{2}\|(1 - \mu)e\|_2^2 = \frac{d}{2}(1 - \mu)^2.$$

Taking the supremum proves (691). When $\varepsilon \leq 1/2$, its right-hand side is at least $d/8$, which gives the claim about K . $\square$

Ground-state concentration controls state-weighted averages of the potential, not its supremum over a fixed domain. A scalar shift (a global phase) does not restore the claimed scaling either. If $z(\mu) \in \Omega_D$, then $f_\mu(z(\mu)) = 0$, and for every real scalar c the triangle inequality gives

$$\sup_{z \in \Omega_D} |f_\mu(z) - c| \geq \frac{1}{2}|f_\mu(e) - f_\mu(z(\mu))| = \frac{d}{4}(1 - \mu)^2.$$

A second, independent error concerns the time rescaling. Let $\mu : [0, 1] \rightarrow [\varepsilon, 1]$ be continuous. After division by $h(t)\mu(t)$, the evolution equation is

$$i\eta \partial_t \Psi = \left[-\frac{\theta(t)}{2} \nabla^2 + \frac{f_{\mu(t)}}{\mu(t)h(t)} \right] \Psi, \quad \theta(t) = \frac{h(t)}{\mu(t)} = a\mu(t). \quad (692)$$

The linear substitution used in the QCPM proof treats θ as constant, although it varies with t . The clock that normalizes the kinetic term is instead

$$\tau(t) = \frac{1}{\eta} \int_0^t \theta(u) du = \frac{a}{\eta} \int_0^t \mu(u) du. \quad (693)$$

Indeed, $dt/d\tau = \eta/\theta(t)$, so (692) becomes

$$i\partial_\tau \tilde{\Psi} = \left[-\frac{1}{2} \nabla^2 + \frac{f_{\mu(t(\tau))}}{h(t(\tau))^2} \right] \tilde{\Psi}. \quad (694)$$

Combining (689)–(693) yields the exact corrected identity

$$\Lambda := \|V\|_{\infty,1} = \frac{1}{a\eta} \int_0^1 \sup_{z \in \Omega_D} \frac{f_{\mu(t)}(z)}{\mu(t)^3} dt. \quad (695)$$

This identity does not use a lower-bound argument. The clock correction by itself works in the authors' favor: it lowers the norm weight from the constant- θ argument's μ^{-4} to μ^{-3} . The ε^{-3} blow-up below is driven entirely by the false global supremum bound.

The schedule in Algorithm 1 is

$$g(t) = c_e^{-1} \int_0^t \exp\left[-\frac{1}{\tau(1 - \tau)}\right] d\tau, \quad \mu(t) = \varepsilon + (1 - \varepsilon)(1 - g(t)), \quad (696)$$

where c_e normalizes $g(1) = 1$. The bump integrand is extended by zero at both endpoints. It is continuous, nonnegative, and positive on $(0, 1)$, so $c_e > 0$.

Lemma 15.2 (Width of the flat endpoint). *For $0 < \varepsilon \leq \min\{1/4, c_e\}$, put $u_\varepsilon = 1/(2 \log(1/\varepsilon))$. Then $\mu(t) \leq 2\varepsilon$ for every $t \in [1 - u_\varepsilon, 1]$.*

Proof. For $0 < u < 1$, substitute $v = 1 - \tau$ in the tail of (696). Since $(1 - v)v \leq v$,

$$1 - g(1 - u) \leq \frac{1}{c_e} \int_0^u e^{-1/v} dv \leq \frac{u}{c_e} e^{-1/u}. \quad (697)$$

At $u = u_\varepsilon$, the exponential equals ε^2 . The last quantity in (697) is at most ε : $\varepsilon \leq 1/4$ implies $0 < u_\varepsilon \leq 1/2$, and $u_\varepsilon \varepsilon \leq c_e$ follows from $\varepsilon \leq c_e$. For $t \geq 1 - u_\varepsilon$, monotonicity then gives $\mu(t) \leq \varepsilon + (1 - \varepsilon)\varepsilon \leq 2\varepsilon$. $\square$

Theorem 15.3 (Corrected QCPM simulator-norm scale). *For the choices in Algorithm 1 and $0 < \varepsilon \leq \min\{1/4, c_e\}$,*

$$\Lambda \geq \frac{d}{128a\eta \varepsilon^3 \log(1/\varepsilon)}. \quad (698)$$

If $\Omega_D = [0, D]^d$, the following global upper bound holds:

$$\Lambda \leq \frac{F_D}{a\eta \varepsilon^3}, \quad F_D = \frac{d}{2} [D(D\|M\|_\infty + \|q\|_\infty) + 1]^2. \quad (699)$$

Here $\|M\|_\infty$ is the maximum absolute row sum.

Proof. On the interval supplied by lemma 15.2, take $\varepsilon \leq 1/4$. Proposition 15.1 gives $\sup_{\Omega_D} f_{\mu(t)} \geq d/8$, while $\mu(t)^{-3} \geq (2\varepsilon)^{-3}$. The interval has length $1/(2 \log(1/\varepsilon))$. Substitution in (695) proves (698).

For the upper bound, every $z \in [0, D]^d$ satisfies

$$|s_i(z)| \leq D\|M\|_\infty + \|q\|_\infty.$$

Hence $|F_i(z) - \mu|$ is at most $D(D\|M\|_\infty + \|q\|_\infty) + 1$ for $0 < \mu \leq 1$, and $f_\mu(z) \leq F_D$. Since $\mu(t) \geq \varepsilon$, the exact identity (695) gives (699). $\square$

Since $a^{-1} = 2R_1 C_{d,\delta}/\gamma^2$, equation (698) gives the simulator-norm lower bound

$$\Omega \left(\frac{R_1 d C_{d,\delta}}{\gamma^2 \eta \varepsilon^3 \log(1/\varepsilon)} \right). \quad (700)$$

By contrast, the bound claimed in v2 is

$$\|V\|_{\infty,1} \leq \frac{K\gamma^2}{\theta\eta}, \quad O\left(\frac{K\gamma^2}{a\eta\varepsilon}\right) = O\left(\frac{R_1 C_{d,\delta}}{\eta\varepsilon}\right) \quad (K = O(1)).$$

Thus γ^2 cancels against $1/a$. If all other hypotheses of the simulation theorem hold, (699) gives the norm-based upper bound

$$O\left(\frac{F_D R_1 C_{d,\delta}}{\gamma^2 \eta \varepsilon^3}\right)$$

excluding polylogarithmic factors and the cost of evaluating the potential.

The same calculation bounds every Lipschitz schedule.

Corollary 15.4 (Norm-speed tradeoff). *Let $\mu : [0, 1] \rightarrow [\varepsilon, 1]$ be L -Lipschitz, with $\mu(0) = 1$, $\mu(1) = \varepsilon$, and $0 < L < \infty$. In particular, a differentiable schedule with $|\dot{\mu}| \leq L$ satisfies this hypothesis by the mean value theorem. Retain $h = a\mu^2$. For $\varepsilon \leq 1/4$,*

$$\Lambda \geq \frac{d}{64a\eta L \varepsilon^2}. \quad (701)$$

Proof. The endpoint values imply $L \geq 1 - \varepsilon$, hence $\varepsilon/L \leq 1$. On the terminal interval $[1 - \varepsilon/L, 1]$, the Lipschitz bound gives $\mu(t) \leq \varepsilon + L(1 - t) \leq 2\varepsilon$. Throughout that interval, [proposition 15.1](#) gives $\sup_{\Omega_D} f_\mu \geq d/8$, and $\mu^{-3} \geq (2\varepsilon)^{-3}$. Equation (695) proves the result. $\square$

No monotonicity assumption is needed for this bound.

A large L weakens (701), but L then enters the derivative, smoothness, and adiabatic estimates. The corollary does not optimize this tradeoff, and it does not rule out a theorem that rigorously exploits localization.

As a numerical check, adaptive quadrature gives $c_e = 0.007029858406609657$. For

$$J(\varepsilon) = \varepsilon^3 \log(1/\varepsilon) \int_0^1 (1 - \mu(t))^2 \mu(t)^{-3} dt,$$

the computed values are

ε	$J(\varepsilon)$
10^{-2}	0.77120646
10^{-3}	0.89055595
10^{-4}	0.95036732

The sampled values are consistent with a limit of one, but the computation is not part of any proof. Keeping only the lower bound $f_\mu(e) = d(1 - \mu)^2/2$ on the supremum gives

$$\Lambda \geq \Lambda_e := \frac{d}{2a\eta} \int_0^1 (1 - \mu(t))^2 \mu(t)^{-3} dt \geq \frac{d}{128a\eta\varepsilon^3 \log(1/\varepsilon)} \quad (0 < \varepsilon \leq \min\{1/4, c_e\}).$$

The proof shows $J(\varepsilon) \geq 1/64$ in this range; we do not claim that this constant is sharp. The quadrature does not estimate the full supremum.

Formal verification of the norm correction. The companion Lean development in `formal/QipmFormal/QCPM/` checks the finite-dimensional potential and initialization identity, the compact-domain supremum bound, robustness under scalar shifts, the corrected clock and its inverse, and the integral change of variables. It also checks the normalized bump schedule itself, the explicit endpoint threshold, both lower-bound constants, and the upper bound on a fixed box. The main bounds use the spatial supremum of the stated potential; their continuity and integrability are proved rather than assumed as extra properties of that supremum. The initialization row equations $Me + q = e$ and inclusion of the required points in the fixed domain are explicit hypotheses. The formal statements follow Algorithm 1 of v2, whose normalization differs from its equation (23). The verification does not formalize Schrödinger evolution, the external simulation or adiabatic theorems, or an unconditional query lower bound.

The norm and clock errors invalidate the stated complexity derivation, not the QCPM idea itself, and (700) is not an unconditional query lower bound for every implementation of the evolution. Chakrabarti et al. analyze missing error controls in the pseudo-spectral real-space simulation [40]. Gribling et al. cite that analysis and separately identify the absence of a suitable directly applicable infinite-dimensional adiabatic theorem [70]. We do not claim those broader criticisms as new. Even after the norm is corrected, the following steps remain unproved:

- The schedule in (696) is nonconstant and C^∞ -flat, but it is not analytic through either endpoint. An analytic function with its zero endpoint Taylor series would be locally constant. The quoted exponential adiabatic theorem assumes analytic continuation through the closed interval.
- The adiabatic theorem must be applied to the generator in (692); the preprint’s discussion of derivatives analyzes a differently scaled Hamiltonian and does not establish uniform derivative constants for time-dependent $\hbar$.

- The exact potential (688) is generally quartic. The displayed Gaussian state and oscillator gap concern its quadratic approximation. No transfer to the exact ground state and spectral gap, uniform in the dimension, μ , and the input, is established.
- The preprint’s proposition and algorithm use different choices of h ; the proposition’s choice does not depend on γ or δ , so by itself it cannot achieve an arbitrary neighborhood width γ and failure probability δ .
- The initial exact ground state is assumed rather than prepared. Knowing its minimizer e does not prepare the ground state of a quartic Hamiltonian.
- The real-space theorem assumes a smooth periodic potential and periodic boundary conditions. Restriction of (688) to a box neither makes it periodic nor controls the boundary and truncation errors.
- The box size must retain its solution-radius dependence unless an input-size promise bounds the central path.

A repair based on localization or a clipped potential would need leakage, spurious-state, uniform-gap, boundary, and comparison estimates; concentration alone does not provide them.

15.2 Sparse metric, dense inverse metric

The proposal of Gribling, Apers, Nieuwboer, and Walter replaces the Euclidean Laplacian by the Laplace–Beltrami operator of the Hessian metric [70]. If $G(x) = \nabla^2 \phi(x)$, its principal part is

$$L\psi = \frac{1}{\sqrt{\det G}} \sum_{i,j} \partial_i \left((G^{-1})_{ij} \sqrt{\det G} \partial_j \psi \right). \quad (702)$$

Even when the constraints are sparse, the coefficients $(G^{-1})_{ij}$ in standard coordinates need not be entrywise sparse.

Theorem 15.5 (Path-simplex inverse metric). *Let*

$$\mathcal{D}_n = \{x \in \mathbb{R}^n : 0 < x_1 < \cdots < x_n < 1\}$$

and use the logarithmic barrier

$$\phi(x) = -\log x_1 - \sum_{i=1}^{n-1} \log(x_{i+1} - x_i) - \log(1 - x_n). \quad (703)$$

Every inequality contains at most two variables, and the constraint-intersection graph is a path. At the analytic center $\bar{x}_i = i/(n+1)$, the Hessian $G = \nabla^2 \phi(\bar{x})$ is tridiagonal, but

$$(G^{-1})_{ij} = \frac{\min(i, j)(n+1 - \max(i, j))}{(n+1)^3} > 0 \quad (704)$$

for every i, j . For $i, j \in [(n+1)/4, \lfloor 3(n+1)/4 \rfloor]$, these entries are $\Theta(1/n)$, so $\Theta(n^2)$ coefficients are of comparable order.

Proof. Write the $n+1$ slacks as $s_0 = x_1$, $s_i = x_{i+1} - x_i$ for $1 \leq i < n$, and $s_n = 1 - x_n$. At $\bar{x}$, every slack is $1/(n+1)$. If B is the signed path-incidence Jacobian of this affine slack map, then

$$G = B^\top \text{diag}(s^{-2}) B = (n+1)^2 T_n,$$

where T_n has diagonal entries two and adjacent off-diagonal entries minus one. The discrete Green identity is

$$(T_n^{-1})_{ij} = \frac{\min(i, j)(n + 1 - \max(i, j))}{n + 1}.$$

Scaling gives (704). In the middle half, both factors in its numerator are bounded above by n and below by a fixed positive multiple of n , proving the final assertion. $\square$

Thus a direct finite-difference stencil for the principal symbol in (702), written in standard coordinates and in nondivergence form, has $\Omega(n^2)$ mixed-derivative directions at the grid point at the analytic center, although the input has only $O(n)$ nonzeros. This is a warning about densification, not a lower bound on simulation cost or queries. Indeed, at an arbitrary interior point put $r_k = s_k^2$, $R = \sum_{k=0}^n r_k$, and let $i \leq j$. Weighted path inversion gives

$$(G(x)^{-1})_{ij} = \frac{(\sum_{k=0}^{i-1} r_k)(\sum_{k=j}^n r_k)}{R}. \quad (705)$$

Every strict off-diagonal block is rank one, and prefix sums give an $O(n)$ matrix–vector product. The example therefore refutes only the inference from a sparse Hessian to an entrywise sparse inverse metric tensor. It does not rule out implicit, transform-based, or coordinate-adapted simulation.

Abe and Nagai parameterize their Riemannian Hamiltonian simulation by the kinetic-matrix sparsity and separately assume sparse-access oracles for that matrix; they note that constructing those black-box oracles is generally nontrivial [2]. Theorem 15.5 shows only that input-level $O(n)$ sparsity is lost: the direct stencil has $\Theta(n^2)$ terms. On a tensor grid with q points per coordinate it still has at most $2n^2 + 1$ neighbors per row, sparse relative to the q^n -dimensional grid. Thus the theorem does not rule out sparse-Hamiltonian access or refute the separately assumed Abe–Nagai oracles. Gribling et al. likewise leave the simulation and qubit-realization questions open.

15.3 A hidden-sign start-oracle separation

We reuse the path-sparse hidden-sign LP (683) from section 14.5, without repeating its construction. Recall that $p_0 = 1$, $p_i = \prod_{j=1}^i \varsigma_j$, and $q_i = u_i + v_i$.

Theorem 15.6 (Value, output, and central-start separation). *For the family (683):*

1. *It is strictly primal–dual feasible and has the unique optimum*

$$\text{OPT} = N + 1 + \alpha p_N. \quad (706)$$

Estimating its shifted value $\text{OPT} - (N + 1)$ to additive error less than α requires $\Omega(N)$ bounded-error raw sparse queries.

2. *Every classical nonnegative output $\hat{x}$ satisfying $\|A_\varsigma \hat{x} - b\|_1 < 1/2$ determines p_N and therefore also requires $\Omega(N)$ queries.*
3. *The primal central path has the public support-two orthonormal null basis stated in section 14.5, and its reduced log-barrier Hessian has condition number one for every $\mu > 0$.*
4. *At $\mu = 3/4$, one lookup in the final cell (u_N, v_N) of the exact primal central point, through a reusable random-access coordinate oracle, reveals p_N . If setup and that lookup use respectively Q_{setup} and Q_{lookup} raw queries, then $Q_{\text{setup}} + Q_{\text{lookup}} = \Omega(N)$. In particular, if lookups after preprocessing use no queries, then $Q_{\text{setup}} = \Omega(N)$.*

Proof. The feasibility argument and central-path calculations are given throughout [section 14.5](#), with the displayed formulas in (684)–(686). In particular, feasibility fixes $d_i = p_i$ and leaves $q_i \geq 1$, so the objective is $\sum_i q_i + \alpha p_N$ and has the unique minimum (706). The shifted value reveals parity.

For an approximate output define $r_0 = \hat{d}_0 - 1$ and $r_i = \hat{d}_i - s_i \hat{d}_{i-1}$. Telescoping gives

$$\hat{d}_N - p_N = \sum_{j=0}^N \left(\prod_{k=j+1}^N s_k \right) r_j, \quad (707)$$

so $|\hat{d}_N - p_N| < 1/2$ under the asserted residual bound. The sign of $\hat{d}_N = \hat{u}_N - \hat{v}_N$ is therefore p_N .

Finally, (685) gives $q = 2$ at $\mu = 3/4$. The final pair is $(3/2, 1/2)$ or its swap according to p_N , so one coordinate-oracle lookup determines parity even with coordinate error below $1/2$. Since quantum parity has query complexity $\Theta(N)$, each claimed raw-query lower bound, including the combined setup–lookup bound, follows from the local sign-oracle reduction used throughout [section 9](#). $\square$

This theorem counts the cost of a feasible start in the original form. It does not apply to infeasible-start or homogeneous self-dual methods, which can use a different, universal start; nor does it say that the scalar reduced Newton solve itself is hard.

15.4 Dead ends and false shortcuts

First inverse versus filtering. A small value of $\|H\| \|H^{-1}f\| / \|f\|$ does not imply a fast DLS filtering solve. Filtering depends on the second inverse through $\|H\| \|H^{-2}f\| / \|H^{-1}f\|$, and active–inactive coupling can restore the full condition-number exponent. The exact identities and a sparse pair of examples showing sharpness are in [sections 5.1](#) and [5.2](#).

Sparse support versus inverse access. Sparse Hessian or KKT support does not imply a sparse inverse, a cheap block encoding after elimination, or low circuit depth on bounded-range hardware. The results on factor normalization and inverse compilation in [section 12.1](#) prove the access cost; the bounded-range depth claim also needs the geometric input placement with bounded occupancy of [section 14.5](#). Separately, [section 15.2](#) shows that a sparse barrier Hessian can have a dense inverse in standard coordinates. None of these statements says that every structured inverse is expensive.

Condition one versus the surrounding maps. A condition-one reduced Newton matrix does not make the affine feasible offset, coordinate gauge, transformed right-hand side, recovery map, or requested output free. The LP separations in [sections 9.4](#) and [9.5.1](#) place parity in the affine offset while leaving the reduced geometry public. The full-KKT and preconditioner dichotomies in [sections 9.6](#) and [12.3](#) show how input dependence can move between these objects.

Exact Schur preconditioning. An exact Schur transformation can make the abstract product constantly conditioned while moving input dependence into its construction, the loading of the transformed right-hand side, or recovery. The full cost accounting is proved in [sections 9.6](#) and [12.1](#). Supplying the preconditioned product itself as a unit-cost oracle is a strictly stronger input model, not a consequence of sparse access to the original system.

State output versus coordinate access. A normalized solution state is not a reusable coordinate oracle. It does not carry the norm or the global phase of the solution, and building a coherent diagonal from it also needs controlled and adjoint access and costs a normalization factor.

The tight state-to-diagonal product appears in [section 12.4](#); the implicit-coordinate approach in [section 13.3](#) states the stronger interface that it constructs.

One-step bounds versus a fixed trajectory. Reapplying a one-step lower bound at every point of one fixed LP trajectory is invalid without freshness, a memory restriction, an adaptivity/causality restriction, or mandatory explicit output. The fresh-oracle direct sum is [theorem 12.14](#); the surrounding discussion shows how caching a static input can erase later raw-query cost.

Residual bounds versus projective variation. Ordinary short-step neighborhood and relative OSS residual bounds do not imply finite projective variation. [Section 13.1](#) gives a sparse cycling counterexample and the additional hypotheses needed for reuse. [Section 13.2](#) also corrects a sharpness family proposed in an earlier version: even though its leakage and transverse-forcing sums converge, its realized variation grows linearly. A separate family with oscillating displacements shows that adding the $O(\mu_t)$ tail bound on face leakage does not help either. Finite realized variation is therefore assumed as an independent hypothesis rather than derived from these sums.

Expanderization. Replacing a path by an expander does not evade the bounded-degree local-linear mixture or cut/resistance obstruction. These obstructions are proved in [sections 10.1](#) and [10.3](#). In the favorable well-spread regime, [section 13.4](#) gives the matching classical sampling shortcut; making rare columns influential instead moves cost into normalization or one-time search.

Scalar gap versus density visibility. A constant scalar objective gap can coexist with a trace-normalized central density that is only $O(1/G)$ from a public state. Thus an output lower bound must state its accuracy and normalization requirements. The zero-query boundary and the mass-visibility inequalities are proved in [sections 6.5](#) and [6.6](#); the dilution example is [remark 7.1](#).

Pairwise distance versus a common decoder. Pairwise distinguishability of states indexed by an unknown winner set does not provide one public measurement that decodes every set. The collision shortcut loses a factor $1/k$ for k winners. [Section 8.1](#) repairs this only by measuring a proposed label and verifying that label with fresh input queries; the verifier, not pairwise trace distance alone, supplies the common decision procedure.

15.5 Status of remaining algorithmic candidates

We record the status of algorithmic ideas that remain incomplete or are already known. To state the factorized rectangular observation precisely, let B_μ have full row rank, let $f = B_\mu h$ lie in $\text{range}(B_\mu)$, and let $\alpha_{B_\mu} \geq \|B_\mu\|$ be its block-encoding normalization. Define

$$\rho_{B_\mu}(f) = \alpha_{B_\mu} \frac{\|(B_\mu B_\mu^\top)^{-1} f\|_2}{\sqrt{f^\top (B_\mu B_\mu^\top)^{-1} f}}. \quad (708)$$

This pseudoinverse parameter is distinct from the nonsingular filtering parameter in [\(86\)](#). In the mixed, strictly complementary asymptotic regime with exact centering, assume $B_\mu B_\mu^\top = \mu^{-1}C + \mu D + o(\mu)$, $f = b \in U := \text{range } C$, and the generic active-inactive coupling of [section 5.1](#), so that

$$(B_\mu B_\mu^\top)^{-1} b = \mu(p, -g) + o(\mu), \quad b^\top (B_\mu B_\mu^\top)^{-1} b = \Theta(\mu).$$

With the essential normalization $\alpha_{B_\mu} = \Theta(\mu^{-1/2})$, substitution in [\(708\)](#) proves $\rho_{B_\mu}(b) = \Theta(1)$.

Here $D_\mu = (XS^{-1})^{1/2}$, $B_\mu = AD_\mu$, and $h = (XS)^{-1/2}r_c$ is the scaled complementarity right-hand side; thus $f = B_\mu h$ is the row-space right-hand side, abbreviated Bh . The two scaled direction components are

$$u = \Pi_{\ker B_\mu} h, \quad v = B_\mu^+ f = B_\mu^+(B_\mu h),$$

with $u + v = h$. They recover the physical directions through $\Delta x = D_\mu u$ and $\Delta s = D_\mu^{-1}v$. The constant value of (708) is not an end-to-end QIPM speedup. Six steps remain unproved: a DLS-quality rank-deficient rectangular filter theorem with its cutoff and error law; preparation of $f = B_\mu h$; coherent construction, norm control, and cancellation-safe physical recovery of u and v ; inexact-neighborhood propagation; maintenance of the changing diagonal oracles; and a full output-cost accounting with a comparison against strong classical methods. Li’s instance-dependent rectangular minimum-norm algorithm already uses closely related second-inverse geometry [94], so this primitive also overlaps substantially with prior work.

A further candidate, the augmented spectral Newton sketch, instead samples the augmented matrix $[D_N^{1/2}A^\top, v_N]$, where D_N and v_N denote the sketch’s Newton scaling and regression target. Under its stated row-access assumptions, a $(1 \pm \varepsilon)$ embedding of the augmented Gram matrix gives a valid least-squares embedding lemma at $\tilde{O}(\sqrt{nm}/\varepsilon)$ row-query scale; the infeasible-step analysis, the implicit update, and the end-to-end comparison remain open, and the direction overlaps the factorized rectangular candidate above.

State-to-diagonal lifting is a useful but known primitive, not a new result: Rattew and Rebentrost give the amplitude-to-diagonal block encoding, with the phase-aligned coherent access assumptions recorded in section 12.4 [121]. Likewise, the candidate combining projective reuse with implicit coordinates on expanders is settled by the dichotomy in section 13.4: bounded contributions admit a comparable classical sketch, while rare influential columns incur normalization or one-time search cost.

16 Discussion and Open Problems

We asked whether QIPMs beat classical IPMs on sparse problems. Instead of a single yes or no, the results separate three levels, each with its own obstructions and ways around them, that informal complexity claims often combine: the geometry of the central path, the model of access to the resulting linear-algebra objects, and the interface through which one iteration produces the next.

16.1 What the paper establishes

Geometry. For barrier-Newton methods, part of the condition-number growth is forced by the optimization problem. Theorem 3.3 bounds the condition number of the reduced barrier Hessian from below at a fixed objective gap, for every self-concordant barrier, and theorem 3.5 shows that the log and log-det barriers attain the resulting order up to instance-dependent constants. Theorem 3.4 extends this to a geometric description of the whole spectrum. In the stated minimax sense, changing the barrier cannot in general remove the lower bound given by sublevel chords.

This is not the folklore claim that every central path has $\kappa = \Theta(\mu^{-2})$; the rate depends on the geometry of the sublevel sets. A positive-dimensional optimal face gives growth of order g^{-2} in the objective gap g ; a curved unique optimum can give growth of order g^{-1} ; at a unique LP vertex the condition number can stay bounded; and an attained singularity degree can give fractional exponents (corollaries 3.6 and 3.8). Nor does a large condition number alone determine solver cost: Krylov iteration counts can be polylogarithmic in an arbitrarily large ratio between two eigenvalue clusters (propositions 4.1 and 4.2), and the Newton right-hand side on the path can avoid the small eigenvalues (proposition 4.5 and corollary 4.6).

Access models. The quantum cost depends on how the matrix is supplied, not only on its spectrum. The clustered comparison in [theorem 4.3](#) separates plain normalized block-encoding access from access to a factor G with $H = G^\dagger G$. With plain access, the known worst-case query cost $\Theta(\kappa)$ remains for two-point spectra in sufficiently large dimension. Norm-tight factor access admits a polynomial of square-root cost but an end-to-end solve only under overlap assumptions; a generic factor-access QLS solver of square-root cost would contradict the plain-access lower bound, because the square-root factor of that hard instance is easy to query. By the exact coupling and spectral-dimension laws ([theorems 5.1](#) and [5.4](#)), the distribution of the right-hand side over the spectrum decides whether truncation or filtering can improve on κ .

The same holds for preconditioning. A whitening factor may trade condition number for a larger normalization; a congruence may trade the transformed condition number for sensitivity when the physical solution is recovered; and a perfect preconditioner that depends on the input may move the hard information into setup or RHS preparation ([theorems 12.2, 12.4, 12.7](#) and [12.8](#)). These results bound products of costs or the total cost of all phases. They show not that preconditioning is useless, but that a small transformed condition number is not by itself a complexity bound.

Interfaces. In our hard families the main obstruction is often loading or recovery rather than inversion. The prefix LP has a reduced Hessian of condition number one along its central path, yet preparing a robust primal direction in the original coordinates needs linearly many raw queries ([theorem 9.7](#)). The SDP families distinguish a gap in the optimal value ([theorem 11.1](#)), parity-hard primal and complete-KKT solution states ([theorem 11.2](#)), and a public Schur solve for which loading the original coordinates obeys an accuracy–query tradeoff ([theorems 11.3](#) and [11.4](#)). The canonical KKT block encoding is exact at one raw query per call, yet preparing the root, global, or complete-KKT state still needs linearly many calls to it, and first constructing a block encoding of the tangent projector needs linearly many raw queries ([section 11.1.2](#)). Nor is a normalized state a reusable coordinate oracle: producing the changing diagonal of the next Newton system obeys the state-to-diagonal and state-to-next-solve product bounds in [theorems 12.9](#) and [12.13](#).

In the condensation results, the winner mass controls both the condition number of the reduced Hessian and the trace distance from a public state with no winner ([theorems 6.5](#) and [6.6](#)). The bounded-incidence holonomy SDP has matching query bounds for the value and for a compact output that names the winner ([theorems 6.7](#) and [7.3](#)), while the cost of preparing the central state changes across the phases of condensation ([section 7.7](#) and [equation \(243\)](#)). The decoder that verifies sampled labels turns a trace-close mixed output into a public decision procedure with one-sided error, and parity collapse gives the query cost as a function of the allowed trace-distance error ([theorems 8.1](#) and [8.7](#) and [corollary 8.9](#)). The lower bound holds for a specified accuracy of the output state; a constant primal objective gap alone would not imply it.

In short, conditioning is governed by sublevel geometry, quantum performance by the access actually supplied, and end-to-end performance by loading, recovery, updates, checks, and output between iterations. Sparsity can help at each level, but no single notion of sparsity controls all three.

16.2 What the paper does not establish

The paper proves neither an unconditional sparse-QIPM advantage nor an unconditional lower bound. The Apers–Gribbling result for tall LPs remains the only proved asymptotic quantum–classical separation, in row queries, among the QIPM results considered here [\[10\]](#); our results do not weaken it. Conversely, no parity or holonomy family rules out a method with a different oracle, a weaker but still useful output, an optimization primitive free of the condition number, or a structure-aware preconditioner outside the stated class.

The barrier theorem concerns the reduced Hessian in an orthonormal basis of the tangent space, not automatically an augmented KKT system, primal–dual scaling, or a Hamiltonian central-path method. The clustered QLS theorem depends on the access model and has conditions on the dimension and on overlaps. The LP and SDP loading theorems count raw coefficient queries; a supplied parity, prefix, tangent-projector, or preconditioned-product oracle is a stronger input model. The online direct sum requires fresh independent input in each round, whereas a fixed instance can be cached ([theorem 12.14](#) and [section 12.5](#)). The physical light-cone bound concerns data and output placed locally with bounded occupancy per site, not all-to-all query access ([theorem 14.13](#)).

Likewise, the positive modules are components, not a complete algorithm. Projective reuse of a rescaled Newton direction, as analyzed here, needs finite realized variation and a fixed residual tolerance; equilibration of the active range and face repair need known strict-complementarity tail margins and spectral gaps; compressed dual output needs a retained dimension that is actually smaller and must count the cost of its coordinate compiler; and the block-angular module needs a projected barrier with a small parameter and a local representation kept between iterations ([section 13.7](#)). The counterexamples in [sections 13.1](#), [13.2](#) and [13.4](#) show that the corresponding module fails when one of these conditions is removed.

We propose the five accounting rules of [section 13](#) as a standard for future positive claims, and add the comparison rules used throughout the paper: match the residual and precision requirements, count setup and reuse costs on both sides, and compare with the best classical method for the same structure. A query advantage is not automatically a gate or time advantage, and a comparison of two upper bounds is not a separation.

16.3 Collected open problems

We collect every open-problem environment in [sections 3](#), [4](#), [8](#), [9](#) and [11](#) and the further open questions raised in the technical discussion, grouped by the level at which new information is needed.

16.3.1 Geometry and conditioning

- G1. **Degenerate-vertex constants.** Determine the limiting spectrum, not only its boundedness, of the canonical reduced log-barrier Hessian at a unique degenerate LP vertex from the geometry of the active set ([open problem 3.12](#)).
- G2. **Facial characterization of chord growth.** Characterize the exact growth $D(g)$ of objective-sublevel chords of an SDP from its facial data, and decide when the singularity-degree exponent is attained by the sublevel diameters ([open problem 3.13](#)).
- G3. Relate the neighborhood (175) used by robust condensation, which bounds the blockwise stationarity residual in the barrier norm by η and the trace residual by ζ , to the algorithm-specific inexact Newton residuals of [definition 2.8](#), whose forcing tolerance is a separate η ([section 6.11](#)).

16.3.2 Quantum access and state conversion

- A1. **Exact subnormalization shift.** Decide whether plain access to H permits a block encoding of $I - H$ with unit subnormalization without paying the apparent cost of uniform amplification at the shifted spectral edge ([open problem 4.4](#)). This is the promise-sensitive form of the normalization question of Orsucci and Dunjko [118].
- A2. **General inner predicate.** Prove or refute the $\sqrt{\delta} Q(f) \sqrt{G/k}$ small-success composition lower bound for a general inner predicate f , beyond the proof by parity collapse ([open problem 8.11](#)).

- A3. Determine the minimax output radius and query curve when the public internal states of the losing blocks differ by η_{int} , including the regime below condensation ([section 8.4](#)).
- A4. Resolve the state-conversion range $\delta = o(k/G)$, where a public output with uniform labels may already satisfy part of the output requirement and the winner mass alone does not determine the boundary ([section 8.4](#)).
- A5. **Intrinsically nonabelian word hardness.** Prove a lower bound on local-symbol queries for distinguishing conjugacy classes under a promise on words whose letters do not commute, and transfer it to output of the conjugacy class in the group-holonomy SDP, instead of restricting to a cyclic subgroup ([open problem 11.8](#)).

16.3.3 Hard families and classes of gadgets

- F1. **Bounded-scale full-KKT parity amplifier.** Construct, or rule out, a standard-form LP of linear size with bounded row and column degree; bounded coefficients, each depending on at most one input bit; public Newton right-hand sides that are simple to prepare; primal and slack starting values bounded above and below by positive constants; a complete Newton direction bounded coordinatewise; and a parity-hard primal component with constant mass in the state of that complete direction ([open problem 9.14](#)).
- F2. Determine whether an arbitrary sparse equality LP with bounded range can support a nonlinear decoder of the normalized state with a fixed global residual tolerance without introducing nonlocal coefficient access ([section 10.6](#)).
- F3. Determine whether a nonorthogonal mixed-mode gadget can solve the same bounded-scale complete-direction problem while counting the cost of every input-dependent source and recovery map ([sections 9.9 and 10.6](#)).

16.3.4 Positive algorithmic results

- P1. Exhibit one explicit family and access model that satisfy all six requirements of [section 13.7](#) simultaneously: stable direct normalization and inverse, stable forcing and RHS preparation, finite projective variation or a barrier with a small parameter, cheap static construction, useful compressed output, and a classical lower bound that holds even with randomized preprocessing.
- P2. Find a structured access model that supplies a quantum operator with constant normalization without also supplying a classical sampling distribution with bounded leverage, and realize it on an explicit sparse LP family ([section 13.4](#)).
- P3. Construct a projected block-angular barrier with parameter $\tilde{O}(k + \text{poly}(p))$ whose local center and derivatives can be recomputed from one scenario block and the explicit border ([section 13.5](#)).

The list omits directions that a theorem here already rules out: turning a bounded-degree local-linear parity gadget into an expander does not escape the mixture and resistance bounds, and ordinary residual control in short steps does not imply finite projective variation ([sections 10.6 and 13.1](#)). A new proposal must drop a premise of these theorems, not relabel the same construction.

16.4 Outlook

The most useful next result would be small and complete: one structured family, one explicit access model, one useful output requirement, an accounting of all costs along the path, and a matching classical lower bound or algorithmic comparison. Problem P1 is deliberately hard because each module alone already has a classical or quantum obstacle. Stable spectral geometry is not enough if RHS preparation is hard; cheap coherent access is not enough if a classical algorithm can sample the same averages; and compressed output is not enough if the next iterate must be written out in full.

Among nearer-term targets, Problem A1 would show whether the separation between plain and factor access for clustered QLS is an artifact of the current construction or a property of the access models; Problem A2 would make the state-conversion curve a reusable composition principle rather than a result about parity; Problem A5 would separate matrix-valued holonomy from its abelian restriction; and Problem P3 would connect the border module to a candidate end-to-end QIPM.

The methodological lesson is that a future sparse-QIPM advantage cannot be claimed from one favorable parameter: it must hold for the geometry, the access model, and the output interface at once. This paper proves bounds and counterexamples for each of these, with conditional modules where one can be made cheap. Solving Problem P1 would give the end-to-end answer that this paper does not claim.

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